

Pauli webs of every declared detector

Whole-circuit ZX diagrams of the $d = 3$ and $d = 5$ cultivation circuits

Contents

Each page after the detector lists shows one declared detector. The whole noiseless circuit is one ZX diagram, and the page overlays a closed web whose records are exactly that detector’s records.

	Records	Declared	Pauli webs	Semiwebs with T defects	Pages
$d = 3$	21	20	16	4	5–24
$d = 5$	112	107	93	14	25–132

Why 16 of the 20

At $d = 3$, 16 of the 20 declared detectors are Pauli webs, and at $d = 5$, 93 of the 107. The others, detectors 7, 14, 16 and 18 at $d = 3$, and 7, 19, 20, 23, 69, 89, 91, 93, 95, 97, 99, 101, 103 and 105 at $d = 5$, are Pauli semiwebs with defects on T spiders.

How each web was chosen

Each detector is a closed Pauli web of the S proxy, the circuit with each T replaced by S and each $T^\dagger$ by $S^\dagger$. The proxy has the same nodes and edges, so its webs can be drawn on the T circuit. There they keep every spider’s phase except at T spiders, where an X that turns into Y is a defect of $\frac{\pi}{2}$ at T and $-\frac{\pi}{2}$ at $T^\dagger$. Two webs with the same records differ by one of $2^6 = 64$ record-free webs, which certify the six qubits reset mid-injection in $|+\rangle$. The page shows the web with the fewest defects, then the fewest labelled edges. A detector is phase-independent exactly when that minimum is zero.

Sources

Definitions and drawing styles: Kissinger and van de Wetering, *ZX-Flow*, arXiv:2603.09580. Circuits: the paper’s inputs, SOFT’s $d = 3$ and $d = 5$ cultivation circuits, `data/inputs/clifft_d3_p001.stim` and `data/inputs/soft_cultivation_d5_p0005.stim` in the companion code. They follow the construction of Gidney, Shutty and Jones, arXiv:2409.17595.

Reading a diagram

Time runs to the right and each qubit is a row. Data qubits of the final code have black labels, other qubits grey. Shaded bands mark the stages of the circuit. At $d = 3$ every page shows the whole circuit. At $d = 5$ each page shows the columns its web uses, and wires crossing the edge of the window are cut.

Legend I — X  Y  Z  Z spider  X spider 

$-\langle \frac{\pi}{4} \rangle = T$ ($-\pi/4$: $T^\dagger$); $-\langle \frac{\pi}{2} \rangle = S$;  = CNOT; preparations  = $|0\rangle$ and  = $|+\rangle$. Measurements are effects drawn at outcome 0: $-\langle \frac{\pi}{4} \rangle = \langle +|$ (MX), $-\langle \frac{\pi}{2} \rangle = \langle 0|$ (M). Outcome r_k adds a phase $r_k\pi$ to that spider, and r_k is printed next to it; the detector’s own records are in bold blue. A qubit measured and then used again without reset restarts from a second spider with the same record. A Pauli product measurement joins one spider per qubit to a central spider carrying $r_k\pi$; a Y product has $S^\dagger$ and S around it. A record-controlled Pauli ($d = 5$ corrections) is a two-legged spider of phase $r_k\pi$. A qubit reset while in use ends in $-\langle \frac{\pi}{4} \rangle$, the effect $\langle +|$: the circuit has put it in $|+\rangle$ for every angle at the earlier T gate.

The final MPP Y_L

Both circuits end with a noiseless layer of T gates (T and $T^\dagger$ at $d = 5$) followed by MPP Y_L , and because $T^\dagger Y T = T X T^\dagger = (X + Y)/\sqrt{2}$ this measures the same transversal operator as the last double check, a noiseless projection onto the magic state. It is deterministic in the noiseless circuit, by a statevector run at $d = 3$ and by $a_0 = 1$ and $e_0 = 0$ in the exact series at both distances, and it enters no declared detector.

The identity on each page

Write $D_{\mathbf{r}}$ for the diagram with records $\mathbf{r}$ and open outputs, and R for the records whose spiders the web meets in the opposite colour. At every spider the labels satisfy $w_v|v(\theta_v)\rangle = \lambda_v|v(\theta_v + \delta_v)\rangle$, and the product over the diagram is

$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}, \quad \lambda(\mathbf{r}) = \lambda_0 (-1)^{\sum_{i \in R} r_i}.$$

$D_{\mathbf{r},\delta}$ has the marked phases changed. With no defects, $D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, so $D_{\mathbf{r}} = 0$ unless $\sum_{i \in R} r_i = b$, where $\lambda_0 = (-1)^b$: the records obey the check in every noiseless run. Every web drawn here has $\lambda_0 = 1$. With defects, $D_{\mathbf{r},\delta}$ is a different circuit, and the check needs the T angles.

Numbered violet stars $*_n$ mark defects, listed under each diagram with the gate, qubit and source line. The phases printed in spiders are the original phases. Each TikZ file records every defect in a comment.

$d = 3$: all 20 declared detectors

Det.	Check	Records	Drawn as	Page
0	r_0	r_0 M q_6 , l. 119	Pauli web	5
1	r_1	r_1 M q_{14} , l. 119	Pauli web	6
2	r_2	r_2 M q_8 , l. 119	Pauli web	7
3	r_3	r_3 M q_5 , l. 119	Pauli web	8
4	r_4	r_4 MX q_2 , l. 120	Pauli web	9
5	r_5	r_5 MX q_{11} , l. 120	Pauli web	10
6	r_6	r_6 MX q_4 , l. 120	Pauli web	11
7	$r_5 \oplus r_6 \oplus r_7$	r_5 MX q_{11} , l. 120; r_6 MX q_4 , l. 120; r_7 MX q_7 , l. 154	semiweb, 8 T defects	12
8	r_8	r_8 MX q_{14} , l. 180	Pauli web	13
9	r_9	r_9 MX q_{11} , l. 180	Pauli web	14
10	$r_7 \oplus r_{10}$	r_7 MX q_7 , l. 154; r_{10} MX q_6 , l. 180	Pauli web	15
11	r_{11}	r_{11} MX q_2 , l. 180	Pauli web	16
12	r_{12}	r_{12} MX q_8 , l. 180	Pauli web	17
13	r_{13}	r_{13} MX q_1 , l. 180	Pauli web	18
14	$r_7 \oplus r_{15}$	r_7 MX q_7 , l. 154; r_{15} MPP $X_0X_3X_7X_{10}$, l. 196	semiweb, 16 T defects	19
15	r_{16}	r_{16} MPP $Z_0Z_3Z_7Z_{10}$, l. 196	Pauli web	20
16	$r_7 \oplus r_{17}$	r_7 MX q_7 , l. 154; r_{17} MPP $X_3X_7X_9X_{12}$, l. 196	semiweb, 16 T defects	21
17	r_{18}	r_{18} MPP $Z_3Z_7Z_9Z_{12}$, l. 196	Pauli web	22
18	$r_7 \oplus r_{19}$	r_7 MX q_7 , l. 154; r_{19} MPP $X_7X_{10}X_{12}X_{13}$, l. 196	semiweb, 16 T defects	23
19	r_{20}	r_{20} MPP $Z_7Z_{10}Z_{12}Z_{13}$, l. 196	Pauli web	24

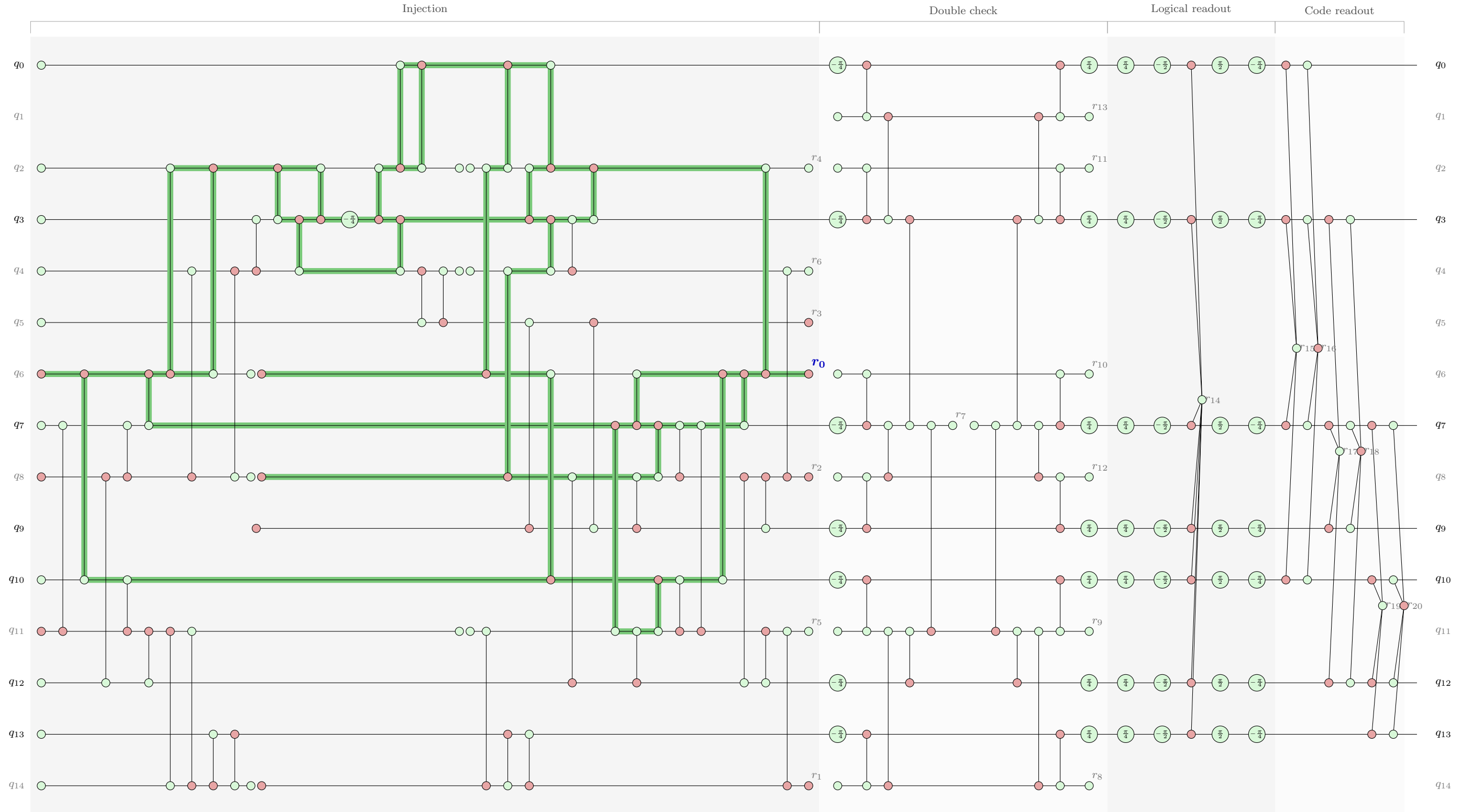
$d = 5$: all 107 declared detectors

Det.	Check	Records	Drawn as	Page
0	r_0	r_0 M q_6 , l. 126	Pauli web	25
1	r_1	r_1 M q_{18} , l. 126	Pauli web	26
2	r_2	r_2 M q_8 , l. 126	Pauli web	27
3	r_3	r_3 M q_5 , l. 126	Pauli web	28
4	r_4	r_4 MX q_2 , l. 127	Pauli web	29
5	r_5	r_5 MX q_{12} , l. 127	Pauli web	30
6	r_6	r_6 MX q_4 , l. 127	Pauli web	31
7	$r_5 \oplus r_6 \oplus r_7$	r_5 MX q_{12} , l. 127; r_6 MX q_4 , l. 127; r_7 MX q_7 , l. 157	semiweb, 8 T defects	32
8	r_8	r_8 MX q_{18} , l. 176	Pauli web	33
9	r_9	r_9 MX q_{12} , l. 176	Pauli web	34
10	$r_7 \oplus r_{10}$	r_7 MX q_7 , l. 157; r_{10} MX q_6 , l. 176	Pauli web	35
11	r_{11}	r_{11} MX q_2 , l. 176	Pauli web	36
12	r_{12}	r_{12} MX q_8 , l. 176	Pauli web	37
13	r_{13}	r_{13} MX q_1 , l. 176	Pauli web	38
14	r_{14}	r_{14} M q_{26} , l. 203	Pauli web	39
15	r_{15}	r_{15} M q_{33} , l. 203	Pauli web	40
16	r_{16}	r_{16} M q_{28} , l. 203	Pauli web	41
17	r_{17}	r_{17} M q_{22} , l. 203	Pauli web	42
18	r_{18}	r_{18} M q_{16} , l. 203	Pauli web	43
19	$r_7 \oplus r_{36}$	r_7 MX q_7 , l. 157; r_{36} MX q_2 , l. 251	semiweb, 8 T defects	44
20	$r_7 \oplus r_{35}$	r_7 MX q_7 , l. 157; r_{35} MX q_4 , l. 251	semiweb, 8 T defects	45
21	r_{27}	r_{27} M q_6 , l. 250	Pauli web	46
22	r_{26}	r_{26} M q_8 , l. 250	Pauli web	47
23	$r_7 \oplus r_{34}$	r_7 MX q_7 , l. 157; r_{34} MX q_{12} , l. 251	semiweb, 8 T defects	48
24	r_{32}	r_{32} MX q_{16} , l. 251	Pauli web	49
25	r_{25}	r_{25} M q_{18} , l. 250	Pauli web	50
26	r_{23}	r_{23} M q_{22} , l. 250	Pauli web	51
27	r_{30}	r_{30} MX q_{26} , l. 251	Pauli web	52
28	r_{29}	r_{29} MX q_{28} , l. 251	Pauli web	53
29	r_{21}	r_{21} M q_{33} , l. 250	Pauli web	54
30	r_{20}	r_{20} M q_{35} , l. 250	Pauli web	55
31	r_{28}	r_{28} MX q_{37} , l. 251	Pauli web	56
32	r_{19}	r_{19} M q_{41} , l. 250	Pauli web	57
33	$r_{36} \oplus r_{54}$	r_{36} MX q_2 , l. 251; r_{54} MX q_2 , l. 310	Pauli web	58
34	$r_{35} \oplus r_{53}$	r_{35} MX q_4 , l. 251; r_{53} MX q_4 , l. 310	Pauli web	59
35	$r_{26} \oplus r_{27} \oplus r_{45}$	r_{26} M q_8 , l. 250; r_{27} M q_6 , l. 250; r_{45} M q_6 , l. 309	Pauli web	60
36	$r_{27} \oplus r_{44}$	r_{27} M q_6 , l. 250; r_{44} M q_8 , l. 309	Pauli web	61
37	$r_{34} \oplus r_{52}$	r_{34} MX q_{12} , l. 251; r_{52} MX q_{12} , l. 310	Pauli web	62
38	r_{51}	r_{51} MX q_{14} , l. 310	Pauli web	63
39	$r_{32} \oplus r_{50}$	r_{32} MX q_{16} , l. 251; r_{50} MX q_{16} , l. 310	Pauli web	64
40	r_{43}	r_{43} M q_{18} , l. 309	Pauli web	65
41	r_{42}	r_{42} M q_{20} , l. 309	Pauli web	66
42	r_{41}	r_{41} M q_{22} , l. 309	Pauli web	67
43	r_{49}	r_{49} MX q_{24} , l. 310	Pauli web	68
44	$r_{30} \oplus r_{48}$	r_{30} MX q_{26} , l. 251; r_{48} MX q_{26} , l. 310	Pauli web	69
45	$r_{29} \oplus r_{47}$	r_{29} MX q_{28} , l. 251; r_{47} MX q_{28} , l. 310	Pauli web	70
46	r_{40}	r_{40} M q_{31} , l. 309	Pauli web	71
47	r_{39}	r_{39} M q_{33} , l. 309	Pauli web	72
48	$r_{21} \oplus r_{38}$	r_{21} M q_{33} , l. 250; r_{38} M q_{35} , l. 309	Pauli web	73
49	$r_{28} \oplus r_{46}$	r_{28} MX q_{37} , l. 251; r_{46} MX q_{37} , l. 310	Pauli web	74
50	r_{37}	r_{37} M q_{41} , l. 309	Pauli web	75
51	$r_{54} \oplus r_{72}$	r_{54} MX q_2 , l. 310; r_{72} MX q_2 , l. 362	Pauli web	76
52	$r_{53} \oplus r_{71}$	r_{53} MX q_4 , l. 310; r_{71} MX q_4 , l. 362	Pauli web	77
53	$r_{44} \oplus r_{45} \oplus r_{63}$	r_{44} M q_8 , l. 309; r_{45} M q_6 , l. 309; r_{63} M q_6 , l. 361	Pauli web	78
54	$r_{45} \oplus r_{62}$	r_{45} M q_6 , l. 309; r_{62} M q_8 , l. 361	Pauli web	79
55	$r_{52} \oplus r_{70}$	r_{52} MX q_{12} , l. 310; r_{70} MX q_{12} , l. 362	Pauli web	80
56	$r_{51} \oplus r_{69}$	r_{51} MX q_{14} , l. 310; r_{69} MX q_{14} , l. 362	Pauli web	81
57	$r_{50} \oplus r_{68}$	r_{50} MX q_{16} , l. 310; r_{68} MX q_{16} , l. 362	Pauli web	82
58	$r_{42} \oplus r_{61}$	r_{42} M q_{20} , l. 309; r_{61} M q_{18} , l. 361	Pauli web	83
59	$r_{41} \oplus r_{43} \oplus r_{60}$	r_{41} M q_{22} , l. 309; r_{43} M q_{18} , l. 309; r_{60} M q_{20} , l. 361	Pauli web	84

Det.	Check	Records	Drawn as	Page
60	$r_{42} \oplus r_{59}$	r_{42} M q_{20} , l. 309; r_{59} M q_{22} , l. 361	Pauli web	85
61	$r_{49} \oplus r_{67}$	r_{49} MX q_{24} , l. 310; r_{67} MX q_{24} , l. 362	Pauli web	86
62	$r_{48} \oplus r_{66}$	r_{48} MX q_{26} , l. 310; r_{66} MX q_{26} , l. 362	Pauli web	87
63	$r_{47} \oplus r_{65}$	r_{47} MX q_{28} , l. 310; r_{65} MX q_{28} , l. 362	Pauli web	88
64	$r_{39} \oplus r_{40} \oplus r_{58}$	r_{39} M q_{33} , l. 309; r_{40} M q_{31} , l. 309; r_{58} M q_{31} , l. 361	Pauli web	89
65	$r_{38} \oplus r_{40} \oplus r_{57}$	r_{38} M q_{35} , l. 309; r_{40} M q_{31} , l. 309; r_{57} M q_{33} , l. 361	Pauli web	90
66	$r_{39} \oplus r_{56}$	r_{39} M q_{33} , l. 309; r_{56} M q_{35} , l. 361	Pauli web	91
67	$r_{46} \oplus r_{64}$	r_{46} MX q_{37} , l. 310; r_{64} MX q_{37} , l. 362	Pauli web	92
68	r_{55}	r_{55} M q_{41} , l. 361	Pauli web	93
69	$r_8 \oplus r_9 \oplus r_{12} \oplus r_{13} \oplus r_{19} \oplus r_{20} \oplus r_{21} \oplus r_{23} \oplus r_{25} \oplus r_{26} \oplus r_{37} \oplus r_{38} \oplus r_{39} \oplus r_{41} \oplus r_{42} \oplus r_{43} \oplus r_{44} \oplus r_{55} \oplus r_{56} \oplus r_{57} \oplus r_{59} \oplus r_{60} \oplus r_{61} \oplus r_{62} \oplus r_{73}$	r_8 MX q_{18} , l. 176; r_9 MX q_{12} , l. 176; r_{12} MX q_8 , l. 176; r_{13} MX q_1 , l. 176; r_{19} M q_{41} , l. 250; r_{20} M q_{35} , l. 250; r_{21} M q_{33} , l. 250; r_{23} M q_{22} , l. 250; r_{25} M q_{18} , l. 250; r_{26} M q_8 , l. 250; r_{37} M q_{41} , l. 309; r_{38} M q_{35} , l. 309; r_{39} M q_{33} , l. 309; r_{41} M q_{22} , l. 309; r_{42} M q_{20} , l. 309; r_{43} M q_{18} , l. 309; r_{44} M q_8 , l. 309; r_{55} M q_{41} , l. 361; r_{56} M q_{35} , l. 361; r_{57} M q_{33} , l. 361; r_{59} M q_{22} , l. 361; r_{60} M q_{20} , l. 361; r_{61} M q_{18} , l. 361; r_{62} M q_8 , l. 361; r_{73} MX q_{25} , l. 415	semiweb, 26 T defects	94
70	r_{74}	r_{74} MX q_{20} , l. 444	Pauli web	96
71	r_{75}	r_{75} MX q_{22} , l. 444	Pauli web	97
72	r_{76}	r_{76} MX q_8 , l. 444	Pauli web	98
73	r_{77}	r_{77} MX q_{10} , l. 444	Pauli web	99
74	r_{78}	r_{78} MX q_2 , l. 444	Pauli web	100
75	r_{79}	r_{79} MX q_6 , l. 444	Pauli web	101
76	$r_{73} \oplus r_{80}$	r_{73} MX q_{25} , l. 415; r_{80} MX q_{24} , l. 444	Pauli web	102
77	r_{81}	r_{81} MX q_{31} , l. 444	Pauli web	103
78	r_{82}	r_{82} MX q_{39} , l. 444	Pauli web	104
79	r_{83}	r_{83} MX q_1 , l. 444	Pauli web	105
80	r_{84}	r_{84} MX q_{12} , l. 444	Pauli web	106
81	r_{85}	r_{85} MX q_{18} , l. 444	Pauli web	107
82	r_{86}	r_{86} MX q_{37} , l. 444	Pauli web	108
83	r_{87}	r_{87} MX q_{33} , l. 444	Pauli web	109
84	r_{88}	r_{88} MX q_{14} , l. 444	Pauli web	110
85	r_{89}	r_{89} MX q_{26} , l. 444	Pauli web	111
86	r_{90}	r_{90} MX q_{28} , l. 444	Pauli web	112
87	r_{91}	r_{91} MX q_{30} , l. 444	Pauli web	113
88	r_{92}	r_{92} MX q_{41} , l. 444	Pauli web	114
89	$r_{72} \oplus r_{93}$	r_{72} MX q_2 , l. 362; r_{93} MPP $X_0X_3X_7X_{11}$, l. 470	semiweb, 8 T defects	115
90	$r_{62} \oplus r_{63} \oplus r_{96}$	r_{62} M q_8 , l. 361; r_{63} M q_6 , l. 361; r_{96} MPP $Z_0Z_3Z_7Z_{11}$, l. 470	Pauli web	116
91	$r_{71} \oplus r_{99}$	r_{71} MX q_4 , l. 362; r_{99} MPP $X_3X_7X_9X_{13}$, l. 470	semiweb, 8 T defects	117
92	$r_{63} \oplus r_{102}$	r_{63} M q_6 , l. 361; r_{102} MPP $Z_3Z_7Z_9Z_{13}$, l. 470	Pauli web	118
93	$r_{70} \oplus r_{73} \oplus r_{105}$	r_{70} MX q_{12} , l. 362; r_{73} MX q_{25} , l. 415; r_{105} MPP $X_7X_{11}X_{13}X_{17}X_{19}X_{25}$, l. 470	semiweb, 12 T defects	119
94	$r_{60} \oplus r_{108}$	r_{60} M q_{20} , l. 361; r_{108} MPP $Z_7Z_{11}Z_{13}Z_{17}Z_{19}Z_{25}$, l. 470	Pauli web	120
95	$r_{69} \oplus r_{94}$	r_{69} MX q_{14} , l. 362; r_{94} MPP $X_9X_{13}X_{15}X_{19}X_{21}X_{27}$, l. 470	semiweb, 12 T defects	121
96	$r_{59} \oplus r_{61} \oplus r_{97}$	r_{59} M q_{22} , l. 361; r_{61} M q_{18} , l. 361; r_{97} MPP $Z_9Z_{13}Z_{15}Z_{19}Z_{21}Z_{27}$, l. 470	Pauli web	122
97	$r_{68} \oplus r_{100}$	r_{68} MX q_{16} , l. 362; r_{100} MPP $X_{15}X_{21}X_{23}X_{29}$, l. 470	semiweb, 8 T defects	123
98	$r_{60} \oplus r_{103}$	r_{60} M q_{20} , l. 361; r_{103} MPP $Z_{15}Z_{21}Z_{23}Z_{29}$, l. 470	Pauli web	124
99	$r_{67} \oplus r_{73} \oplus r_{95}$	r_{67} MX q_{24} , l. 362; r_{73} MX q_{25} , l. 415; r_{95} MPP $X_{17}X_{25}X_{32}X_{36}$, l. 470	semiweb, 8 T defects	125
100	$r_{57} \oplus r_{58} \oplus r_{98}$	r_{57} M q_{33} , l. 361; r_{58} M q_{31} , l. 361; r_{98} MPP $Z_{17}Z_{25}Z_{32}Z_{36}$, l. 470	Pauli web	126
101	$r_{66} \oplus r_{73} \oplus r_{101}$	r_{66} MX q_{26} , l. 362; r_{73} MX q_{25} , l. 415; r_{101} MPP $X_{19}X_{25}X_{27}X_{32}X_{34}X_{38}$, l. 470	semiweb, 12 T defects	127
102	$r_{56} \oplus r_{58} \oplus r_{104}$	r_{56} M q_{35} , l. 361; r_{58} M q_{31} , l. 361; r_{104} MPP $Z_{19}Z_{25}Z_{27}Z_{32}Z_{34}Z_{38}$, l. 470	Pauli web	128
103	$r_{65} \oplus r_{106}$	r_{65} MX q_{28} , l. 362; r_{106} MPP $X_{21}X_{27}X_{29}X_{34}$, l. 470	semiweb, 8 T defects	129
104	$r_{57} \oplus r_{109}$	r_{57} M q_{33} , l. 361; r_{109} MPP $Z_{21}Z_{27}Z_{29}Z_{34}$, l. 470	Pauli web	130
105	$r_{64} \oplus r_{107}$	r_{64} MX q_{37} , l. 362; r_{107} MPP $X_{32}X_{36}X_{38}X_{40}$, l. 470	semiweb, 8 T defects	131
106	r_{110}	r_{110} MPP $Z_{32}Z_{36}Z_{38}Z_{40}$, l. 470	Pauli web	132

$d = 3$: detector 0 $r_0 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

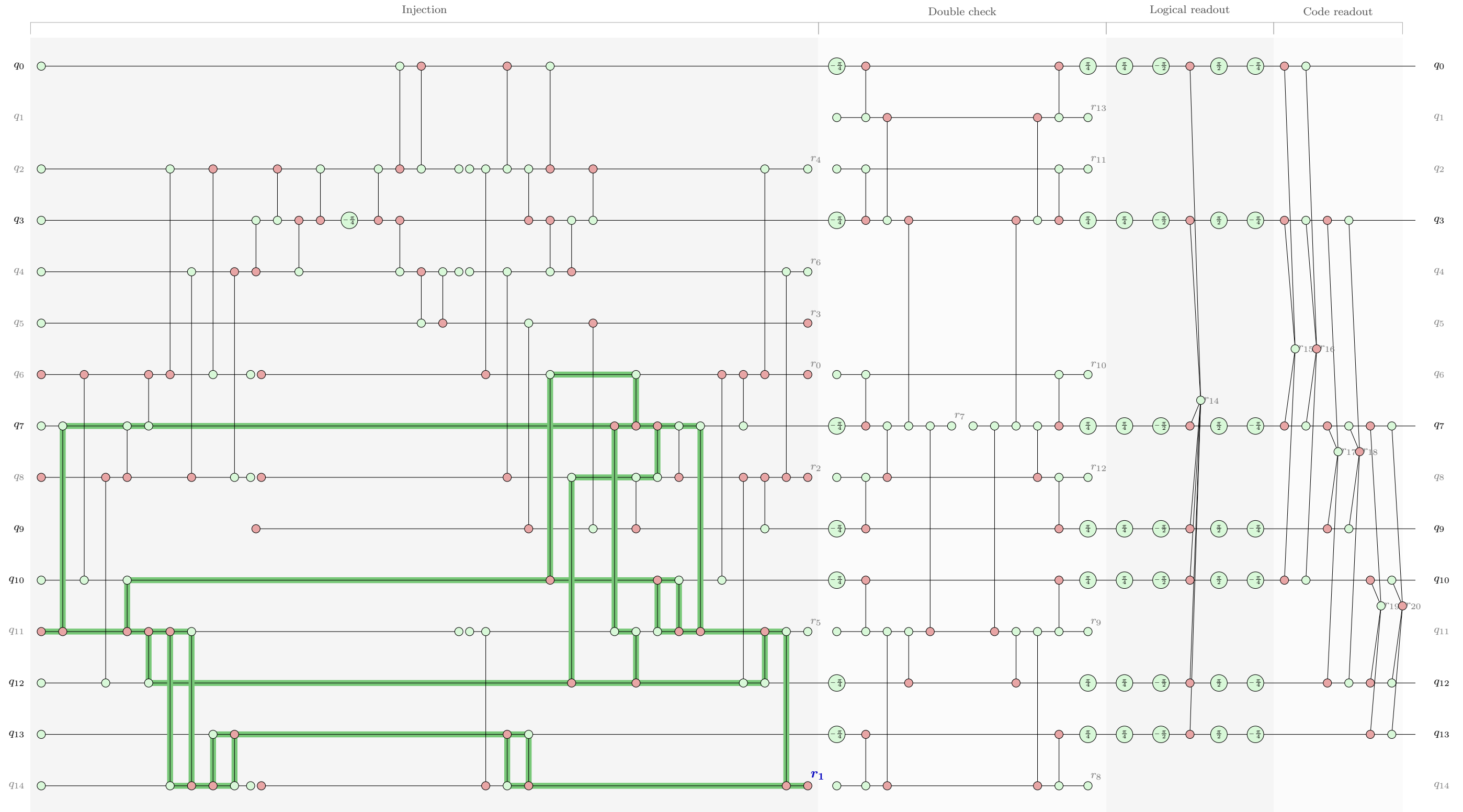


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_0}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_0 = 0.$$

Records: r_0 : M q_6 , line 119.

$d = 3$: detector 1 $r_1 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

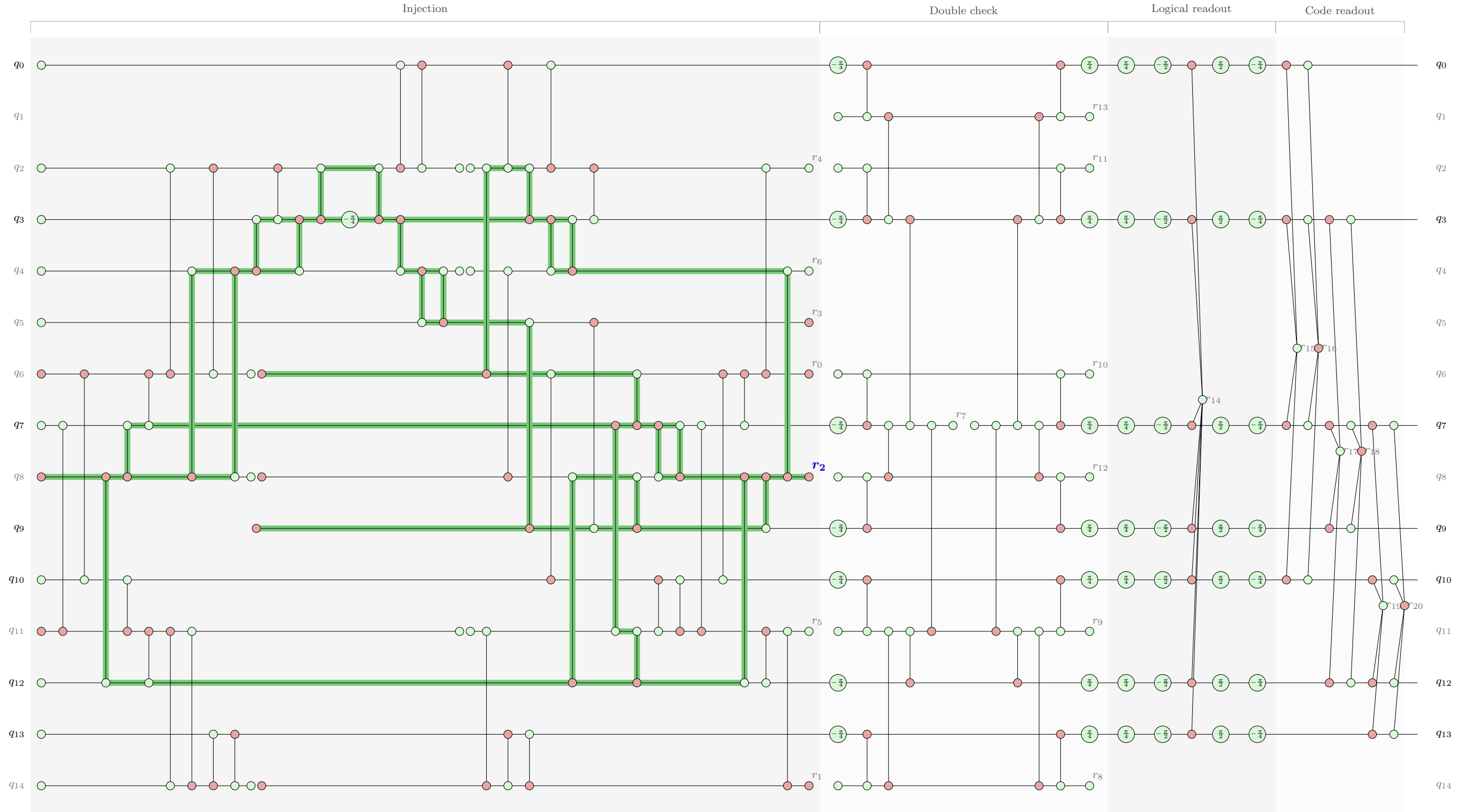


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_1}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_1 = 0.$$

Records: r_1 : M q_{14} , line 119.

$d = 3$: detector 2 $r_2 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

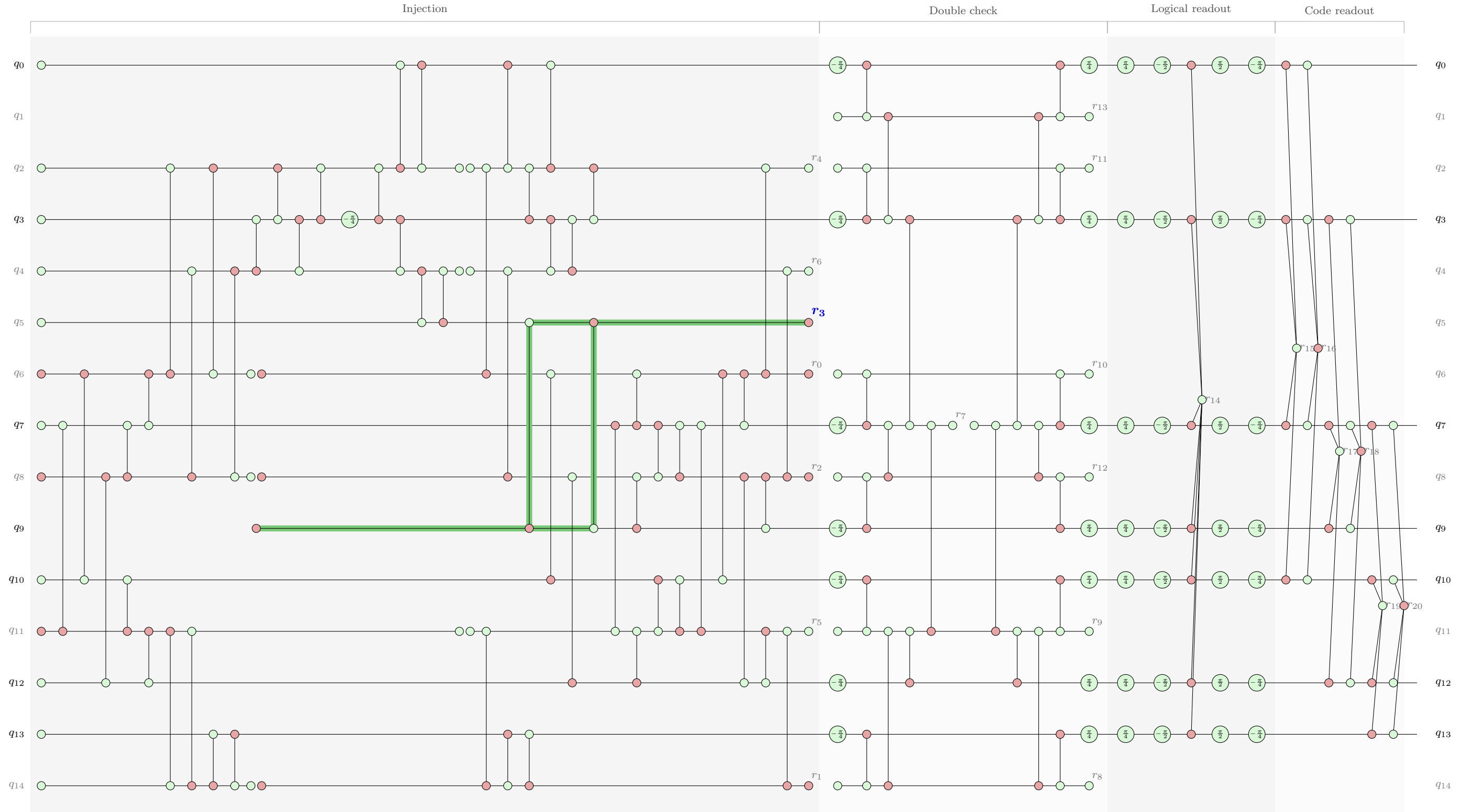


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_2}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_2 = 0.$$

Records: r_2 : M q_8 , line 119.

$d = 3$: detector 3 $r_3 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

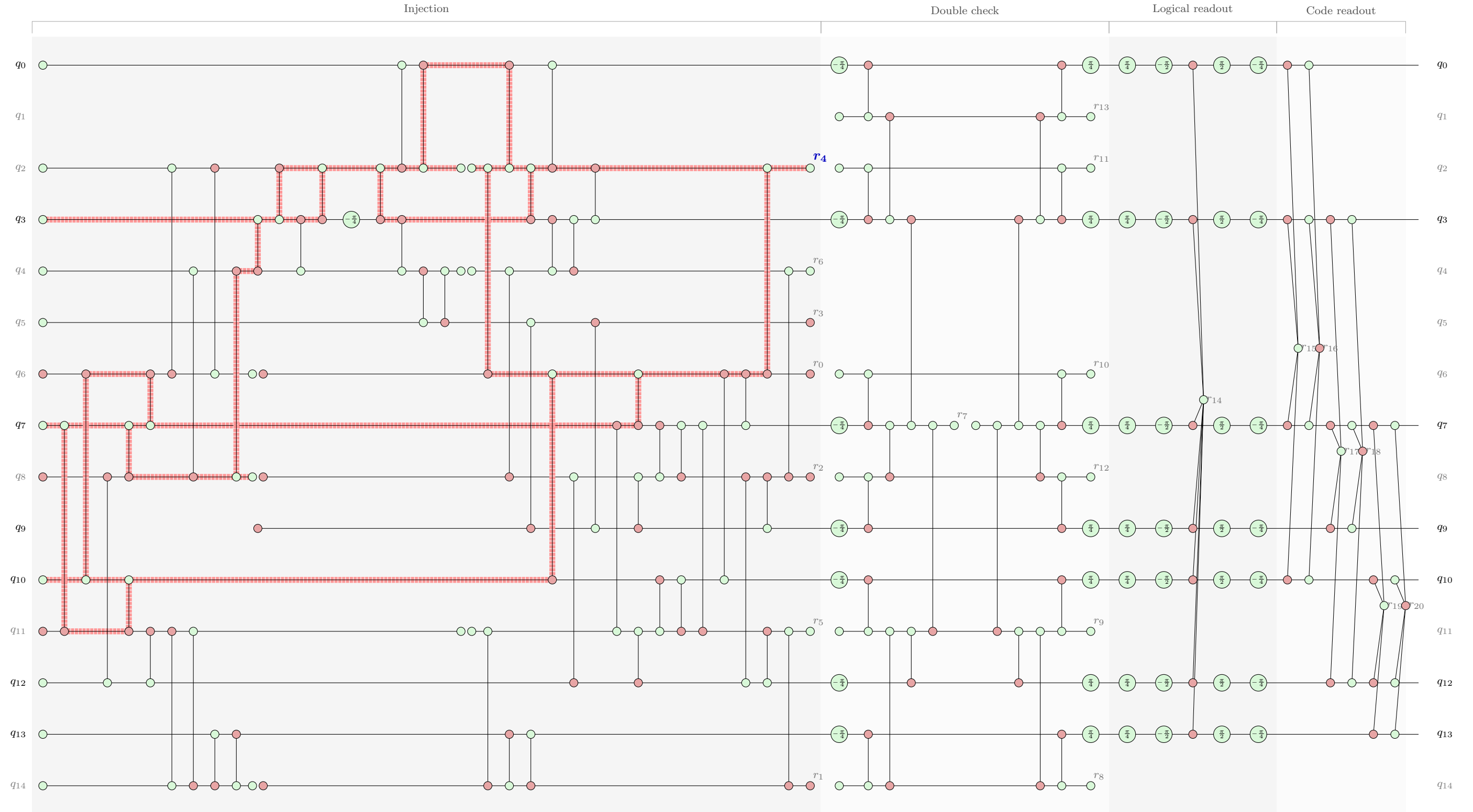


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_3}$. So $D_{\mathbf{r}} = 0$ unless $r_3 = 0$.

Records: r_3 : M q_5 , line 119.

$d = 3$: detector 4 $r_4 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

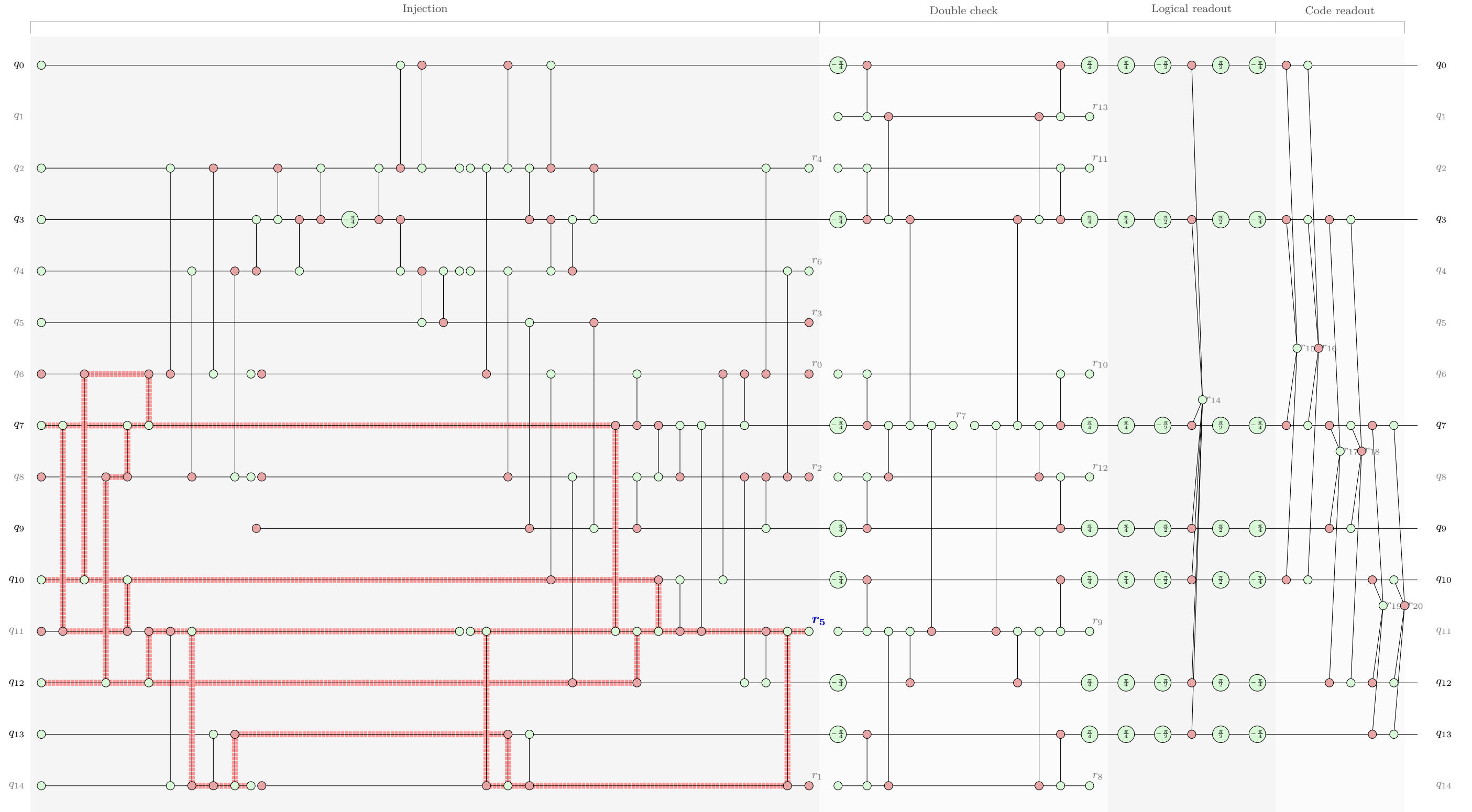


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_4}$. So $D_{\mathbf{r}} = 0$ unless $r_4 = 0$.

Records: r_4 : MX q_2 , line 120.

$d = 3$: detector 5 $r_5 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

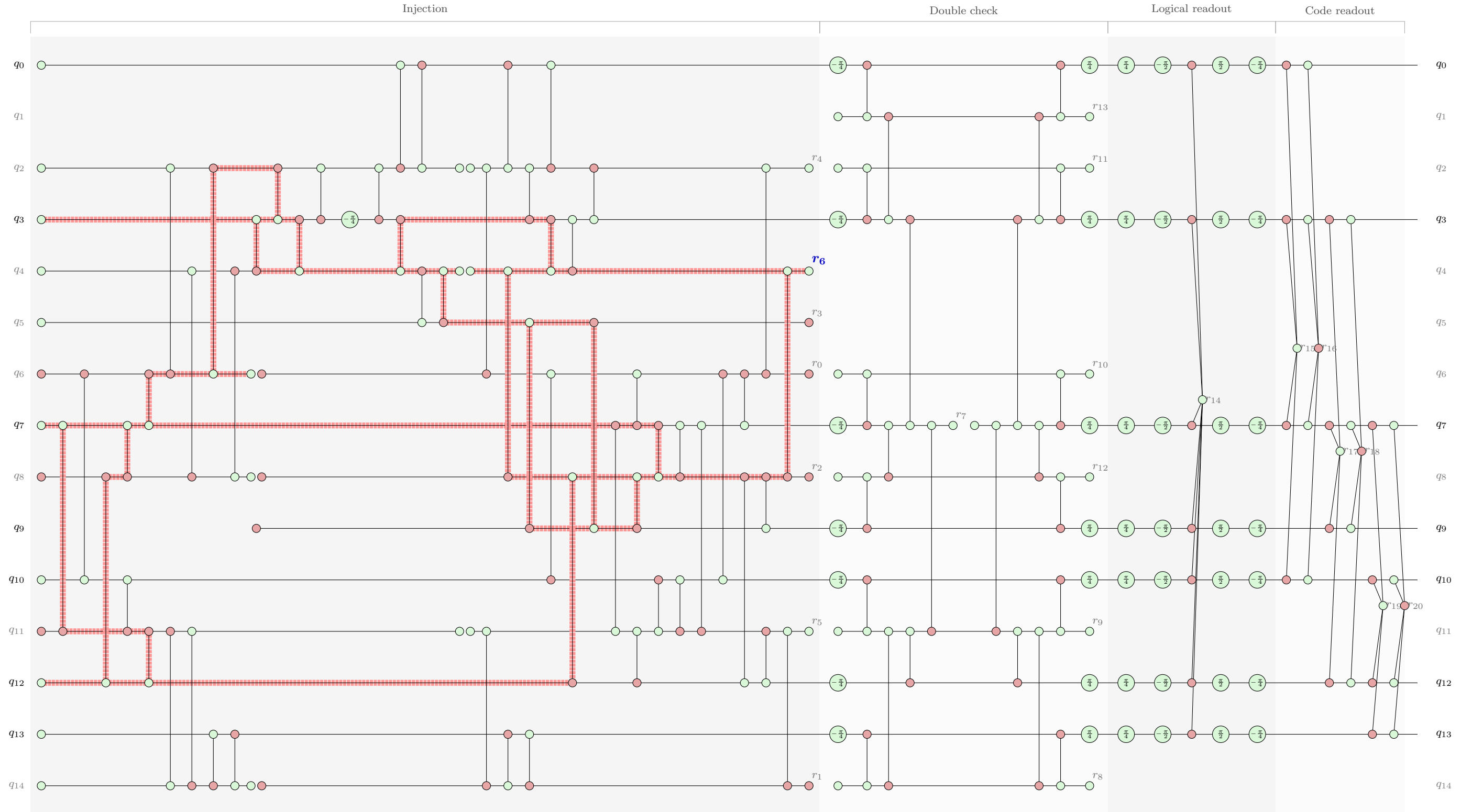


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_5}$. So $D_{\mathbf{r}} = 0$ unless $r_5 = 0$.

Records: r_5 : MX q_{11} , line 120.

$d = 3$: detector 6 $r_6 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

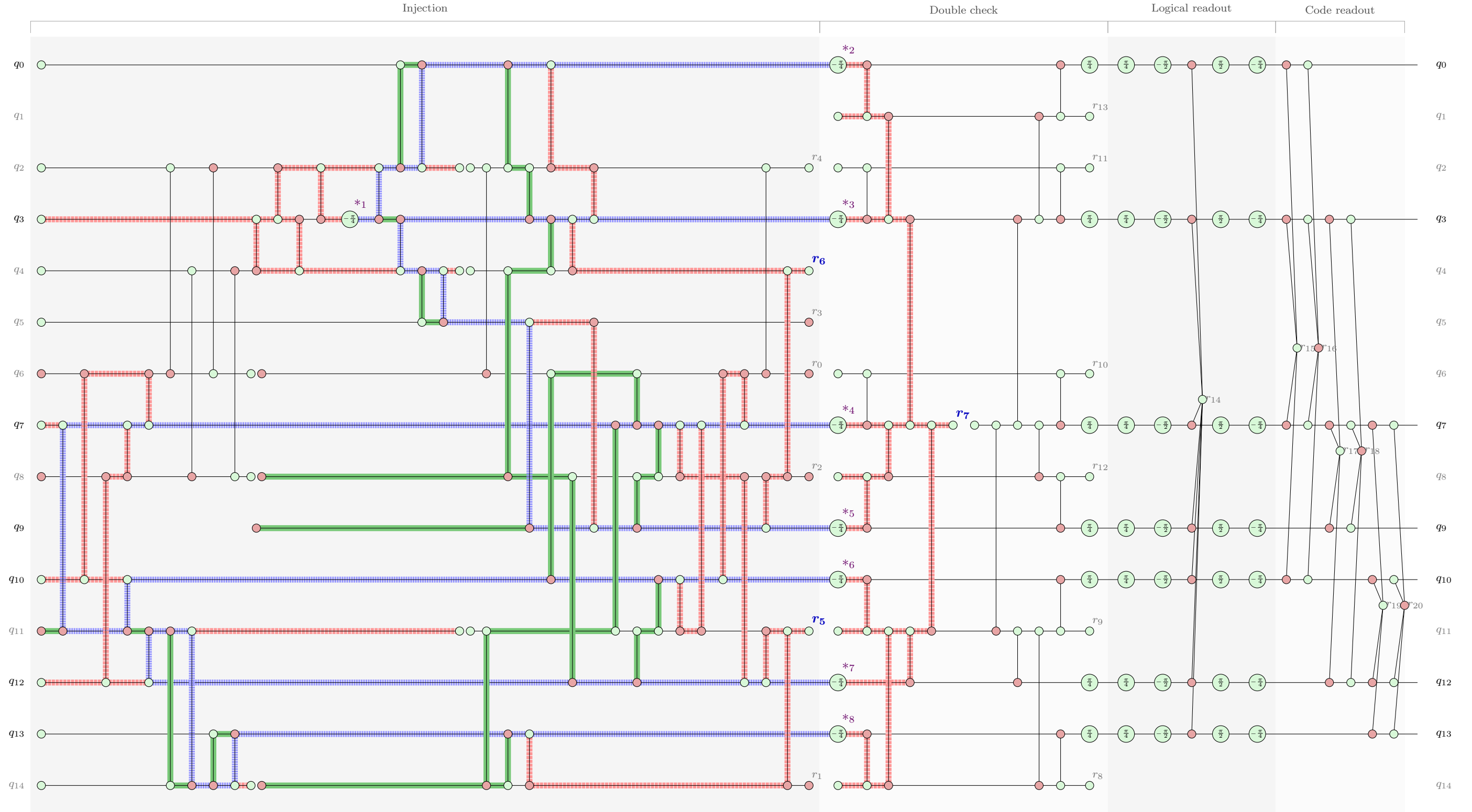


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_6}$. So $D_{\mathbf{r}} = 0$ unless $r_6 = 0$.

Records: r_6 : MX q_4 , line 120.

$d = 3$: detector 7 $r_5 \oplus r_6 \oplus r_7 = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



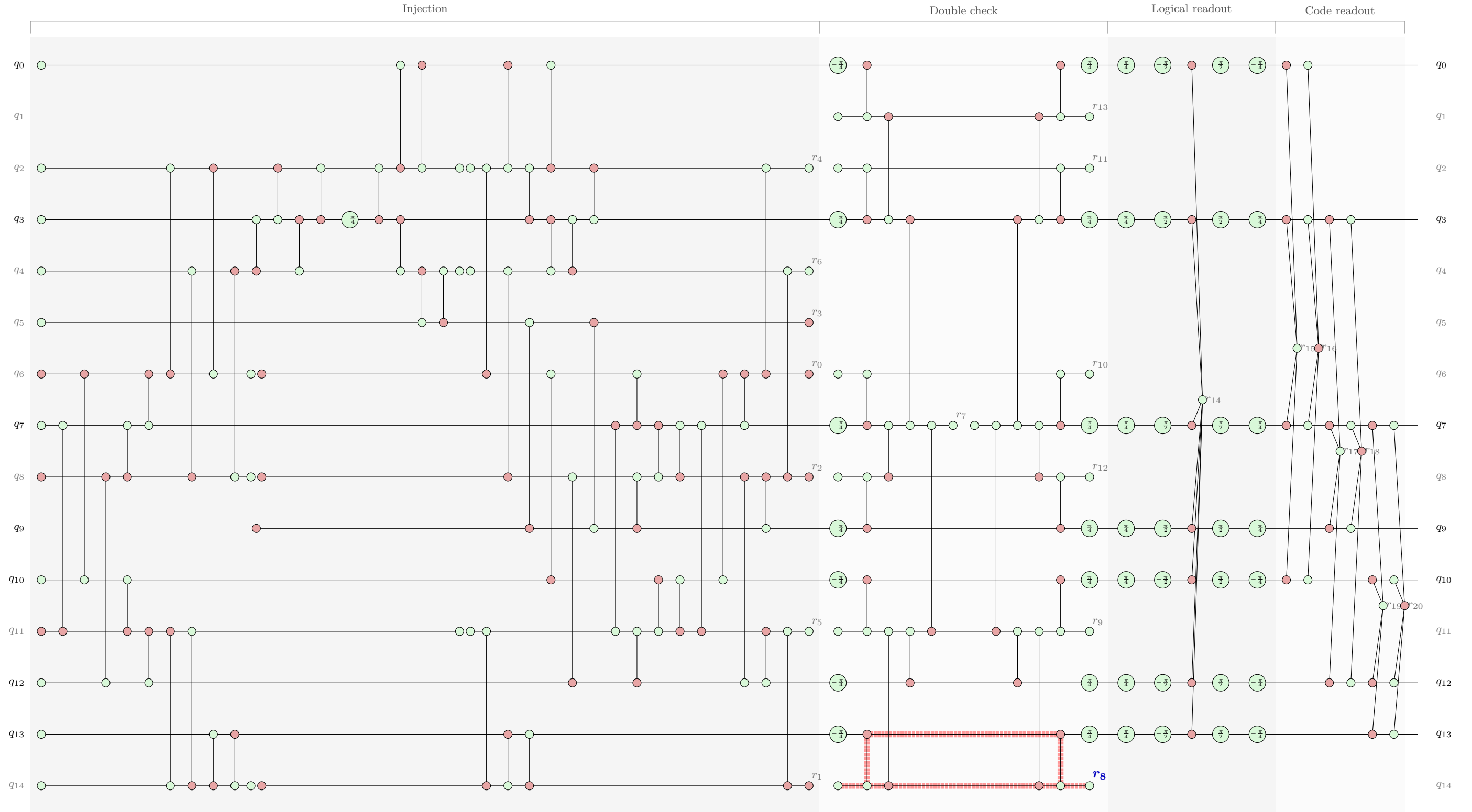
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}, \delta}$, $\lambda(\mathbf{r}) = (-1)^{r_5 + r_6 + r_7}$. $D_{\mathbf{r}, \delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_5 : MX q_{11} , line 120; r_6 : MX q_4 , line 120; r_7 : MX q_7 , line 154.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 62; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_0 , line 135; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 135; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 135; $*_5$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_9 , line 135; $*_6$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{10} , line 135; $*_7$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{12} , line 135; $*_8$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 135.

$d = 3$: detector 8 $r_8 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

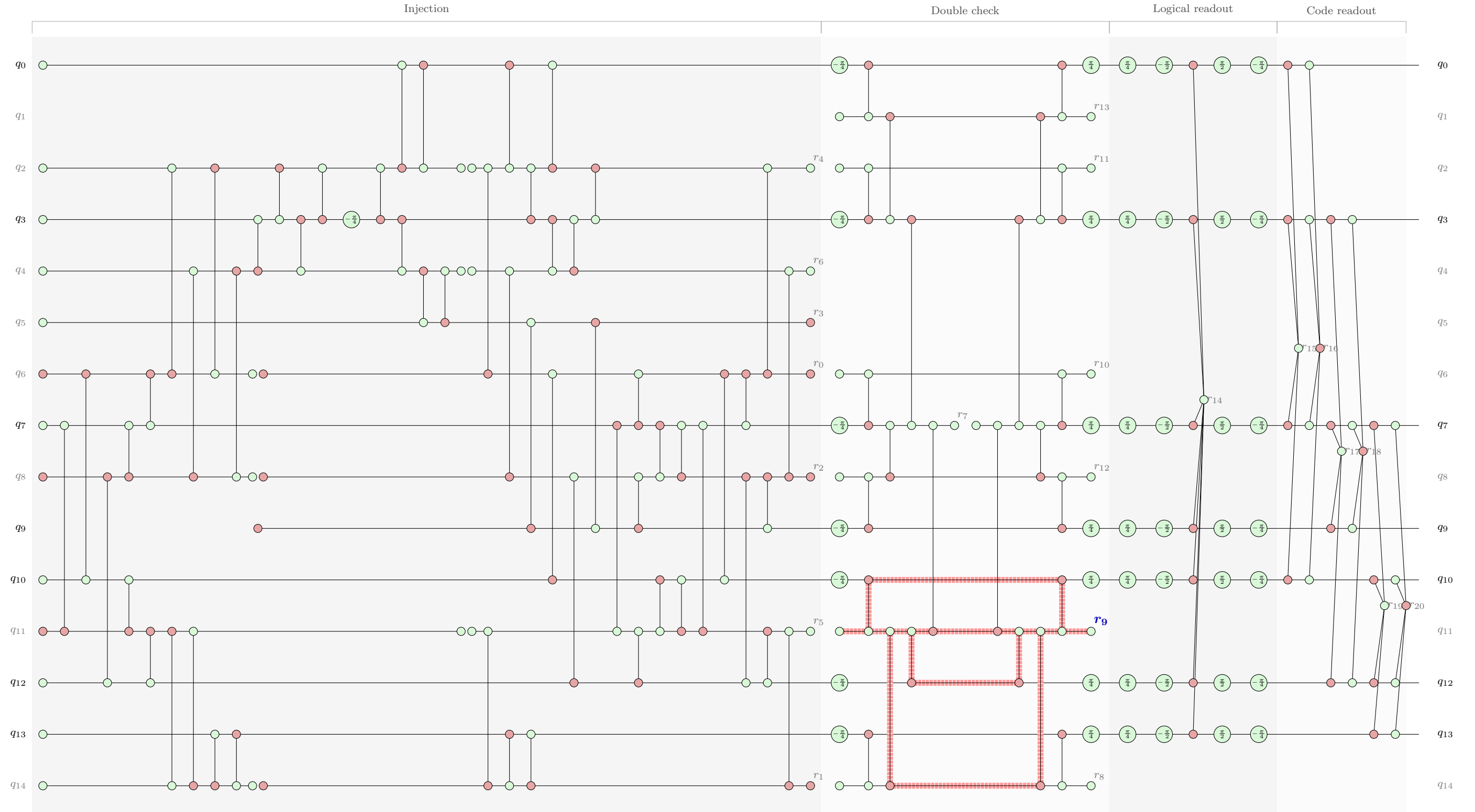


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_8}$. So $D_{\mathbf{r}} = 0$ unless $r_8 = 0$.

Records: r_8 : MX q_{14} , line 180.

$d = 3$: detector 9 $r_9 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

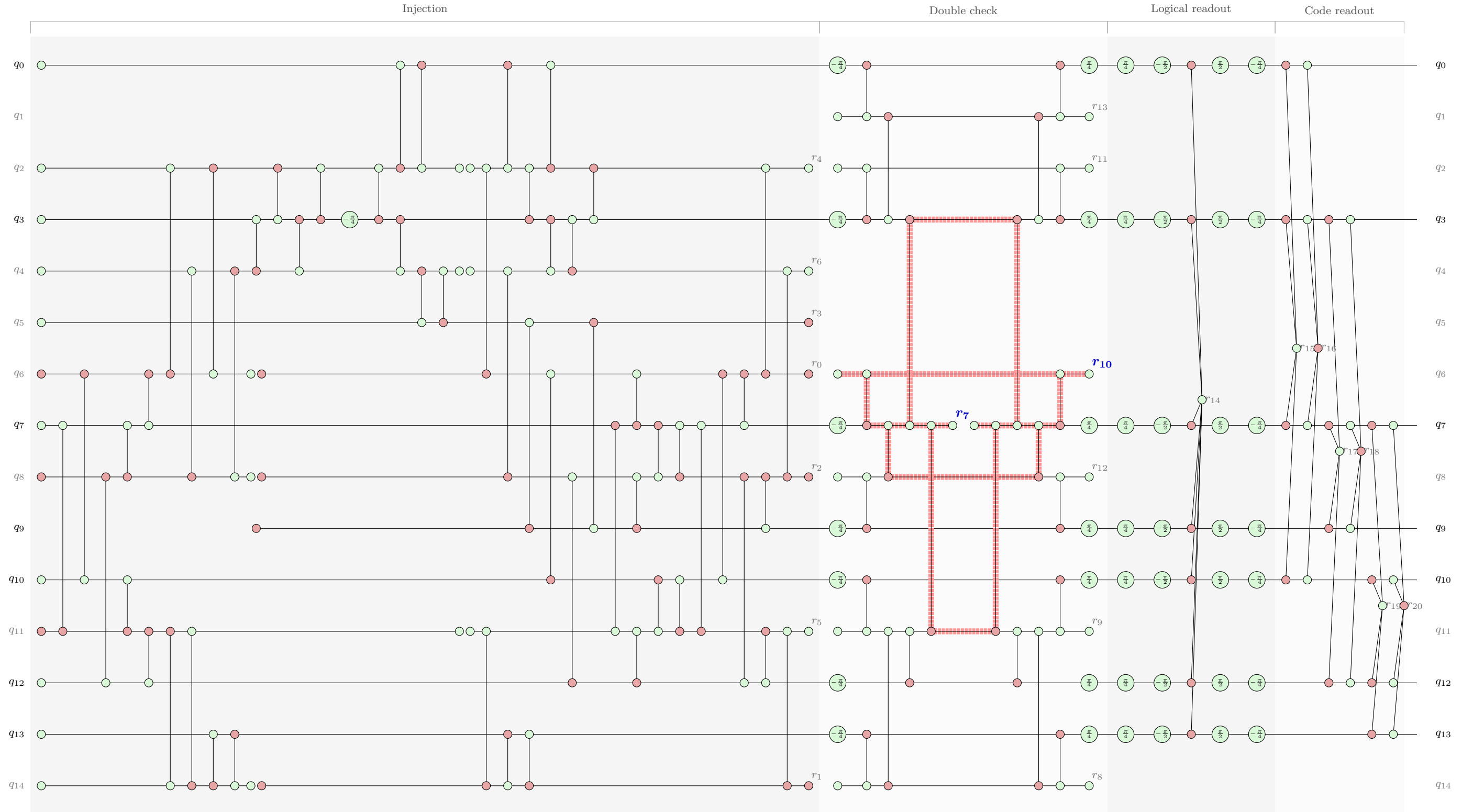


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_9}$. So $D_{\mathbf{r}} = 0$ unless $r_9 = 0$.

Records: r_9 : MX q_{11} , line 180.

$d = 3$: detector 10 $r_7 \oplus r_{10} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

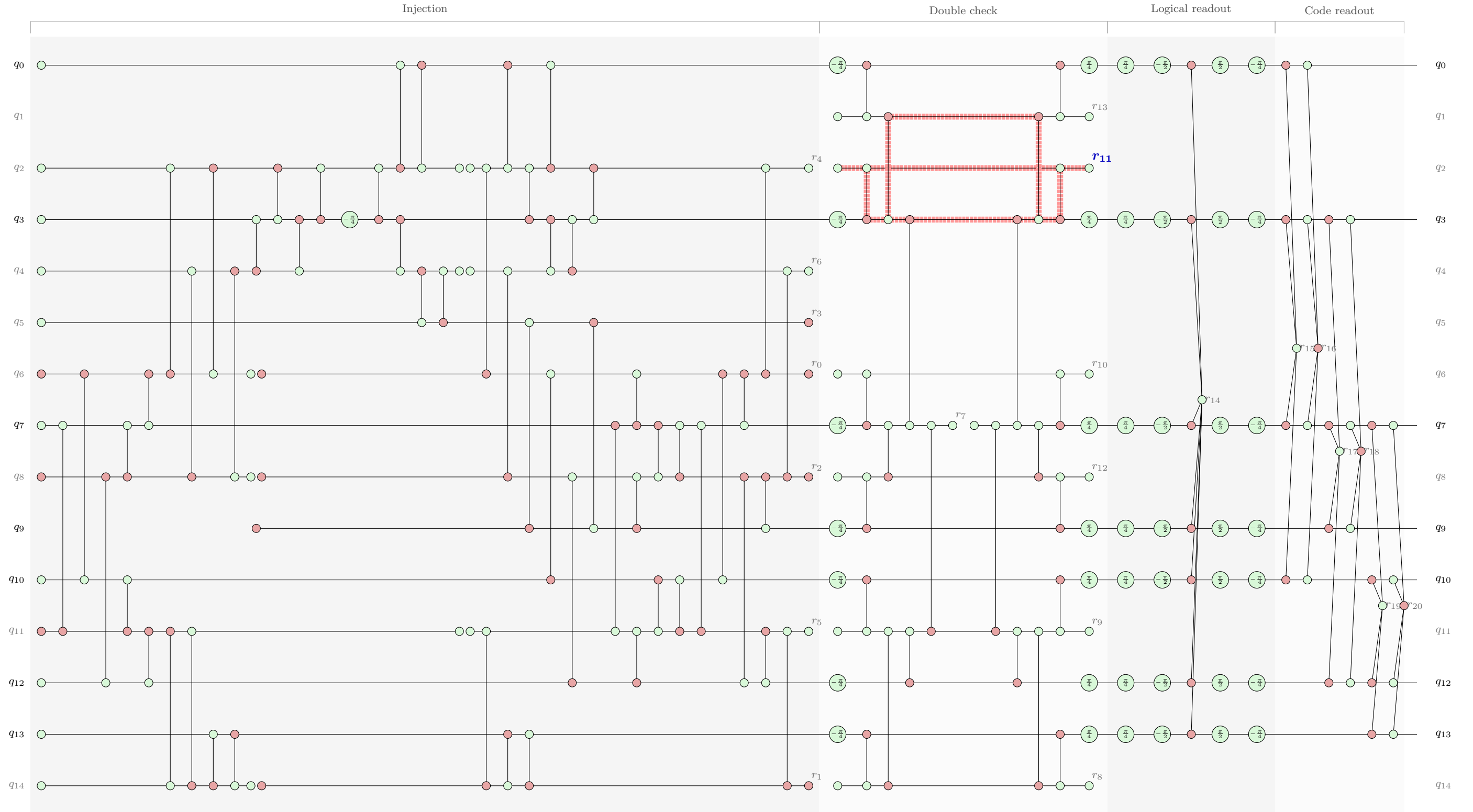


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_7 + r_{10}}$. So $D_{\mathbf{r}} = 0$ unless $r_7 \oplus r_{10} = 0$.

Records: r_7 : MX q_7 , line 154; r_{10} : MX q_6 , line 180.

$d = 3$: detector 11 $r_{11} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

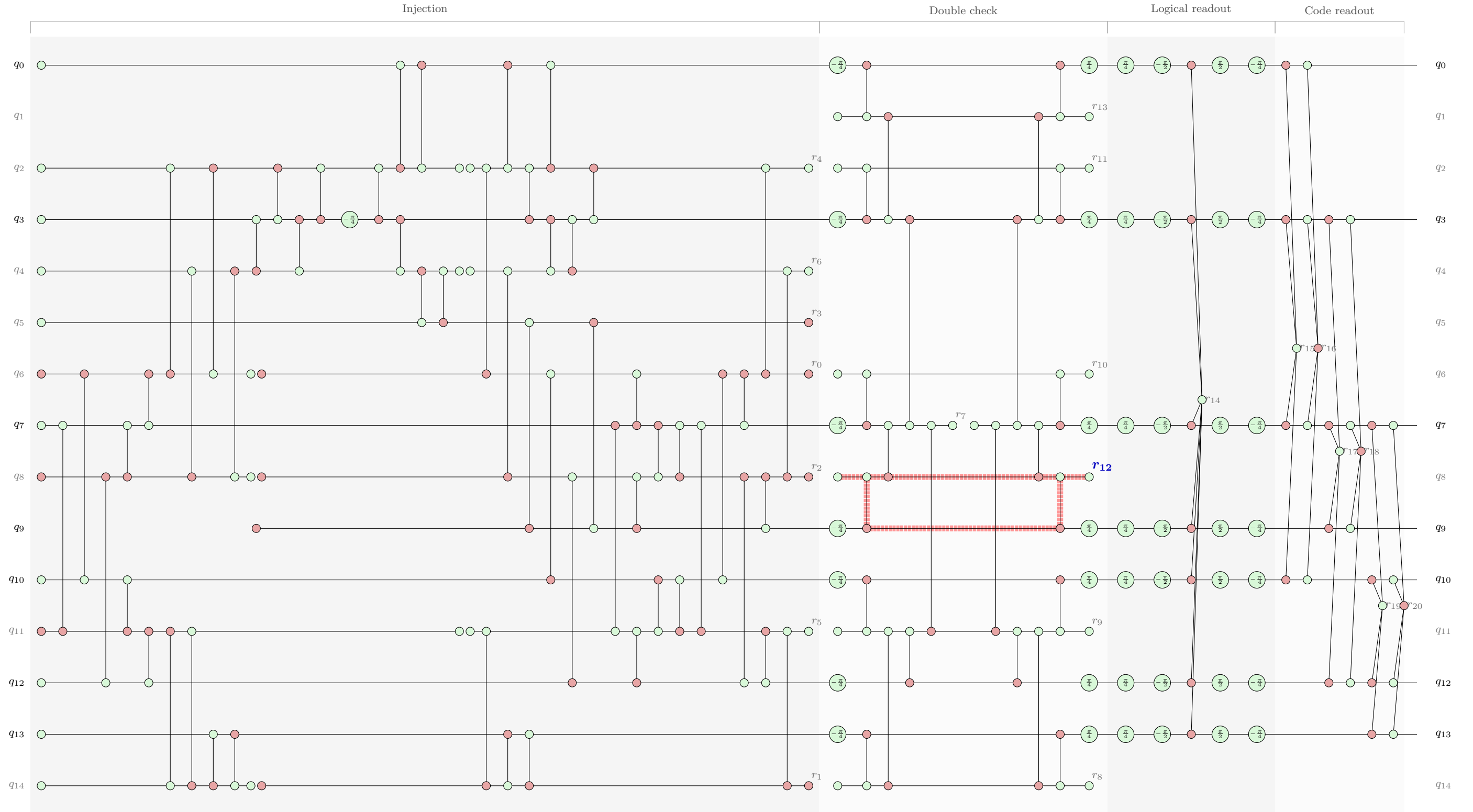


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{11}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{11} = 0.$$

Records: r_{11} : MX q_2 , line 180.

$d = 3$: detector 12 $r_{12} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

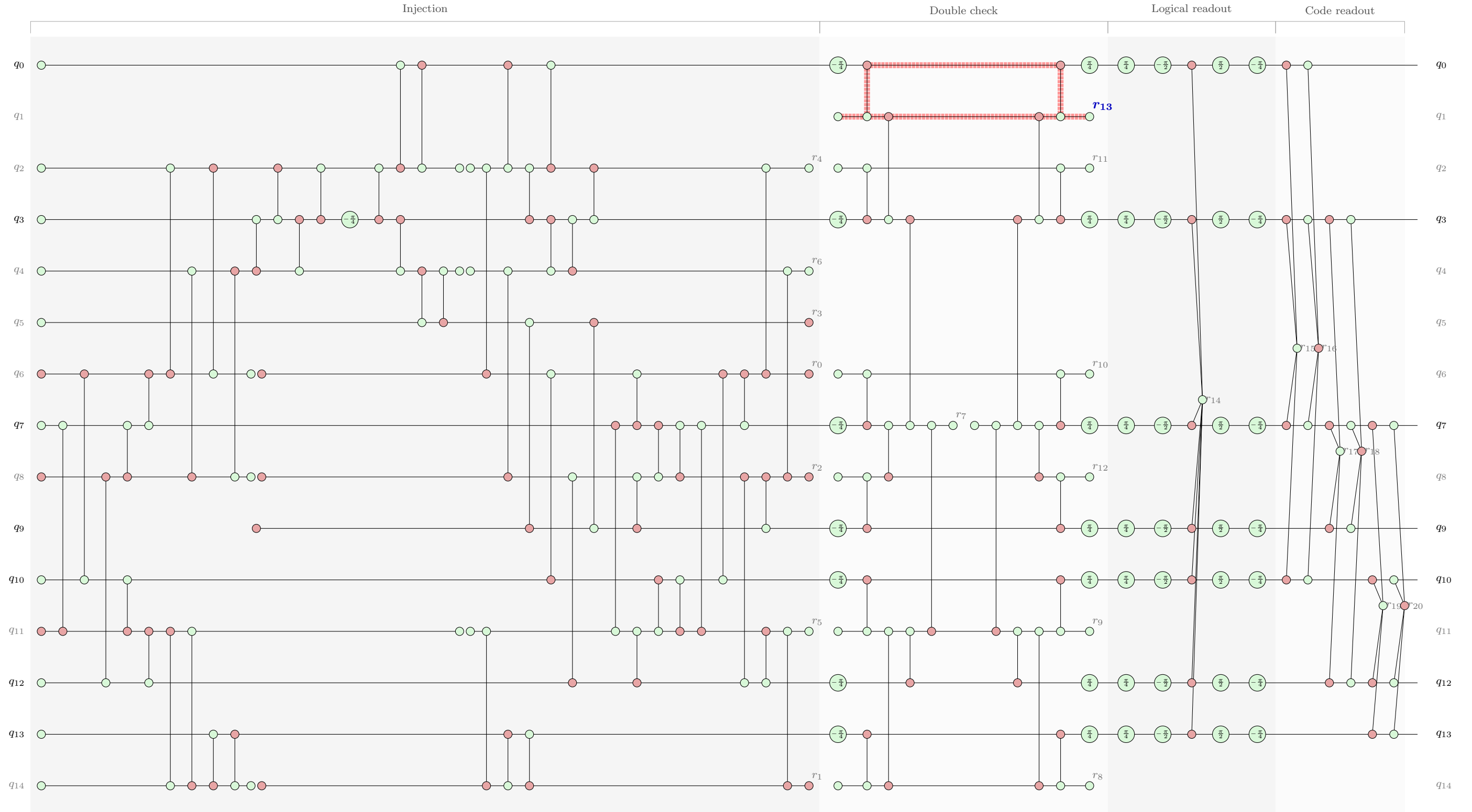


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{12}}$. So $D_{\mathbf{r}} = 0$ unless $r_{12} = 0$.

Records: r_{12} : MX q_8 , line 180.

$d = 3$: detector 13 $r_{13} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

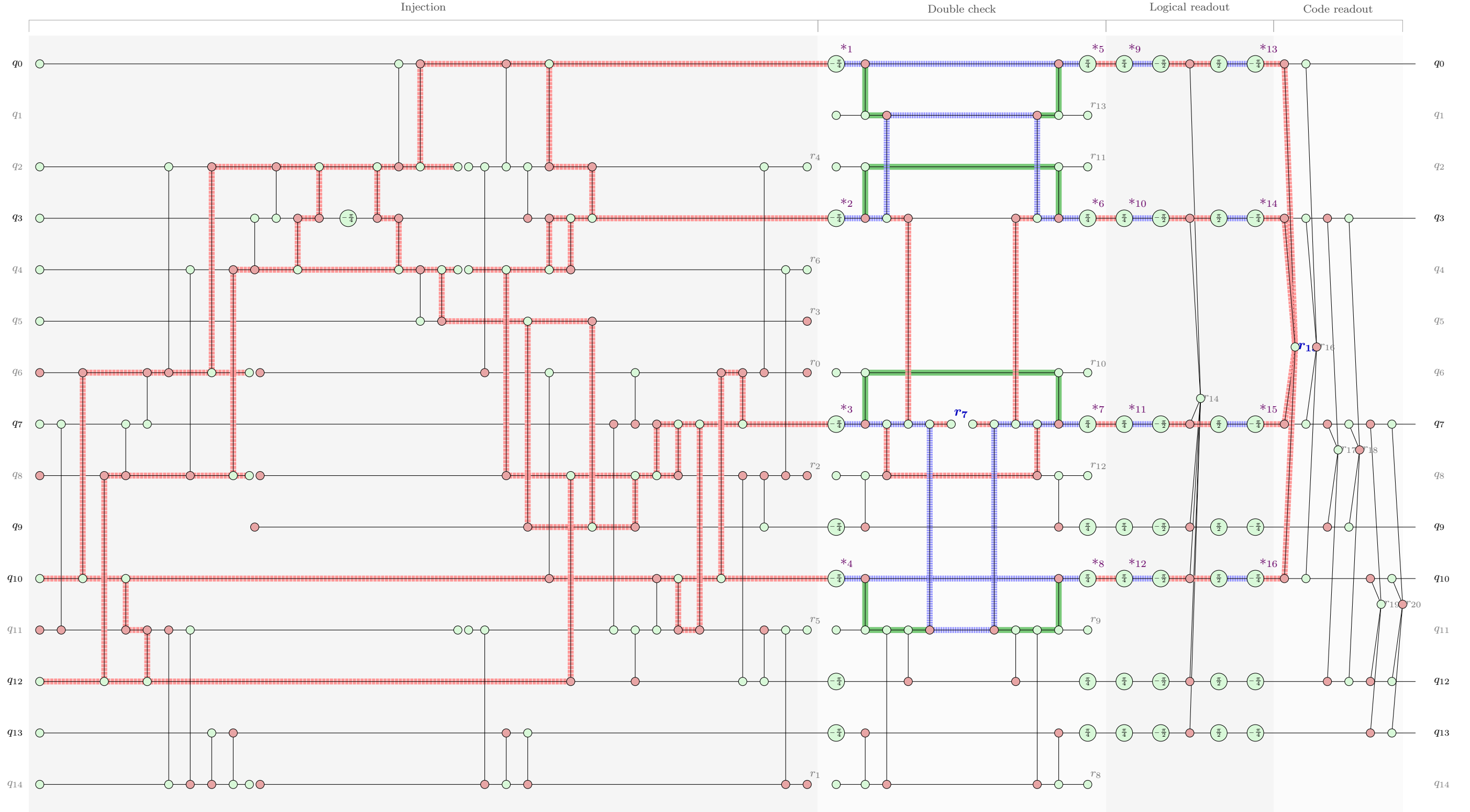


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{13}}.$ So $D_{\mathbf{r}} = 0$ unless $r_{13} = 0$.

Records: r_{13} : MX q_1 , line 180.

$d = 3$: detector 14 $r_7 \oplus r_{15} = 0$

Phase-dependent. Closed semiweb with 16 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



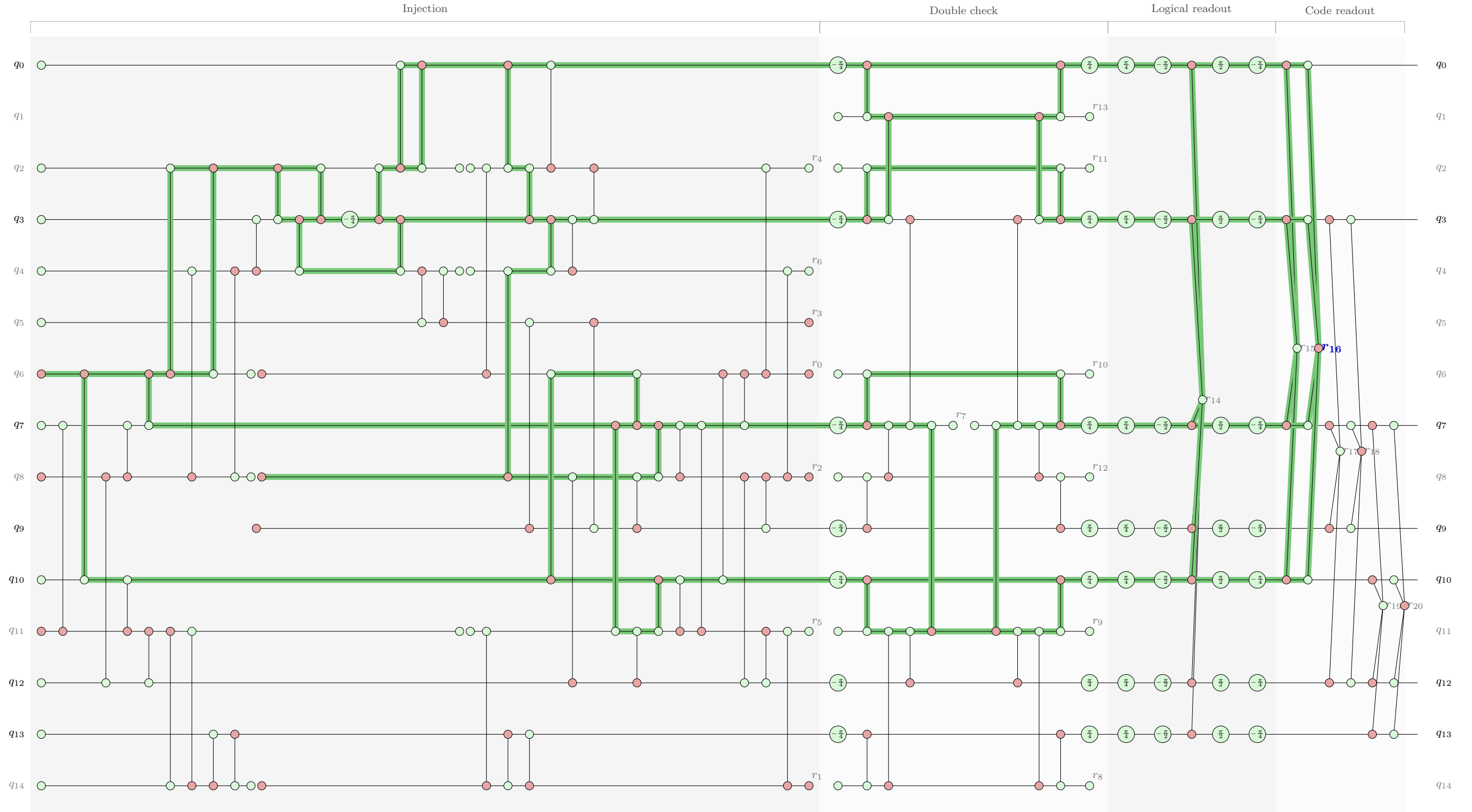
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}, \delta}$, $\lambda(\mathbf{r}) = (-1)^{r_7 + r_{15}}$. $D_{\mathbf{r}, \delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_7 : MX q_7 , line 154; r_{15} : MPP $X_0 X_3 X_7 X_{10}$, line 196.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_0 , line 135; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 135; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 135; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{10} , line 135; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_0 , line 177; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_3 , line 177; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 177; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{10} , line 177; $*_9$: $\delta = \frac{\pi}{2}$ at T on q_0 , line 192; $*_{10}$: $\delta = \frac{\pi}{2}$ at T on q_3 , line 192; $*_{11}$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 192; $*_{12}$: $\delta = \frac{\pi}{2}$ at T on q_{10} , line 192; $*_{13}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_0 , line 194; $*_{14}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 194; $*_{15}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 194; $*_{16}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{10} , line 194.

$d = 3$: detector 15 $r_{16} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

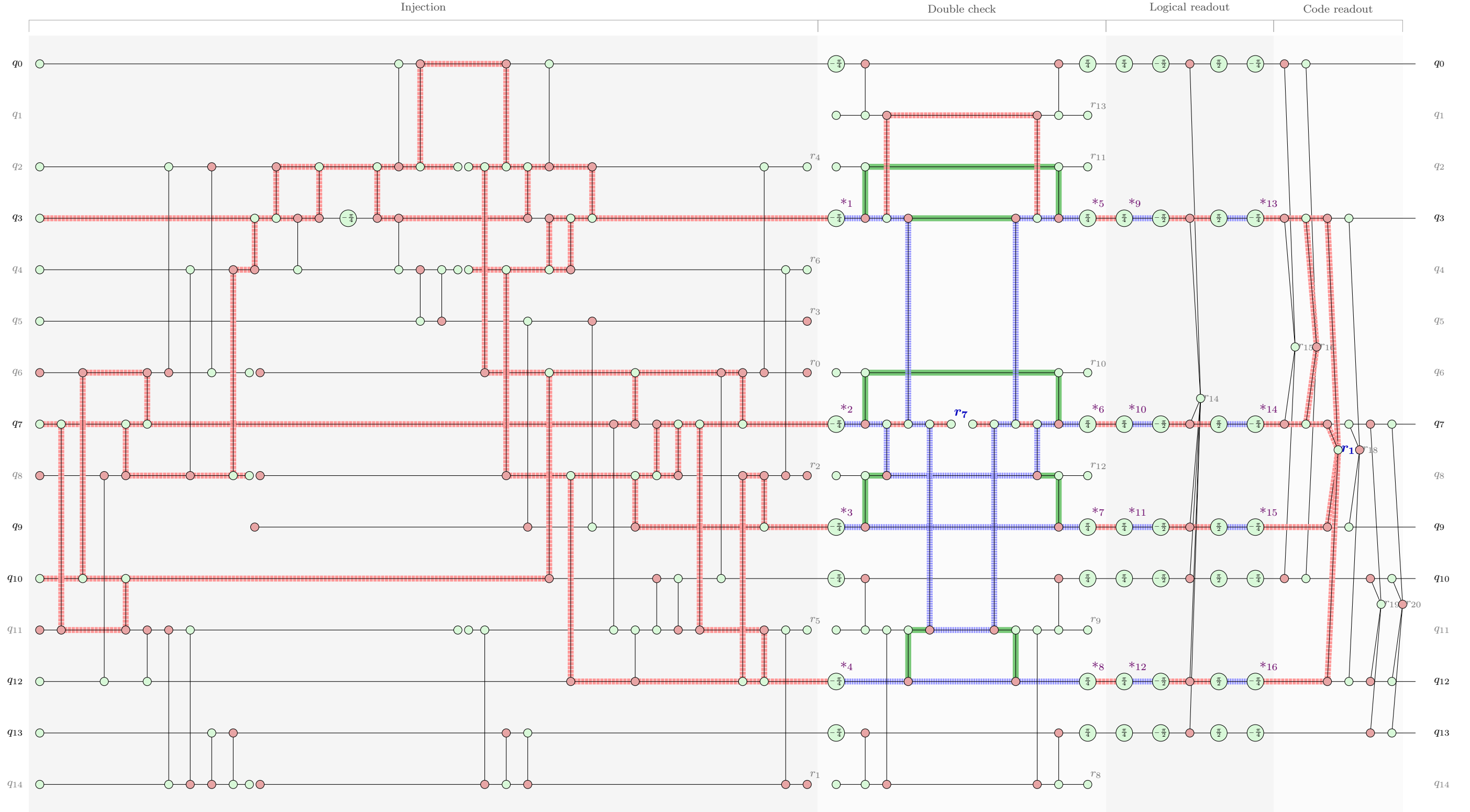


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{16}}$. So $D_{\mathbf{r}} = 0$ unless $r_{16} = 0$.

Records: r_{16} : MPP $Z_0 Z_3 Z_7 Z_{10}$, line 196.

$d = 3$: detector 16 $r_7 \oplus r_{17} = 0$

Phase-dependent. Closed semiweb with 16 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



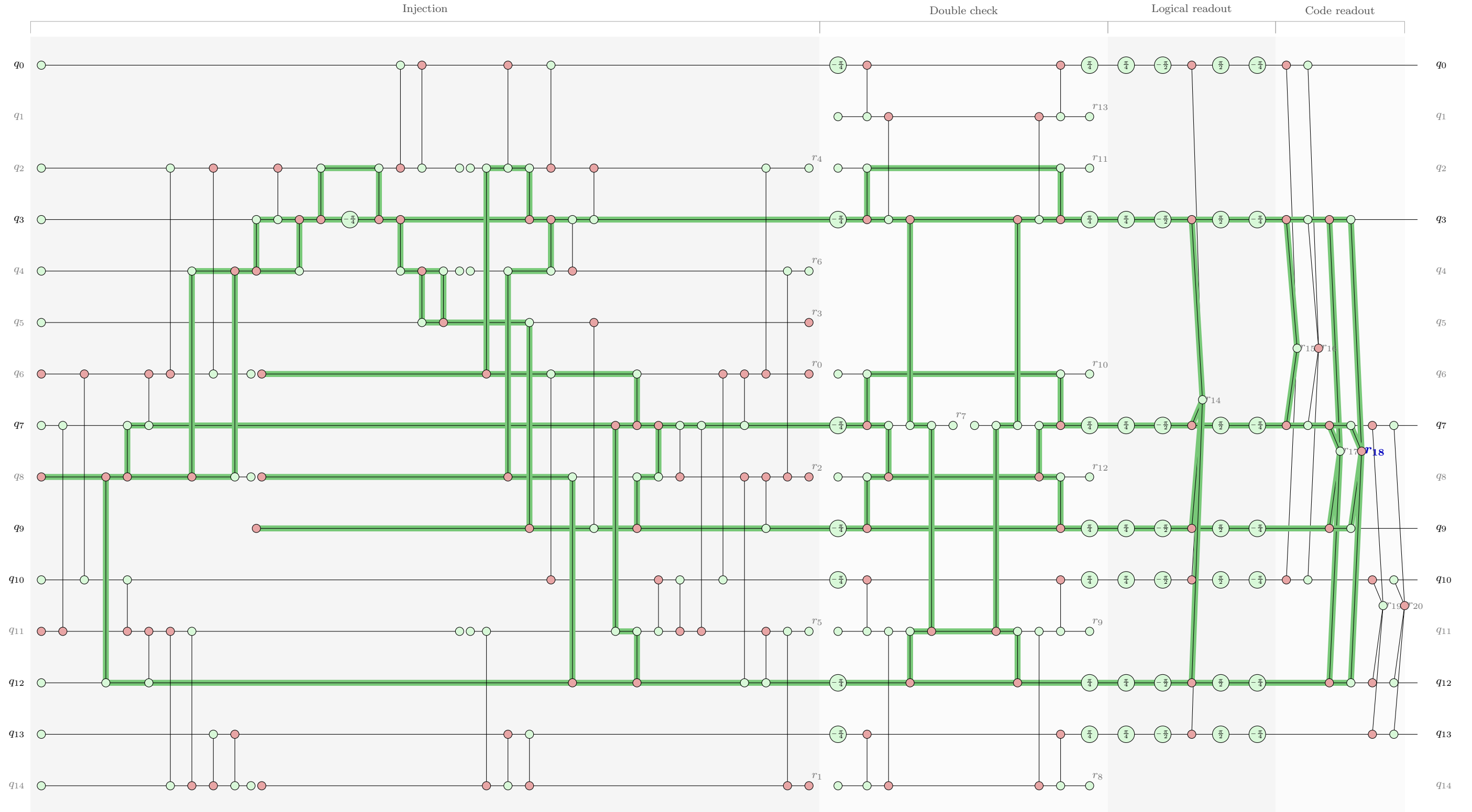
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}, \delta}$, $\lambda(\mathbf{r}) = (-1)^{r_7 + r_{17}}$. $D_{\mathbf{r}, \delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_7 : MX q_7 , line 154; r_{17} : MPP $X_3 X_7 X_9 X_{12}$, line 196.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 135; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 135; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_9 , line 135; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{12} , line 135; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_3 , line 177; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 177; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_9 , line 177; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{12} , line 177; $*_9$: $\delta = \frac{\pi}{2}$ at T on q_3 , line 192; $*_{10}$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 192; $*_{11}$: $\delta = \frac{\pi}{2}$ at T on q_9 , line 192; $*_{12}$: $\delta = \frac{\pi}{2}$ at T on q_{12} , line 192; $*_{13}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 194; $*_{14}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 194; $*_{15}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_9 , line 194; $*_{16}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{12} , line 194.

$d = 3$: detector 17 $r_{18} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

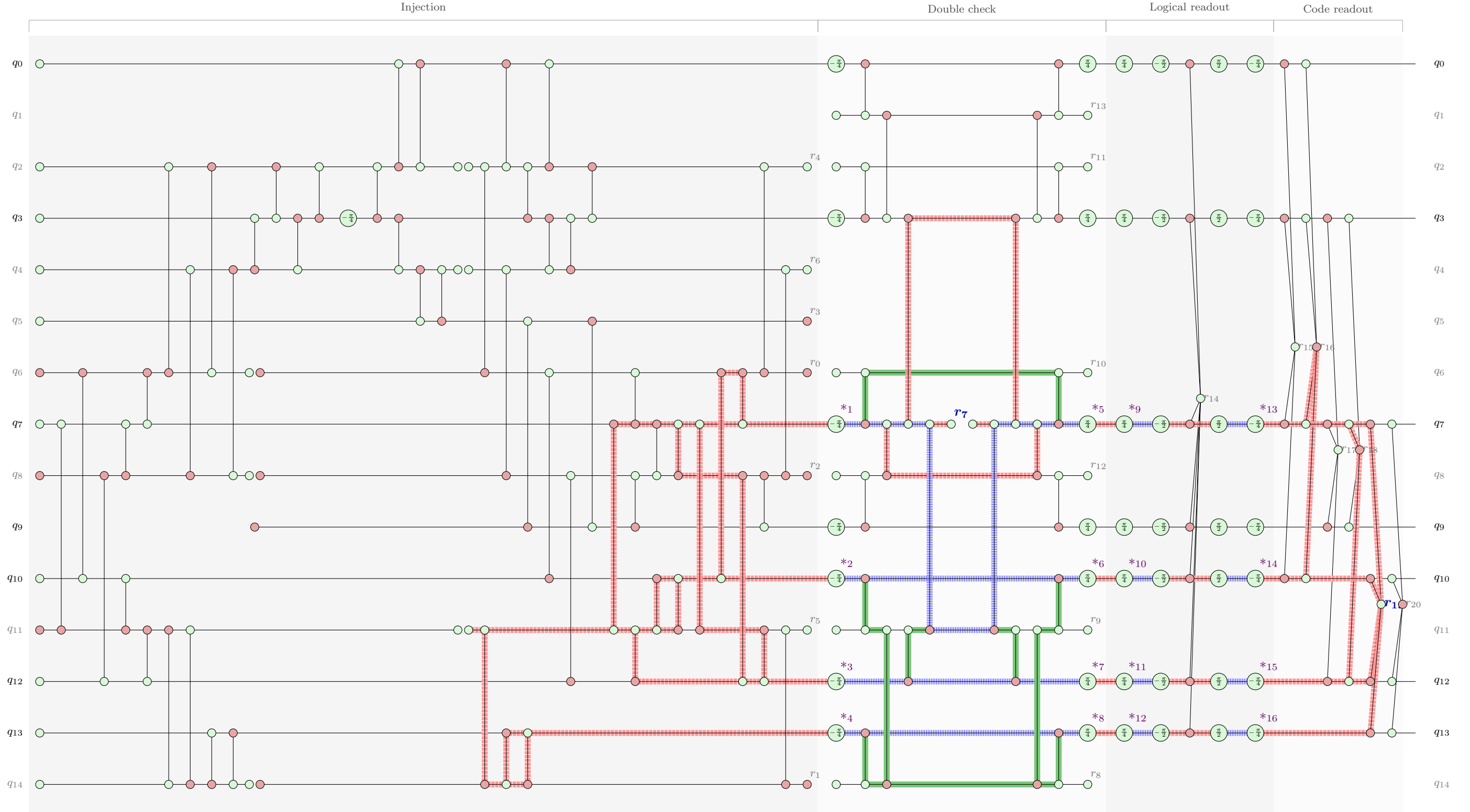


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{18}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{18} = 0.$

Records: r_{18} : MPP $Z_3 Z_7 Z_9 Z_{12}$, line 196.

$d = 3$: detector 18 $r_7 \oplus r_{19} = 0$

Phase-dependent. Closed semiweb with 16 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



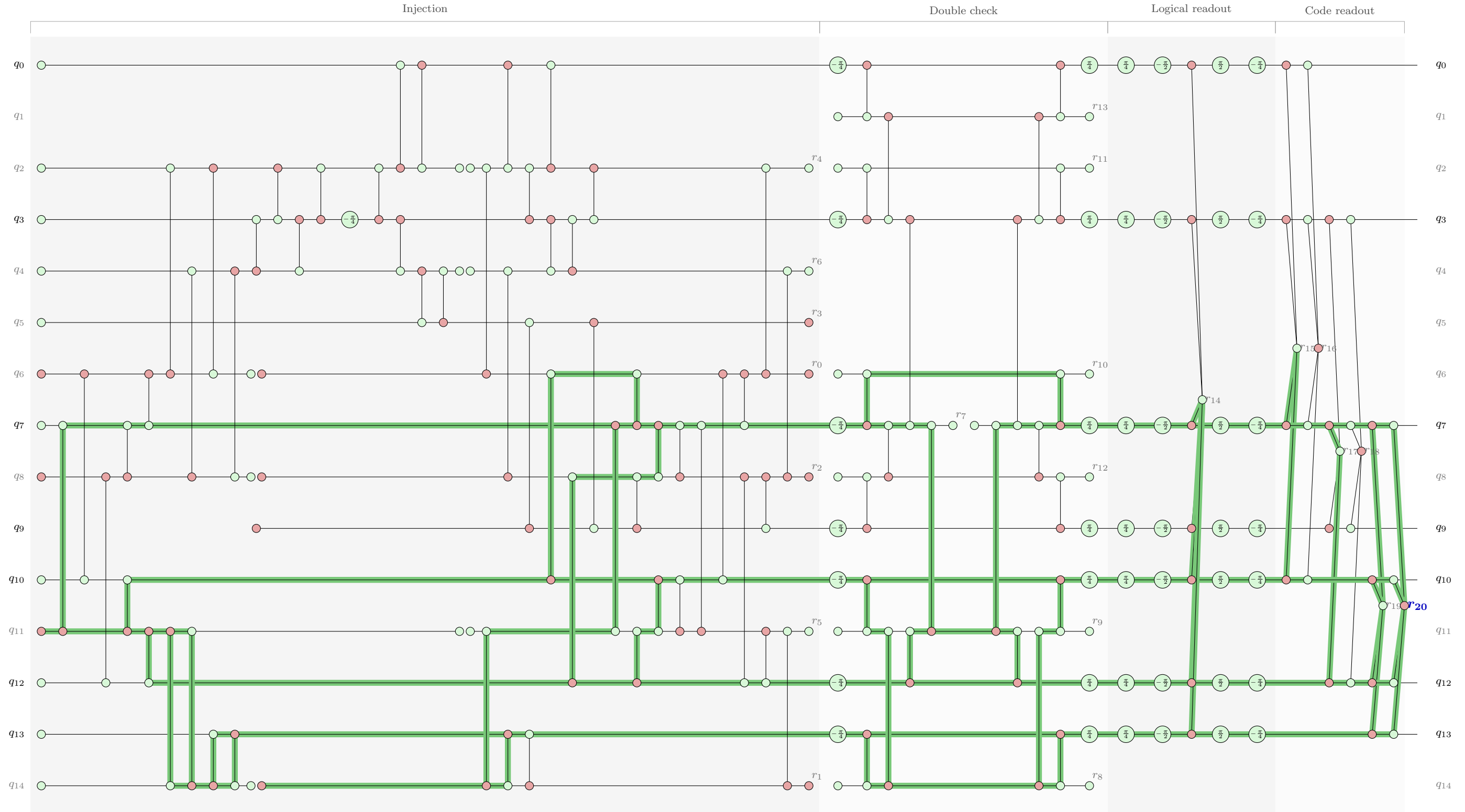
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}, \delta}$, $\lambda(\mathbf{r}) = (-1)^{r_7 + r_{19}}$. $D_{\mathbf{r}, \delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_7 : MX q_7 , line 154; r_{19} : MPP $X_7 X_{10} X_{12} X_{13}$, line 196.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 135; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{10} , line 135; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{12} , line 135; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 135; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 177; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_{10} , line 177; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_{12} , line 177; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{13} , line 177; $*_9$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 192; $*_{10}$: $\delta = \frac{\pi}{2}$ at T on q_{10} , line 192; $*_{11}$: $\delta = \frac{\pi}{2}$ at T on q_{12} , line 192; $*_{12}$: $\delta = \frac{\pi}{2}$ at T on q_{13} , line 192; $*_{13}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 194; $*_{14}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{10} , line 194; $*_{15}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{12} , line 194; $*_{16}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 194.

$d = 3$: detector 19 $r_{20} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

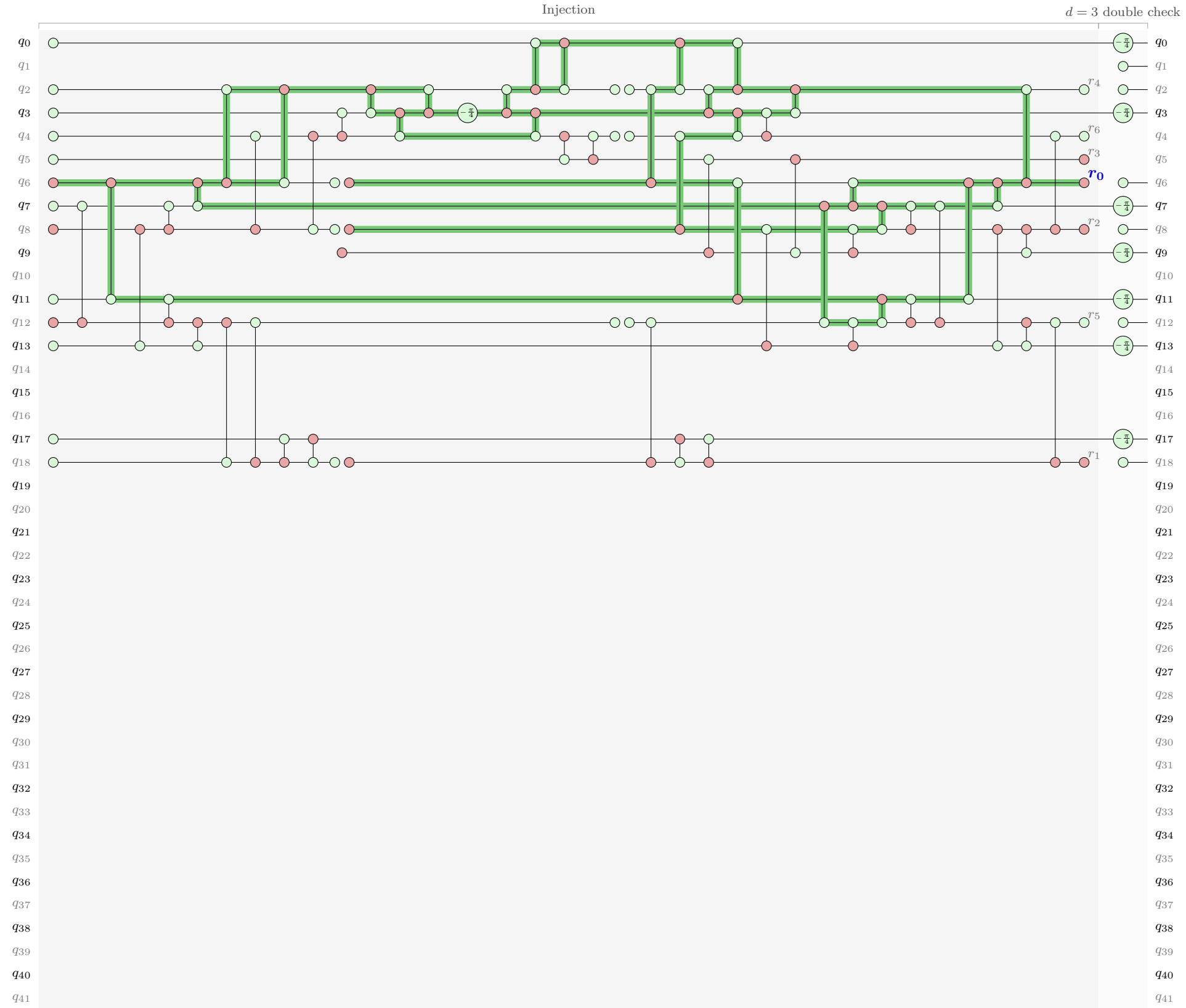


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{20}}$. So $D_{\mathbf{r}} = 0$ unless $r_{20} = 0$.

Records: r_{20} : MPP $Z_7 Z_{10} Z_{12} Z_{13}$, line 196.

$d = 5$: detector 0 $r_0 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

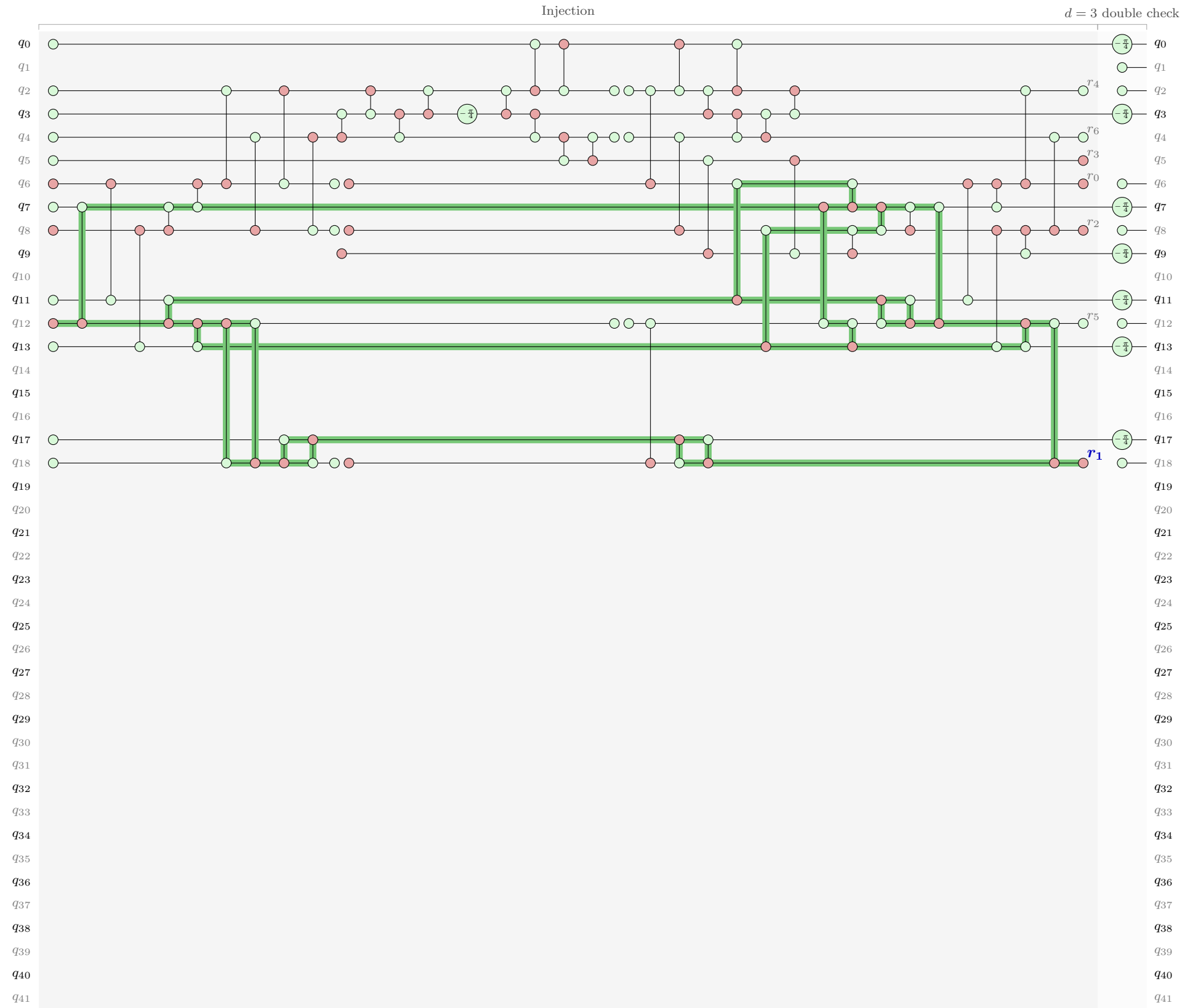


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_0}$. So $D_{\mathbf{r}} = 0$ unless $r_0 = 0$.

Records: r_0 : M q_6 , line 126.

$d = 5$: detector 1 $r_1 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

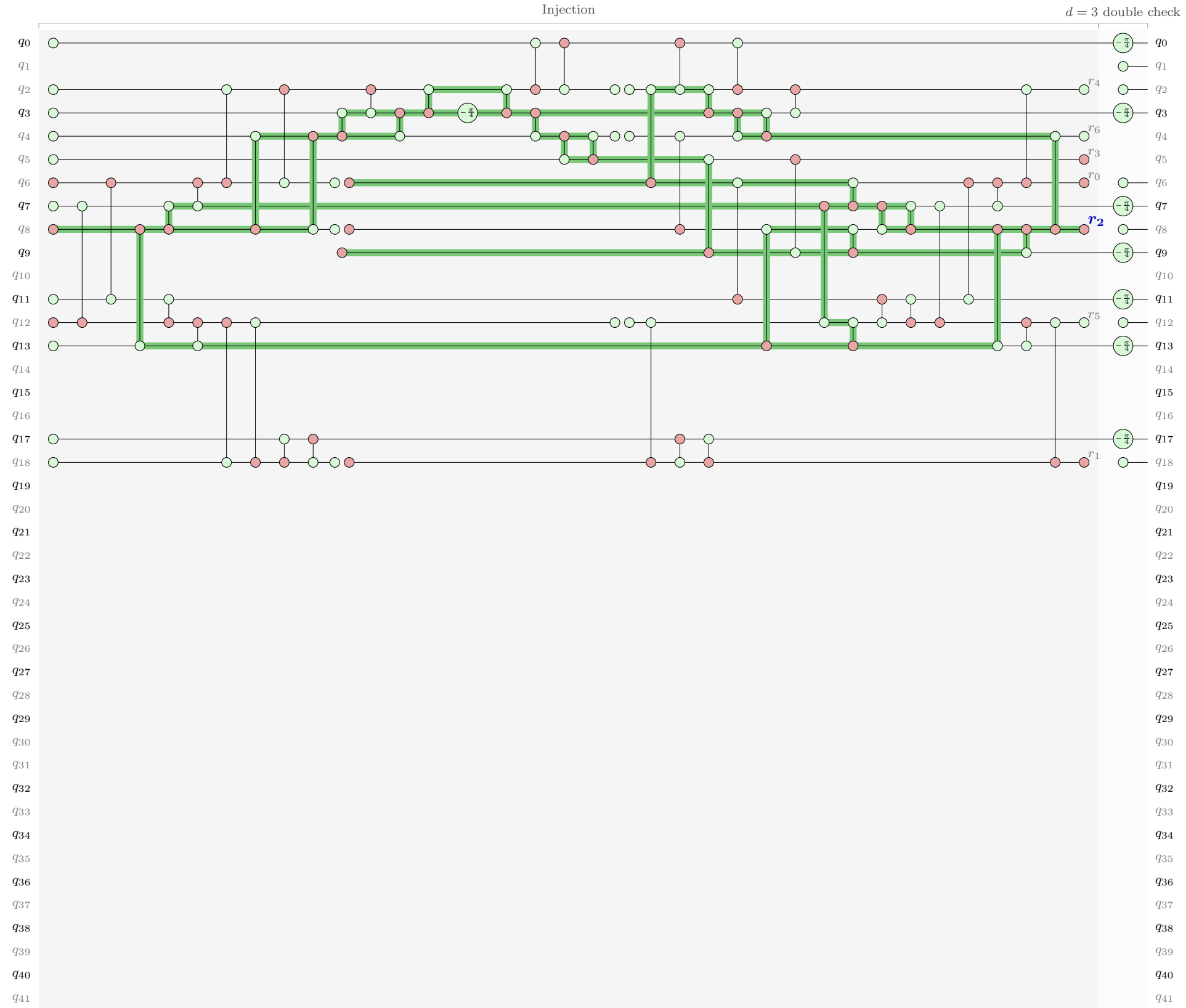


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_1}$. So $D_{\mathbf{r}} = 0$ unless $r_1 = 0$.

Records: r_1 : M q_{18} , line 126.

$d = 5$: detector 2 $r_2 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

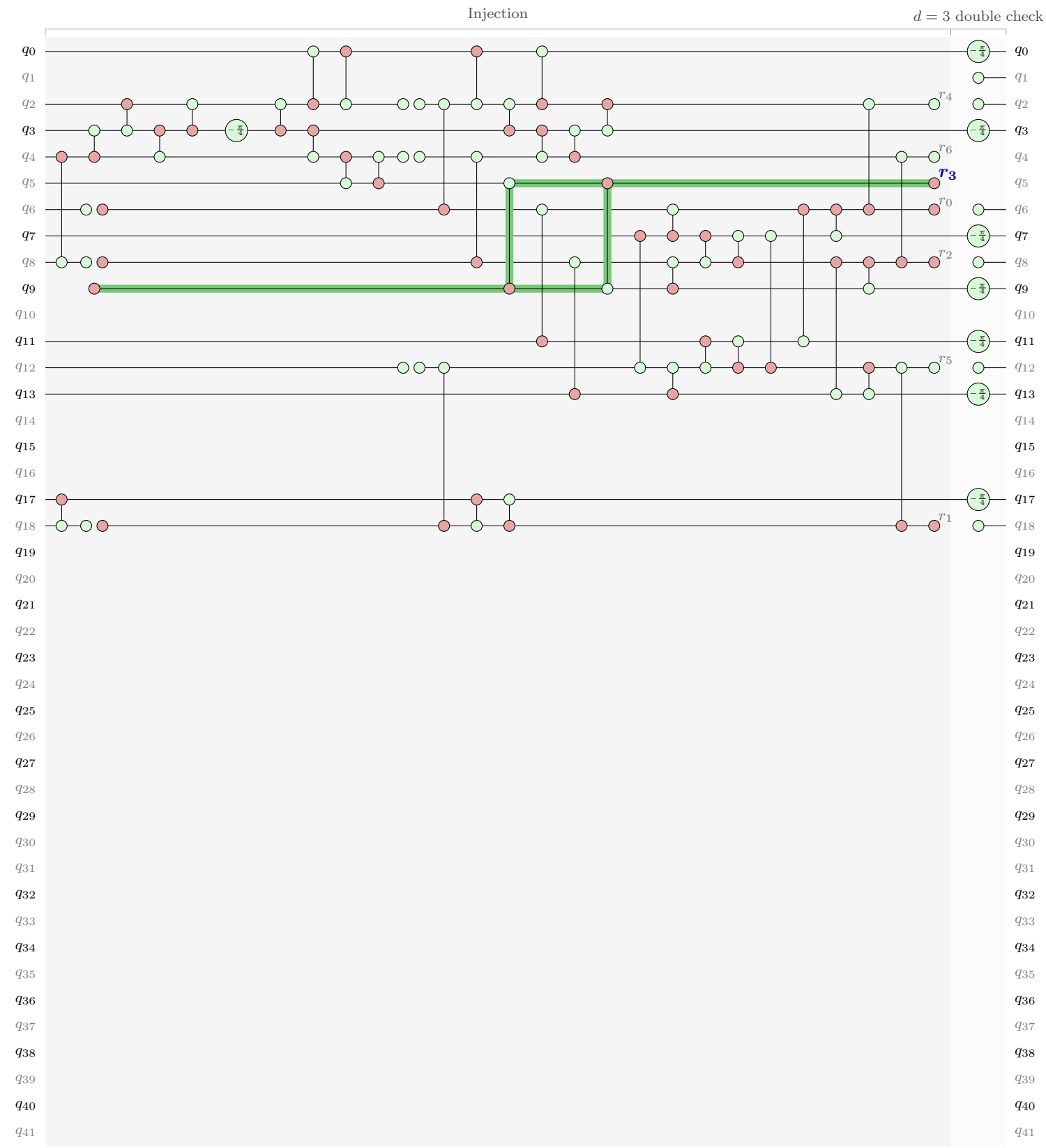


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_2}$. So $D_{\mathbf{r}} = 0$ unless $r_2 = 0$.

Records: r_2 : M q_8 , line 126.

$d = 5$: detector 3 $r_3 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

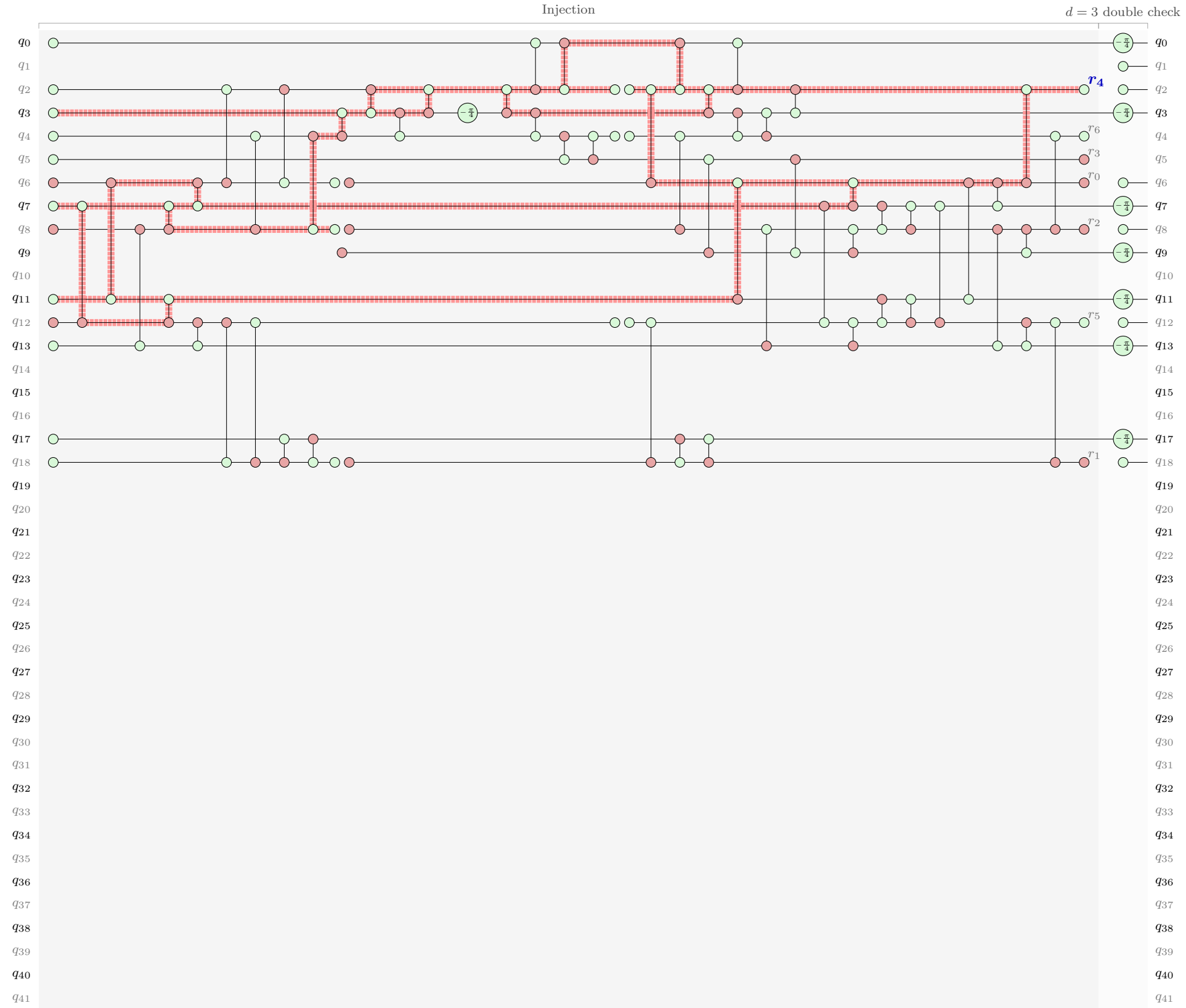


$D_{\boldsymbol{r}} = \lambda(\boldsymbol{r}) D_{\boldsymbol{r}}, \quad \lambda(\boldsymbol{r}) = (-1)^{r_3}. \quad \text{So } D_{\boldsymbol{r}} = 0 \text{ unless } r_3 = 0.$

Records: r_3 : M q_5 , line 126.

$d = 5$: detector 4 $r_4 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

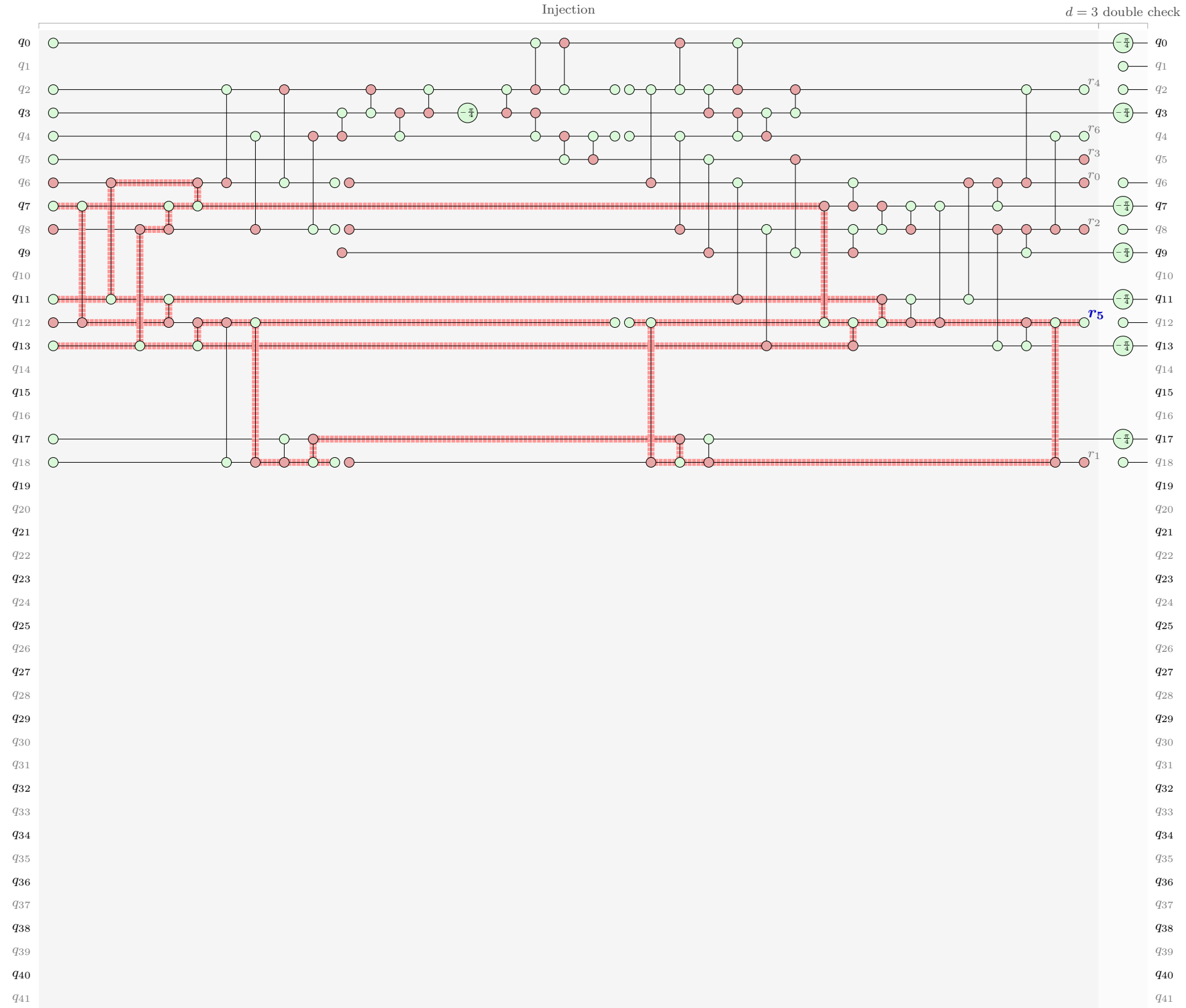


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_4}$. So $D_{\mathbf{r}} = 0$ unless $r_4 = 0$.

Records: r_4 : MX q_2 , line 127.

$d = 5$: detector 5 $r_5 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

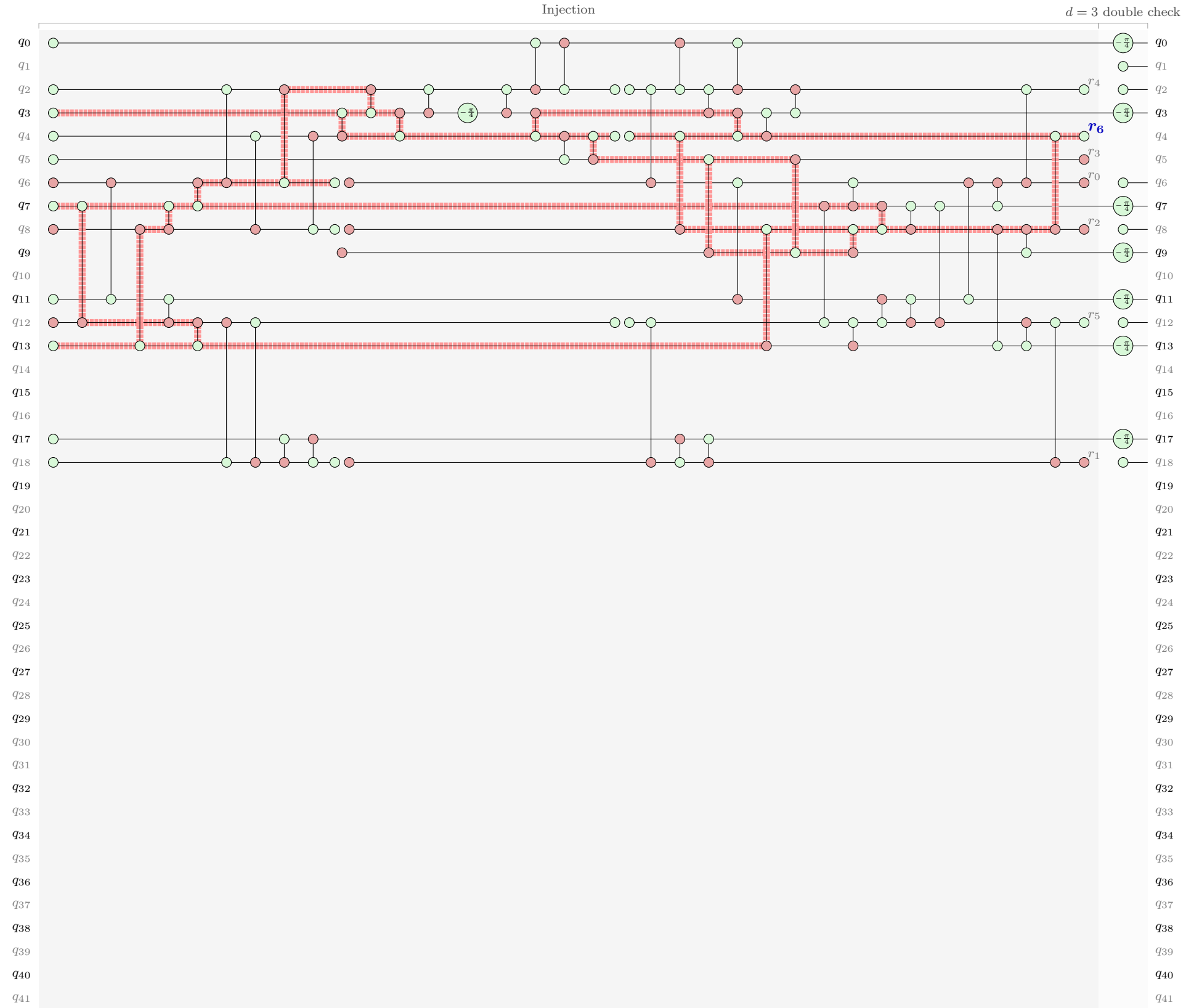


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_5}$. So $D_{\mathbf{r}} = 0$ unless $r_5 = 0$.

Records: r_5 : MX q_{12} , line 127.

$d = 5$: detector 6 $r_6 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

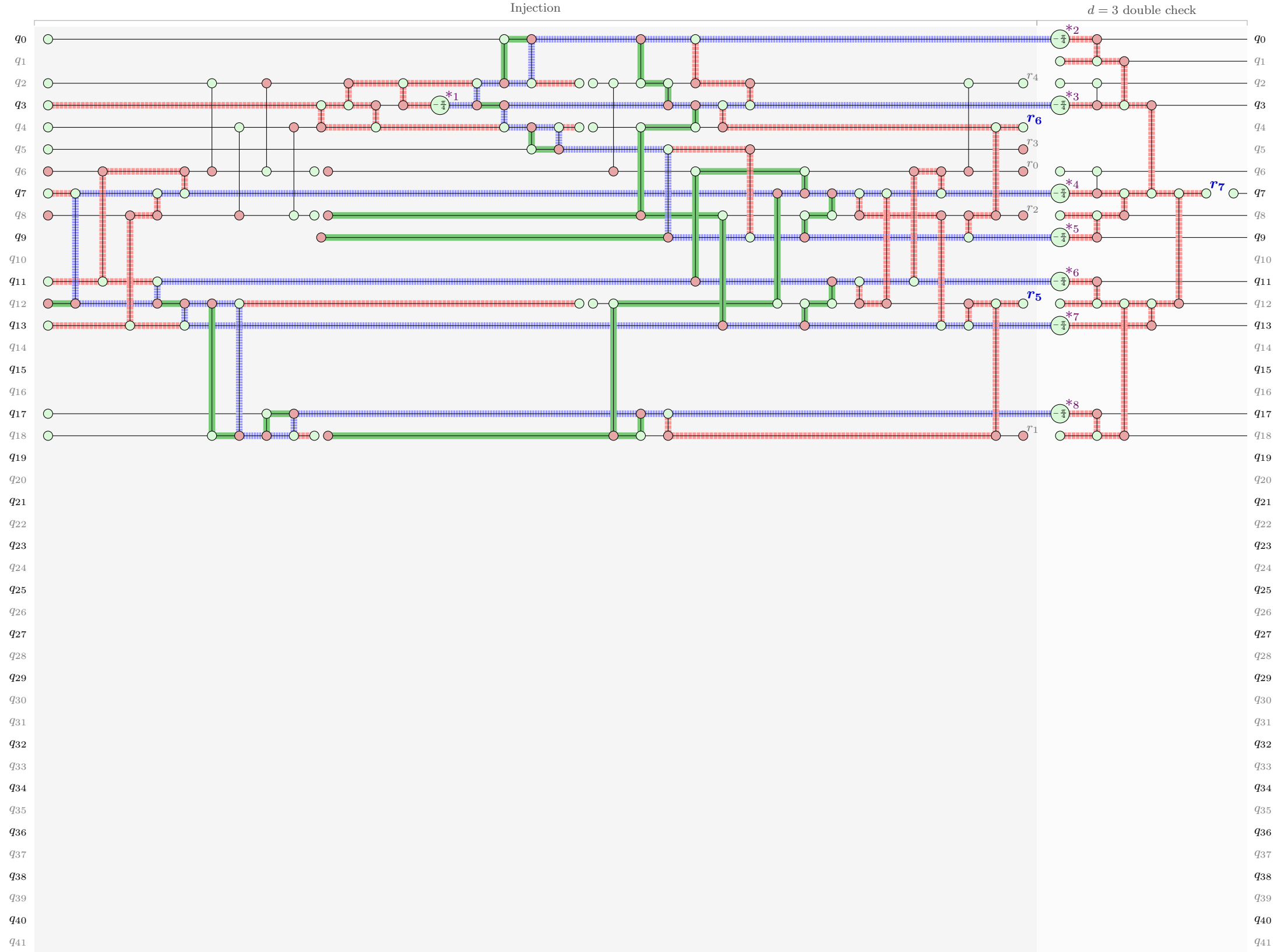


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_6}$. So $D_{\mathbf{r}} = 0$ unless $r_6 = 0$.

Records: r_6 : MX q_4 , line 127.

$d = 5$: detector 7 $r_5 \oplus r_6 \oplus r_7 = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



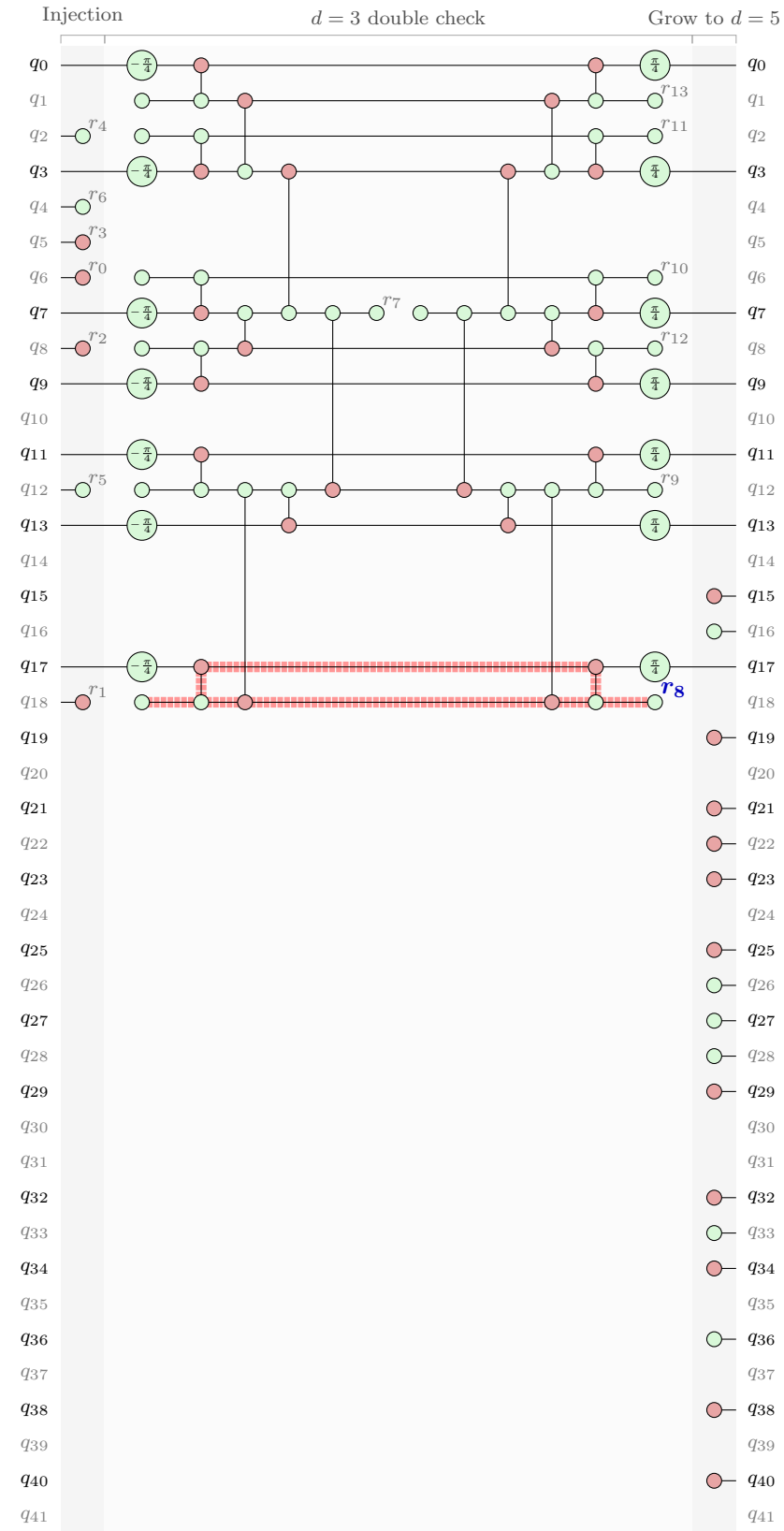
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}, \delta}$, $\lambda(\mathbf{r}) = (-1)^{r_5+r_6+r_7}$. $D_{\mathbf{r}, \delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_5 : MX q_{12} , line 127; r_6 : MX q_4 , line 127; r_7 : MX q_7 , line 157.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 83; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_0 , line 143; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 143; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 143; $*_5$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_9 , line 143; $*_6$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{11} , line 143; $*_7$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 143; $*_8$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{17} , line 143.

$d = 5$: detector 8 $r_8 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

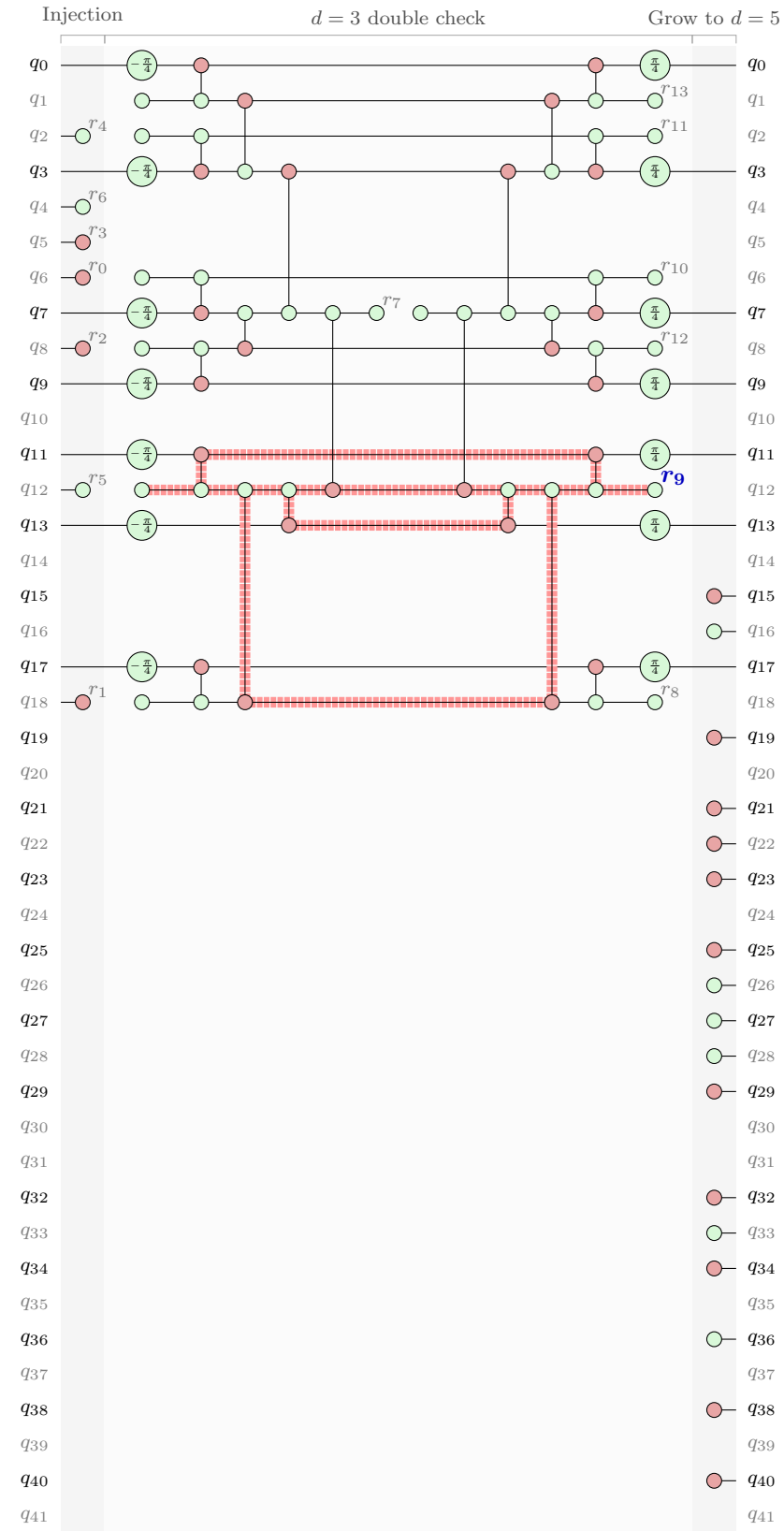


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_8}$. So $D_{\mathbf{r}} = 0$ unless $r_8 = 0$.

Records: r_8 : MX q_{18} , line 176.

$d = 5$: detector 9 $r_9 = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

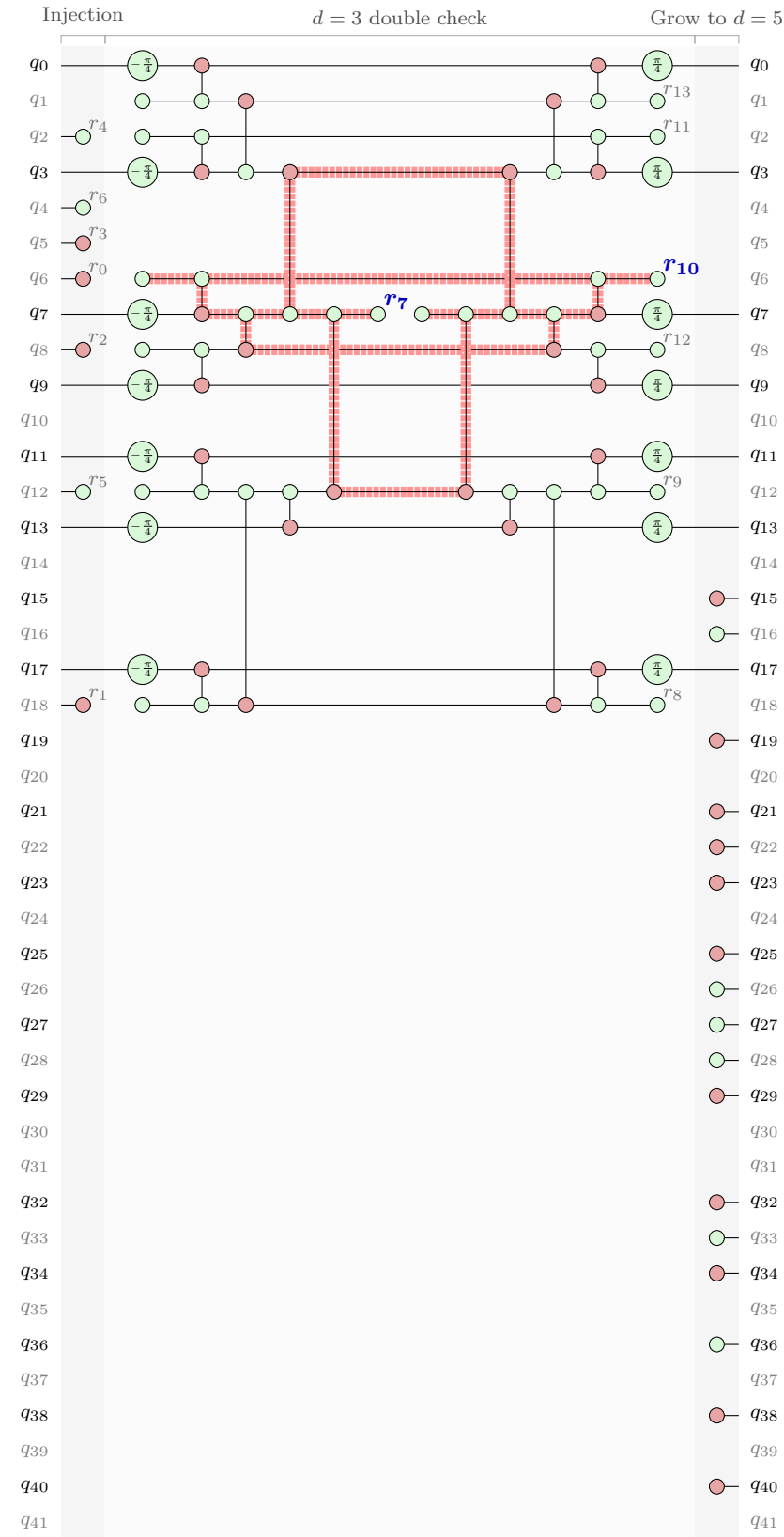


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_9}.$ So $D_{\mathbf{r}} = 0$ unless $r_9 = 0$.

Records: r_9 : MX q_{12} , line 176.

$d = 5$: detector 10 $r_7 \oplus r_{10} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

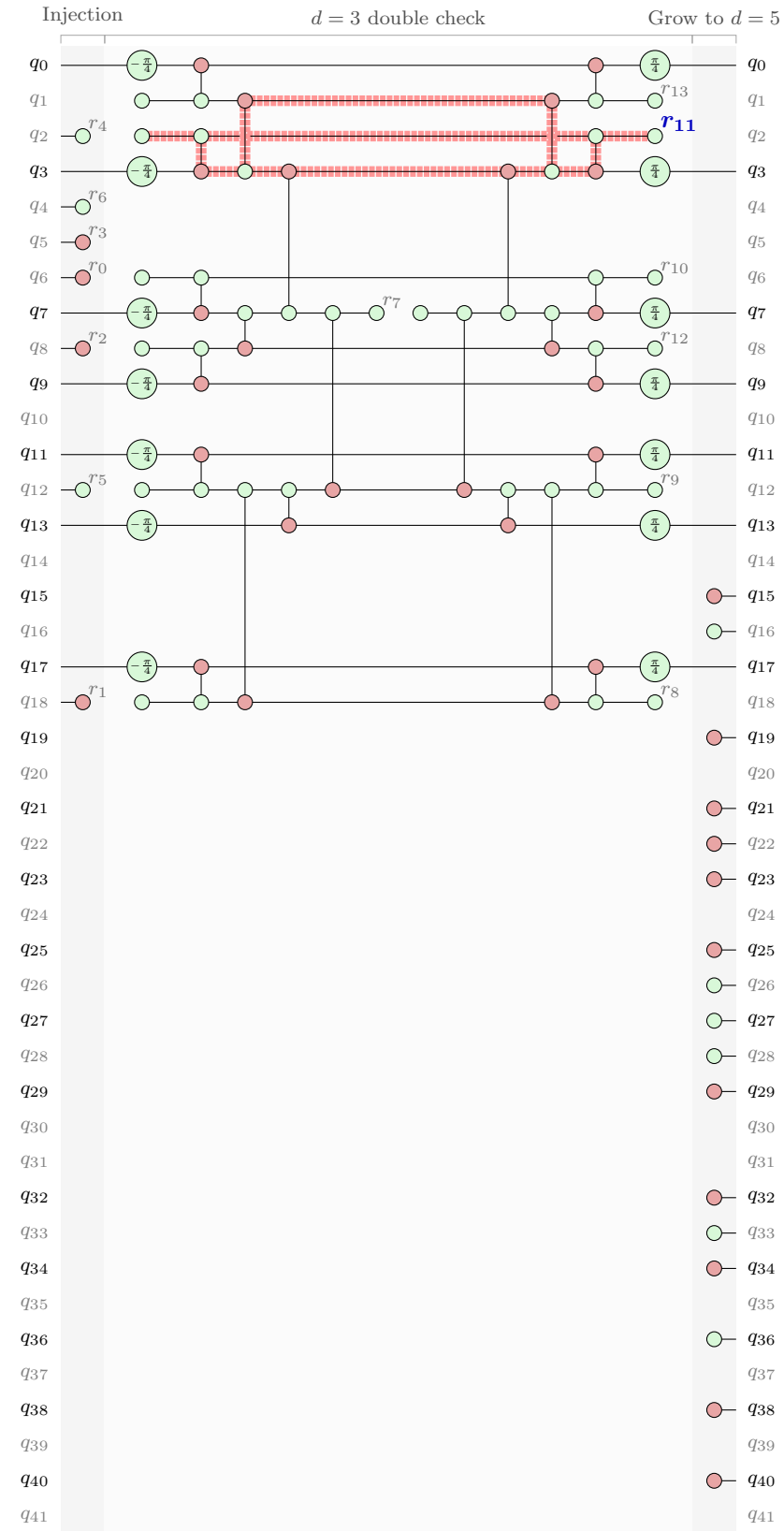


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_7 + r_{10}}$. So $D_{\mathbf{r}} = 0$ unless $r_7 \oplus r_{10} = 0$.

Records: r_7 : MX q_7 , line 157; r_{10} : MX q_6 , line 176.

$d = 5$: detector 11 $r_{11} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

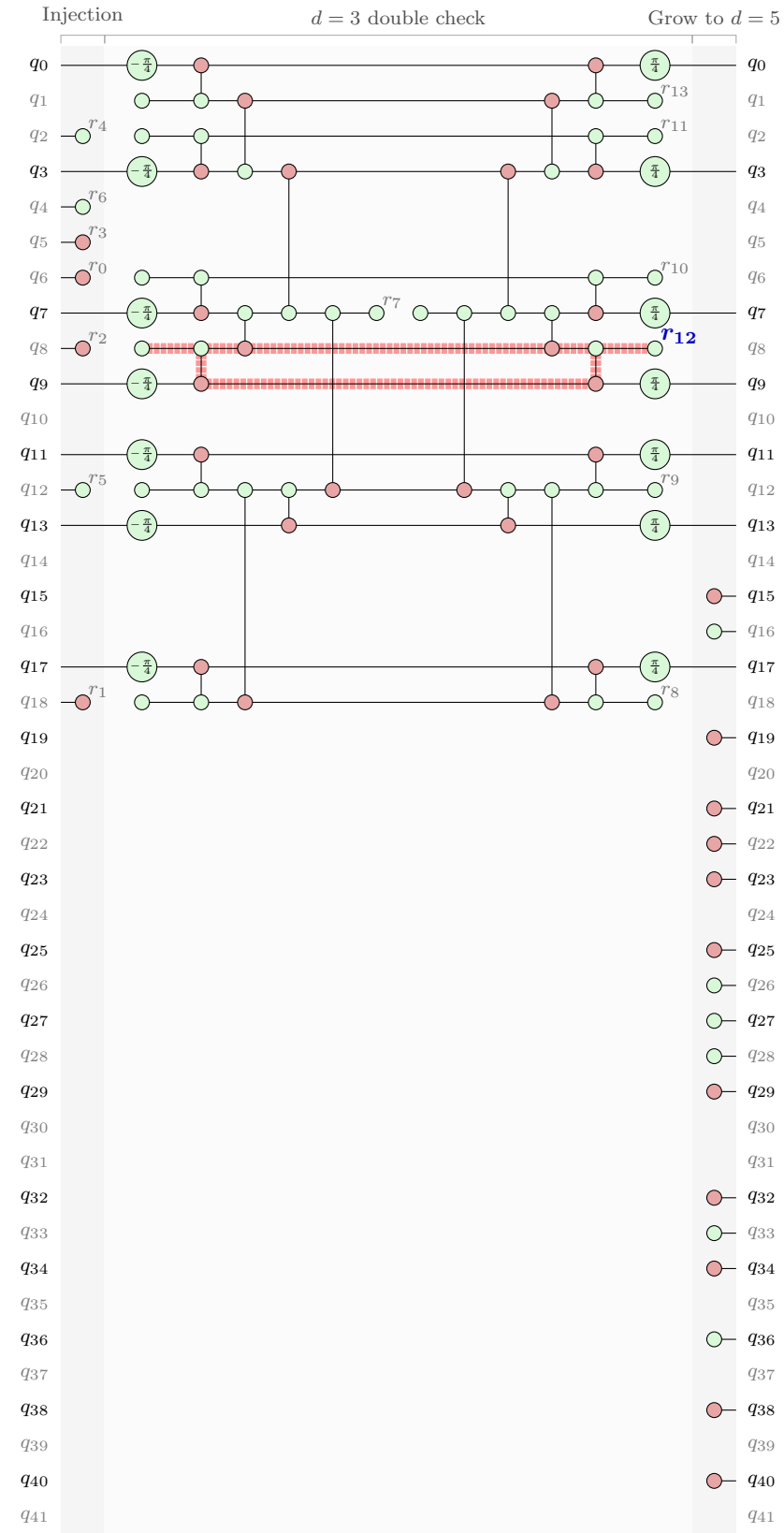


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{11}}$. So $D_{\mathbf{r}} = 0$ unless $r_{11} = 0$.

Records: r_{11} : MX q_2 , line 176.

$d = 5$: detector 12 $r_{12} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

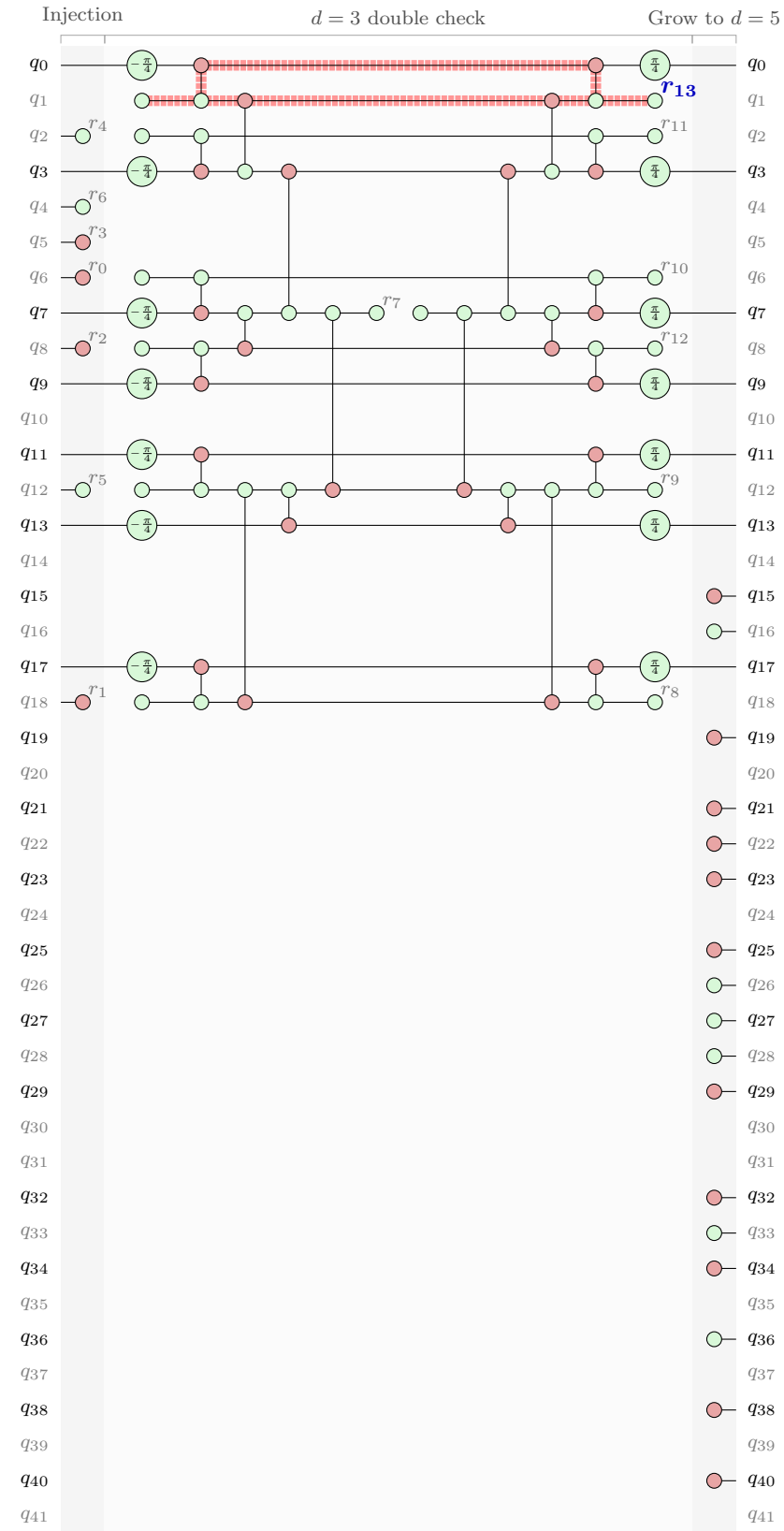


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{12}}$. So $D_{\mathbf{r}} = 0$ unless $r_{12} = 0$.

Records: r_{12} : MX q_8 , line 176.

$d = 5$: detector 13 $r_{13} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

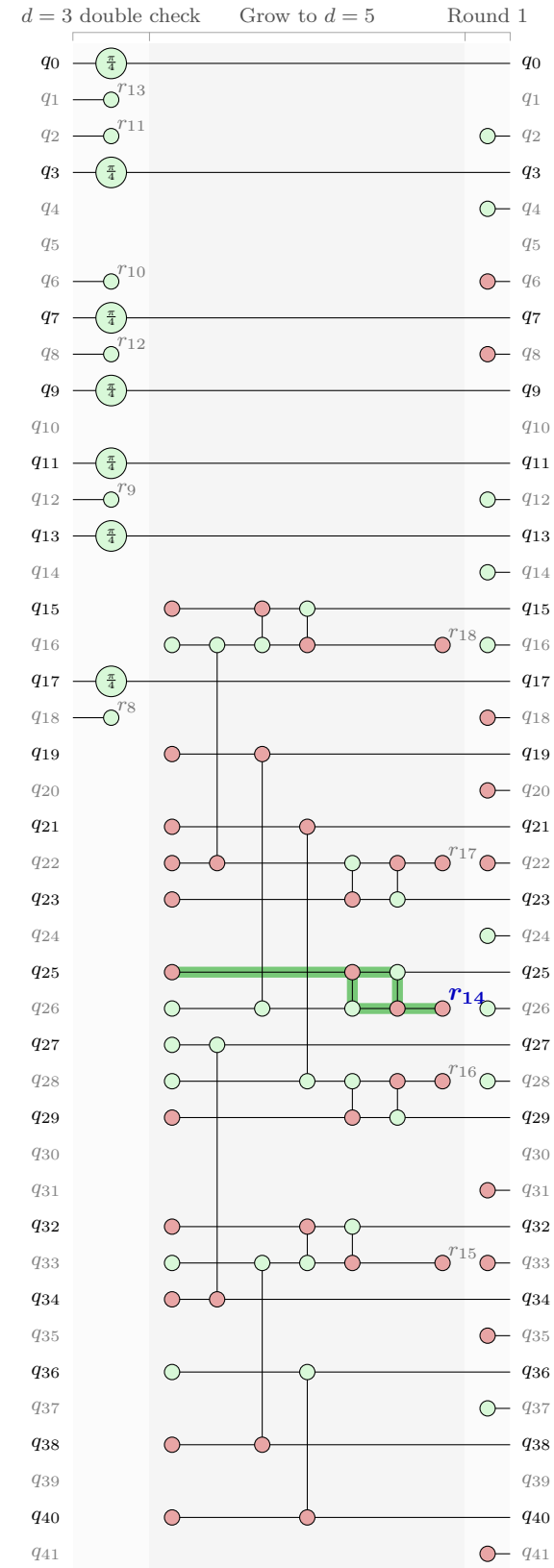


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{13}}$. So $D_{\mathbf{r}} = 0$ unless $r_{13} = 0$.

Records: r_{13} : MX q_1 , line 176.

$d = 5$: detector 14 $r_{14} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

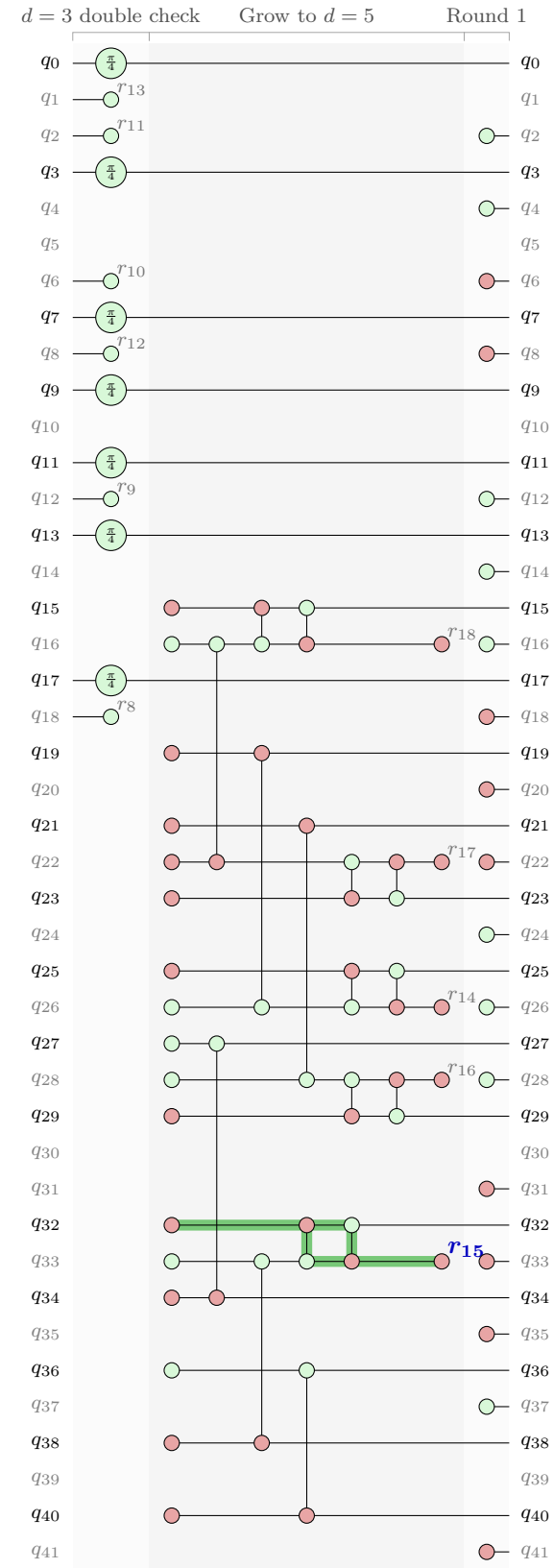


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{14}}$. So $D_{\mathbf{r}} = 0$ unless $r_{14} = 0$.

Records: r_{14} : M q_{26} , line 203.

$d = 5$: detector 15 $r_{15} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

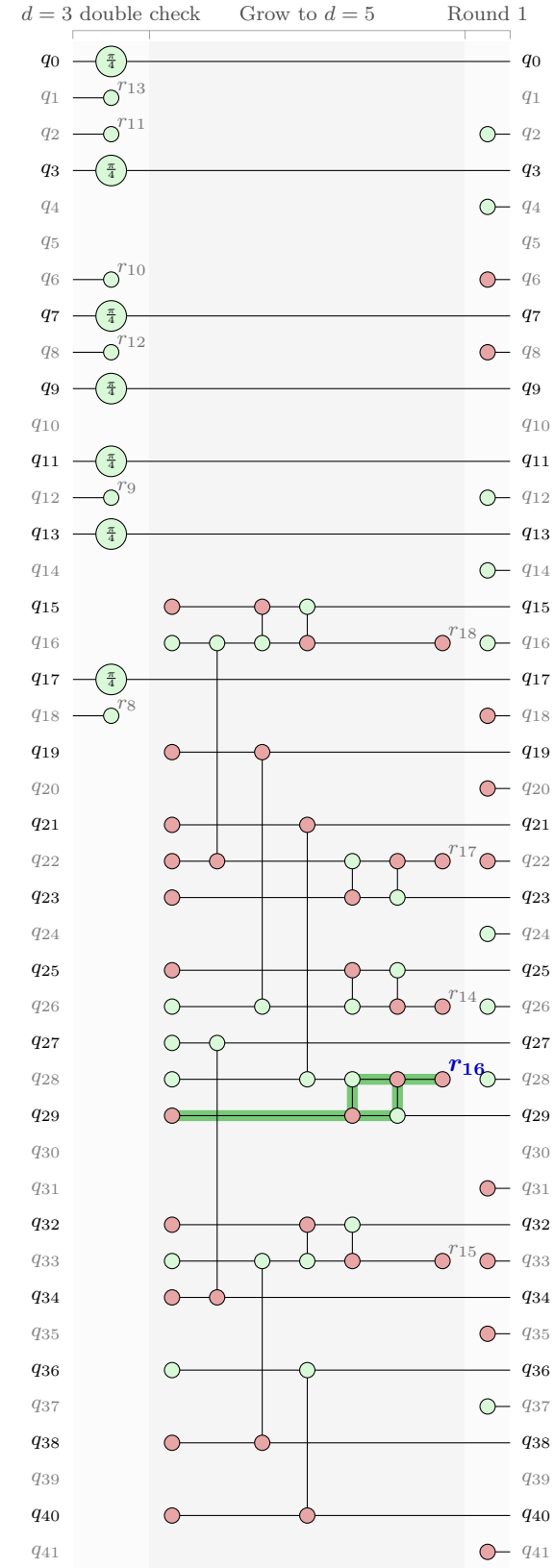


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{15}}$. So $D_{\mathbf{r}} = 0$ unless $r_{15} = 0$.

Records: r_{15} : M q_{33} , line 203.

$d = 5$: detector 16 $r_{16} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

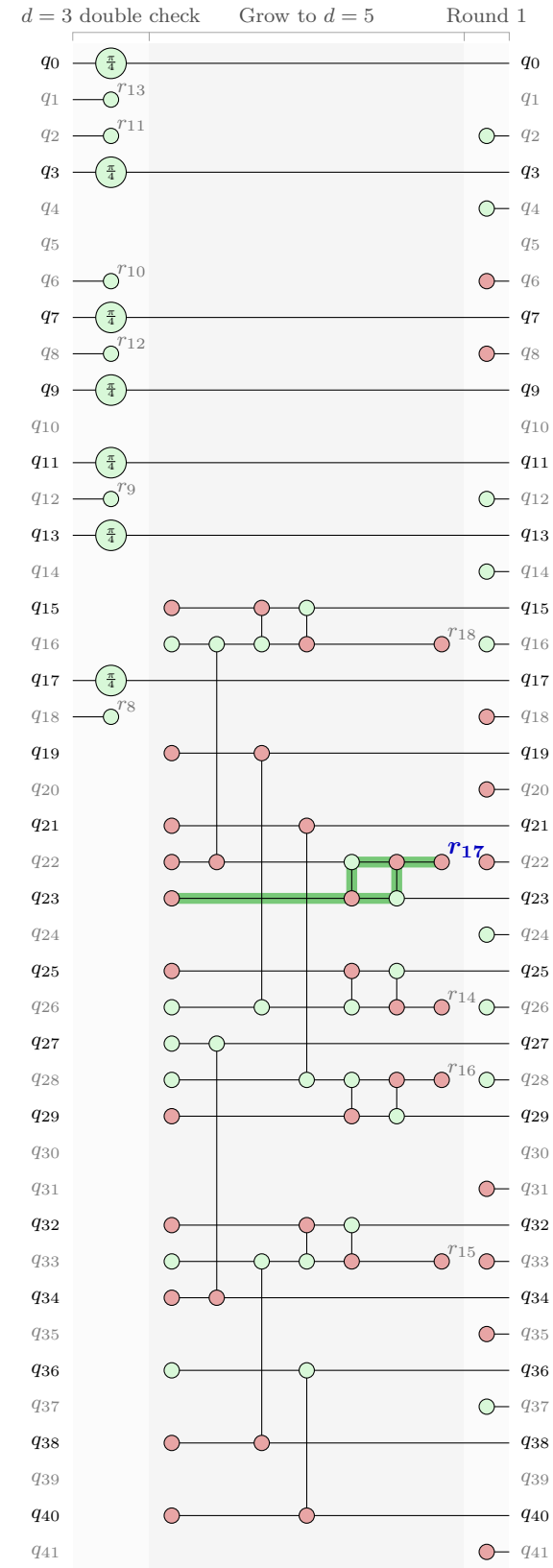


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{16}}$. So $D_{\mathbf{r}} = 0$ unless $r_{16} = 0$.

Records: r_{16} : M q_{28} , line 203.

$d = 5$: detector 17 $r_{17} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

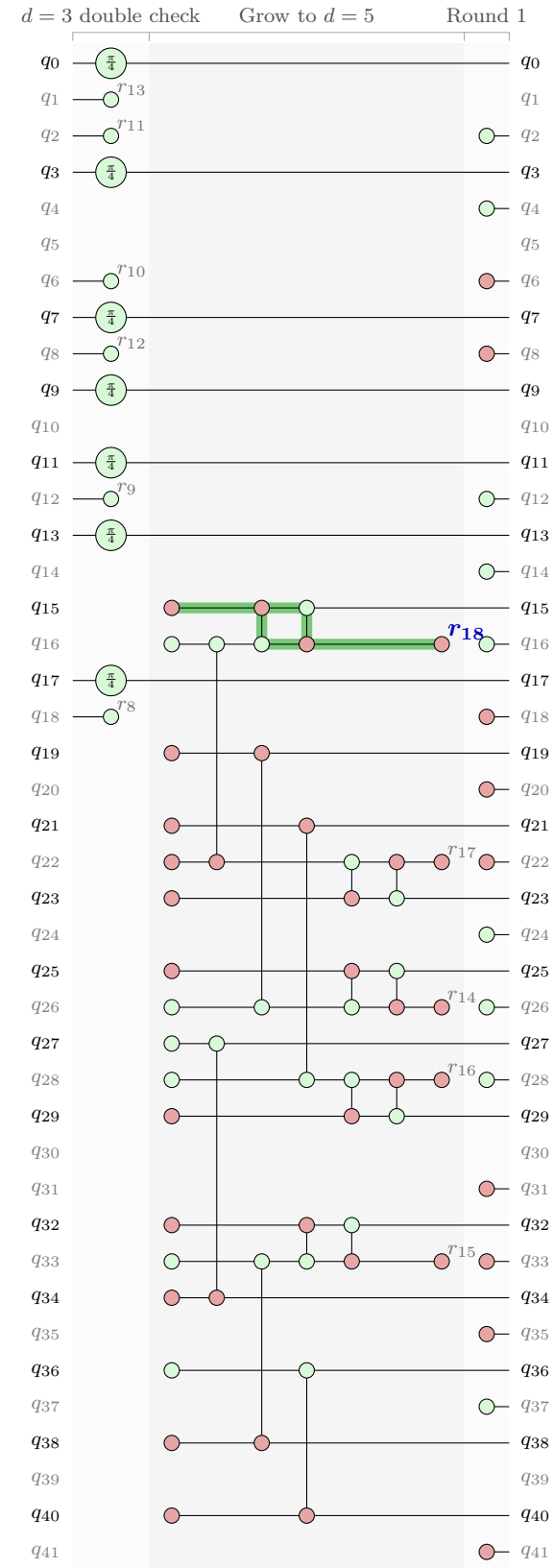


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{17}}$. So $D_{\mathbf{r}} = 0$ unless $r_{17} = 0$.

Records: r_{17} : M q_{22} , line 203.

$d = 5$: detector 18 $r_{18} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

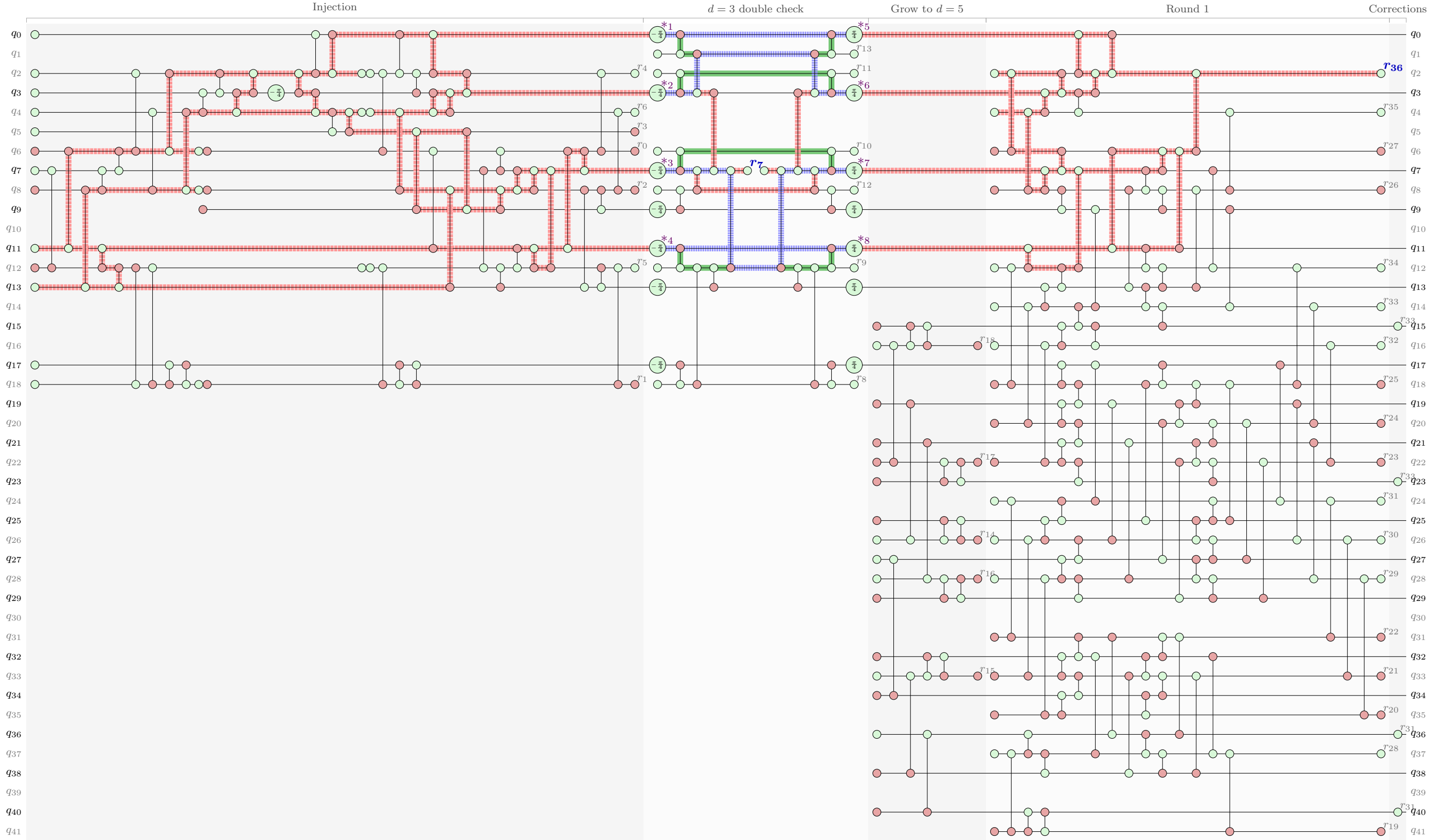


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{18}}$. So $D_{\mathbf{r}} = 0$ unless $r_{18} = 0$.

Records: r_{18} : M q_{16} , line 203.

$d = 5$: detector 19 $r_7 \oplus r_{36} = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



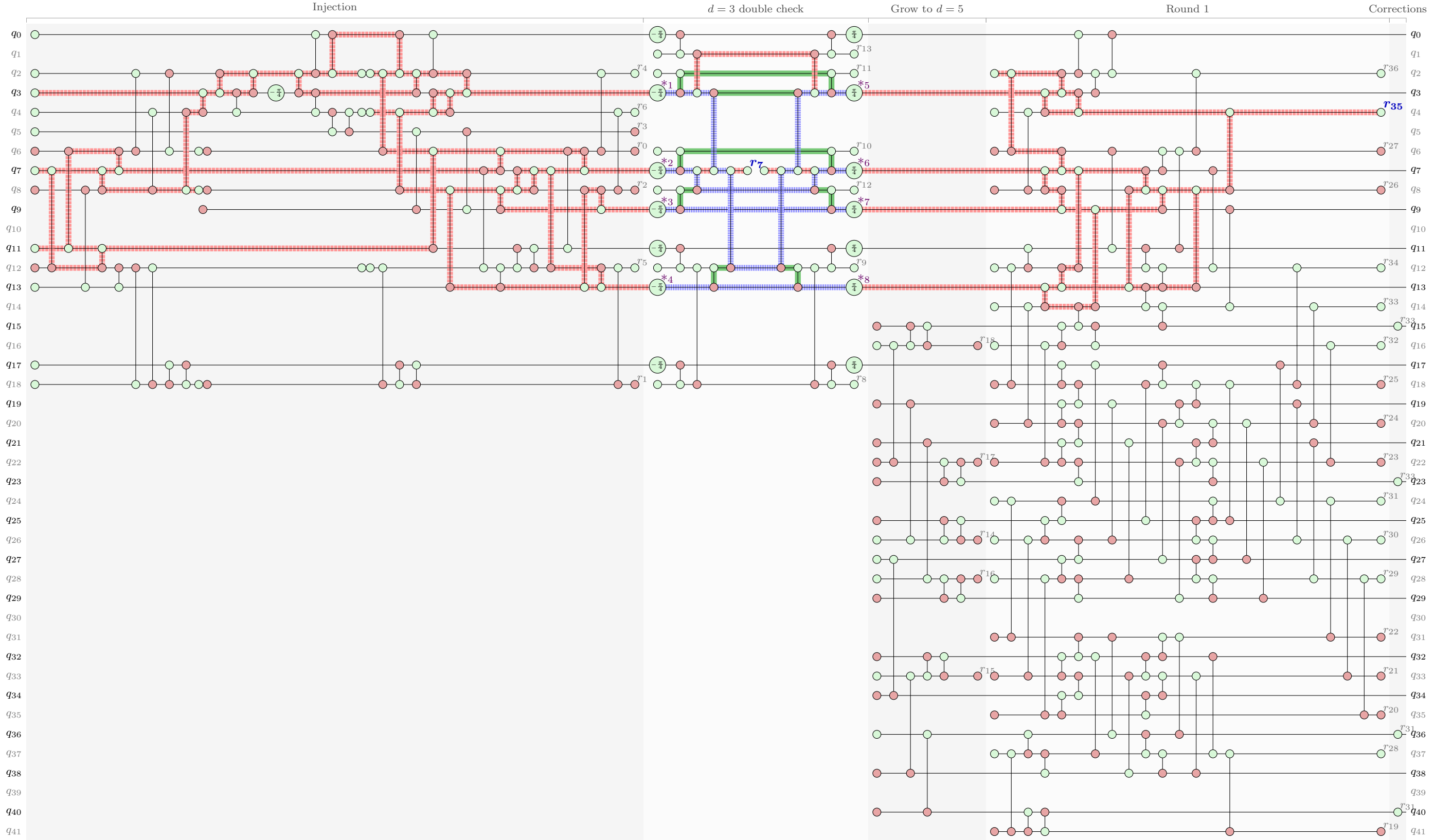
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}, \delta}$, $\lambda(\mathbf{r}) = (-1)^{r_7 + r_{36}}$. $D_{\mathbf{r}, \delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_7 : MX q_7 , line 157; r_{36} : MX q_2 , line 251.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_0 , line 143; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 143; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 143; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{11} , line 143; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_0 , line 174; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_3 , line 174; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 174; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{11} , line 174.

$d = 5$: detector 20 $r_7 \oplus r_{35} = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



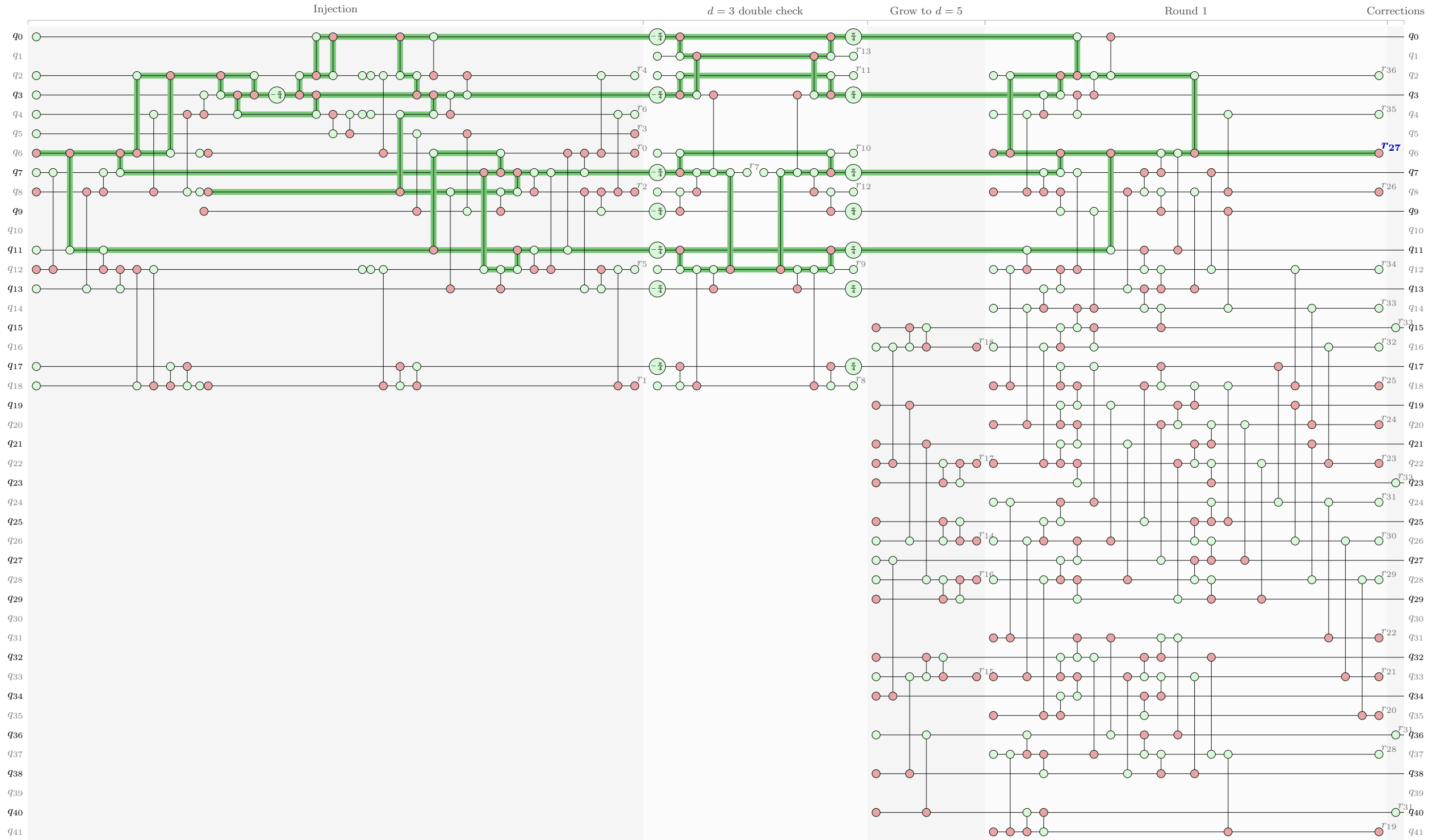
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}, \delta}$, $\lambda(\mathbf{r}) = (-1)^{r_7 + r_{35}}$. $D_{\mathbf{r}, \delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_7 : MX q_7 , line 157; r_{35} : MX q_4 , line 251.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 143; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 143; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_9 , line 143; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 143; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_3 , line 174; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 174; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_9 , line 174; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{13} , line 174.

$d = 5$: detector 21 $r_{27} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

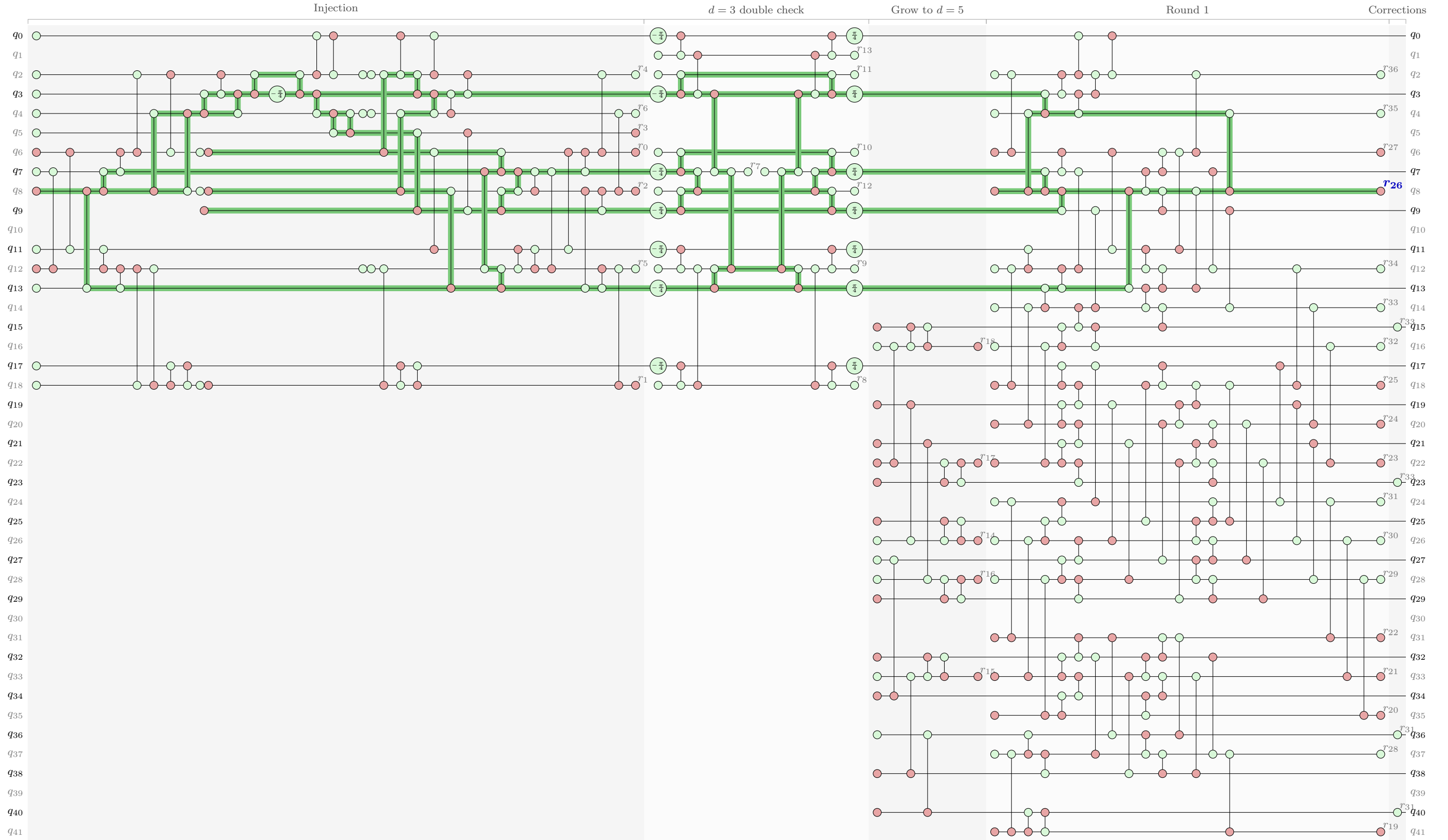


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{27}}$. So $D_{\mathbf{r}} = 0$ unless $r_{27} = 0$.

Records: r_{27} : M q_6 , line 250.

$d = 5$: detector 22 $r_{26} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

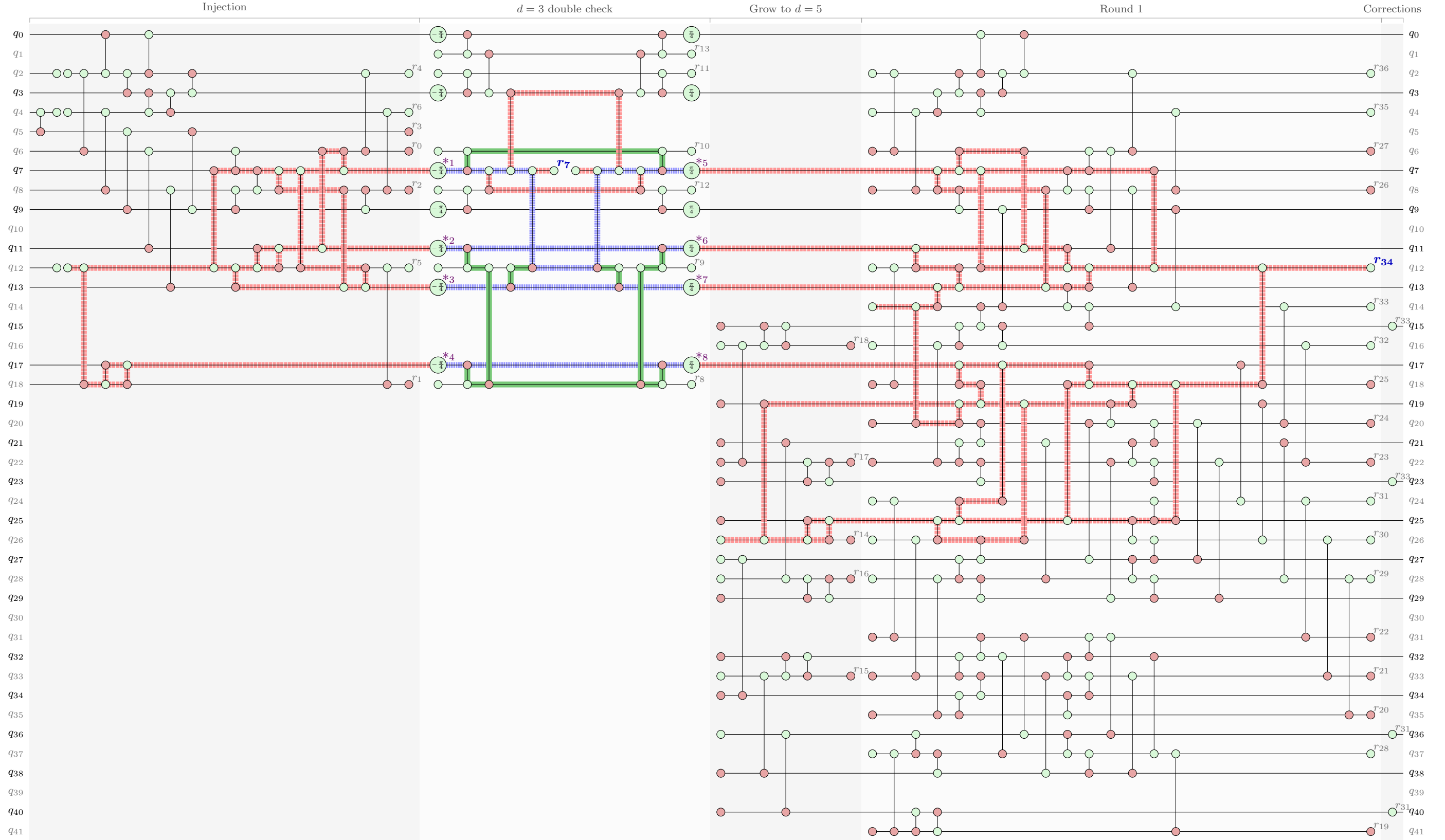


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{26}}$. So $D_{\mathbf{r}} = 0$ unless $r_{26} = 0$.

Records: r_{26} : M q_8 , line 250.

$d = 5$: detector 23 $r_7 \oplus r_{34} = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



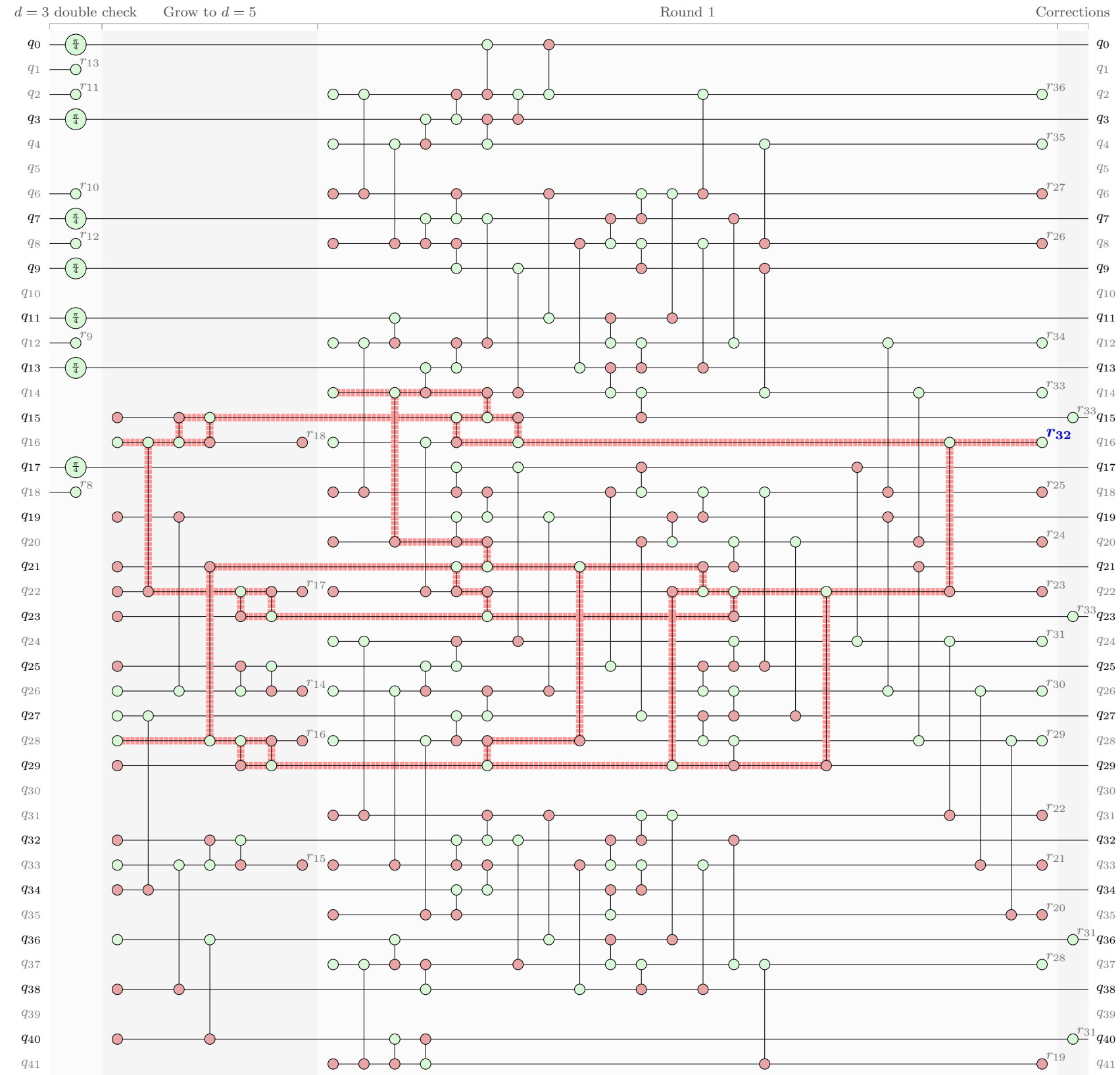
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}, \delta}$, $\lambda(\mathbf{r}) = (-1)^{r_7 + r_{34}}$. $D_{\mathbf{r}, \delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_7 : MX q_7 , line 157; r_{34} : MX q_{12} , line 251.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 143; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{11} , line 143; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 143; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{17} , line 143; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 174; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_{11} , line 174; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_{13} , line 174; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{17} , line 174.

$d = 5$: detector 24 $r_{32} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

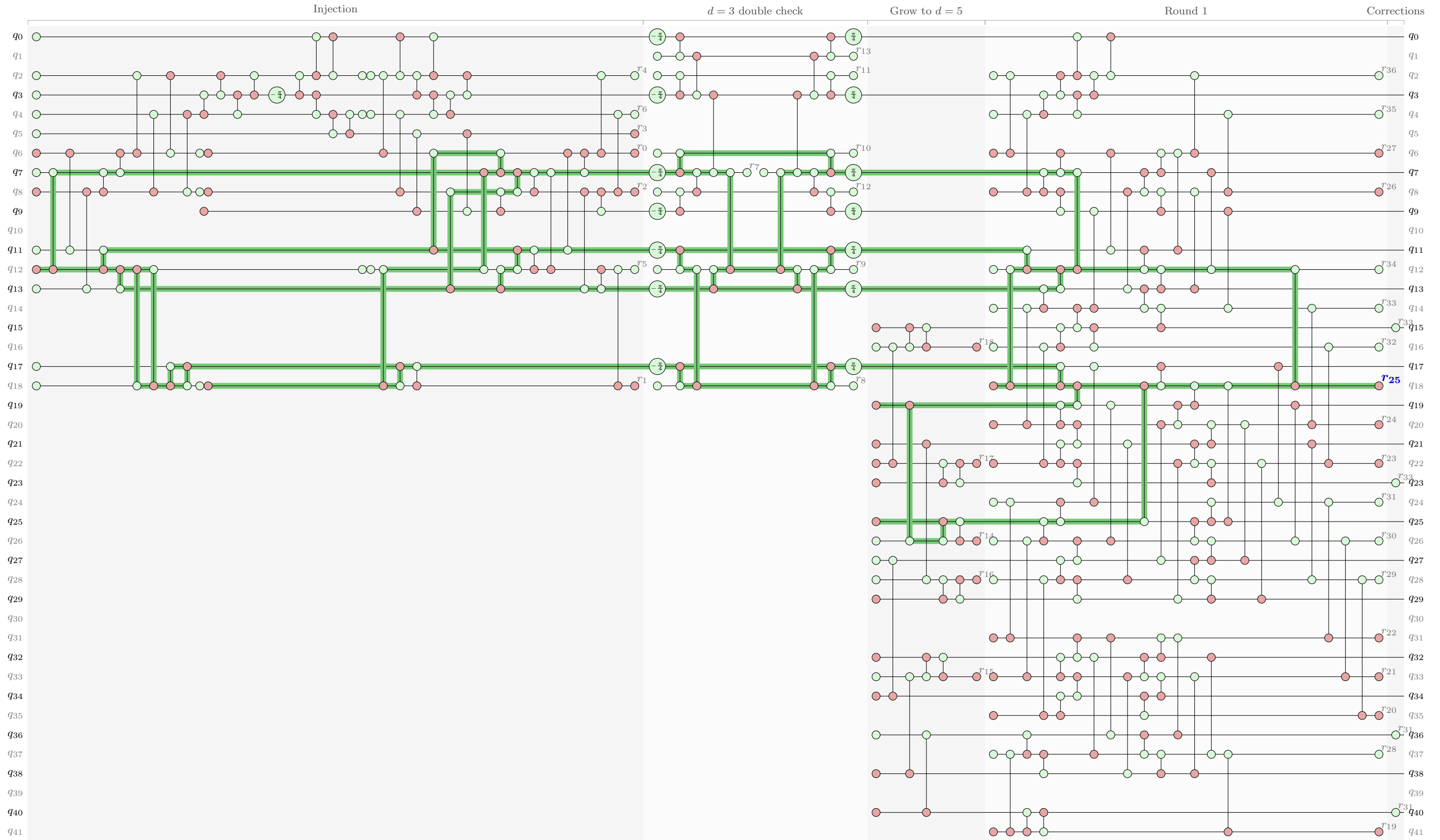


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{32}}$. So $D_{\mathbf{r}} = 0$ unless $r_{32} = 0$.

Records: r_{32} : MX q_{16} , line 251.

$d = 5$: detector 25 $r_{25} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

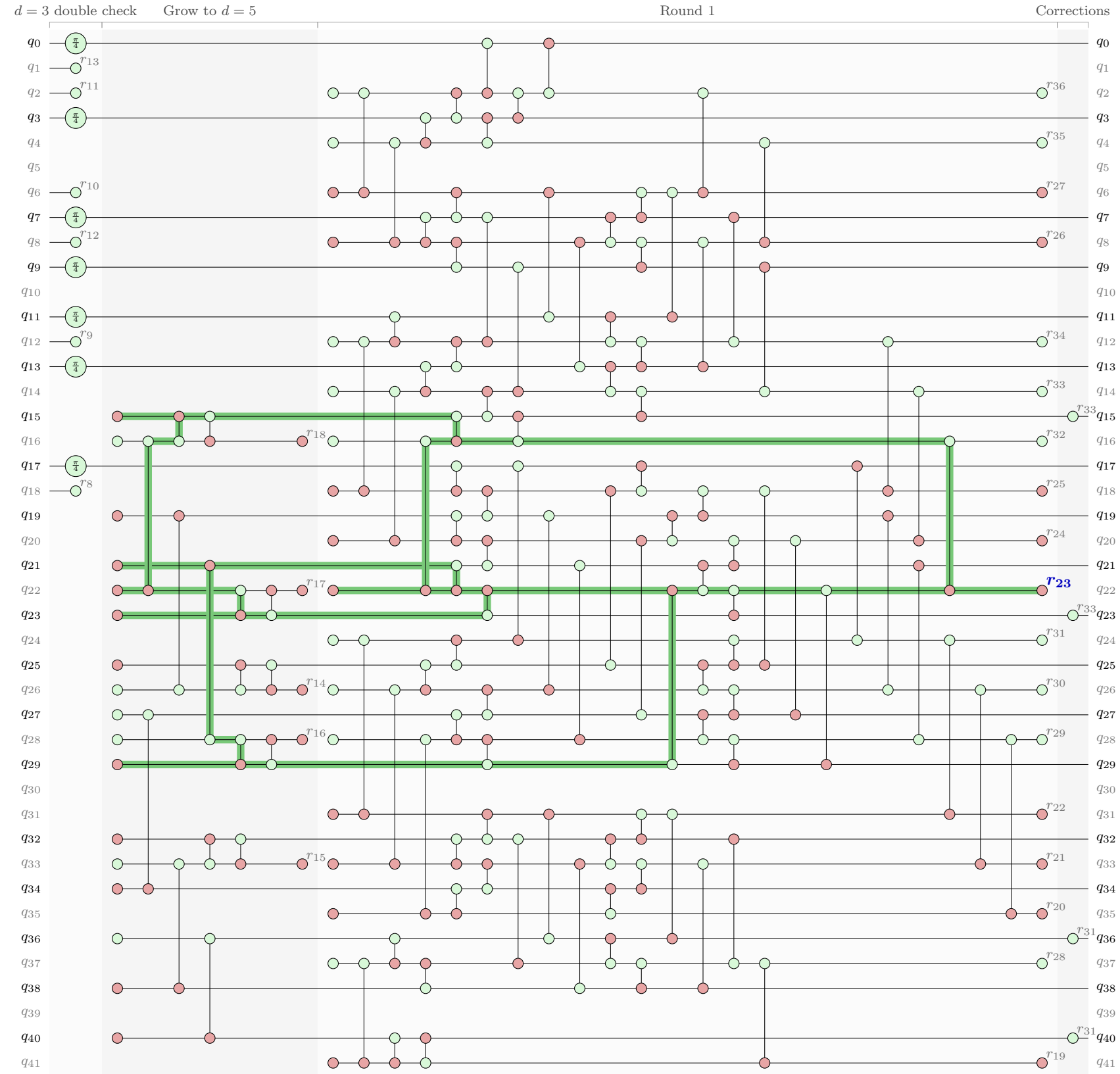


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{25}}$. So $D_{\mathbf{r}} = 0$ unless $r_{25} = 0$.

Records: r_{25} : M q_{18} , line 250.

$d = 5$: detector 26 $r_{23} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

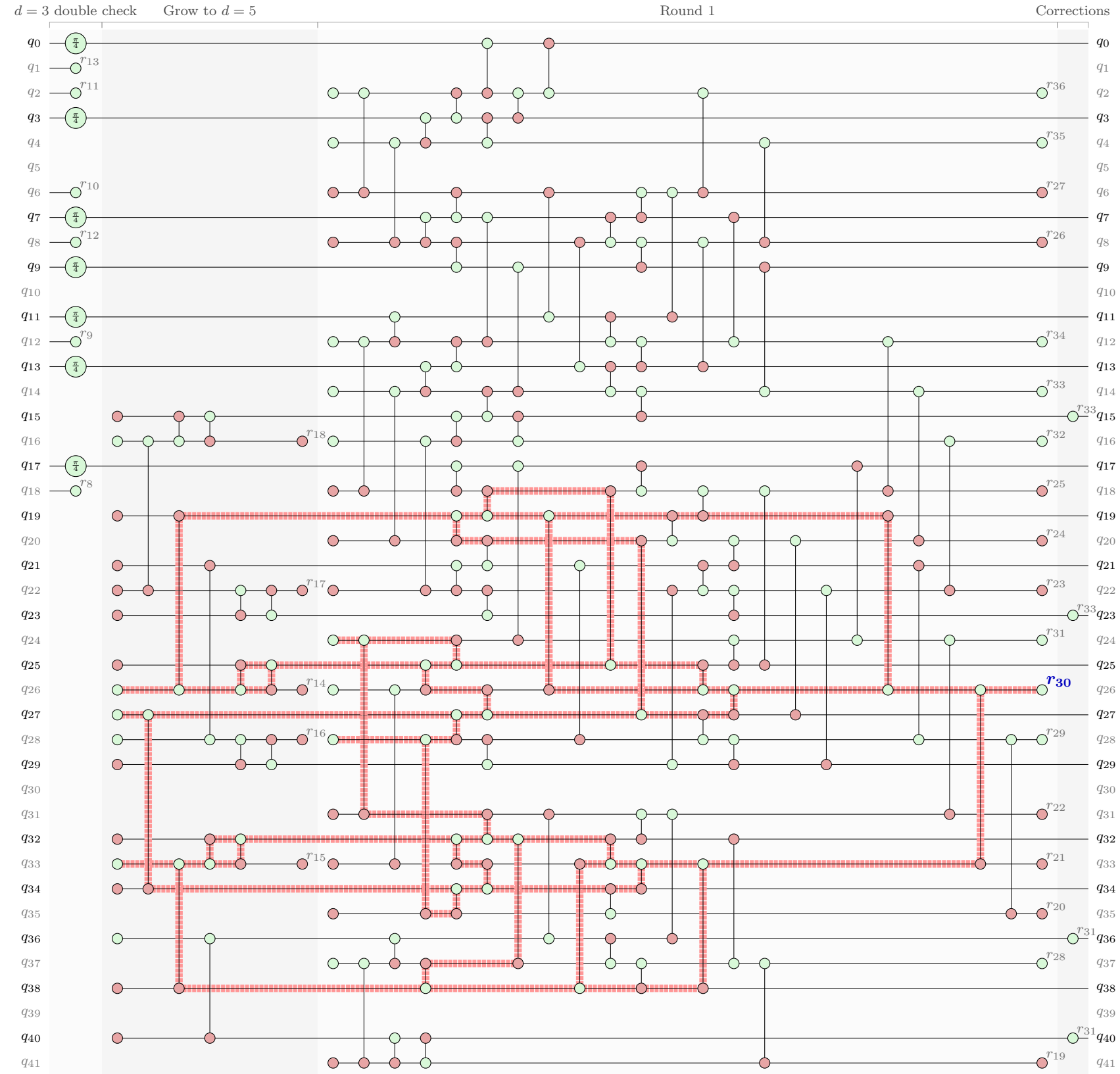


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{23}}$. So $D_{\mathbf{r}} = 0$ unless $r_{23} = 0$.

Records: r_{23} : M q_{22} , line 250.

$d = 5$: detector 27 $r_{30} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

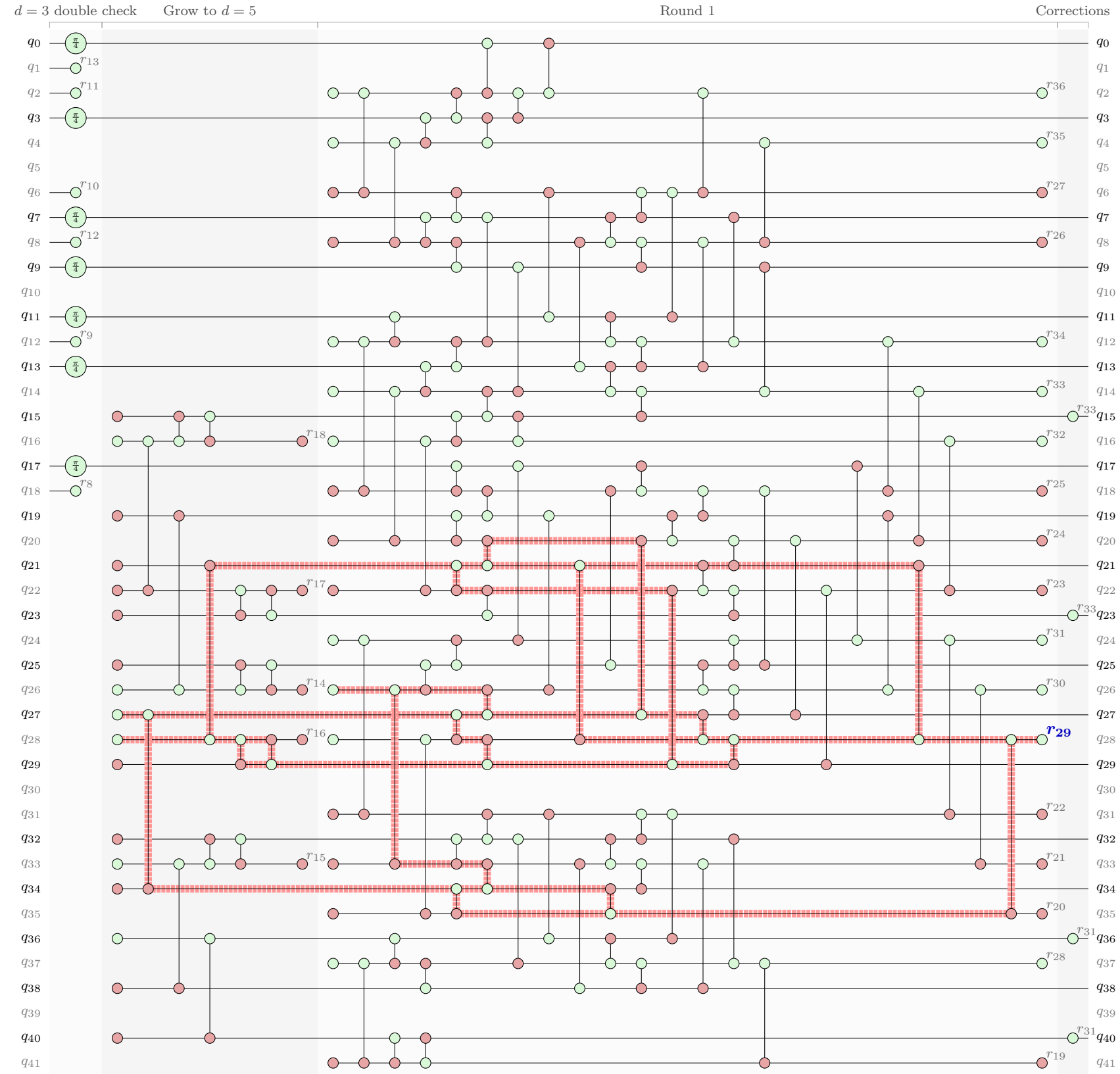


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{30}}$. So $D_{\mathbf{r}} = 0$ unless $r_{30} = 0$.

Records: r_{30} : MX q_{26} , line 251.

$d = 5$: detector 28 $r_{29} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

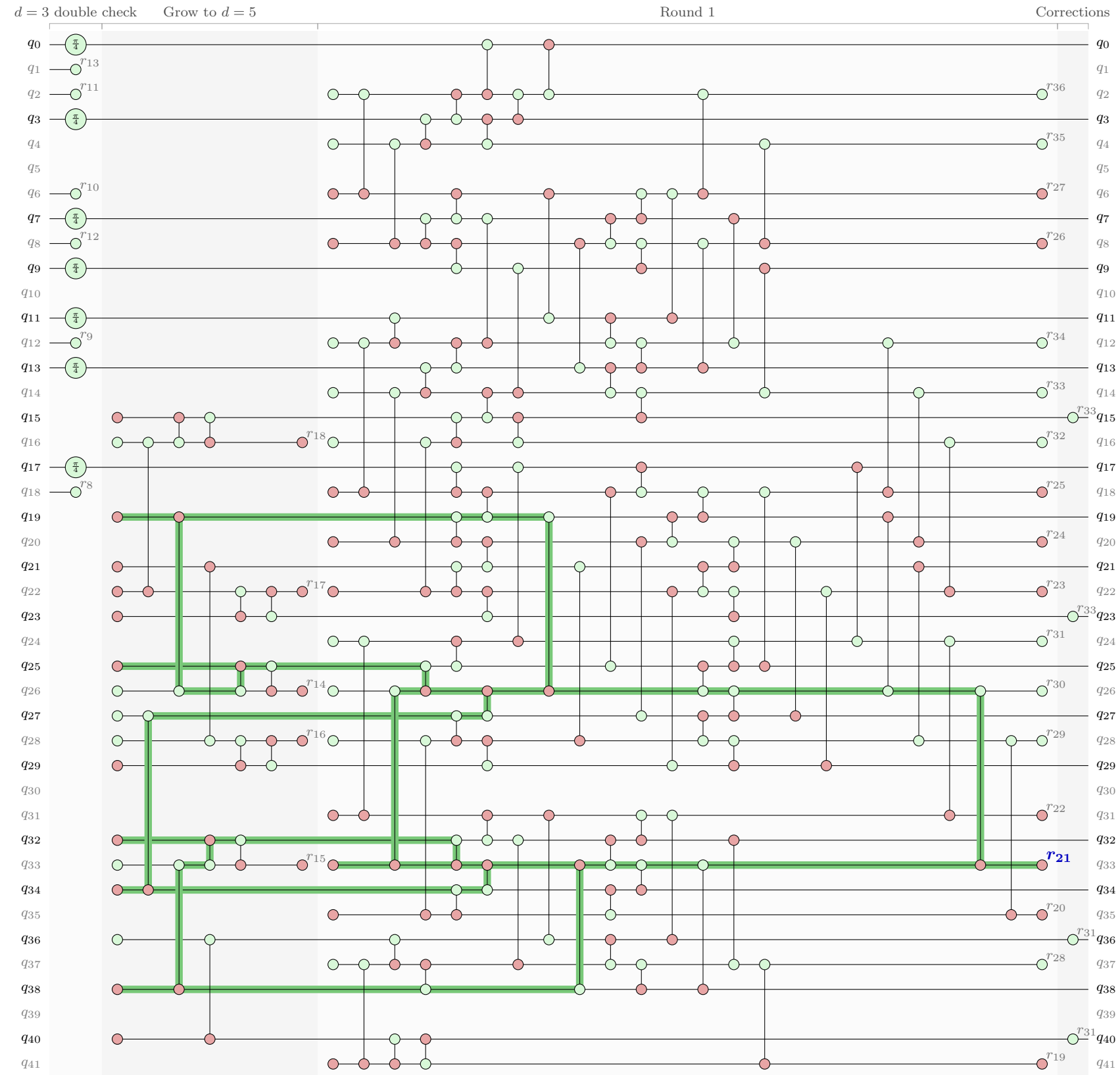


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{29}}$. So $D_{\mathbf{r}} = 0$ unless $r_{29} = 0$.

Records: r_{29} : MX q_{28} , line 251.

$d = 5$: detector 29 $r_{21} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.



$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{21}}$. So $D_{\mathbf{r}} = 0$ unless $r_{21} = 0$.

Records: r_{21} : M q_{33} , line 250.

$d = 5$: detector 30 $r_{20} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

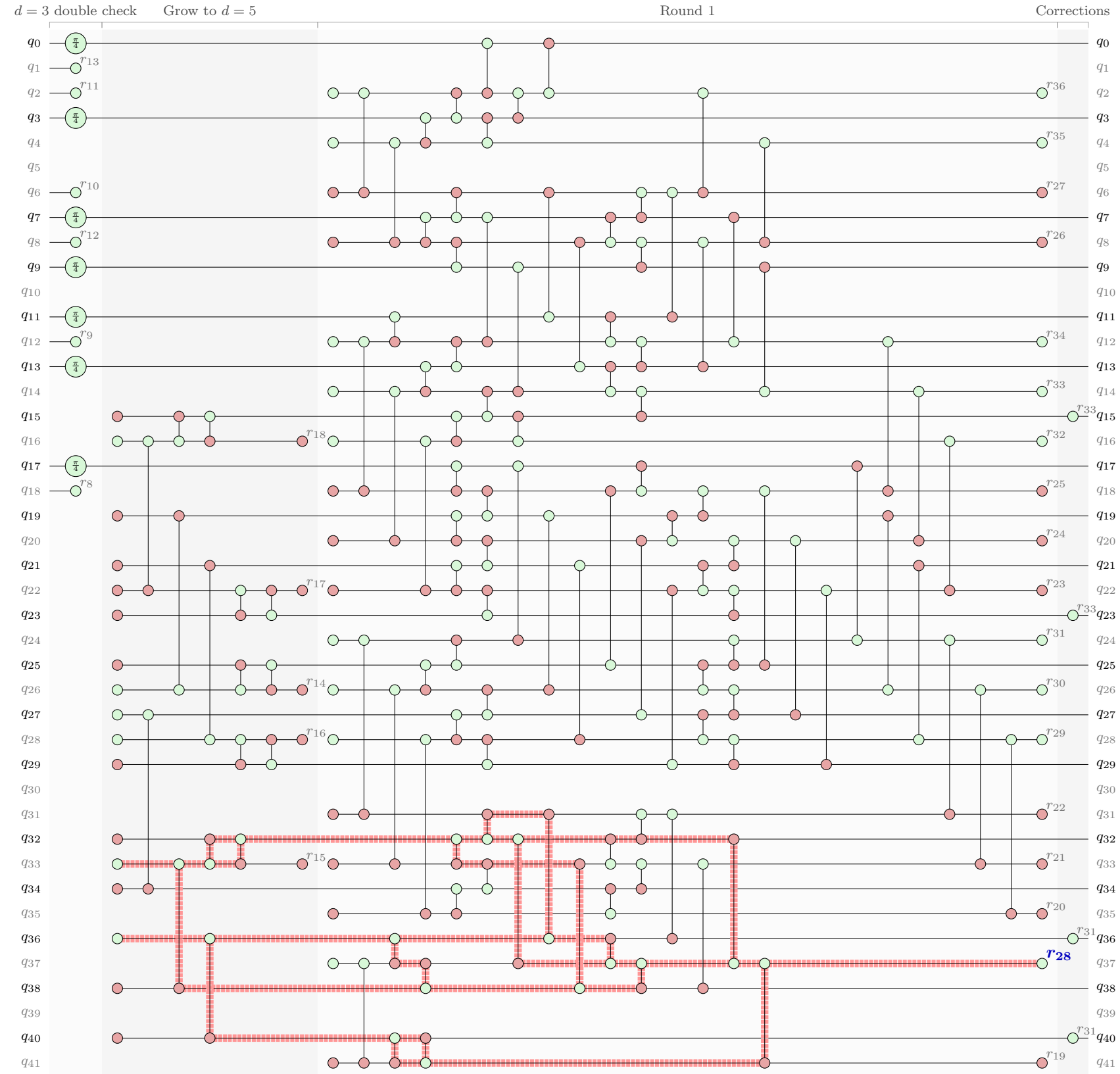


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{20}}$. So $D_{\mathbf{r}} = 0$ unless $r_{20} = 0$.

Records: r_{20} : M q_{35} , line 250.

$d = 5$: detector 31 $r_{28} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

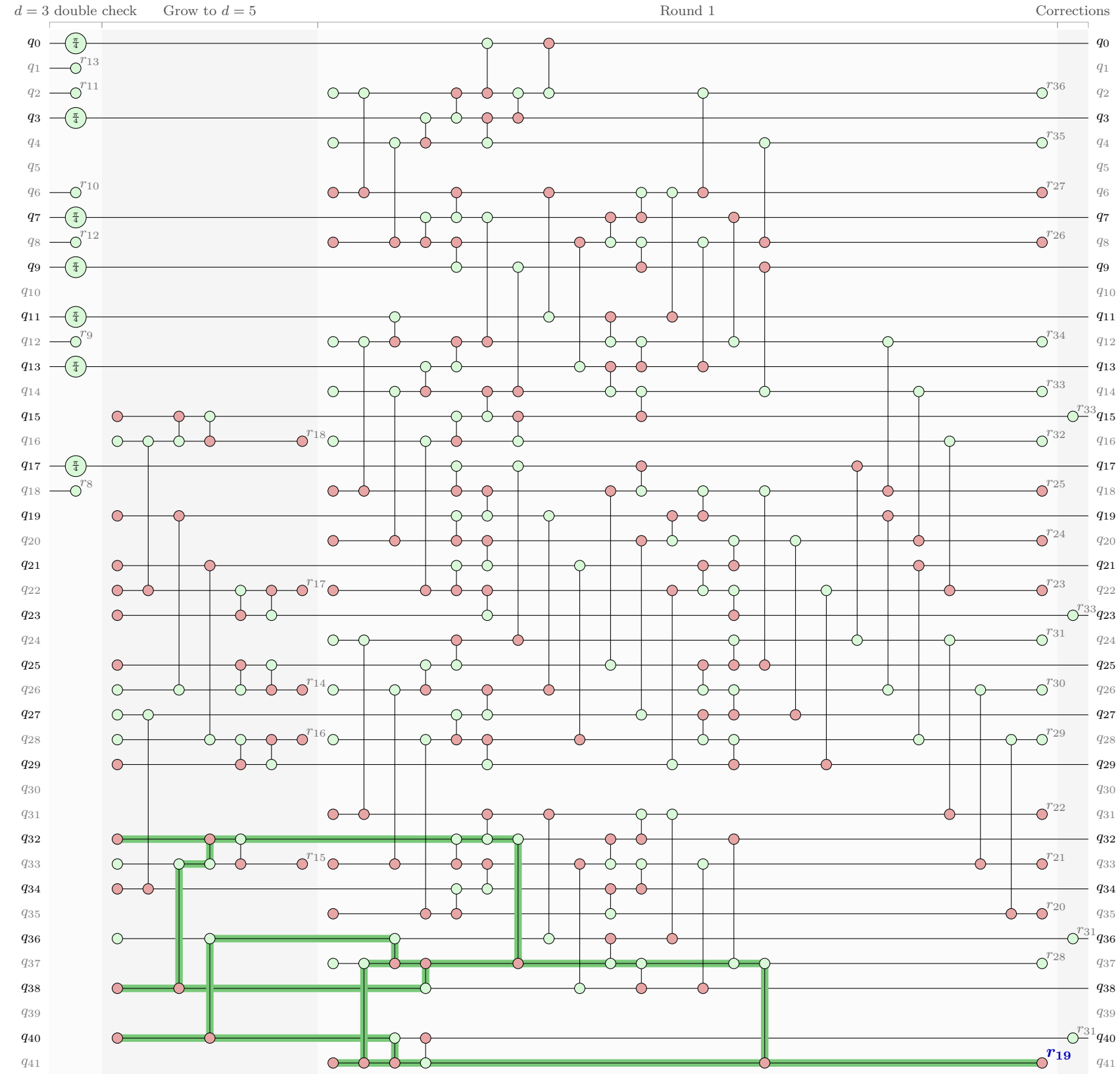


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{28}}$. So $D_{\mathbf{r}} = 0$ unless $r_{28} = 0$.

Records: r_{28} : MX q_{37} , line 251.

$d = 5$: detector 32 $r_{19} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

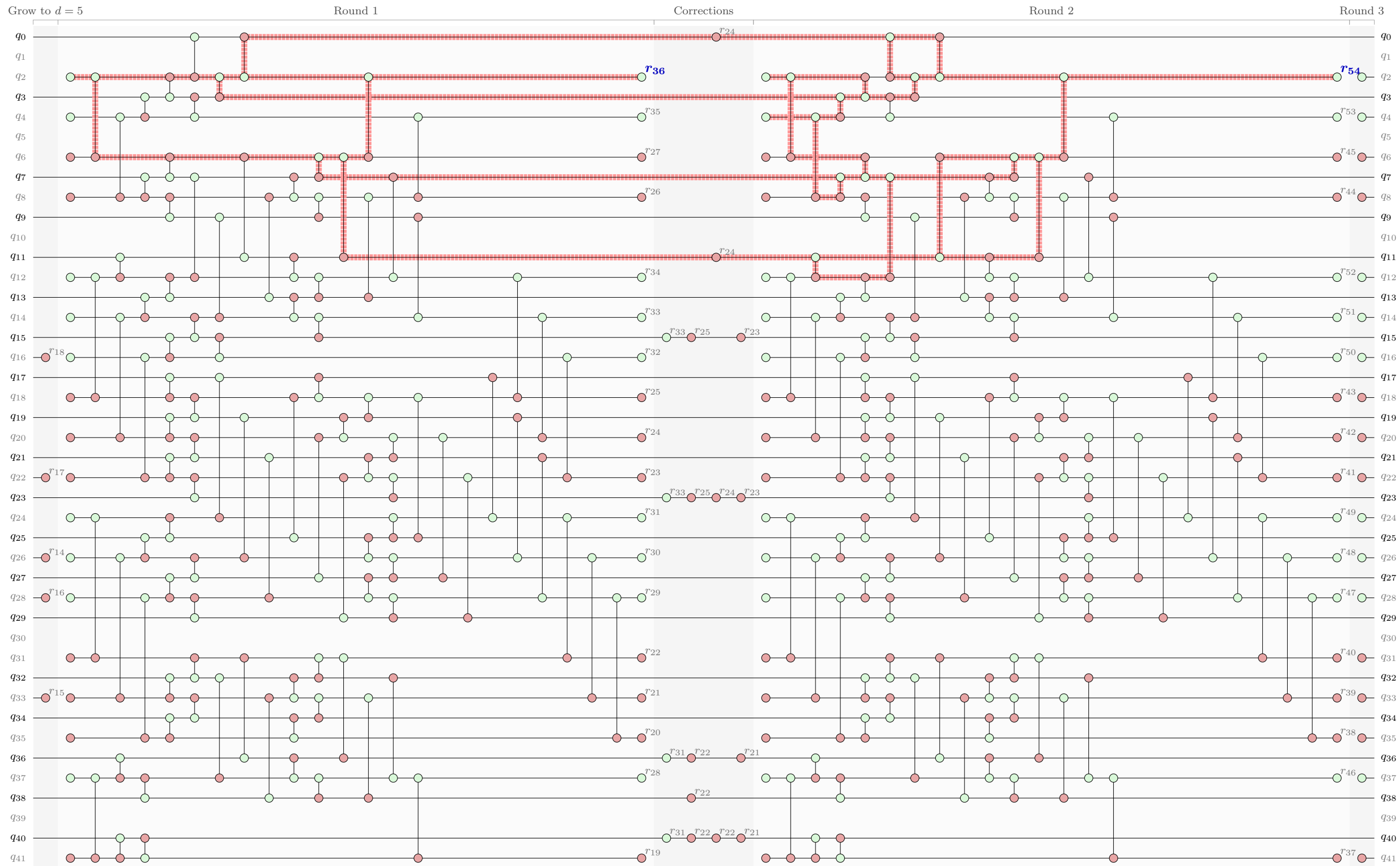


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{19}}$. So $D_{\mathbf{r}} = 0$ unless $r_{19} = 0$.

Records: r_{19} : M q_{41} , line 250.

$d = 5$: detector 33 $r_{36} \oplus r_{54} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

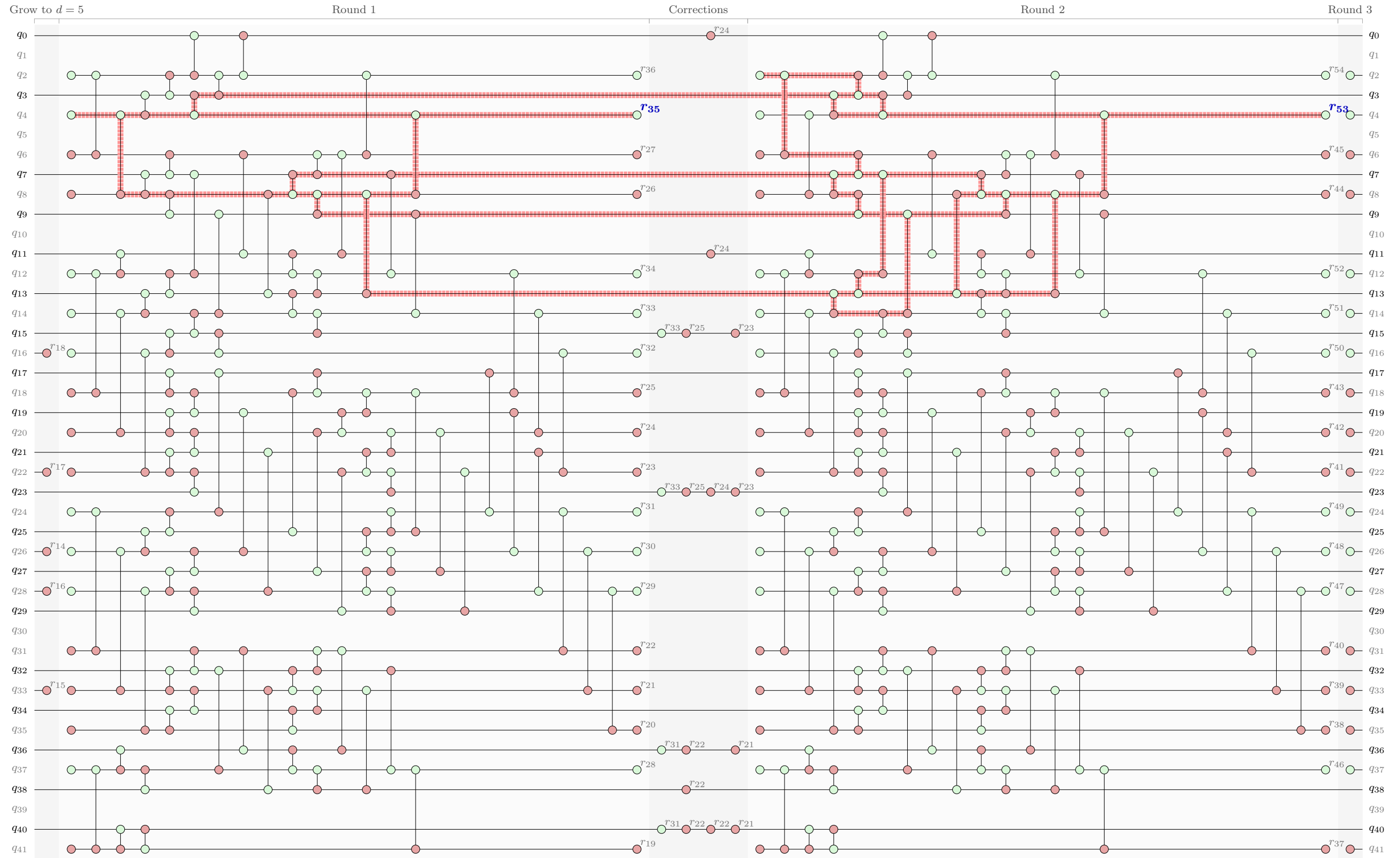


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{36} + r_{54}}$. So $D_{\mathbf{r}} = 0$ unless $r_{36} \oplus r_{54} = 0$.

Records: r_{36} : MX q_2 , line 251; r_{54} : MX q_2 , line 310.

$d = 5$: detector 34 $r_{35} \oplus r_{53} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

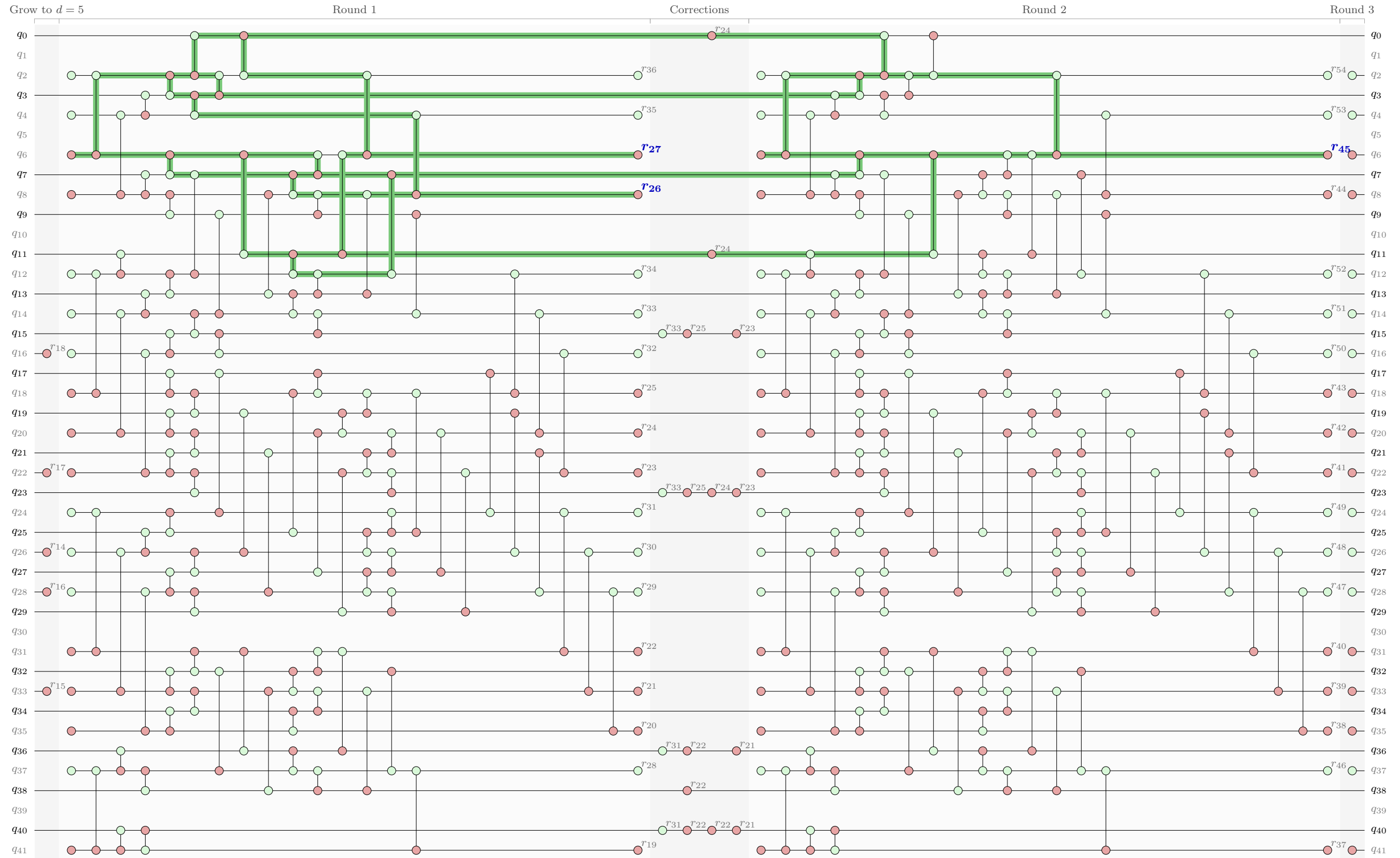


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{35} + r_{53}}$. So $D_{\mathbf{r}} = 0$ unless $r_{35} \oplus r_{53} = 0$.

Records: r_{35} : MX q_4 , line 251; r_{53} : MX q_4 , line 310.

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

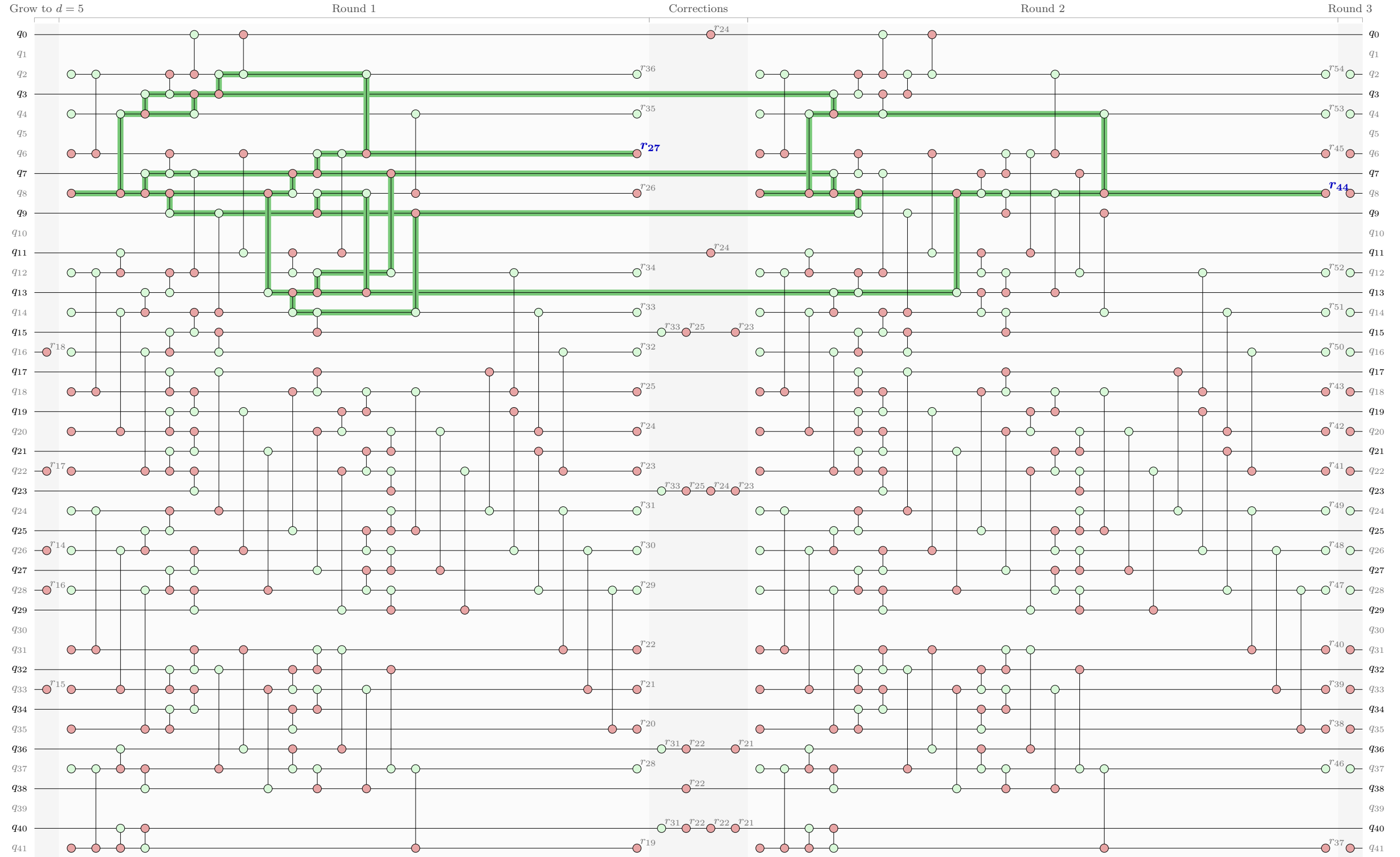
Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{26} + r_{27} + r_{45}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{26} \oplus r_{27} \oplus r_{45} = 0.$$

Records: r_{26} : M q_8 , line 250; r_{27} : M q_6 , line 250; r_{45} : M q_6 , line 309.

$d = 5$: detector 36 $r_{27} \oplus r_{44} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

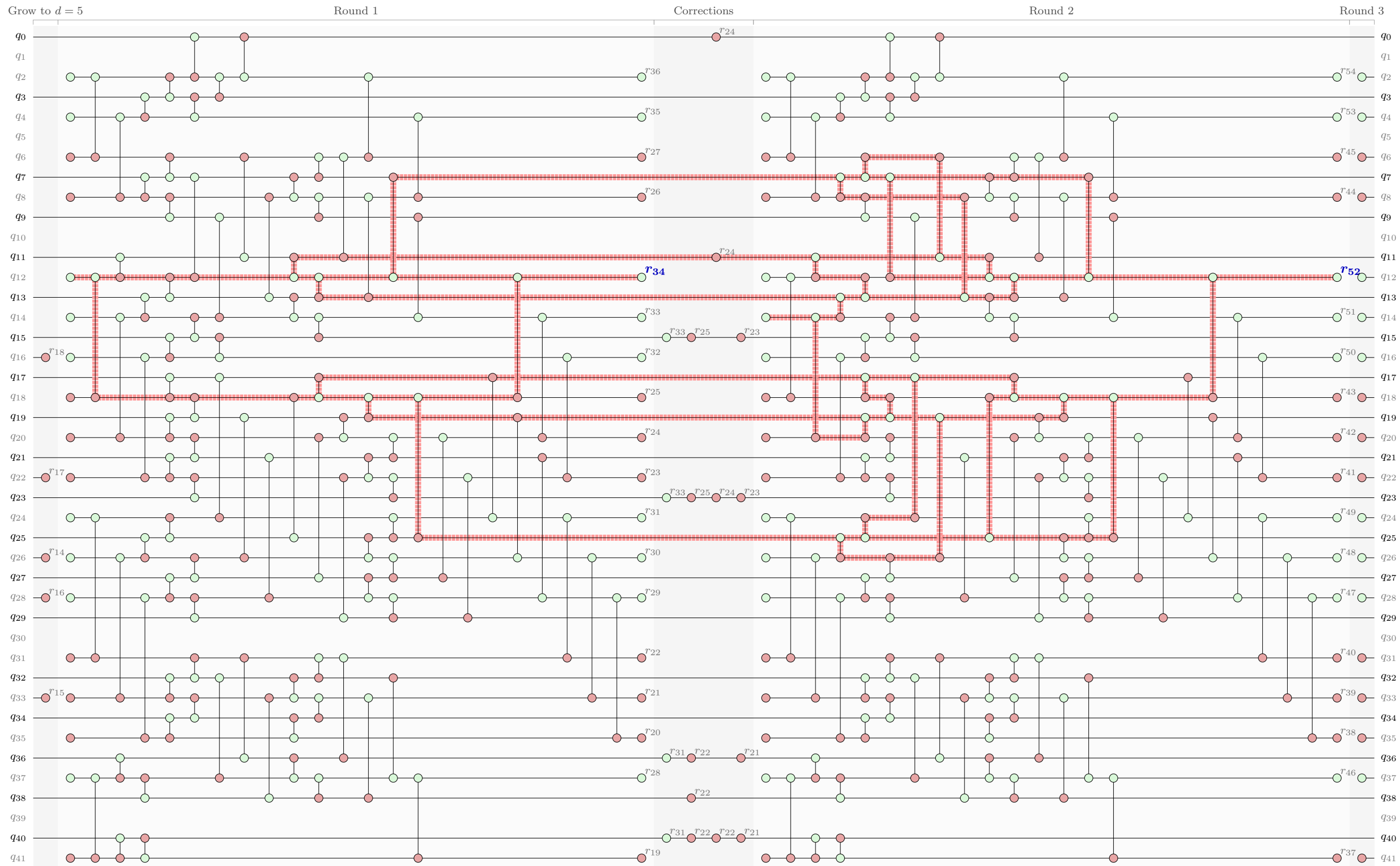


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{27} + r_{44}}$. So $D_{\mathbf{r}} = 0$ unless $r_{27} \oplus r_{44} = 0$.

Records: r_{27} : M q_6 , line 250; r_{44} : M q_8 , line 309.

$d = 5$: detector 37 $r_{34} \oplus r_{52} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

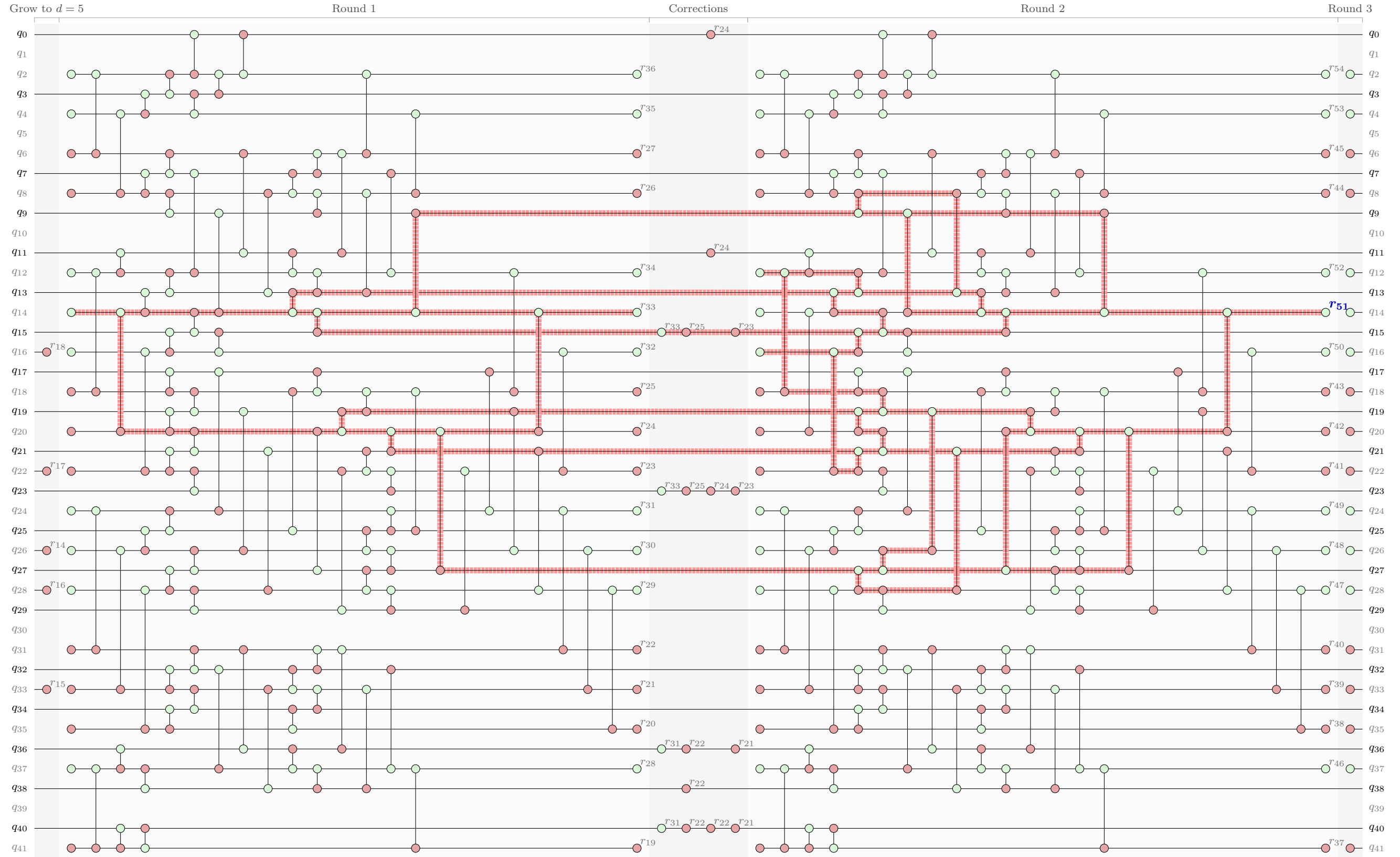


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{34} + r_{52}}$. So $D_{\mathbf{r}} = 0$ unless $r_{34} \oplus r_{52} = 0$.

Records: r_{34} : MX q_{12} , line 251; r_{52} : MX q_{12} , line 310.

$d = 5$: detector 38 $r_{51} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

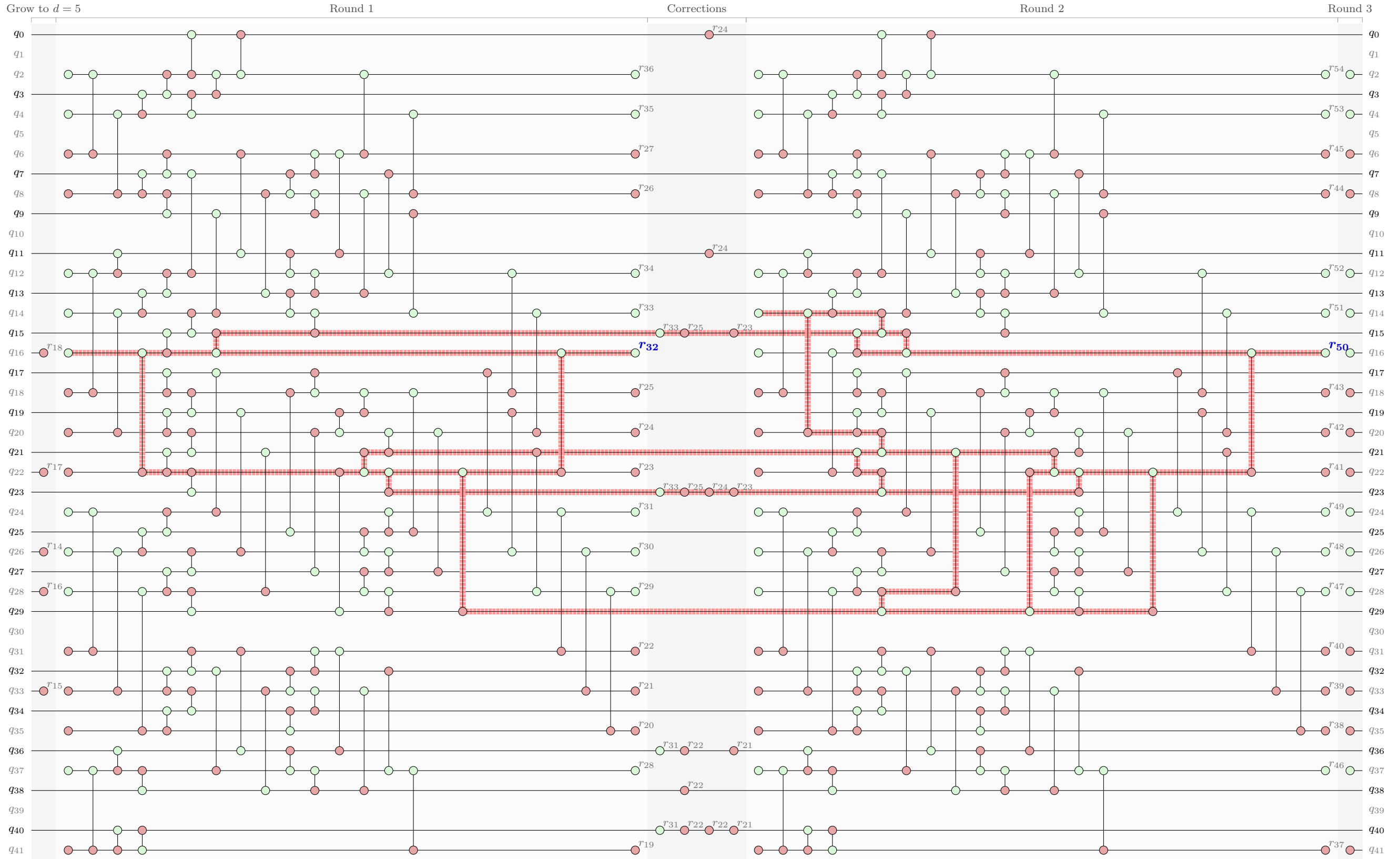


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{51}}$. So $D_{\mathbf{r}} = 0$ unless $r_{51} = 0$.

Records: r_{51} : MX q_{14} , line 310.

$$d = 5: \text{ detector } 39 \quad r_{32} \oplus r_{50} = 0$$

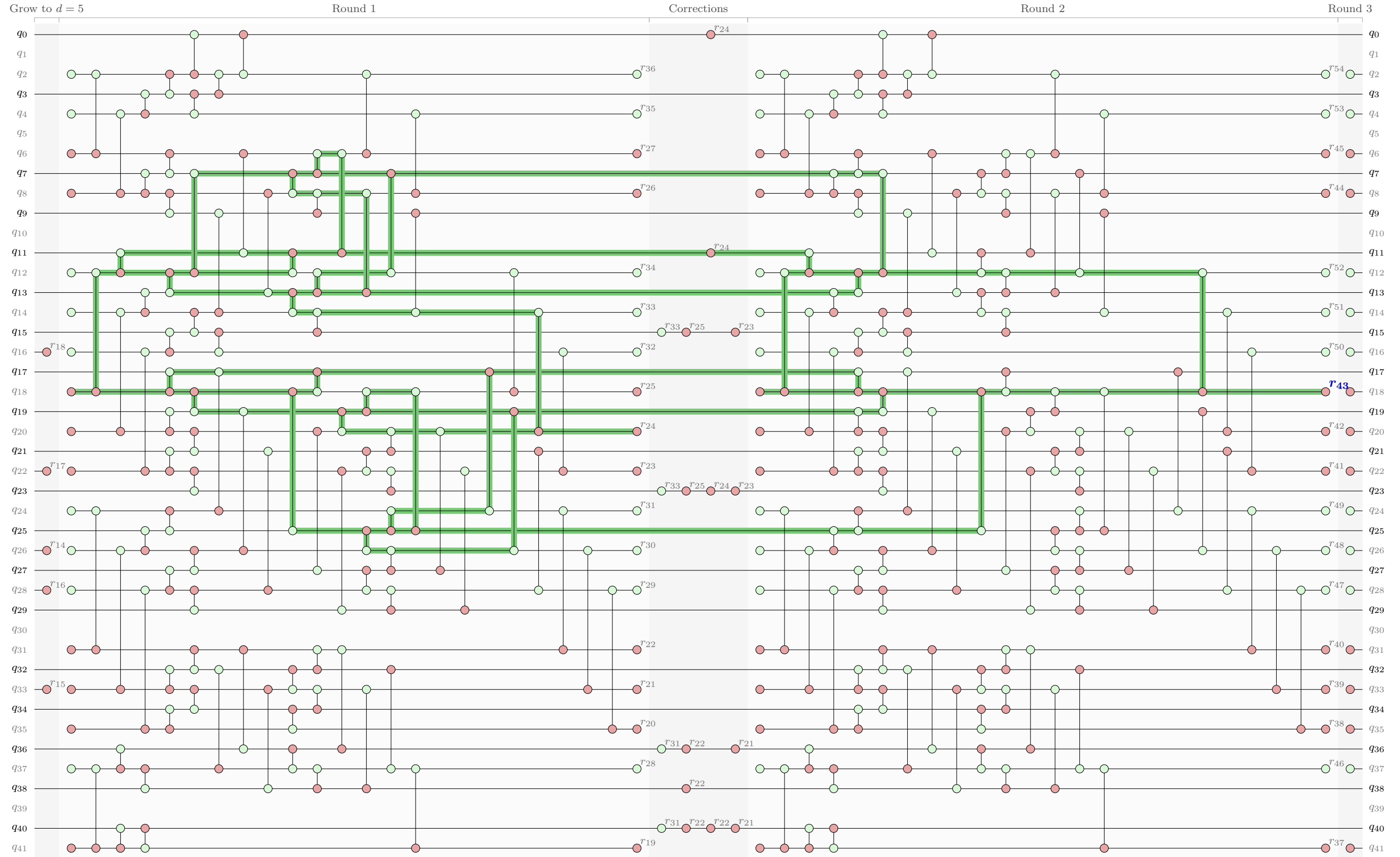
Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{32} + r_{50}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{32} \oplus r_{50} = 0.$$

Records: r_{32} : MX q_{16} , line 251; r_{50} : MX q_{16} , line 310.

$d = 5$: detector 40 $r_{43} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

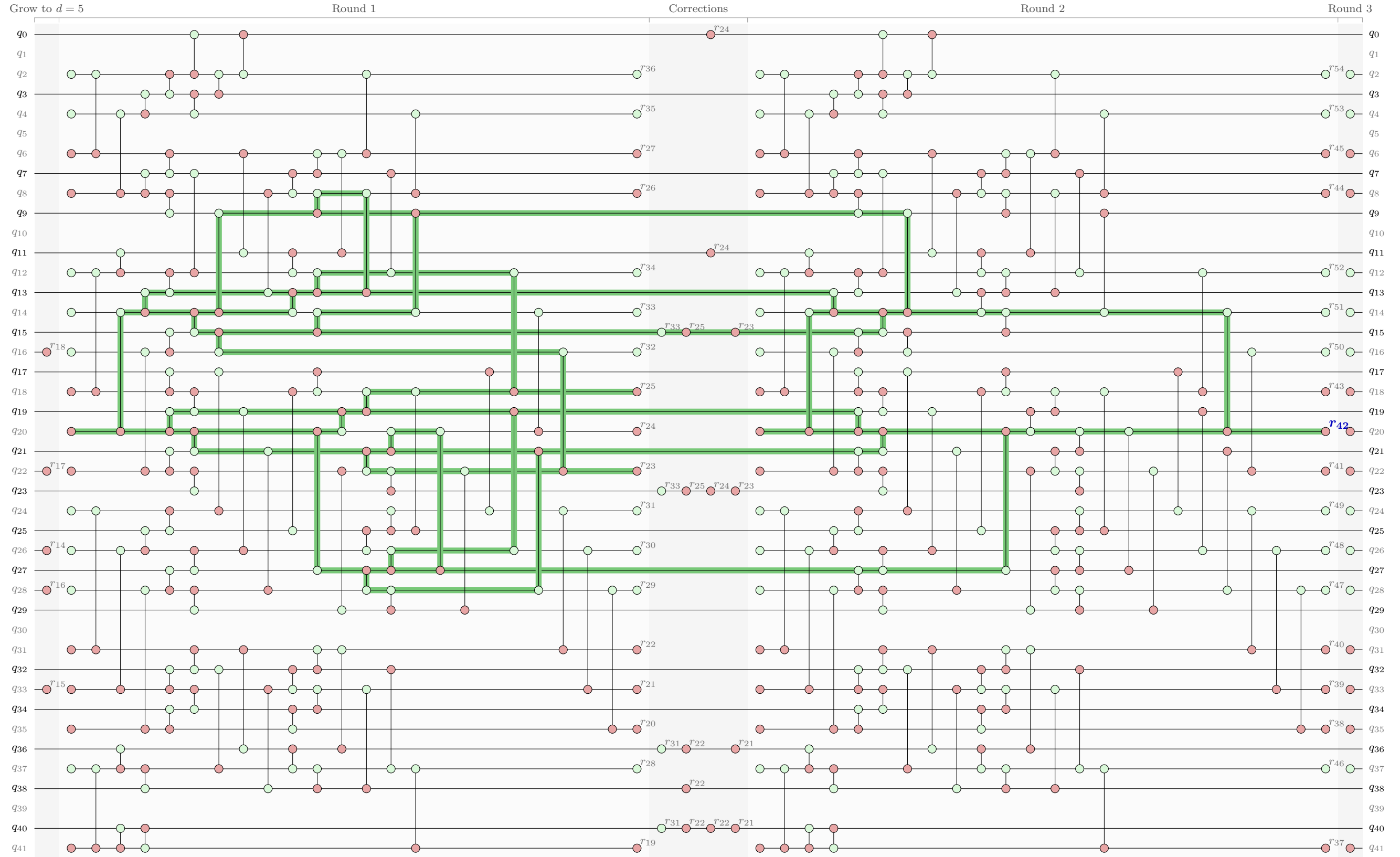


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{43}}$. So $D_{\mathbf{r}} = 0$ unless $r_{43} = 0$.

Records: r_{43} : M q_{18} , line 309.

$d = 5$: detector 41 $r_{42} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

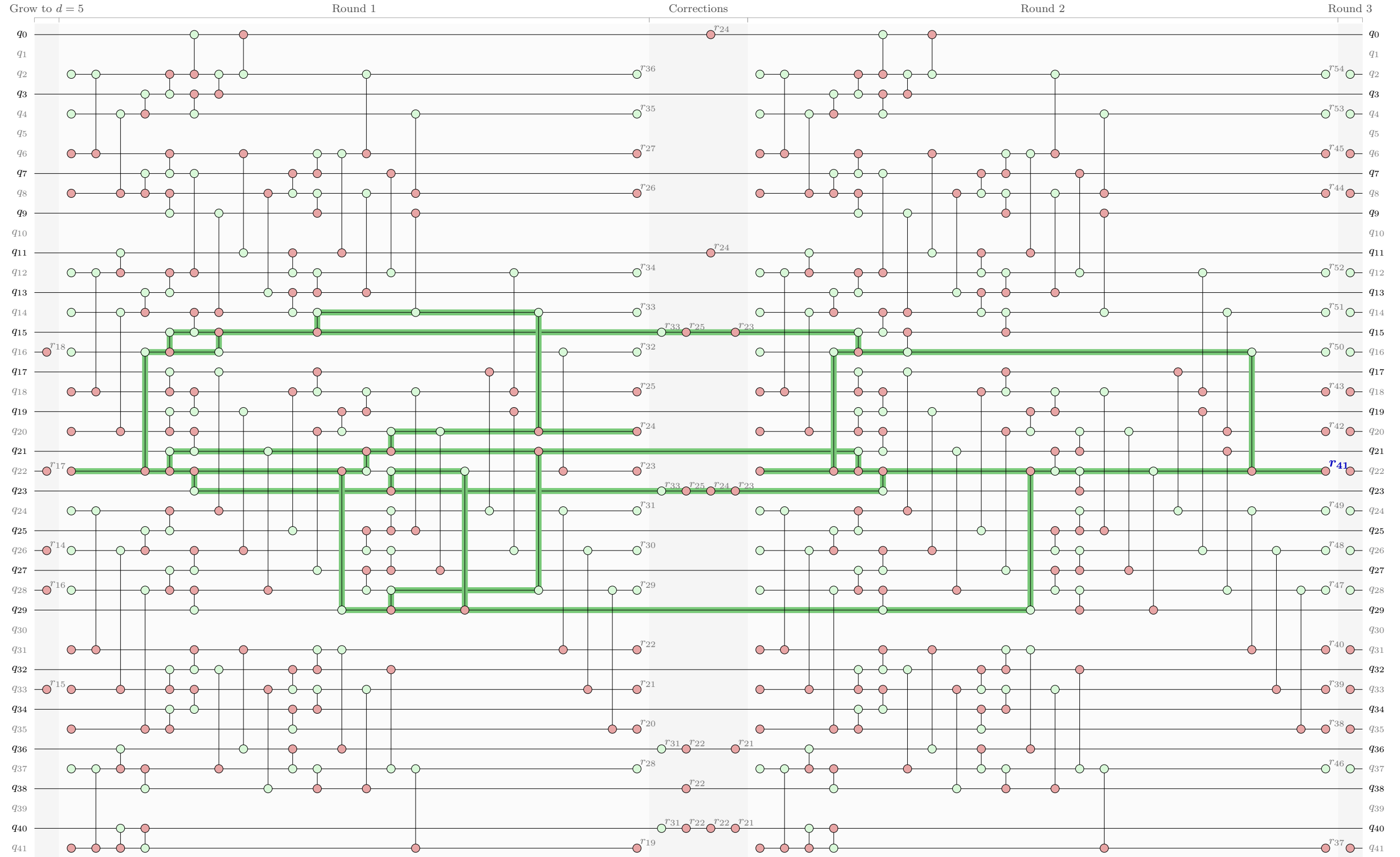


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{42}}$. So $D_{\mathbf{r}} = 0$ unless $r_{42} = 0$.

Records: r_{42} : M q_{20} , line 309.

$d = 5$: detector 42 $r_{41} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

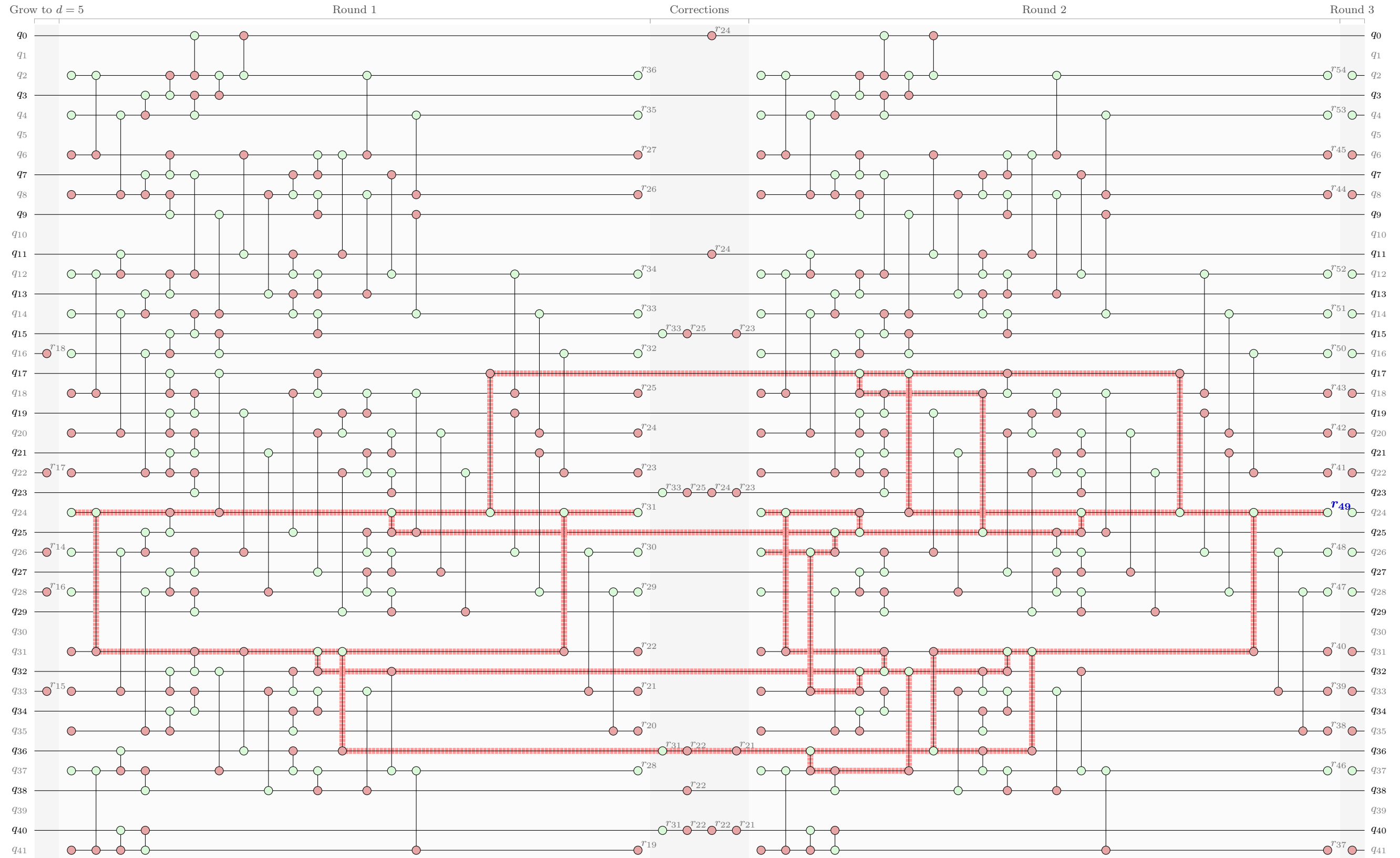


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{41}}$. So $D_{\mathbf{r}} = 0$ unless $r_{41} = 0$.

Records: r_{41} : M q_{22} , line 309.

$d = 5$: detector 43 $r_{49} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

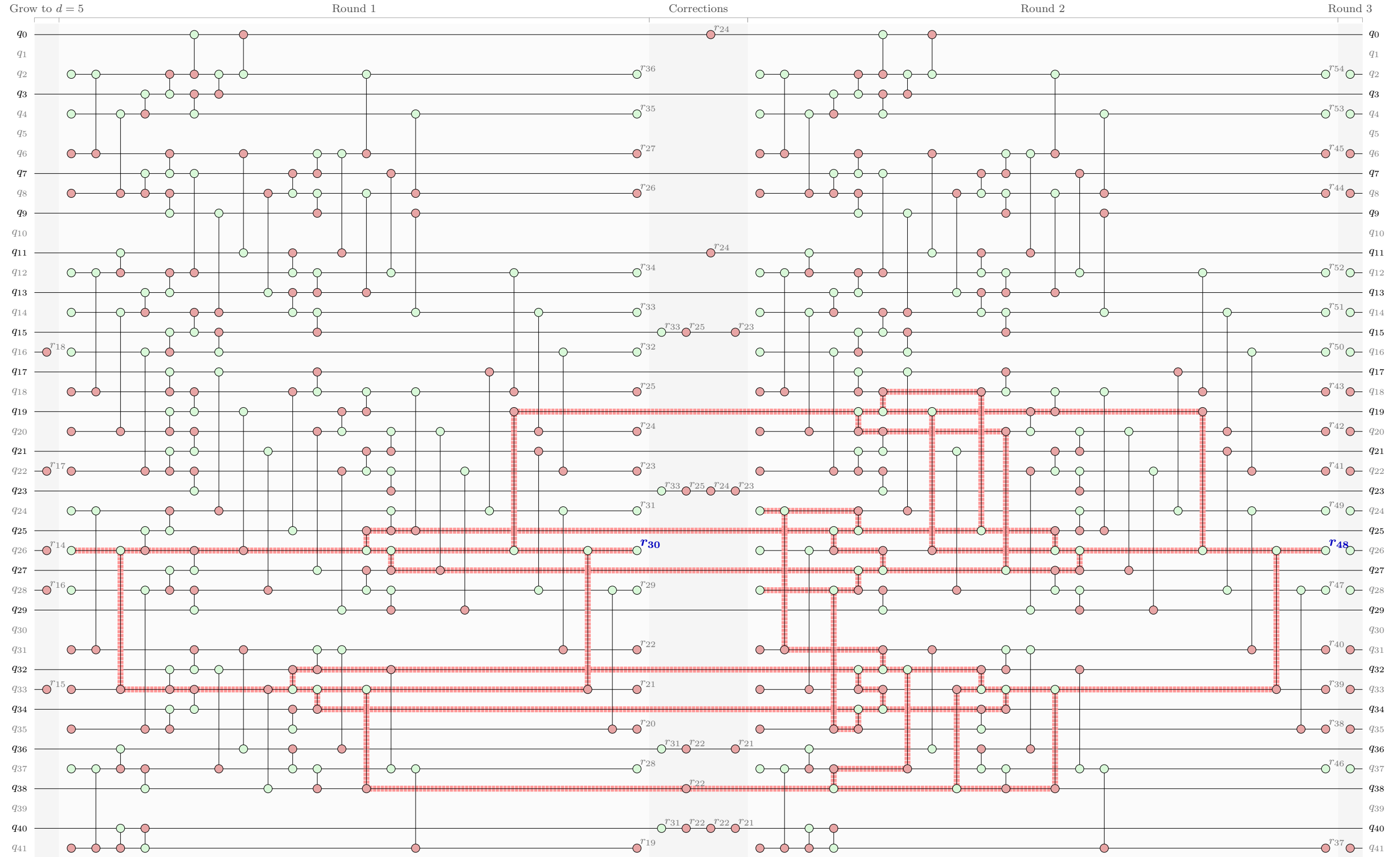


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{49}}$. So $D_{\mathbf{r}} = 0$ unless $r_{49} = 0$.

Records: r_{49} : MX q_{24} , line 310.

$d = 5$: detector 44 $r_{30} \oplus r_{48} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

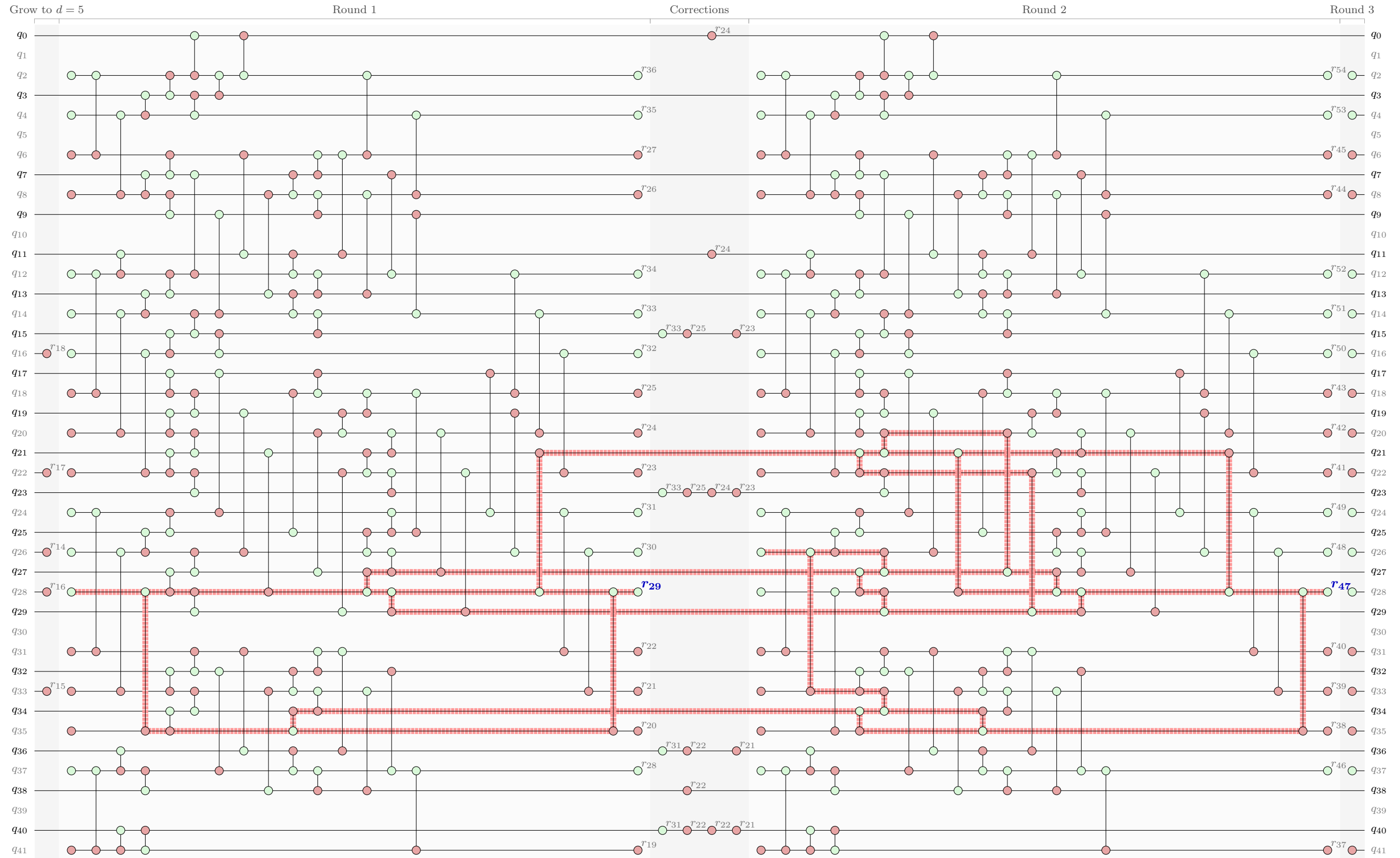


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{30} + r_{48}}$. So $D_{\mathbf{r}} = 0$ unless $r_{30} \oplus r_{48} = 0$.

Records: r_{30} : MX q_{26} , line 251; r_{48} : MX q_{26} , line 310.

$$r_{29} \oplus r_{47} = 0$$

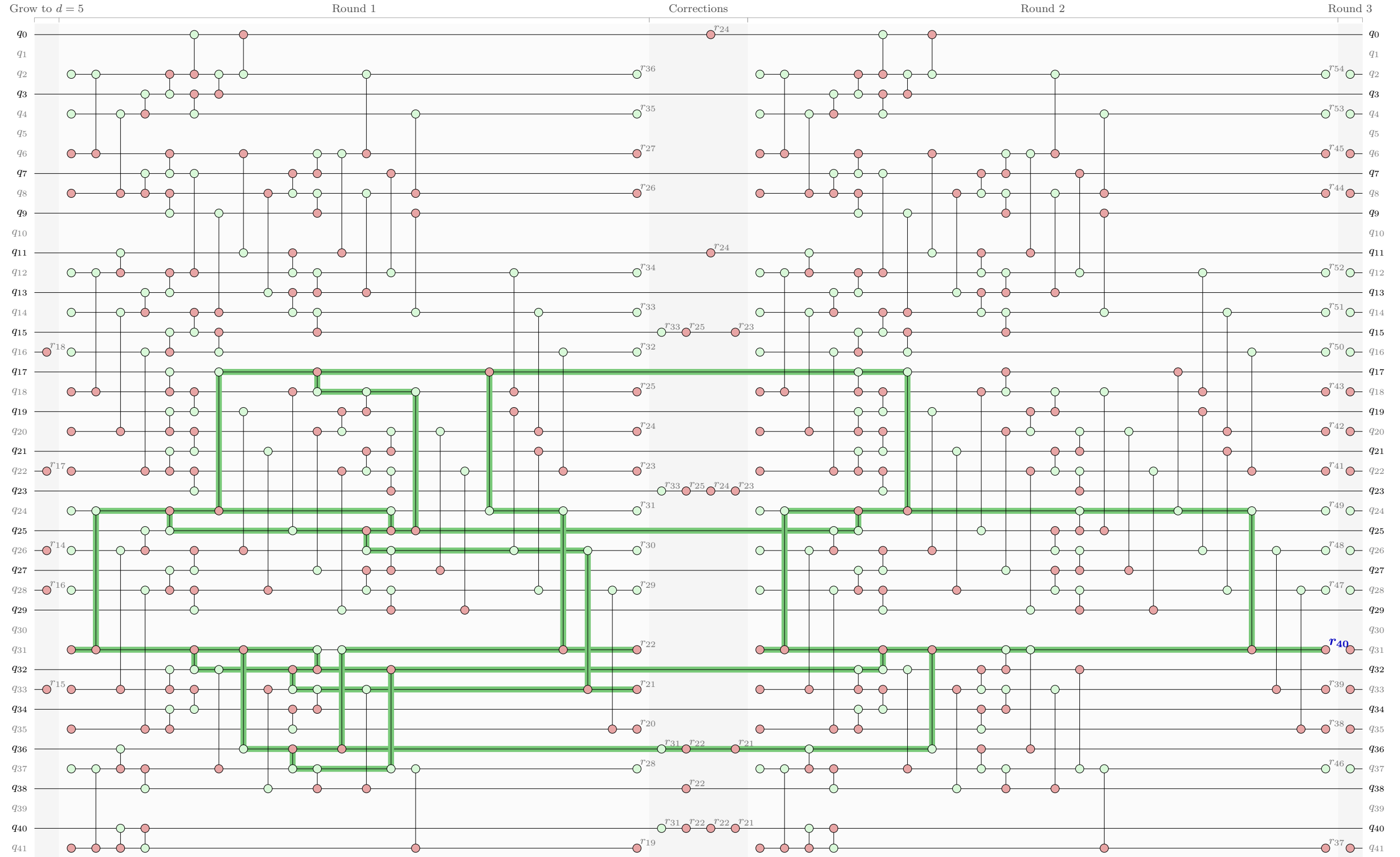
Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{29}+r_{47}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{29} \oplus r_{47} = 0.$$

Records: r_{29} : MX q_{28} , line 251; r_{47} : MX q_{28} , line 310.

$d = 5$: detector 46 $r_{40} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

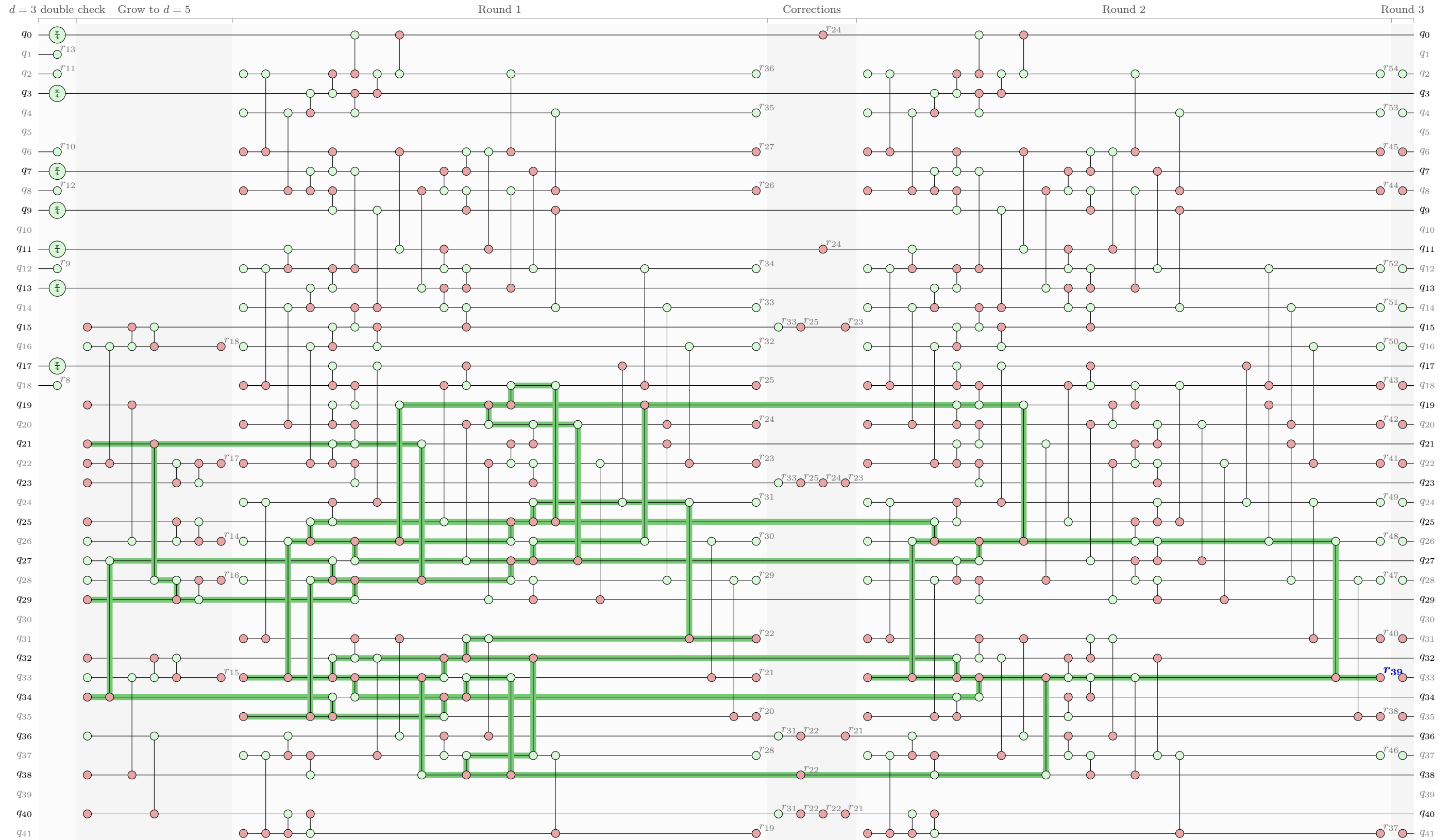


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{40}}$. So $D_{\mathbf{r}} = 0$ unless $r_{40} = 0$.

Records: r_{40} : M q_{31} , line 309.

$d = 5$: detector 47 $r_{39} = 0$

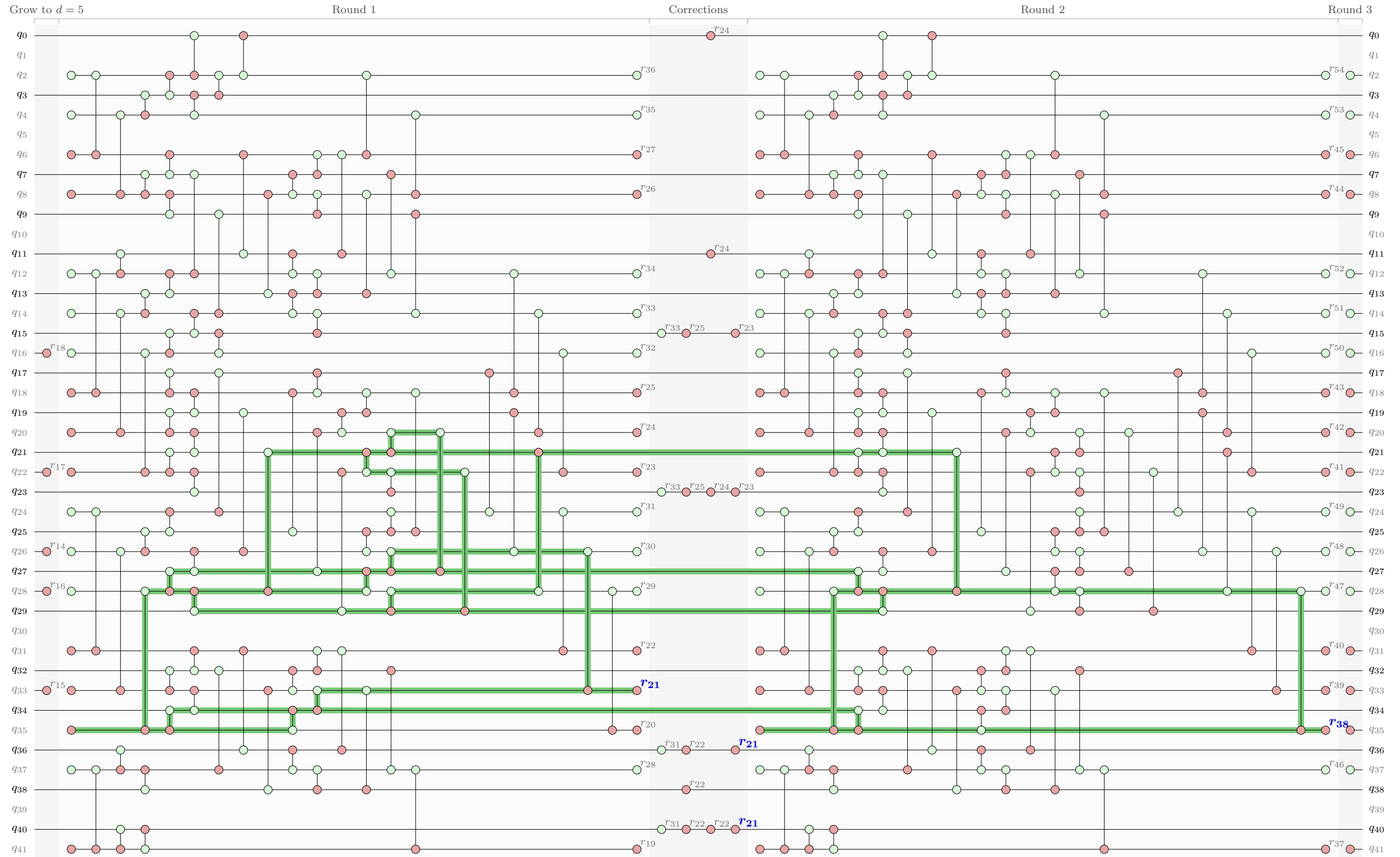
Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{39}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{39} = 0.$$

Records: r_{39} : M q_{33} , line 309.

$d = 5$: detector 48 $r_{21} \oplus r_{38} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

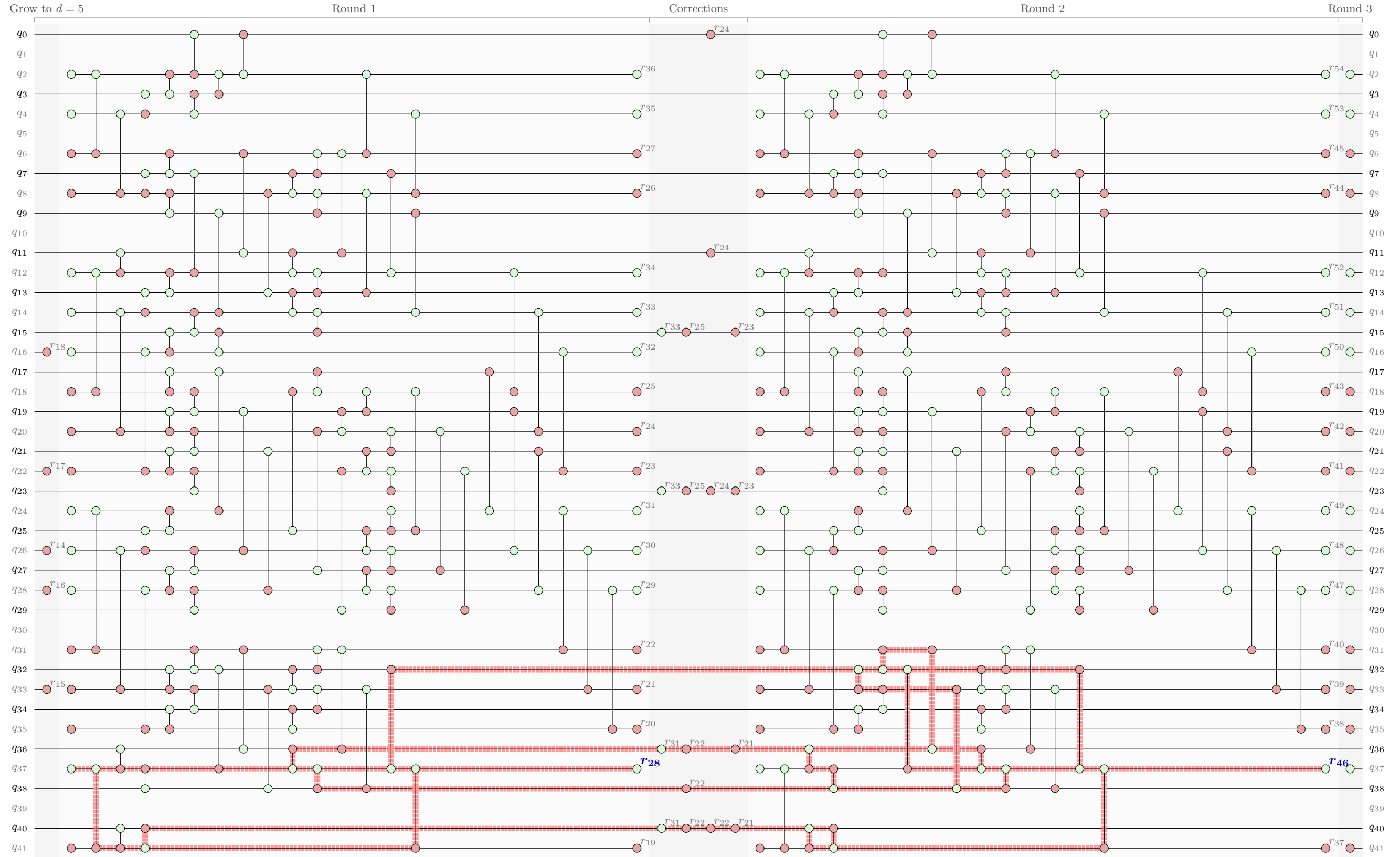


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{21} + r_{38}}$. So $D_{\mathbf{r}} = 0$ unless $r_{21} \oplus r_{38} = 0$.

Records: r_{21} : M q_{33} , line 250; r_{38} : M q_{35} , line 309.

$d = 5$: detector 49 $r_{28} \oplus r_{46} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

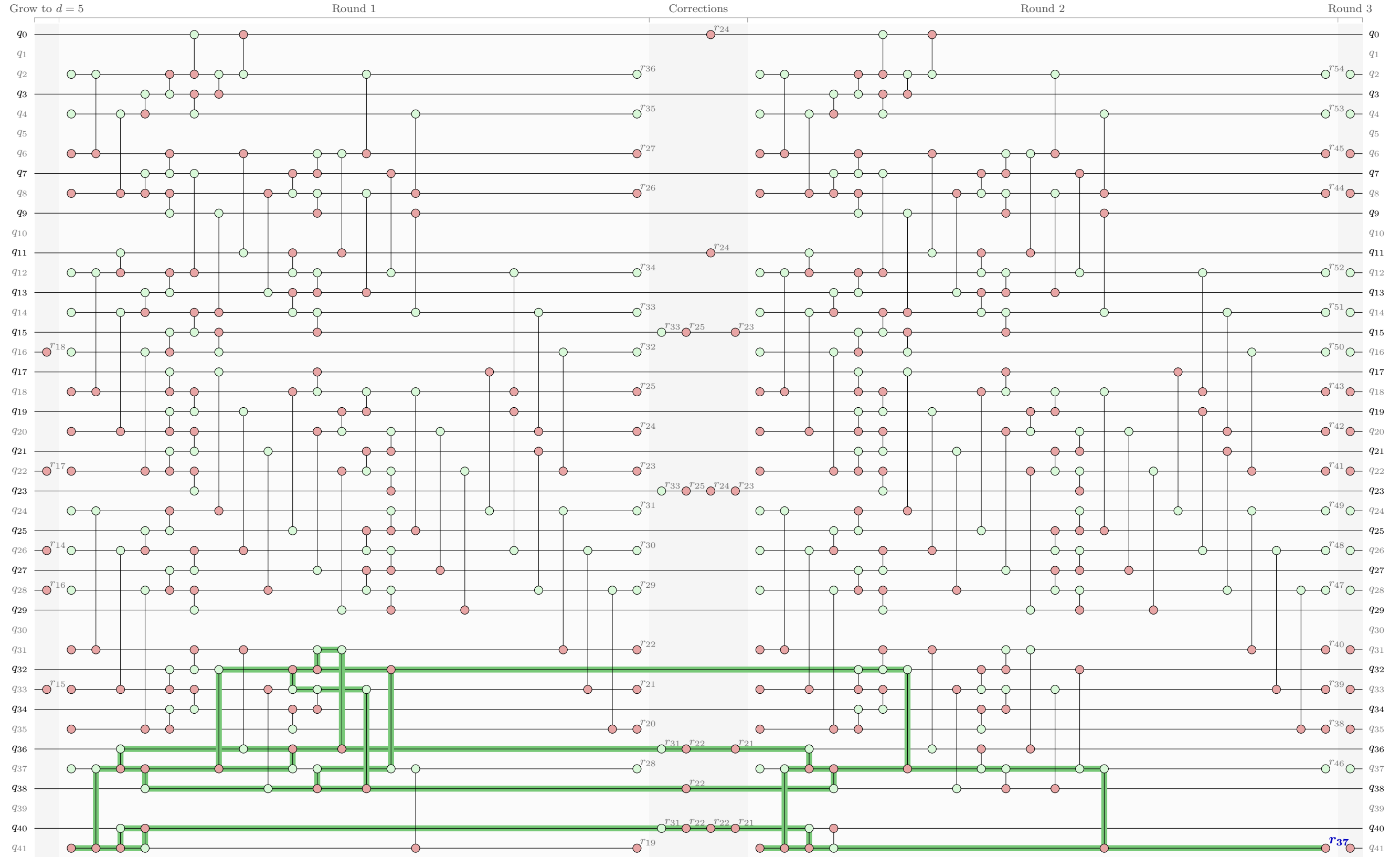


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{28} + r_{46}}$. So $D_{\mathbf{r}} = 0$ unless $r_{28} \oplus r_{46} = 0$.

Records: r_{28} : MX q_{37} , line 251; r_{46} : MX q_{37} , line 310.

$d = 5$: detector 50 $r_{37} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

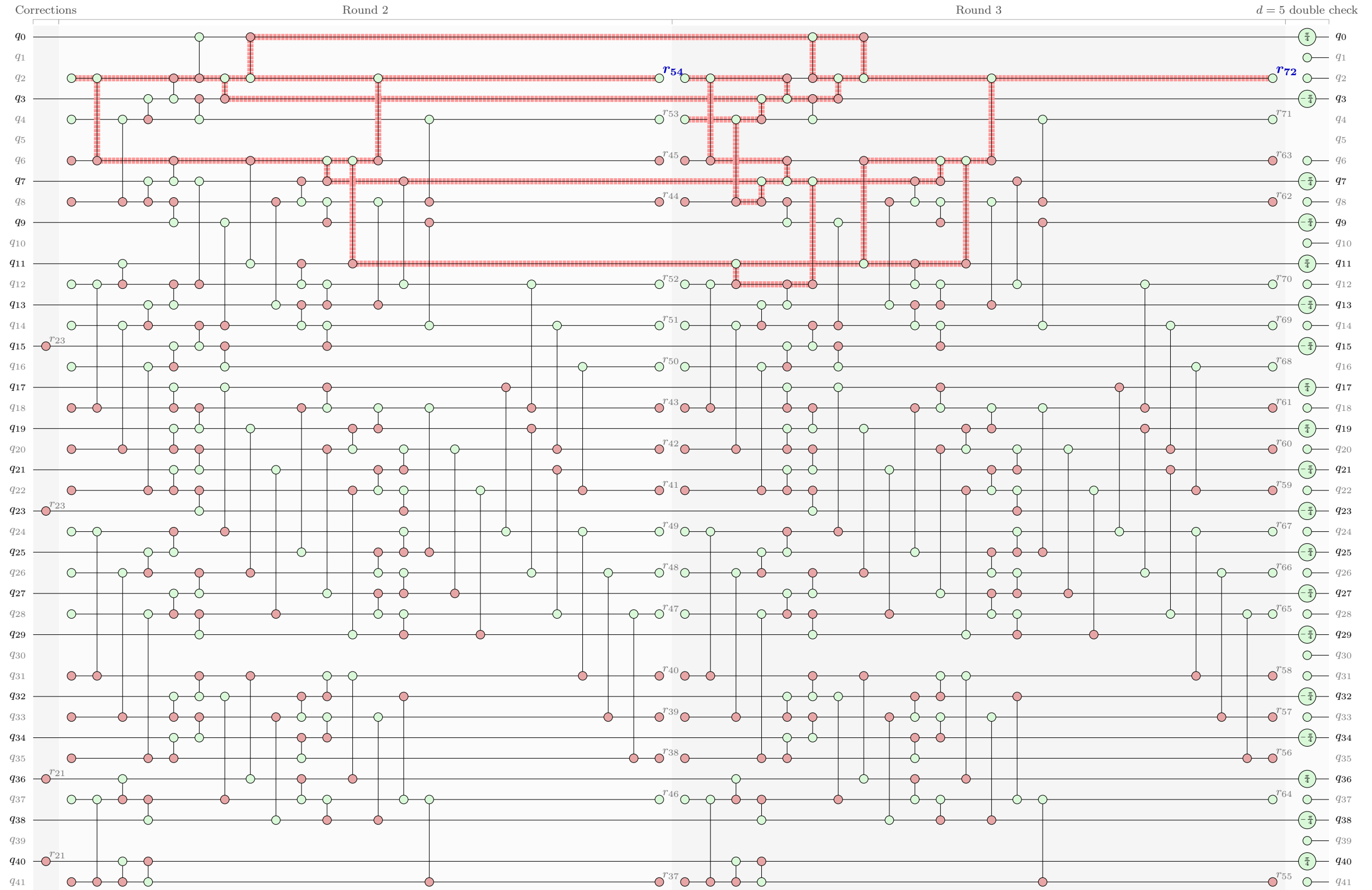


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{37}}$. So $D_{\mathbf{r}} = 0$ unless $r_{37} = 0$.

Records: r_{37} : M q_{41} , line 309.

$d = 5$: detector 51 $r_{54} \oplus r_{72} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

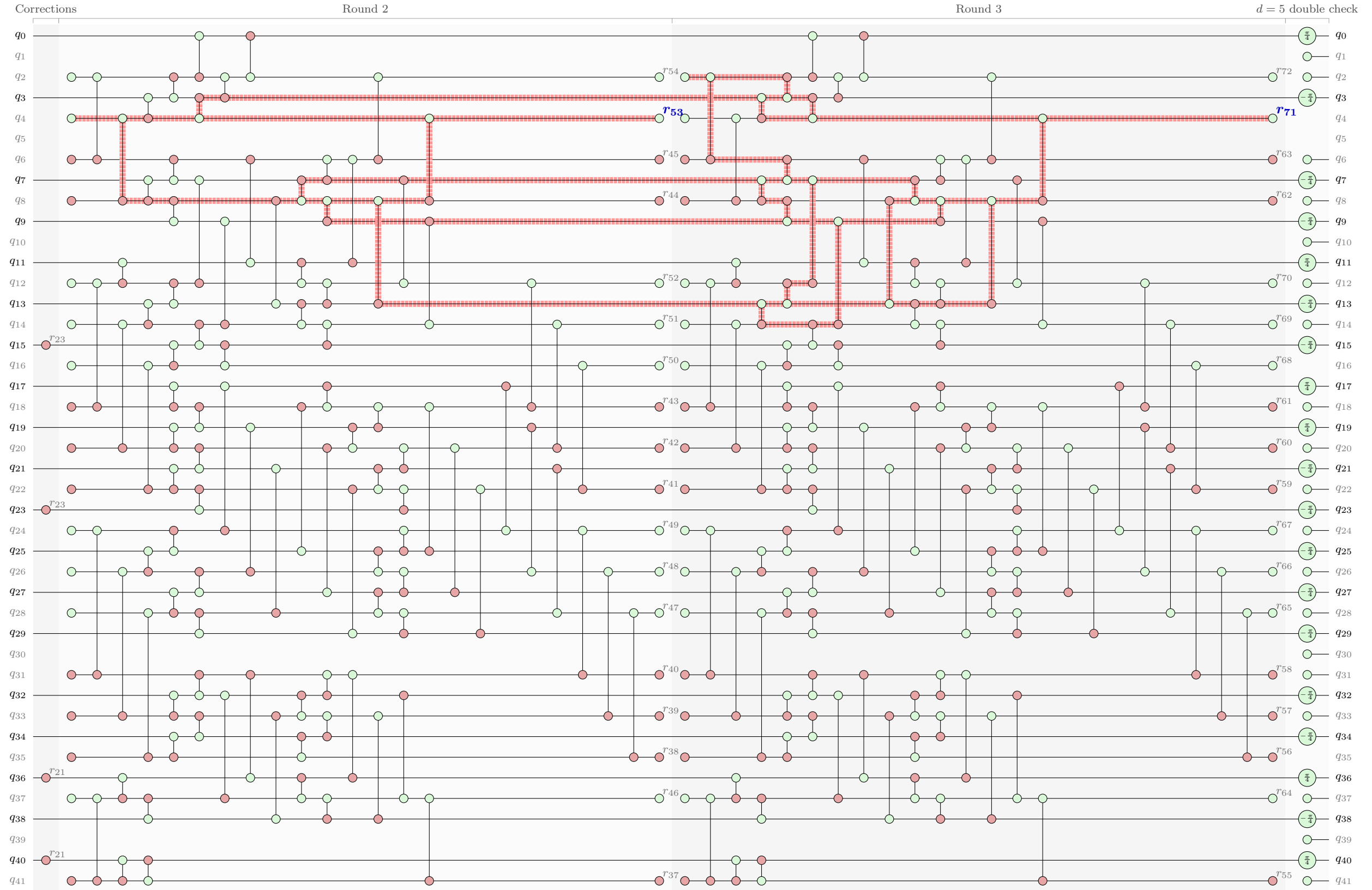


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{54} + r_{72}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{54} \oplus r_{72} = 0.$$

Records: r_{54} : MX q_2 , line 310; r_{72} : MX q_2 , line 362.

$d = 5$: detector 52 $r_{53} \oplus r_{71} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.



$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{53} + r_{71}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{53} \oplus r_{71} = 0.$$

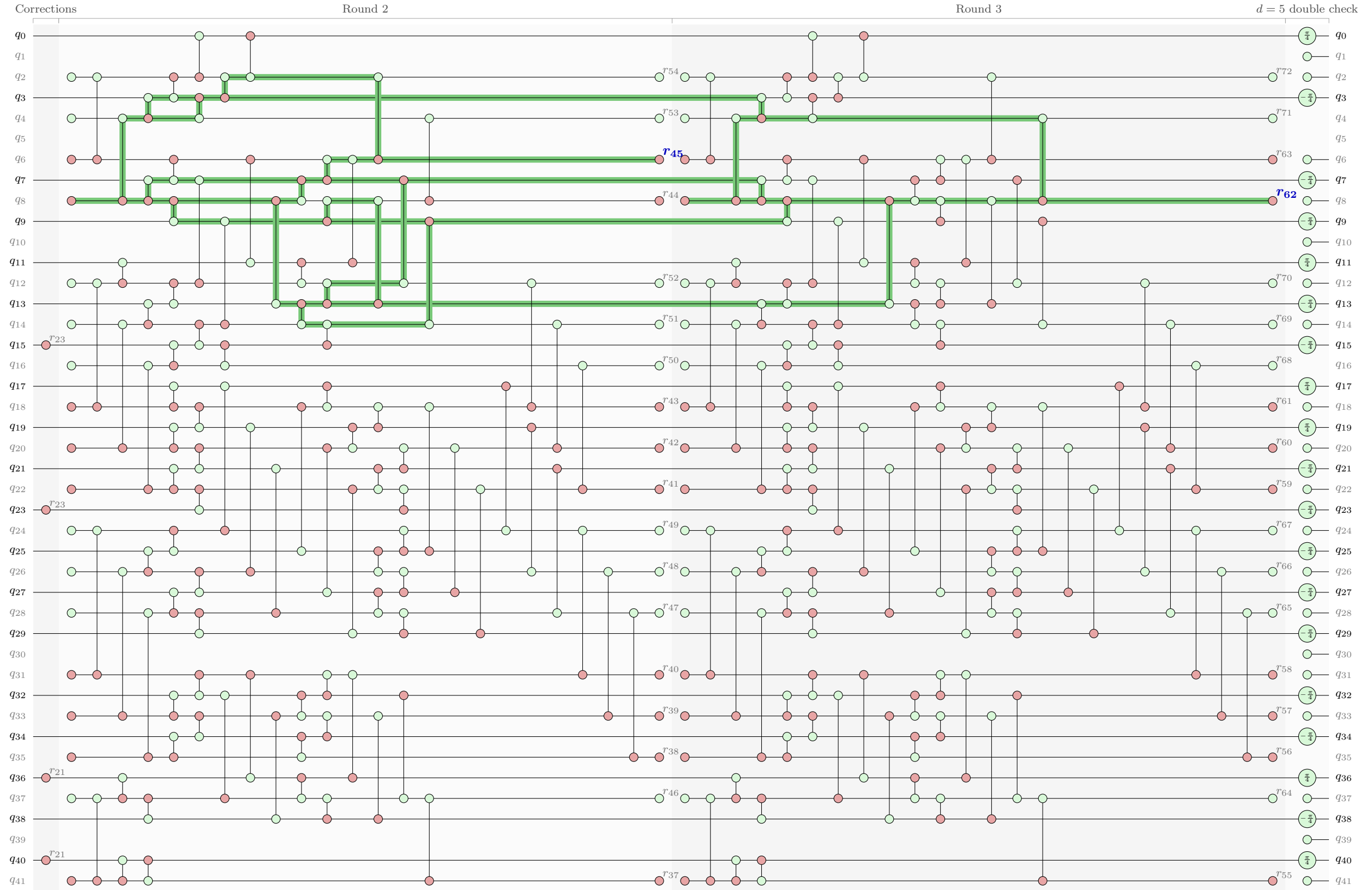
Records: r_{53} : MX q_4 , line 310; r_{71} : MX q_4 , line 362.

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

Records: r_{44} : M q_8 , line 309; r_{45} : M q_6 , line 309; r_{63} : M q_6 , line 361.

$d = 5$: detector 54 $r_{45} \oplus r_{62} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

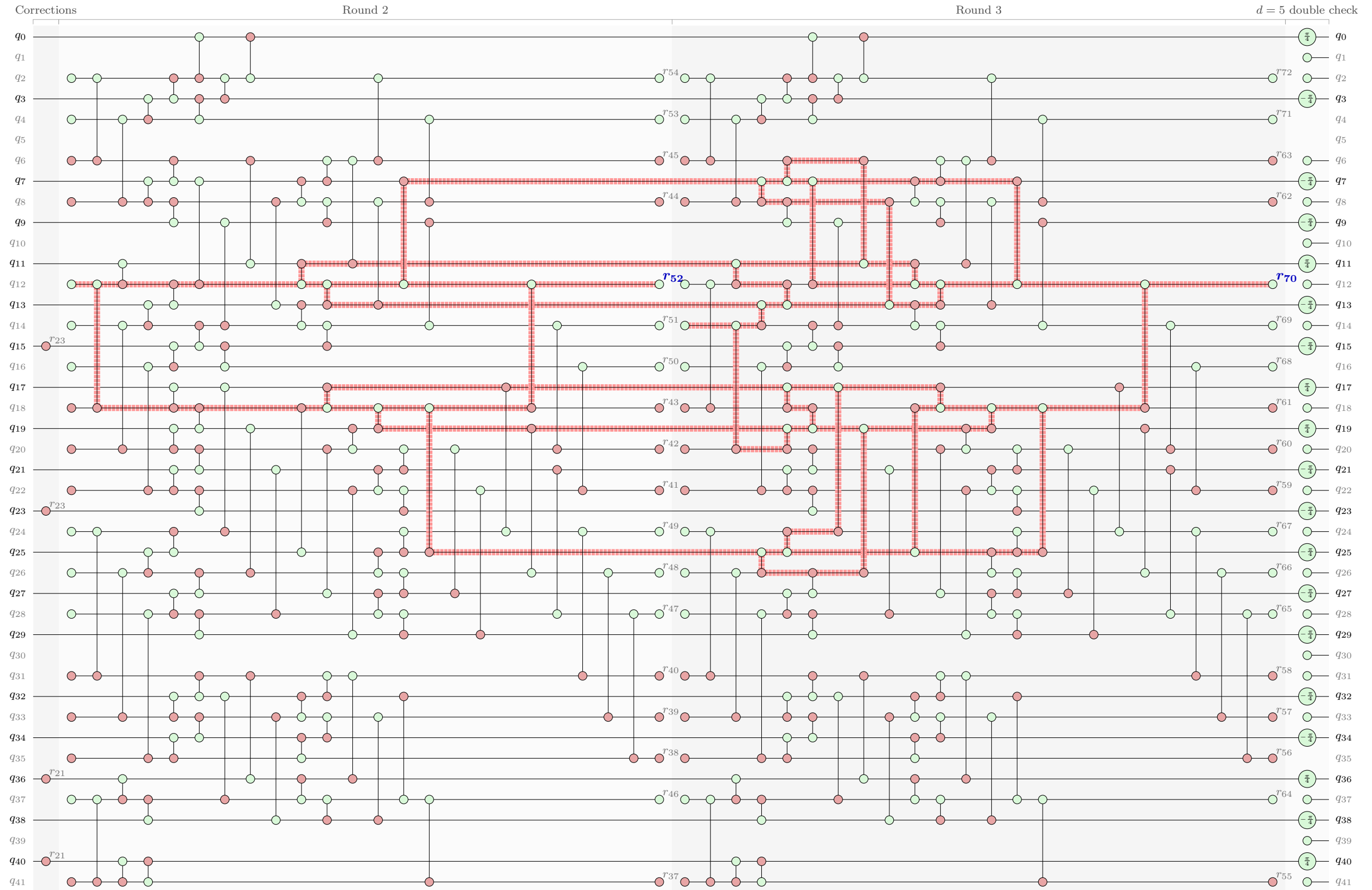


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{45} + r_{62}}$. So $D_{\mathbf{r}} = 0$ unless $r_{45} \oplus r_{62} = 0$.

Records: r_{45} : M q_6 , line 309; r_{62} : M q_8 , line 361.

$d = 5$: detector 55 $r_{52} \oplus r_{70} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

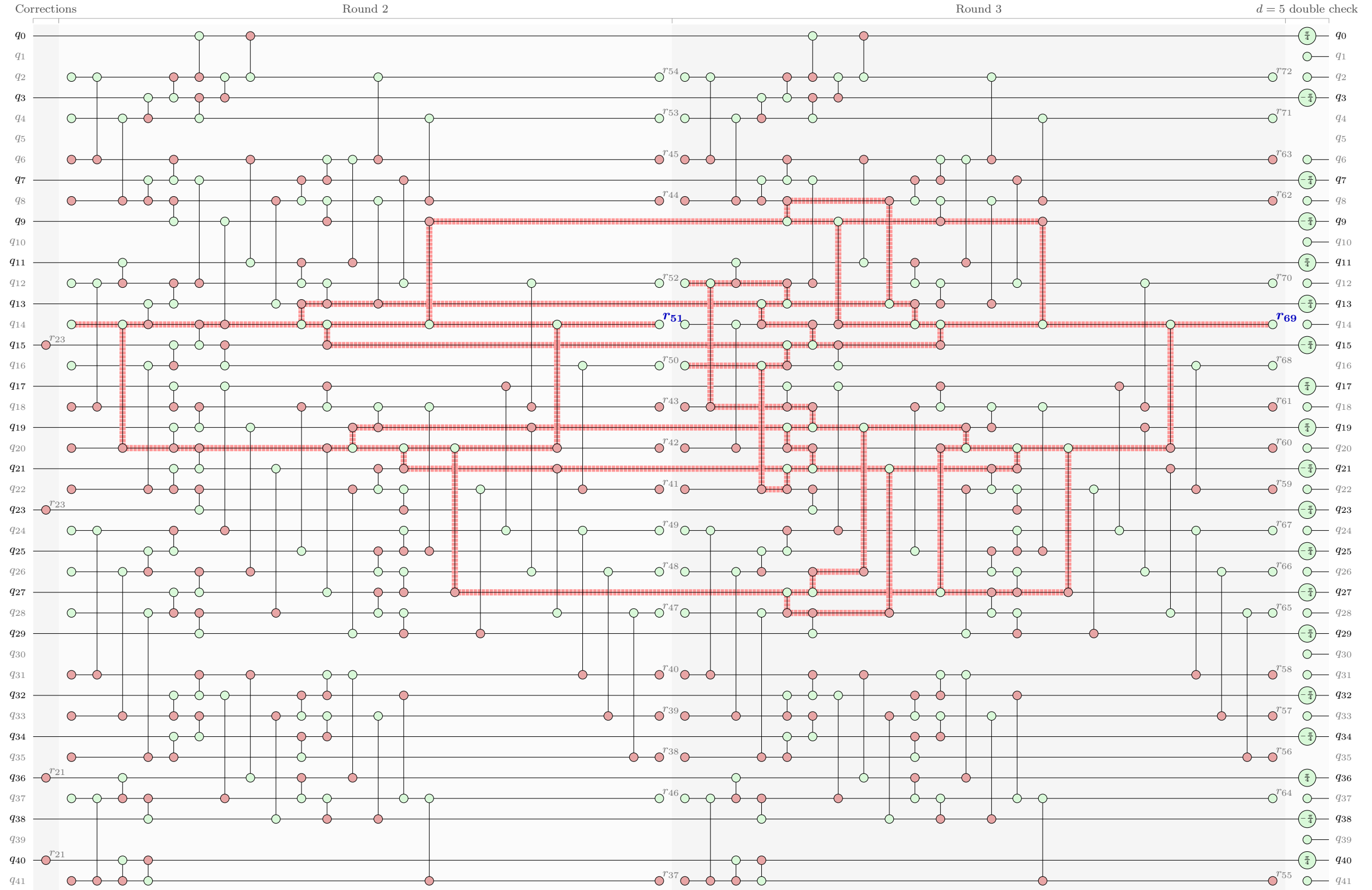


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{52} + r_{70}}$. So $D_{\mathbf{r}} = 0$ unless $r_{52} \oplus r_{70} = 0$.

Records: r_{52} : MX q_{12} , line 310; r_{70} : MX q_{12} , line 362.

$d = 5$: detector 56 $r_{51} \oplus r_{69} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

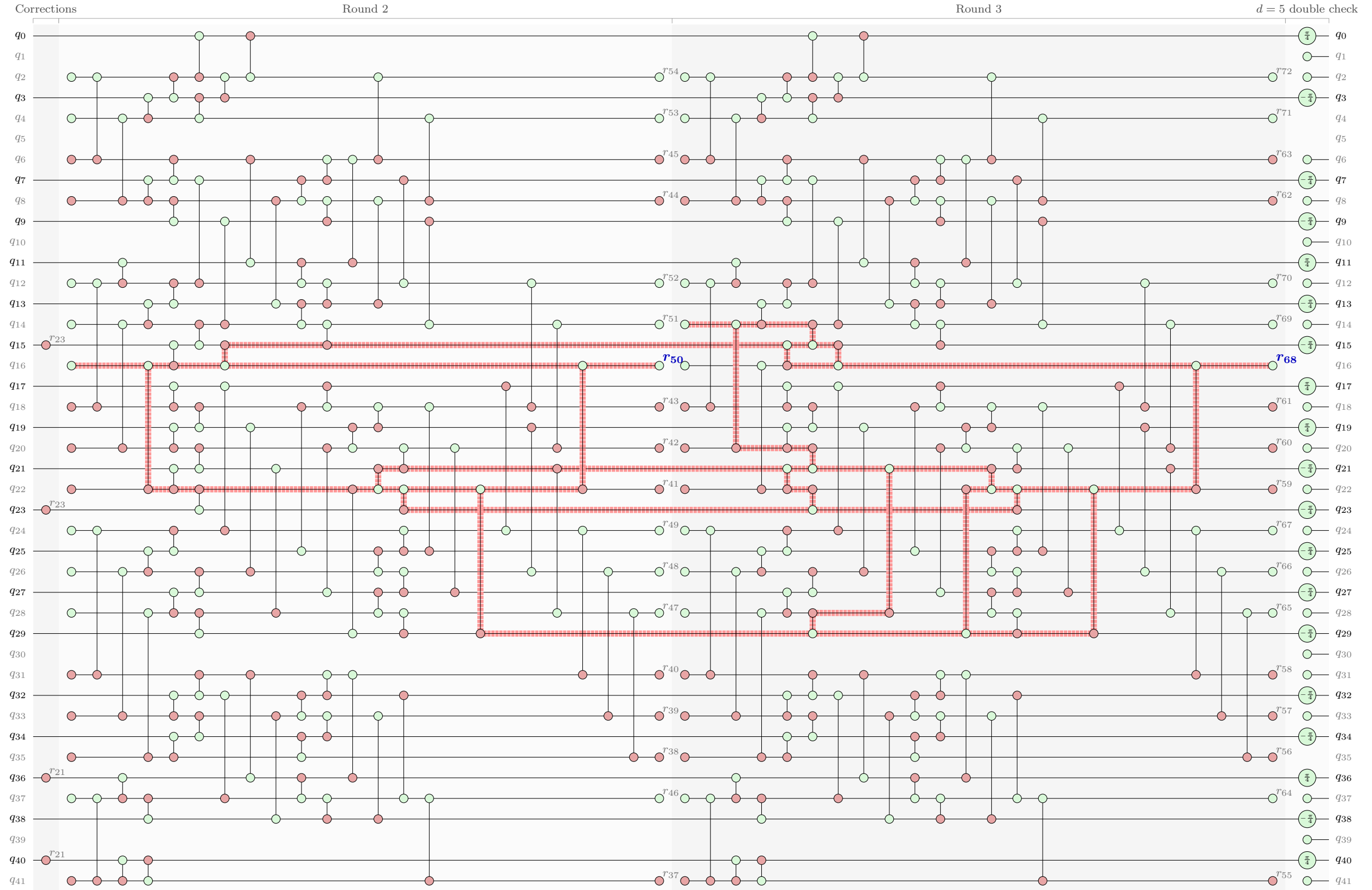


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{51} + r_{69}}$. So $D_{\mathbf{r}} = 0$ unless $r_{51} \oplus r_{69} = 0$.

Records: r_{51} : MX q_{14} , line 310; r_{69} : MX q_{14} , line 362.

$d = 5$: detector 57 $r_{50} \oplus r_{68} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

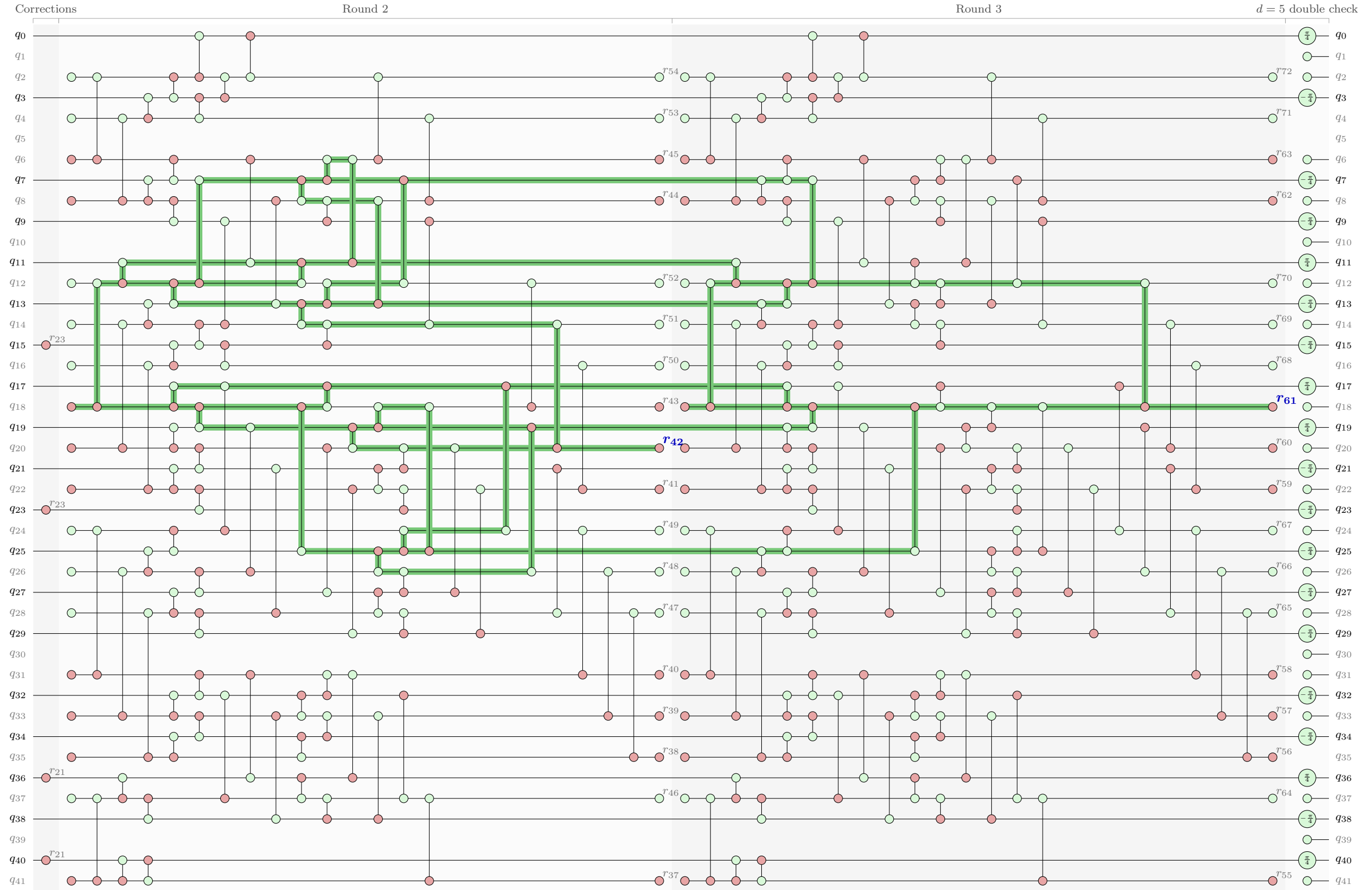


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{50}+r_{68}}$. So $D_{\mathbf{r}} = 0$ unless $r_{50} \oplus r_{68} = 0$.

Records: r_{50} : MX q_{16} , line 310; r_{68} : MX q_{16} , line 362.

$d = 5$: detector 58 $r_{42} \oplus r_{61} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

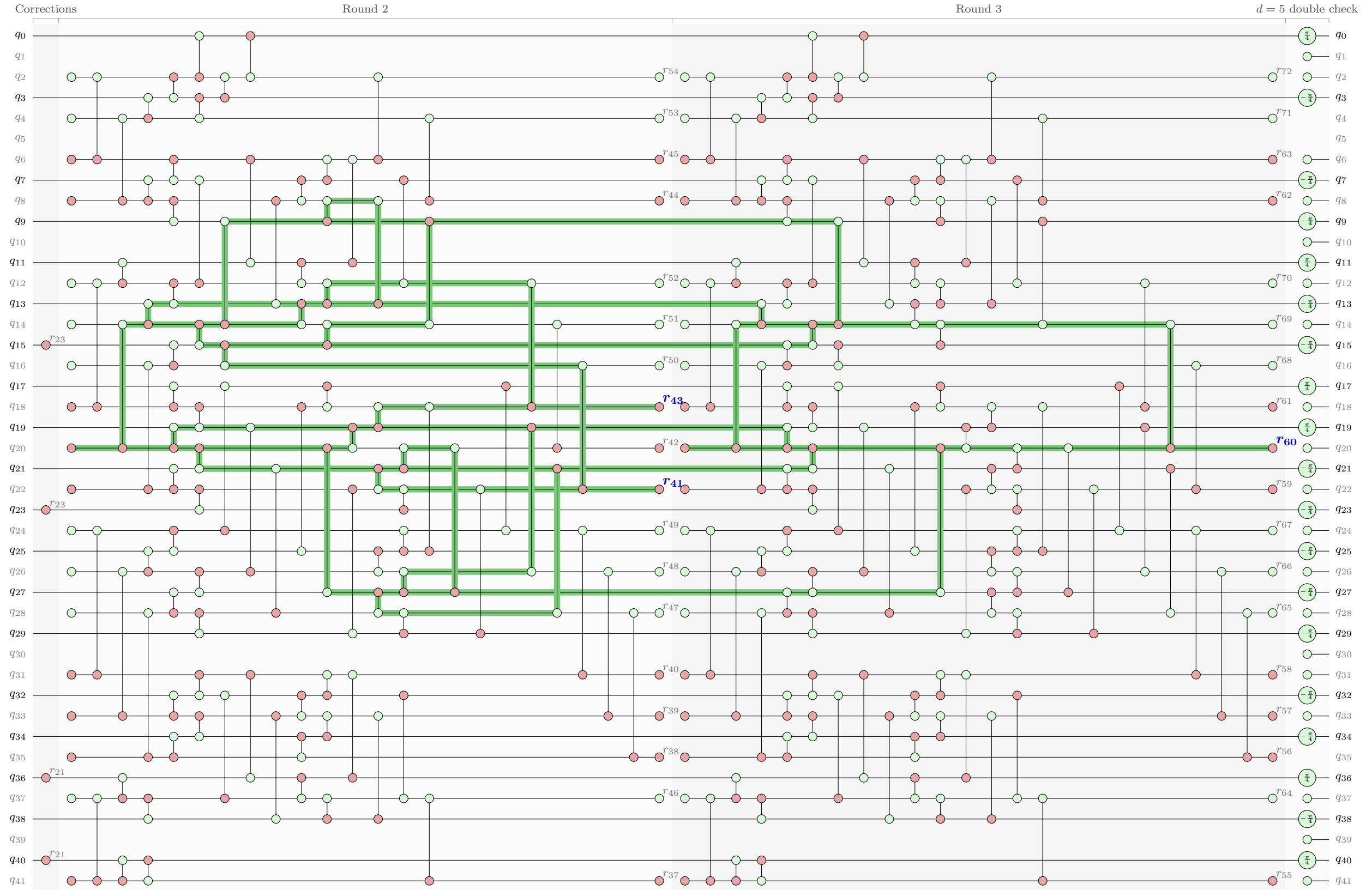


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{42} + r_{61}}$. So $D_{\mathbf{r}} = 0$ unless $r_{42} \oplus r_{61} = 0$.

Records: r_{42} : M q_{20} , line 309; r_{61} : M q_{18} , line 361.

$d = 5$: detector 59 $r_{41} \oplus r_{43} \oplus r_{60} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

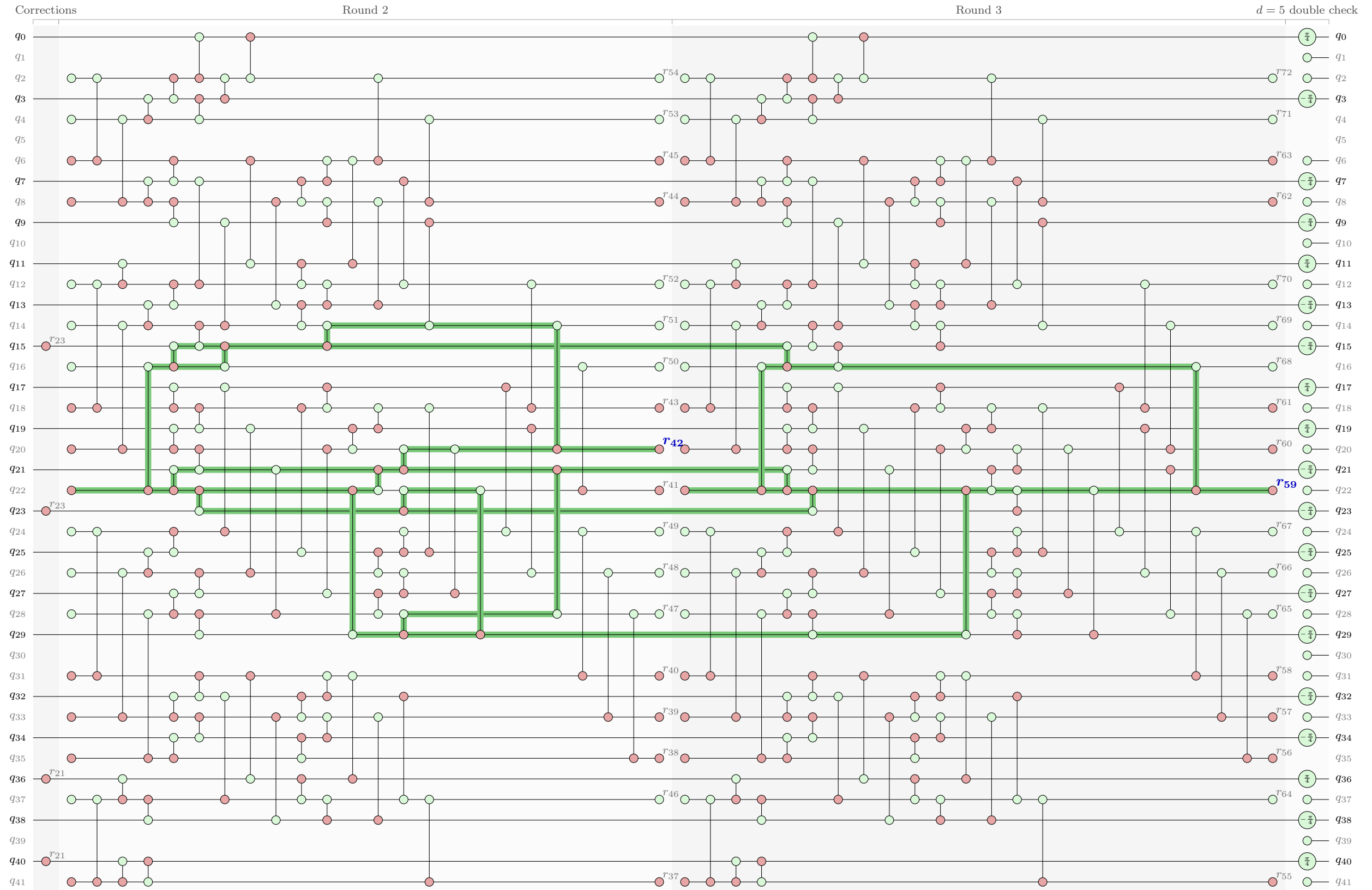


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{41}+r_{43}+r_{60}}$. So $D_{\mathbf{r}} = 0$ unless $r_{41} \oplus r_{43} \oplus r_{60} = 0$.

Records: r_{41} : M q_{22} , line 309; r_{43} : M q_{18} , line 309; r_{60} : M q_{20} , line 361.

$d = 5$: detector 60 $r_{42} \oplus r_{59} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

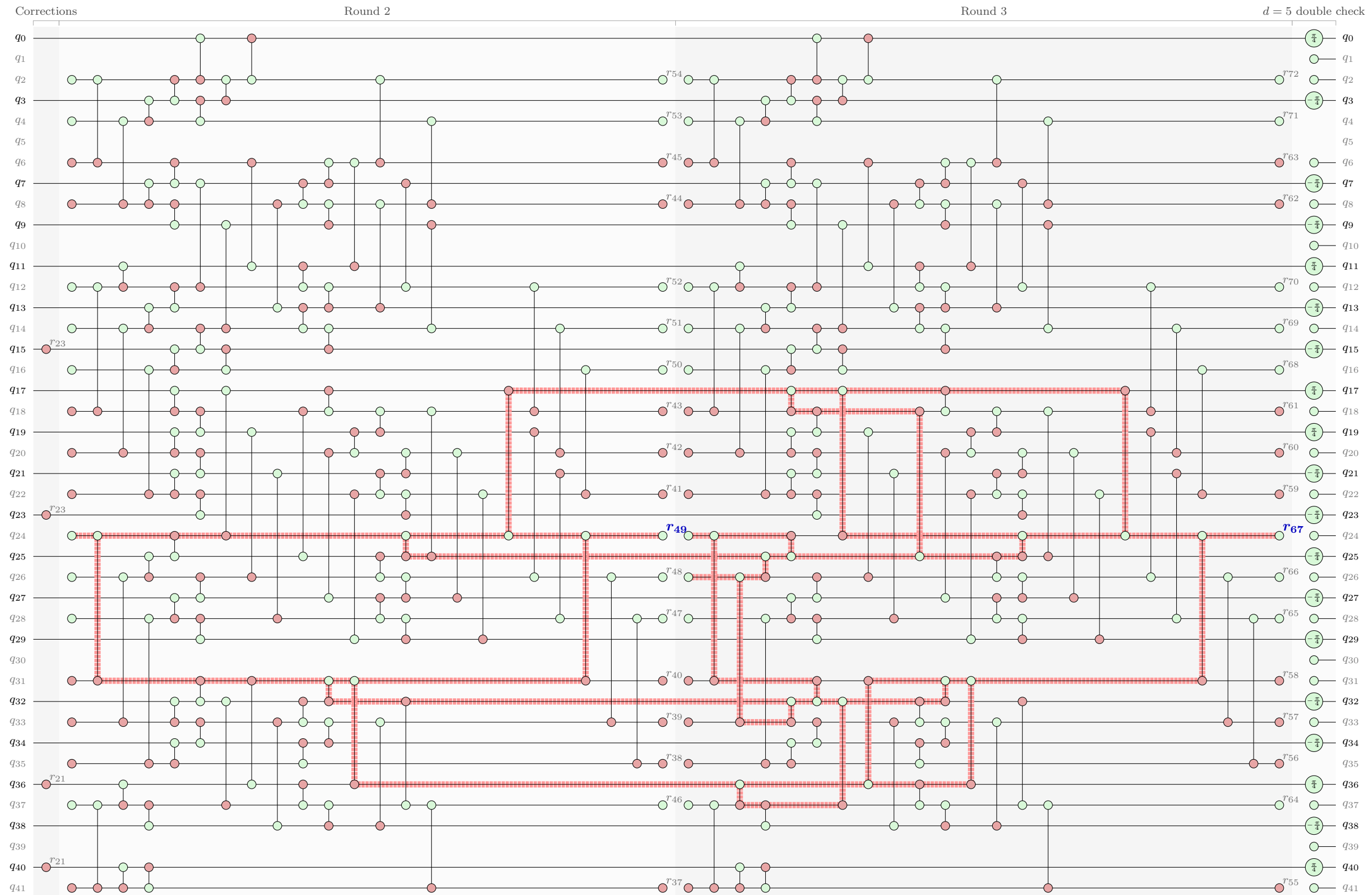


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{42} + r_{59}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{42} \oplus r_{59} = 0.$$

Records: r_{42} : M q_{20} , line 309; r_{59} : M q_{22} , line 361.

$d = 5$: detector 61 $r_{49} \oplus r_{67} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

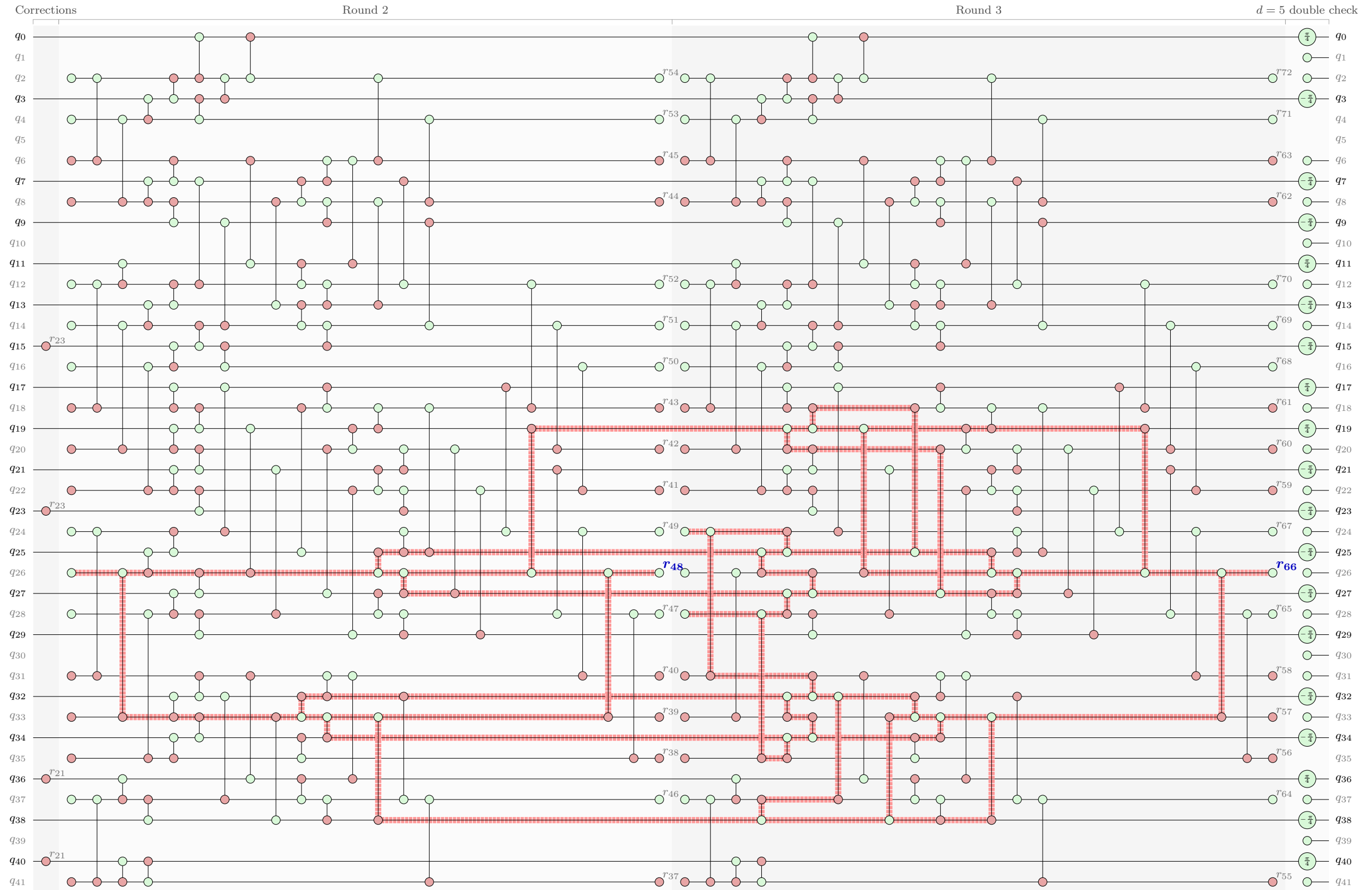


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{49} + r_{67}}$. So $D_{\mathbf{r}} = 0$ unless $r_{49} \oplus r_{67} = 0$.

Records: r_{49} : MX q_{24} , line 310; r_{67} : MX q_{24} , line 362.

$d = 5$: detector 62 $r_{48} \oplus r_{66} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

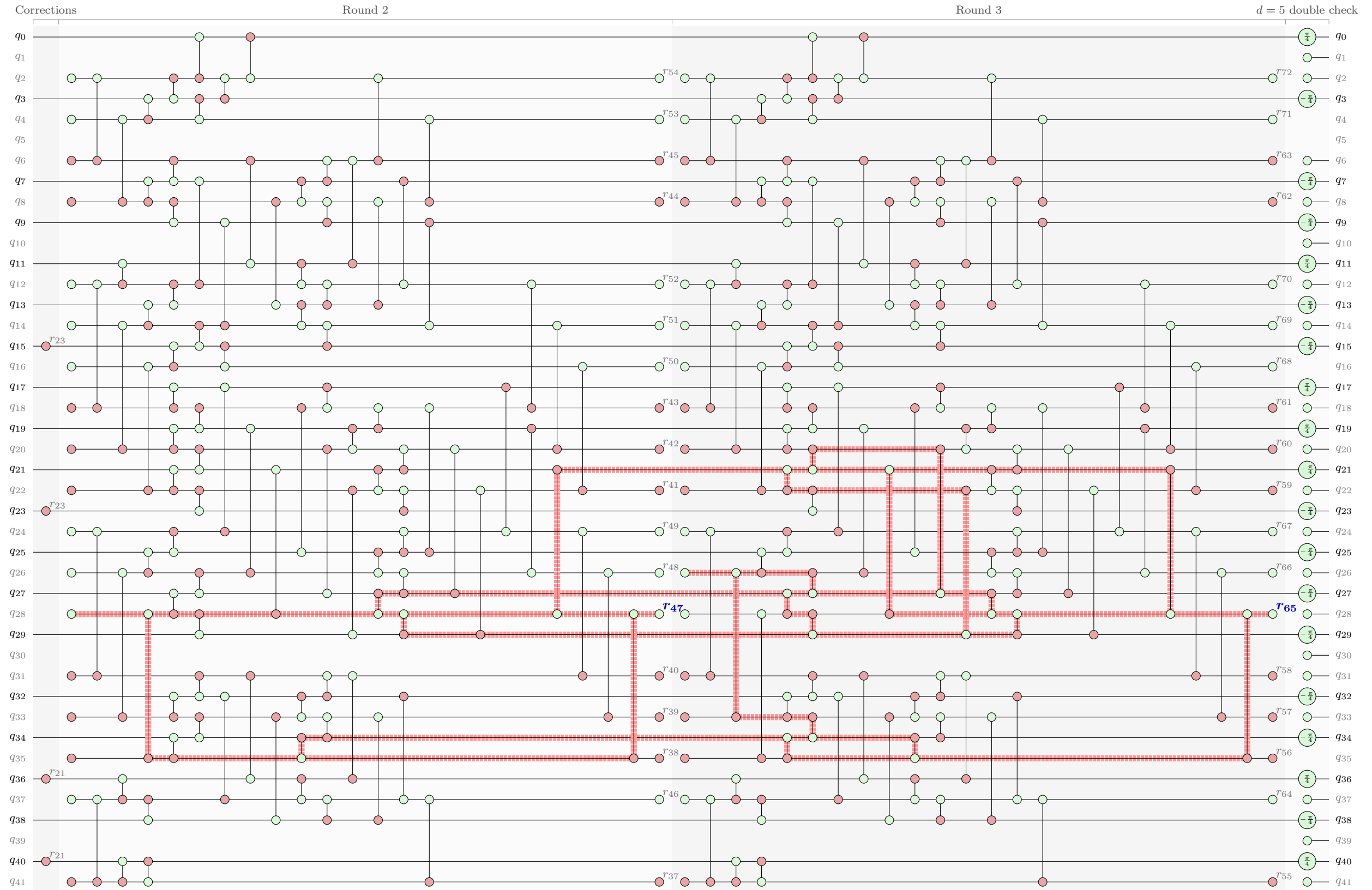


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{48} + r_{66}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{48} \oplus r_{66} = 0.$$

Records: r_{48} : MX q_{26} , line 310; r_{66} : MX q_{26} , line 362.

$d = 5$: detector 63 $r_{47} \oplus r_{65} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

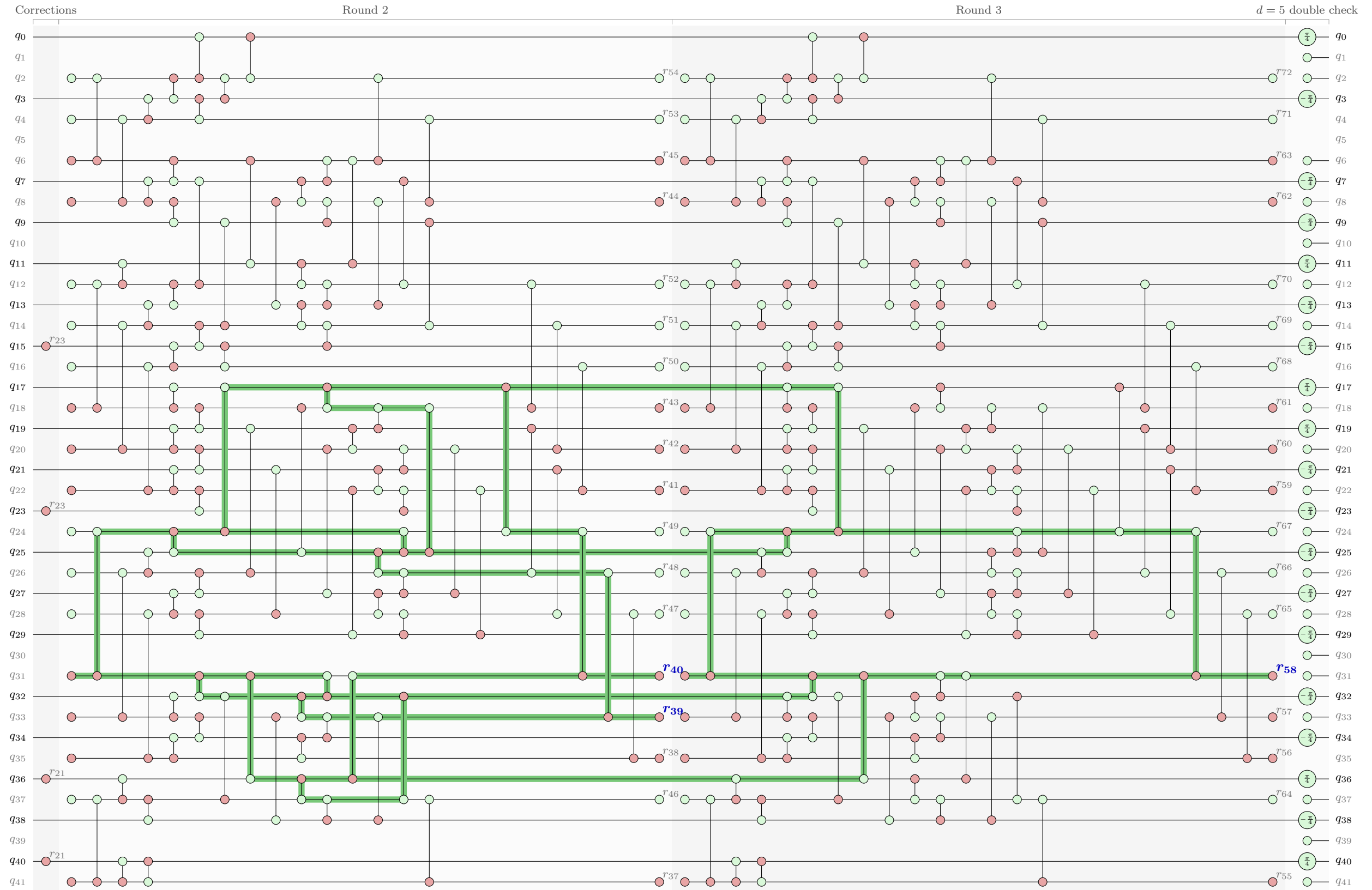


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{47} + r_{65}}$. So $D_{\mathbf{r}} = 0$ unless $r_{47} \oplus r_{65} = 0$.

Records: r_{47} : MX q_{28} , line 310; r_{65} : MX q_{28} , line 362.

$d = 5$: detector 64 $r_{39} \oplus r_{40} \oplus r_{58} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

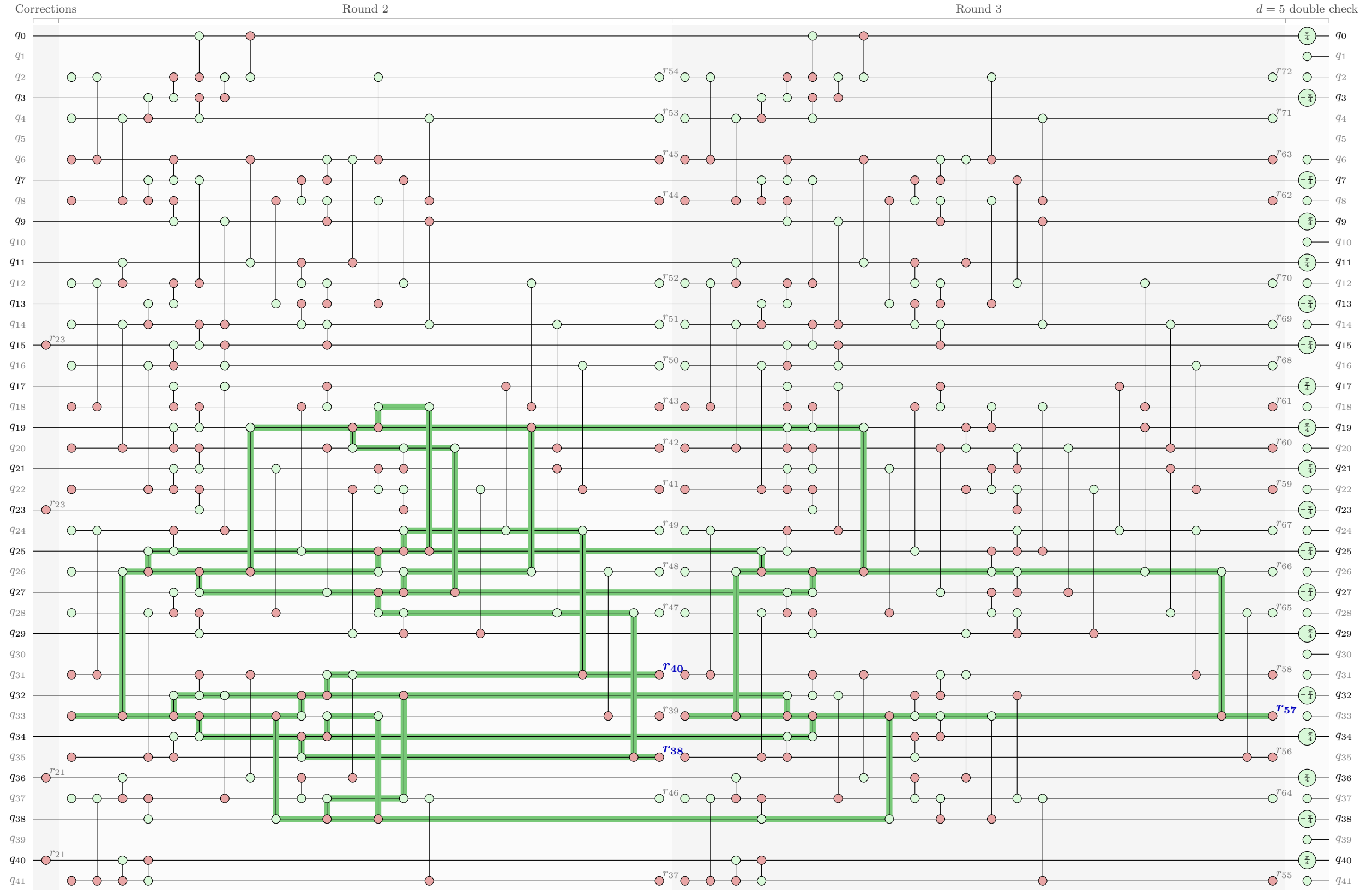


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{39}+r_{40}+r_{58}}$. So $D_{\mathbf{r}} = 0$ unless $r_{39} \oplus r_{40} \oplus r_{58} = 0$.

Records: r_{39} : M q_{33} , line 309; r_{40} : M q_{31} , line 309; r_{58} : M q_{31} , line 361.

$d = 5$: detector 65 $r_{38} \oplus r_{40} \oplus r_{57} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

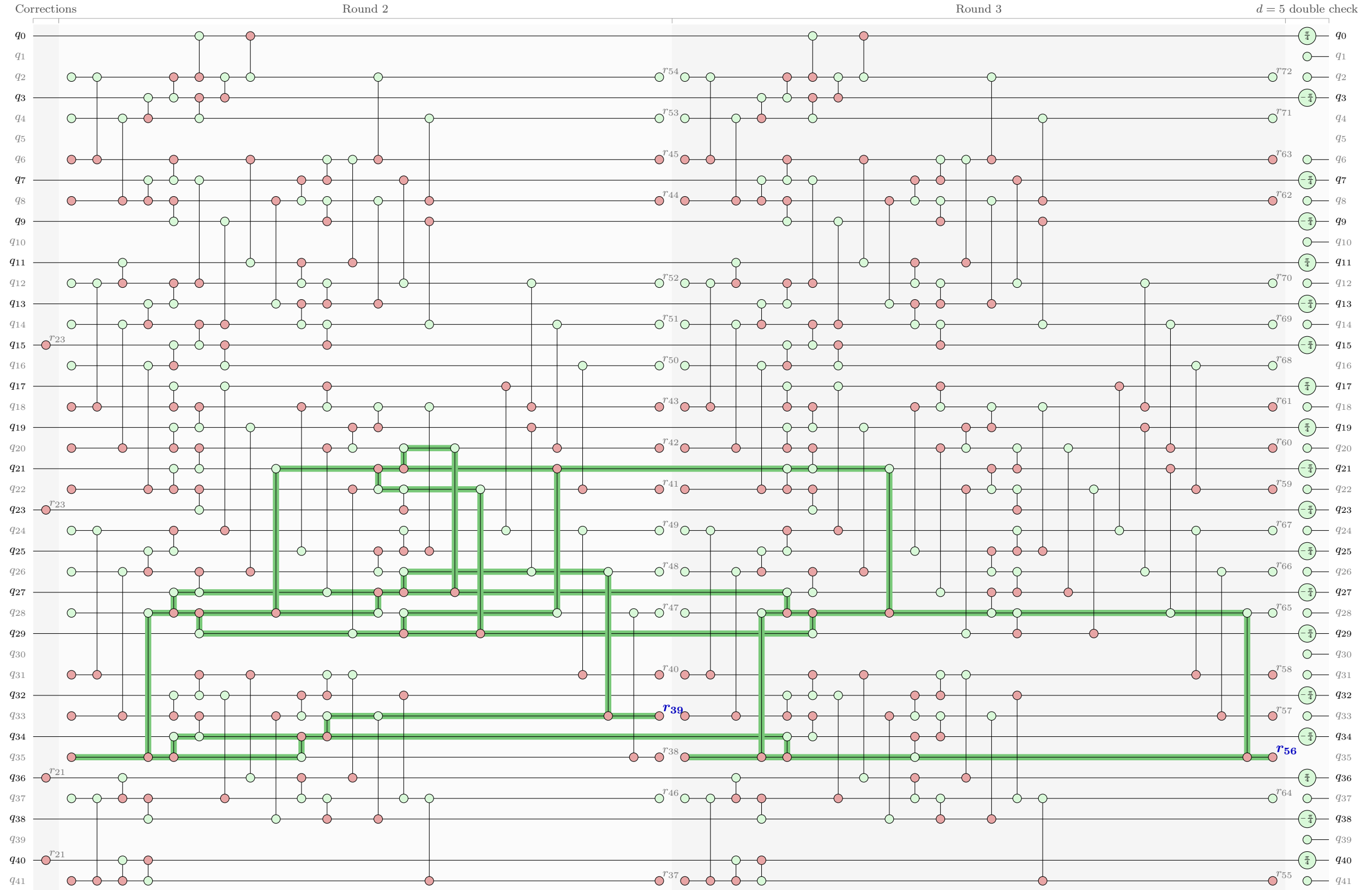


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{38}+r_{40}+r_{57}}$. So $D_{\mathbf{r}} = 0$ unless $r_{38} \oplus r_{40} \oplus r_{57} = 0$.

Records: r_{38} : M q_{35} , line 309; r_{40} : M q_{31} , line 309; r_{57} : M q_{33} , line 361.

$d = 5$: detector 66 $r_{39} \oplus r_{56} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

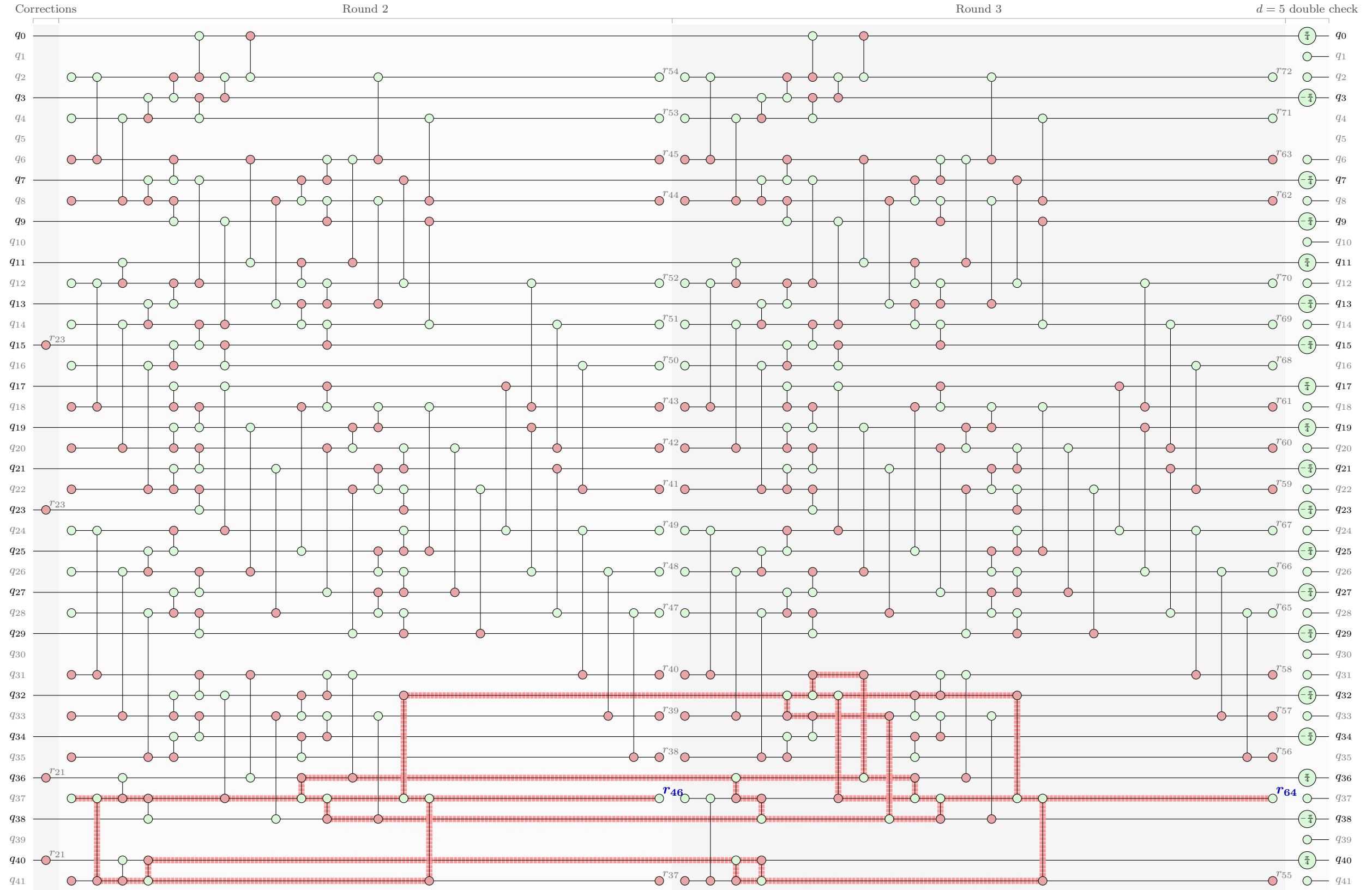


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{39} + r_{56}}$. So $D_{\mathbf{r}} = 0$ unless $r_{39} \oplus r_{56} = 0$.

Records: r_{39} : M q_{33} , line 309; r_{56} : M q_{35} , line 361.

$d = 5$: detector 67 $r_{46} \oplus r_{64} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

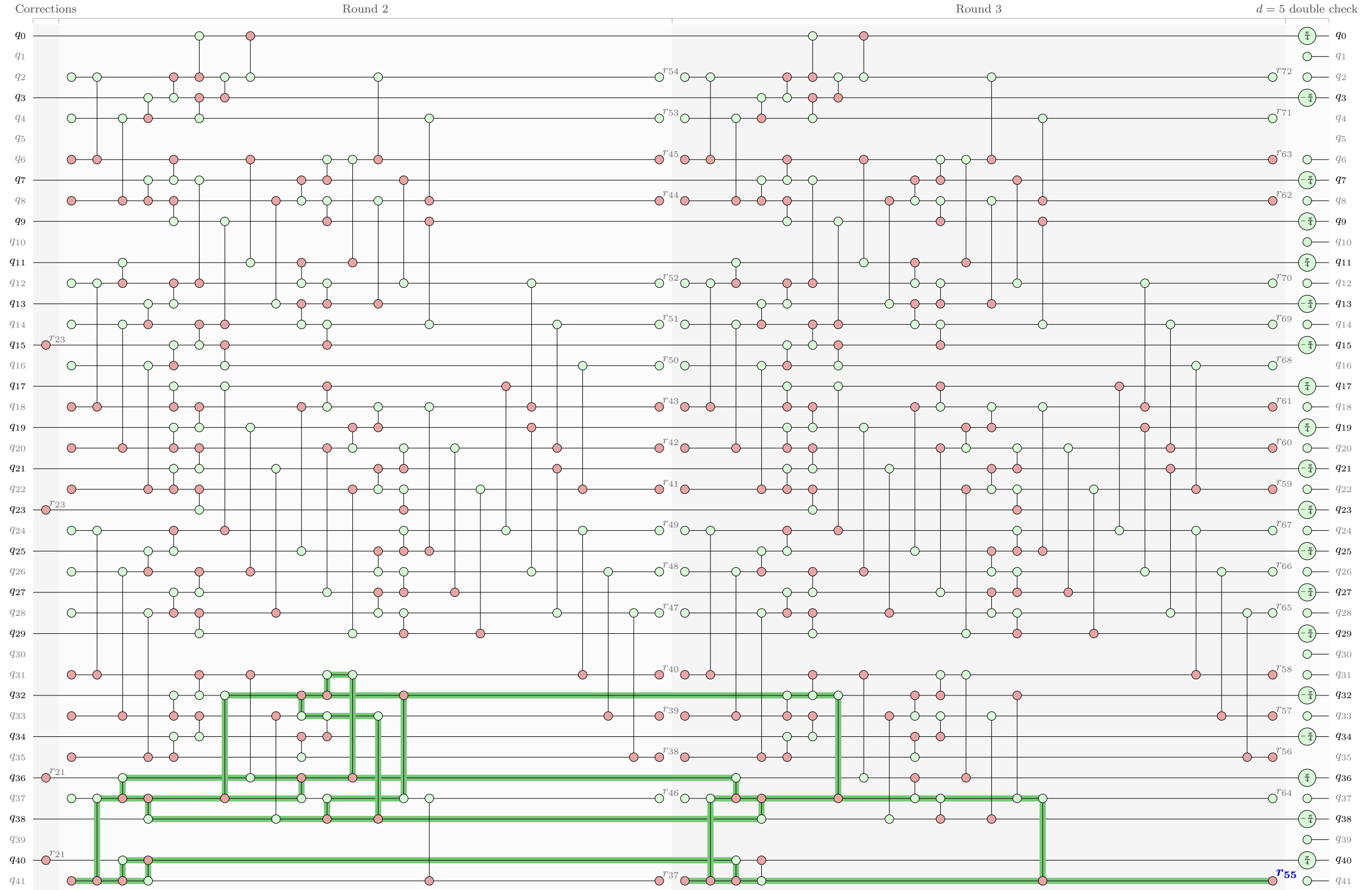


$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{46} + r_{64}}. \quad \text{So } D_{\mathbf{r}} = 0 \text{ unless } r_{46} \oplus r_{64} = 0.$$

Records: r_{46} : MX q_{37} , line 310; r_{64} : MX q_{37} , line 362.

$d = 5$: detector 68 $r_{55} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

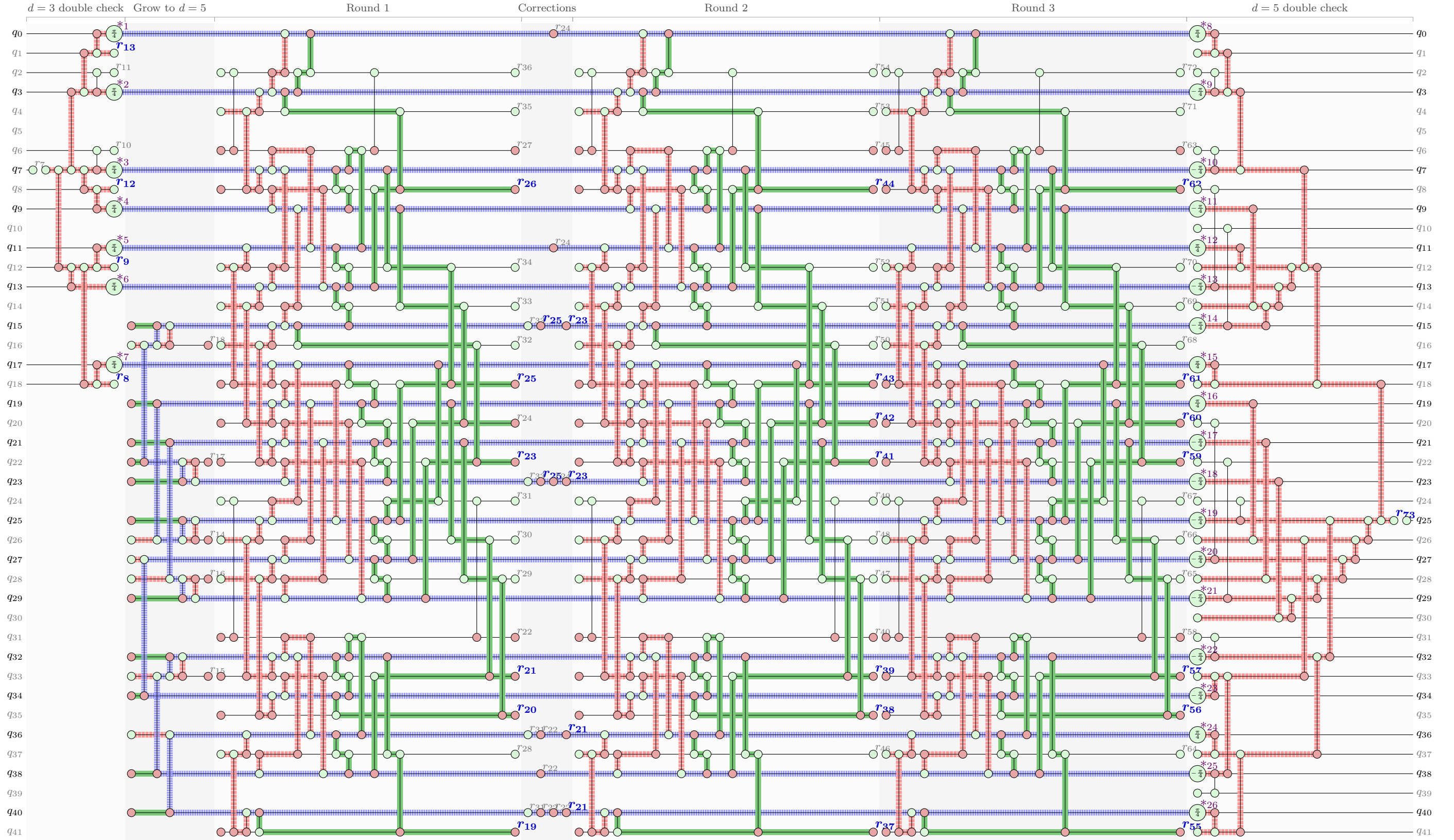


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{55}}$. So $D_{\mathbf{r}} = 0$ unless $r_{55} = 0$.

Records: r_{55} : M q_{41} , line 361.

$d = 5$: detector 69 $r_8 \oplus r_9 \oplus r_{12} \oplus r_{13} \oplus r_{19} \oplus r_{20} \oplus r_{21} \oplus r_{23} \oplus r_{25} \oplus r_{26} \oplus r_{37} \oplus r_{38} \oplus r_{39} \oplus r_{41} \oplus r_{42} \oplus r_{43} \oplus r_{44} \oplus r_{55} \oplus r_{56} \oplus r_{57} \oplus r_{59} \oplus r_{60} \oplus r_{61} \oplus r_{62} \oplus r_{73} = 0$

Phase-dependent. Closed semiweb with 26 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



$$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}, \quad \lambda(\mathbf{r}) = (-1)^{r_8+r_9+r_{12}+r_{13}+r_{19}+r_{20}+r_{21}+r_{23}+r_{25}+r_{26}+r_{37}+r_{38}+r_{39}+r_{41}+r_{42}+r_{43}+r_{44}+r_{55}+r_{56}+r_{57}+r_{59}+r_{60}+r_{61}+r_{62}+r_{73}}.$$

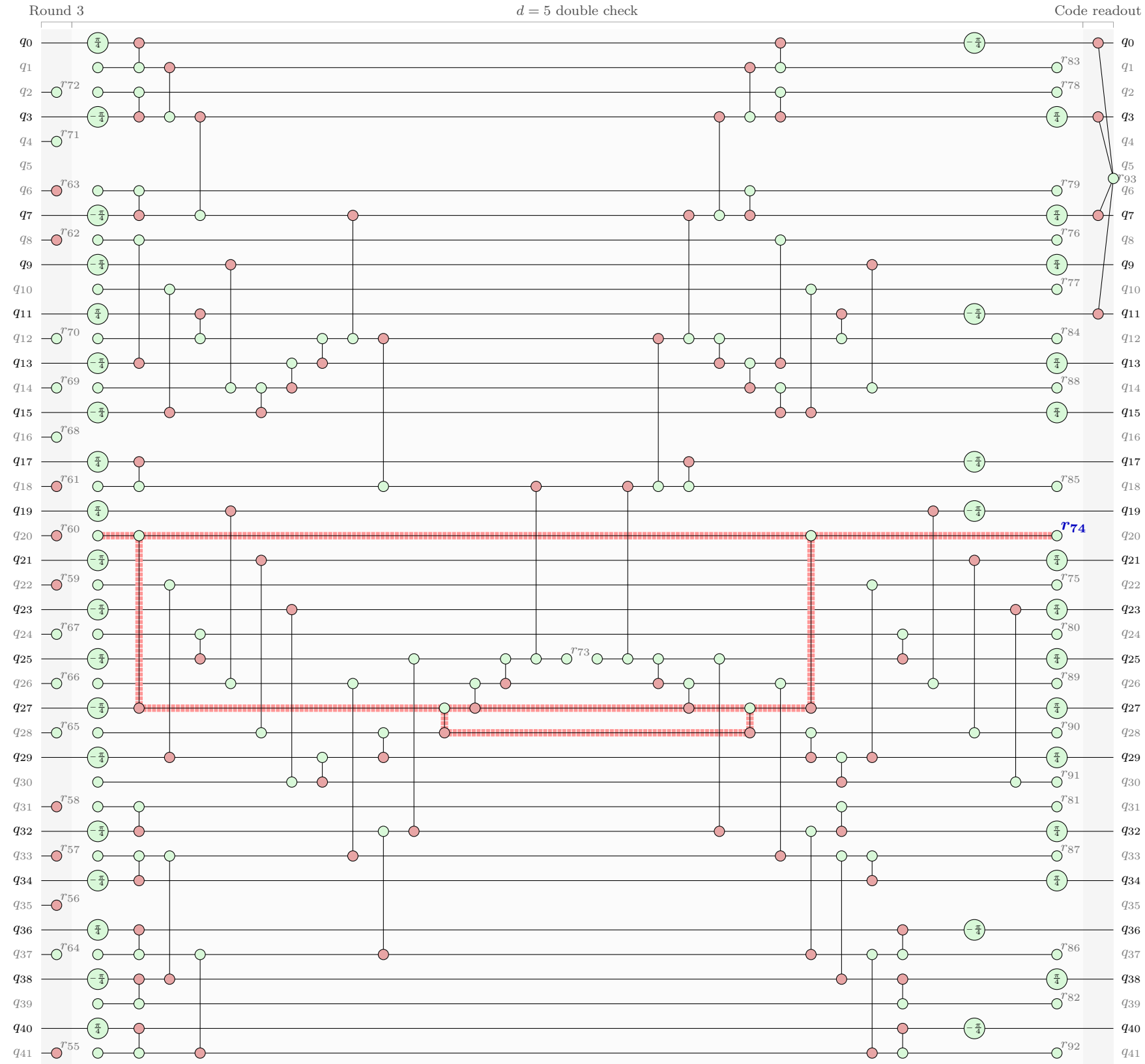
$D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_8 : MX q_{18} , line 176; r_9 : MX q_{12} , line 176; r_{12} : MX q_8 , line 176; r_{13} : MX q_1 , line 176; r_{19} : M q_{41} , line 250; r_{20} : M q_{35} , line 250; r_{21} : M q_{33} , line 250; r_{23} : M q_{22} , line 250; r_{25} : M q_{18} , line 250; r_{26} : M q_8 , line 250; r_{37} : M q_{41} , line 309; r_{38} : M q_{35} , line 309; r_{39} : M q_{33} , line 309; r_{41} : M q_{22} , line 309; r_{42} : M q_{20} , line 309; r_{43} : M q_{18} , line 309; r_{44} : M q_8 , line 309; r_{55} : M q_{41} , line 361; r_{56} : M q_{35} , line 361; r_{57} : M q_{33} , line 361; r_{59} : M q_{22} , line 361; r_{60} : M q_{20} , line 361; r_{61} : M q_{18} , line 361; r_{62} : M q_8 , line 361; r_{73} : MX q_{25} , line 415.

Phase changes: $*_1$: $\delta = \frac{\pi}{2}$ at T on q_0 , line 174; $*_2$: $\delta = \frac{\pi}{2}$ at T on q_3 , line 174; $*_3$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 174; $*_4$: $\delta = \frac{\pi}{2}$ at T on q_9 , line 174; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_{11} , line 174; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_{13} , line 174; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_{17} , line 174; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_0 , line 391; $*_9$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 392; $*_{10}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 392; $*_{11}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_9 , line 392; $*_{12}$: $\delta = \frac{\pi}{2}$ at T on q_{11} , line 391; $*_{13}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 392; $*_{14}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{15} , line 392; $*_{15}$: $\delta = \frac{\pi}{2}$ at T on q_{17} , line 391; $*_{16}$: $\delta = \frac{\pi}{2}$ at T on q_{19} , line 391; $*_{17}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{21} , line 392; $*_{18}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{23} , line 392; $*_{19}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{25} , line 392; $*_{20}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{27} , line 392; $*_{21}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{29} , line 392; $*_{22}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{32} , line 392; $*_{23}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{34} , line 392; $*_{24}$: $\delta = \frac{\pi}{2}$ at T on q_{36} , line 391; $*_{25}$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{38} , line 392; $*_{26}$: $\delta = \frac{\pi}{2}$ at T on q_{40} , line 391.

$d = 5$: detector 70 $r_{74} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

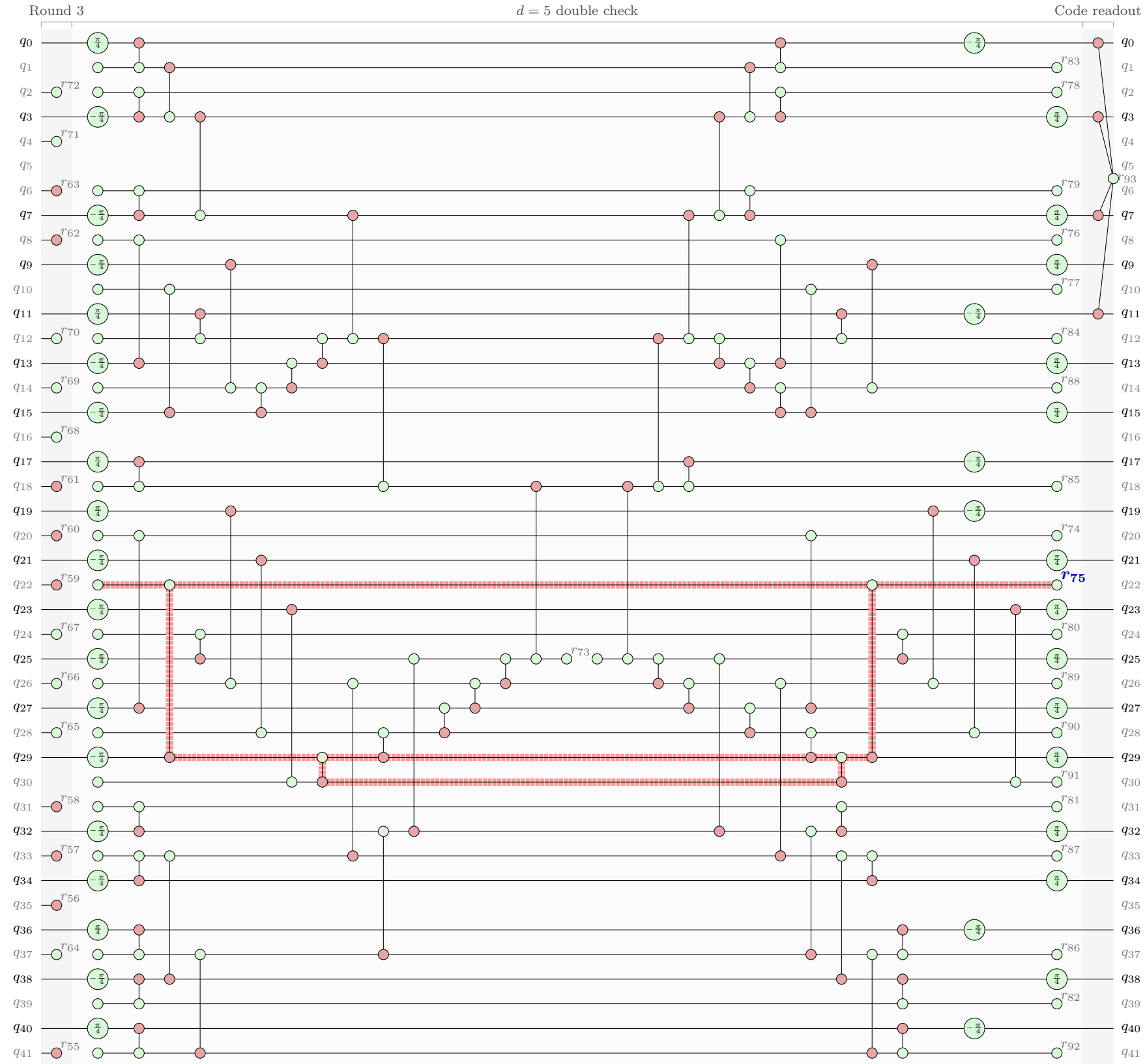


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{74}}$. So $D_{\mathbf{r}} = 0$ unless $r_{74} = 0$.

Records: r_{74} : MX q_{20} , line 444.

$d = 5$: detector 71 $r_{75} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

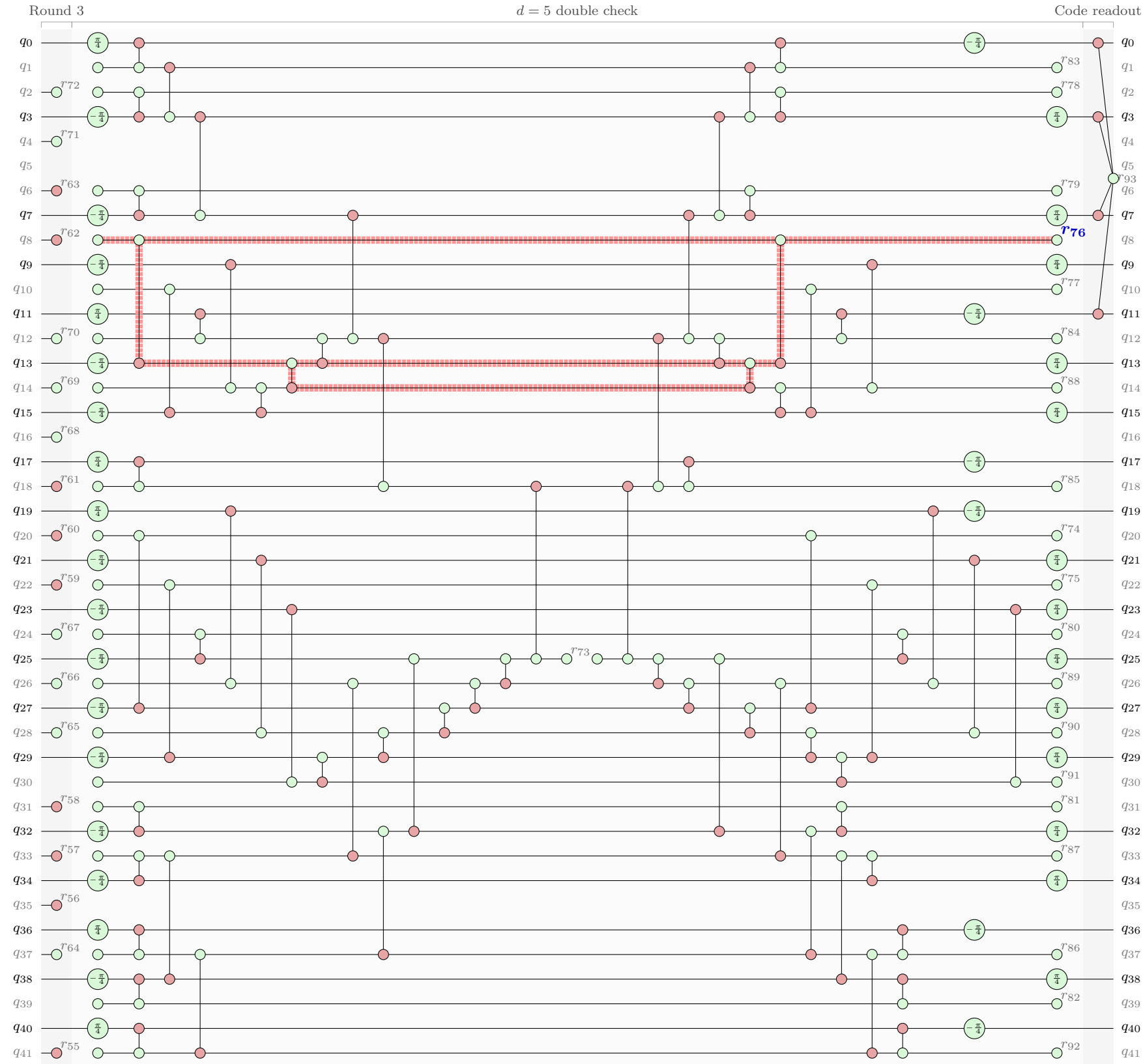


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{75}}$. So $D_{\mathbf{r}} = 0$ unless $r_{75} = 0$.

Records: r_{75} : MX q_{22} , line 444.

$d = 5$: detector 72 $r_{76} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

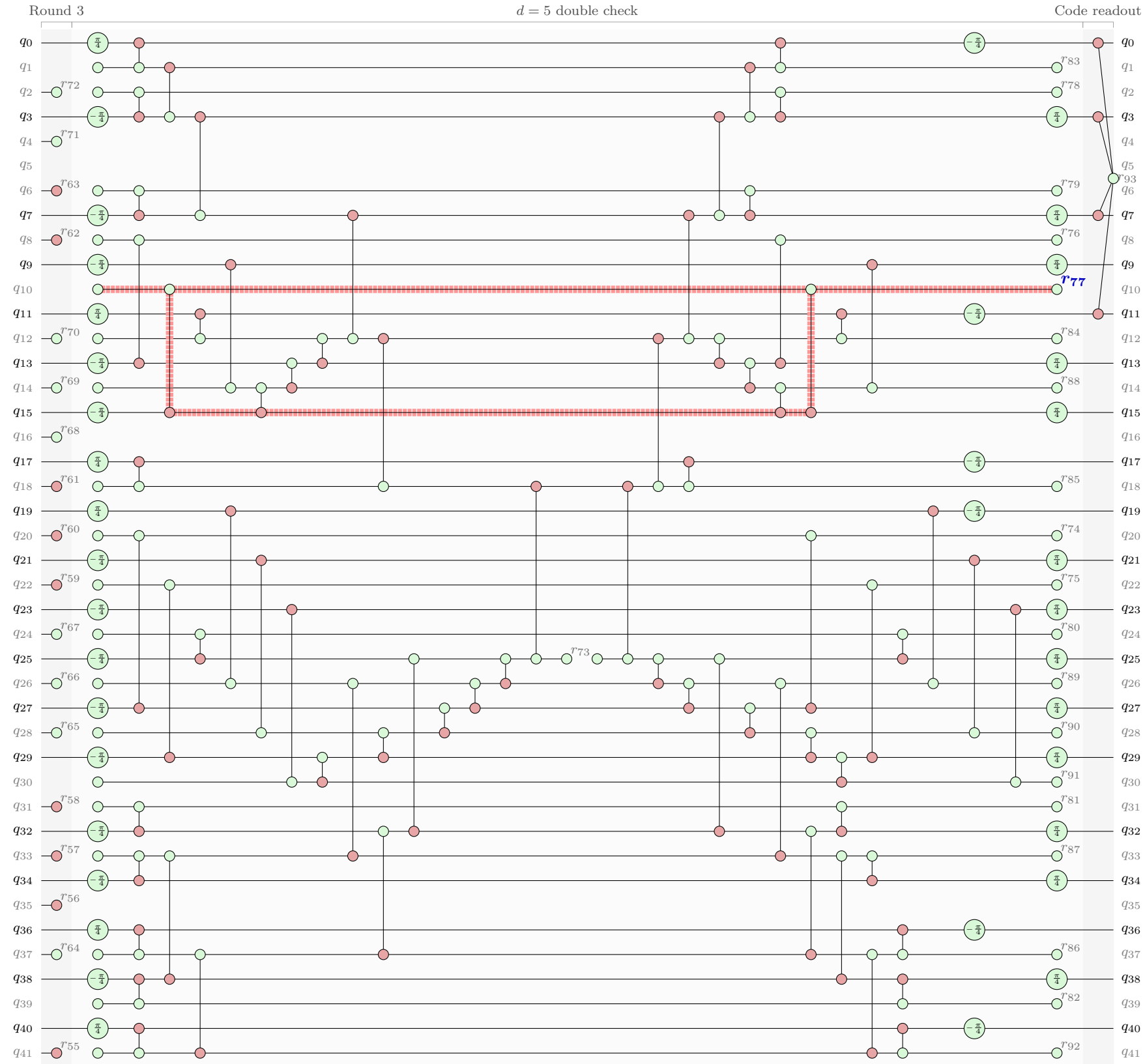


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{76}}$. So $D_{\mathbf{r}} = 0$ unless $r_{76} = 0$.

Records: r_{76} : MX q_8 , line 444.

$d = 5$: detector 73 $r_{77} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

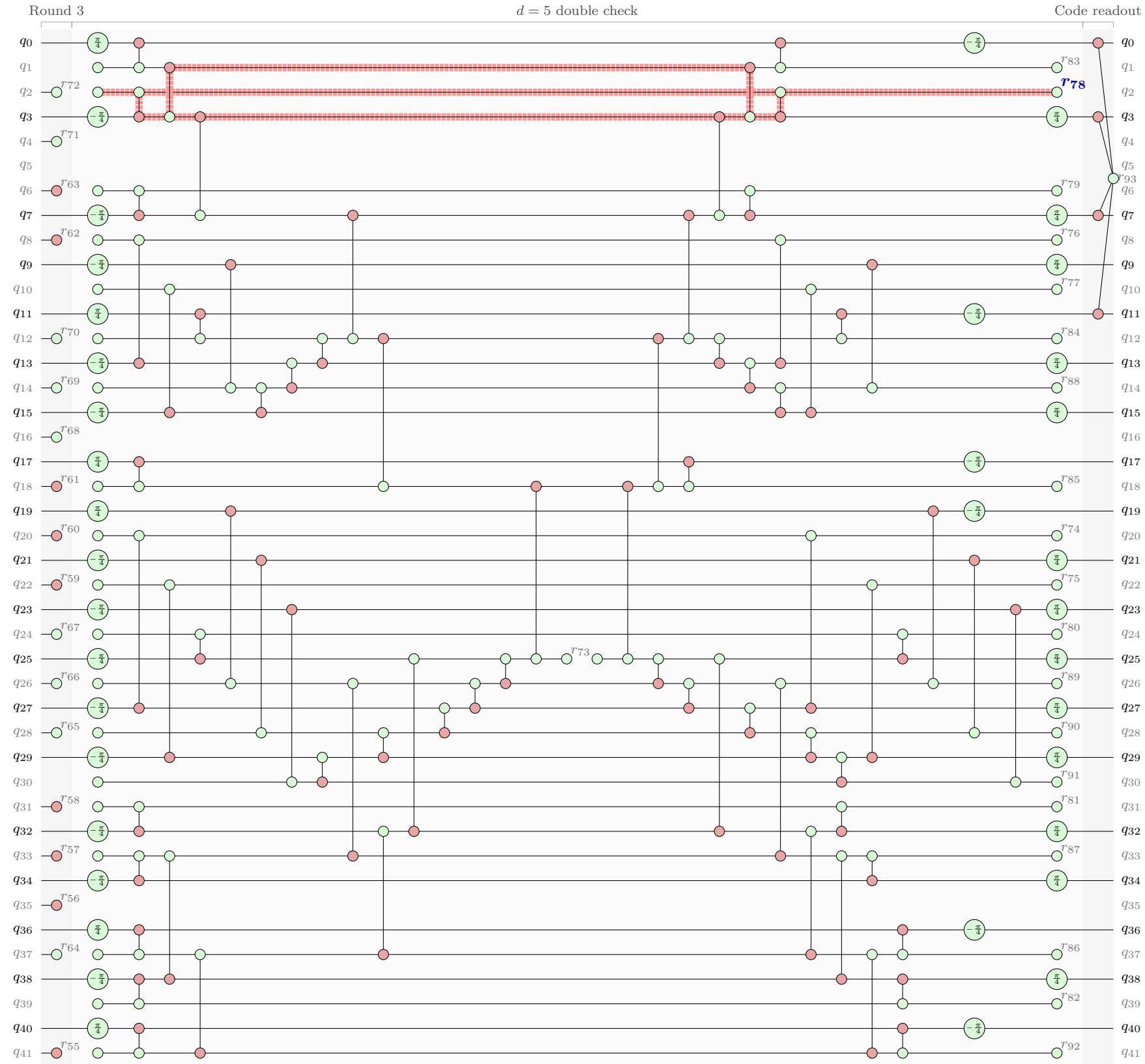


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{77}}$. So $D_{\mathbf{r}} = 0$ unless $r_{77} = 0$.

Records: r_{77} : MX q_{10} , line 444.

$d = 5$: detector 74 $r_{78} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

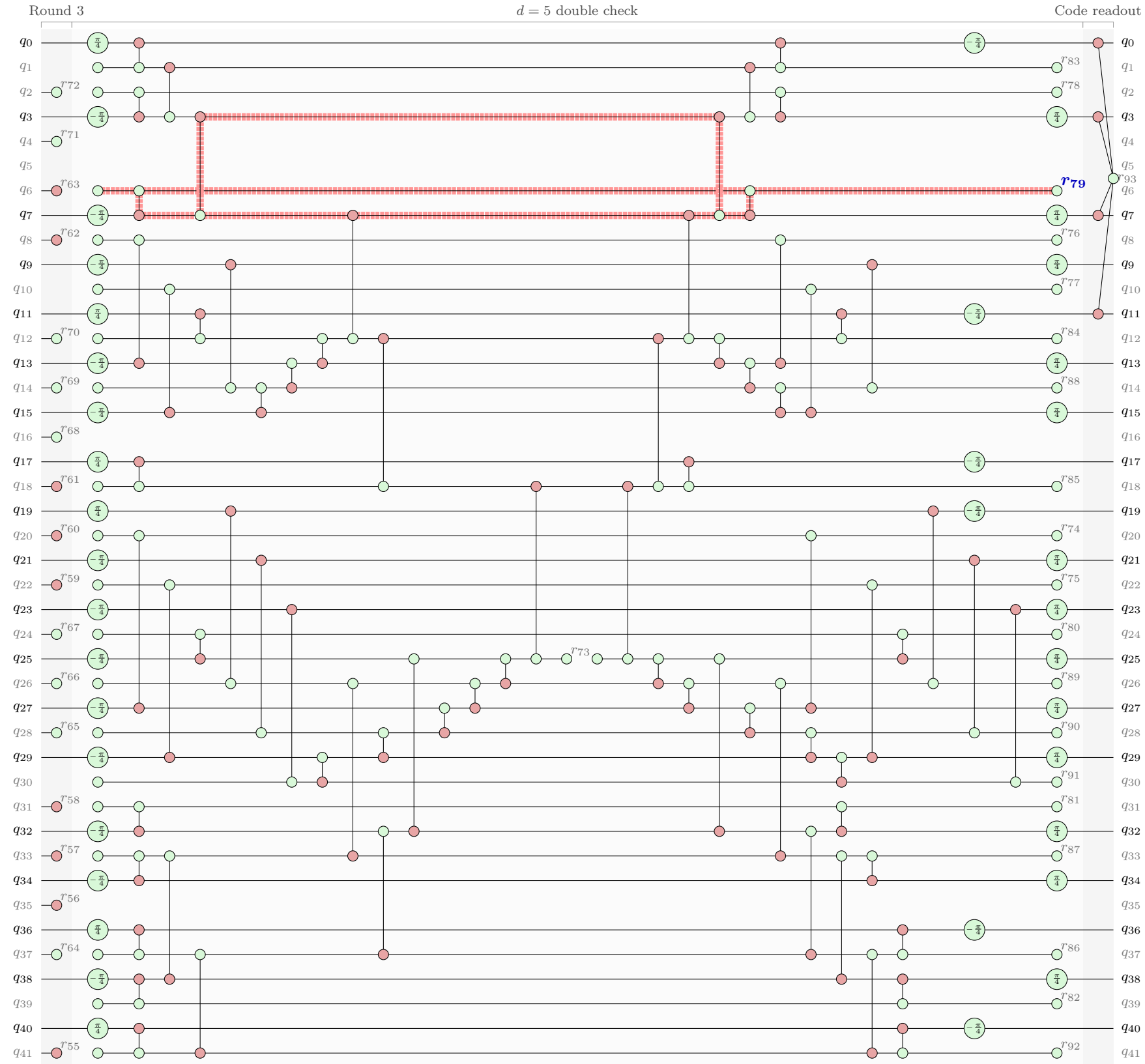


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{78}}$. So $D_{\mathbf{r}} = 0$ unless $r_{78} = 0$.

Records: r_{78} : MX q_2 , line 444.

$d = 5$: detector 75 $r_{79} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

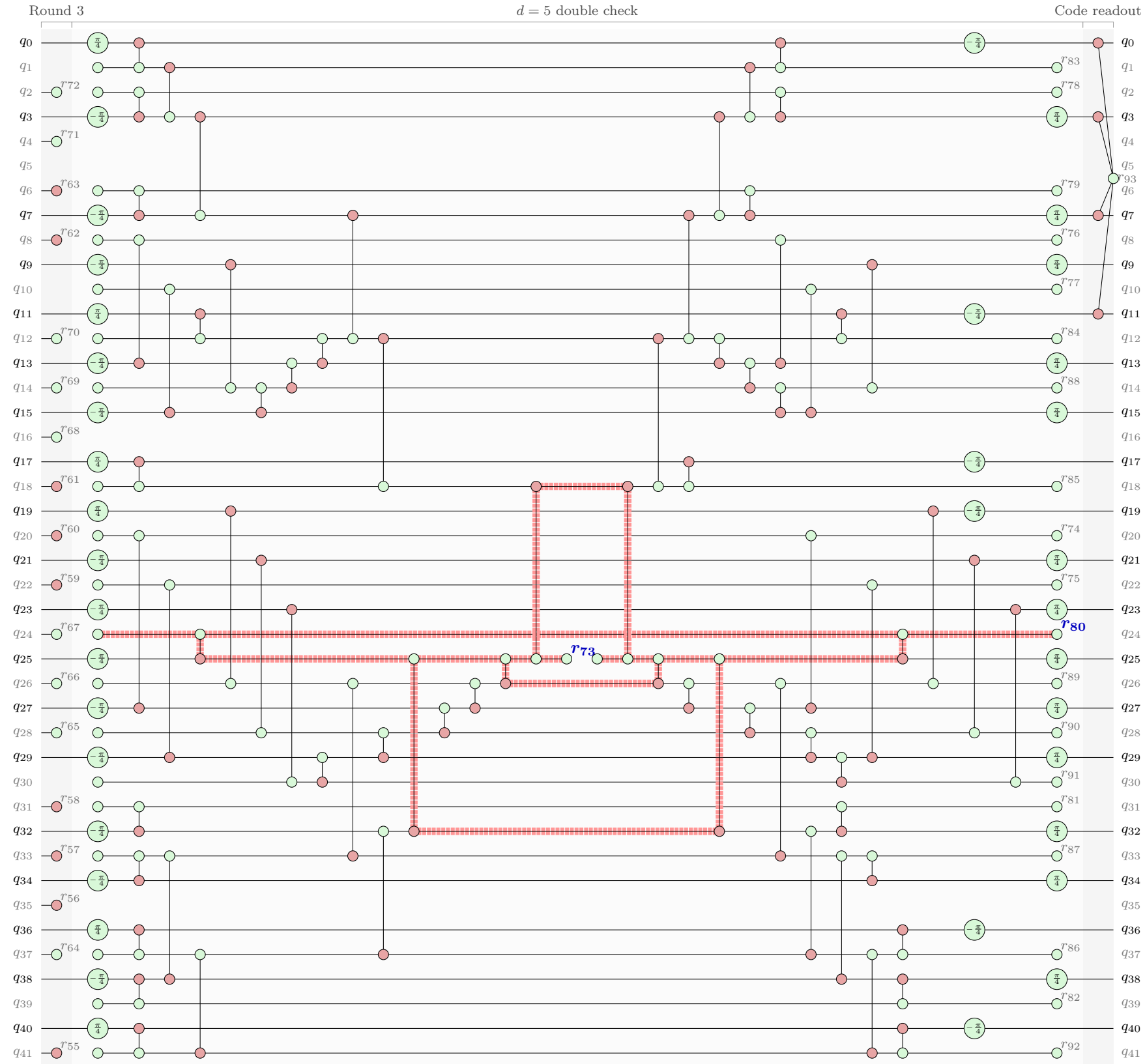


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{79}}$. So $D_{\mathbf{r}} = 0$ unless $r_{79} = 0$.

Records: r_{79} : MX q_6 , line 444.

$d = 5$: detector 76 $r_{73} \oplus r_{80} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

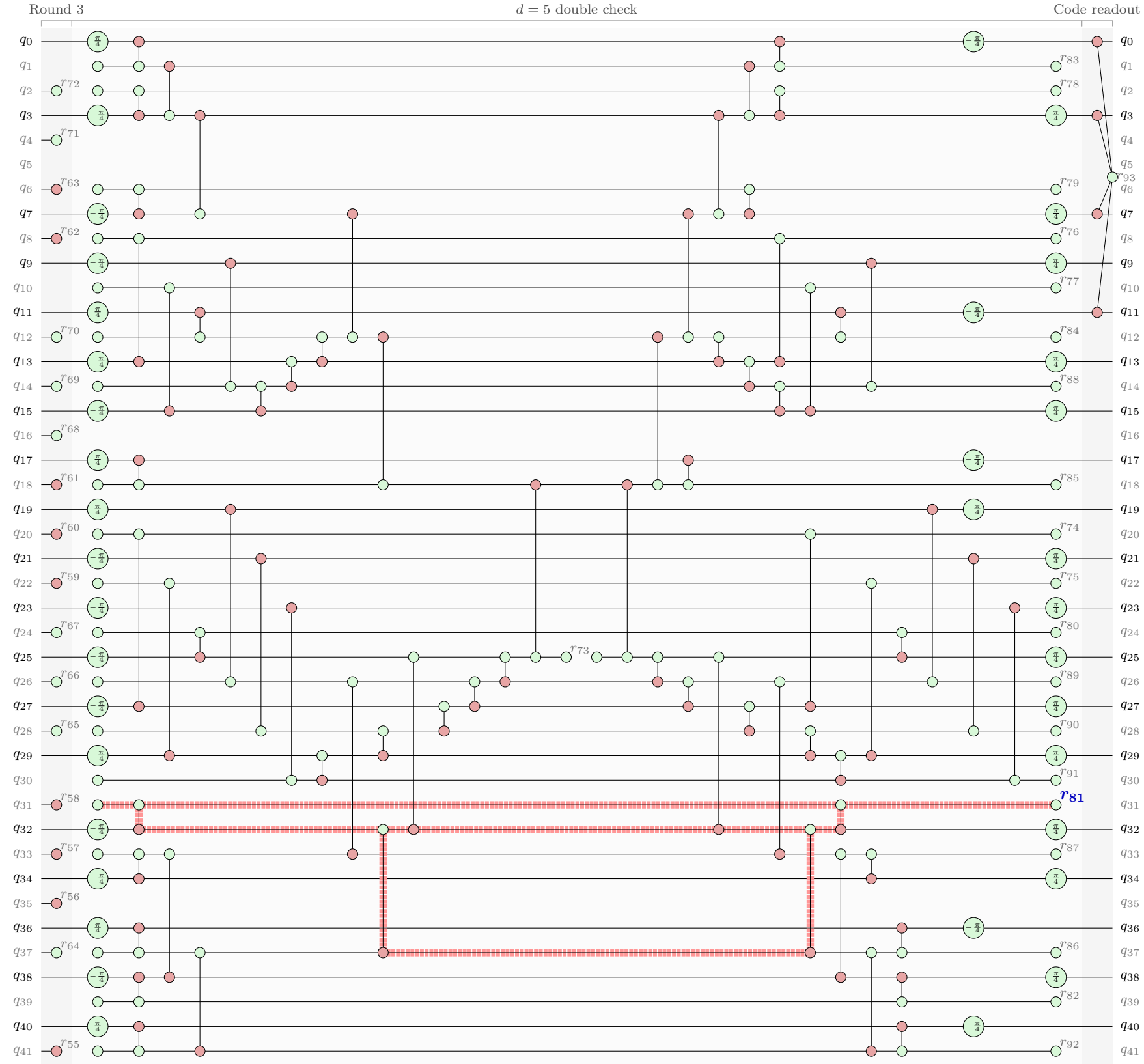


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{73}+r_{80}}$. So $D_{\mathbf{r}} = 0$ unless $r_{73} \oplus r_{80} = 0$.

Records: r_{73} : MX q_{25} , line 415; r_{80} : MX q_{24} , line 444.

$d = 5$: detector 77 $r_{81} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

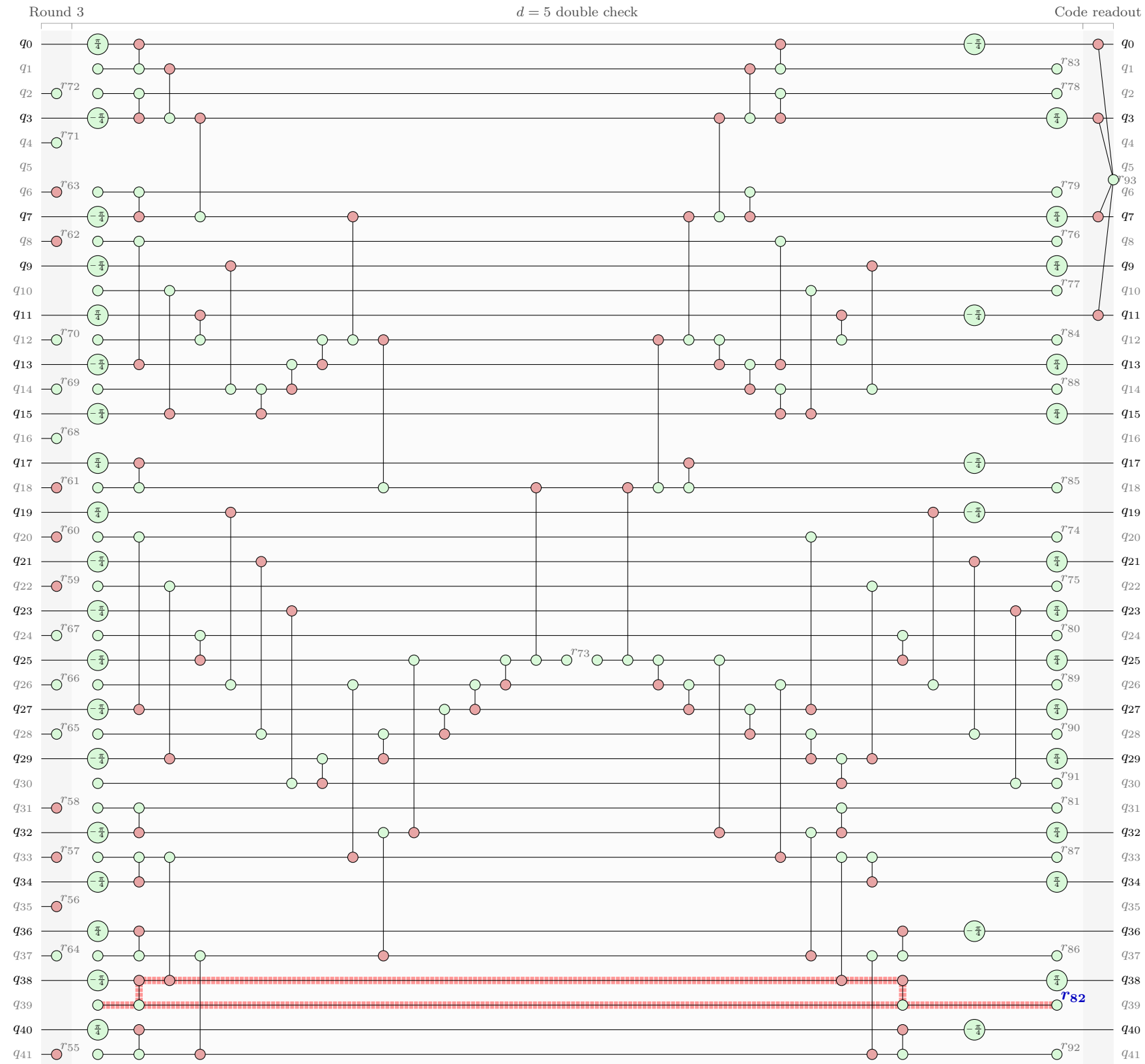


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{81}}$. So $D_{\mathbf{r}} = 0$ unless $r_{81} = 0$.

Records: r_{81} : MX q_{31} , line 444.

$d = 5$: detector 78 $r_{82} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

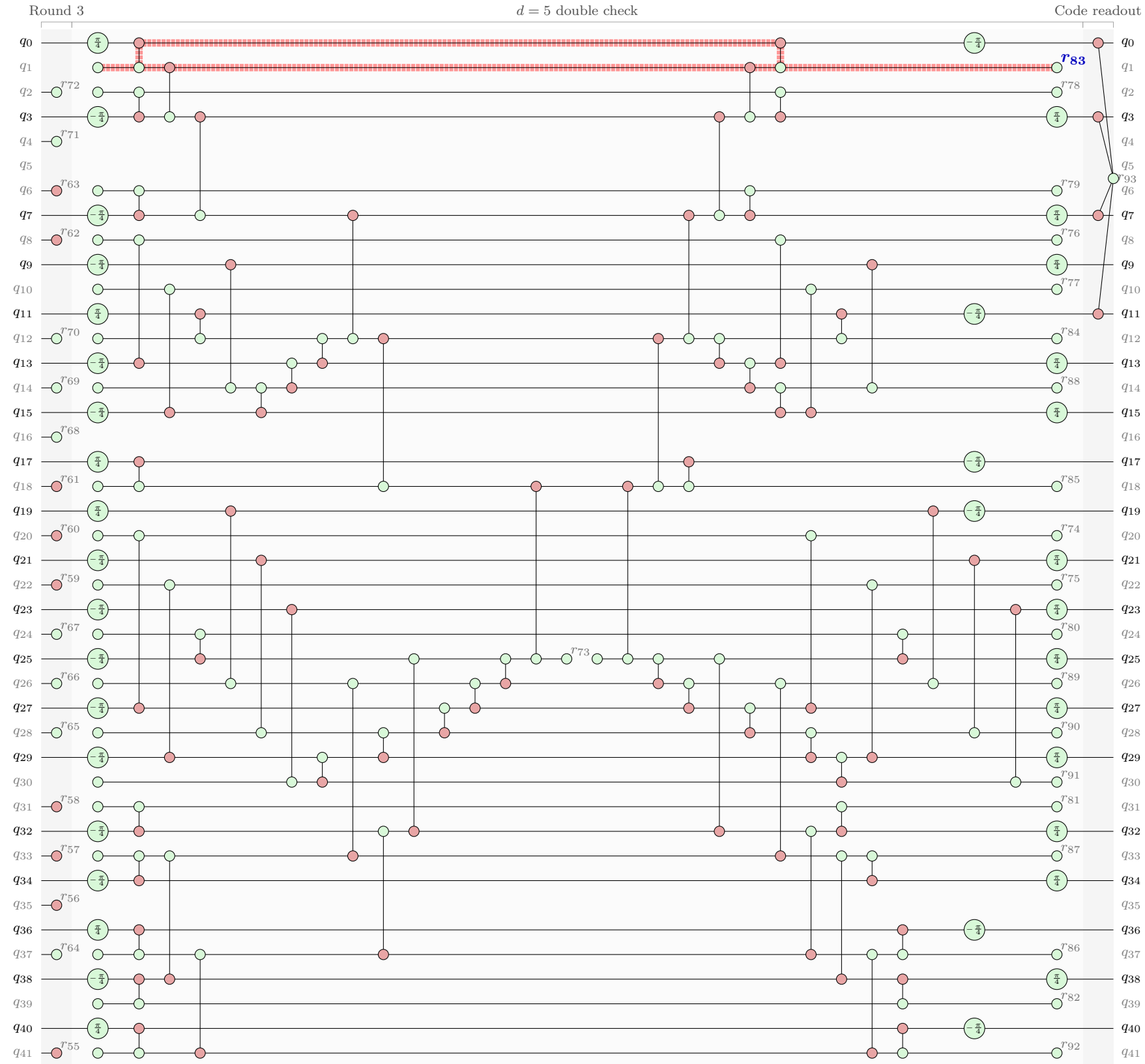


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}, \quad \lambda(\mathbf{r}) = (-1)^{r_{82}}.$ So $D_{\mathbf{r}} = 0$ unless $r_{82} = 0$.

Records: r_{82} : MX q_{39} , line 444.

$d = 5$: detector 79 $r_{83} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

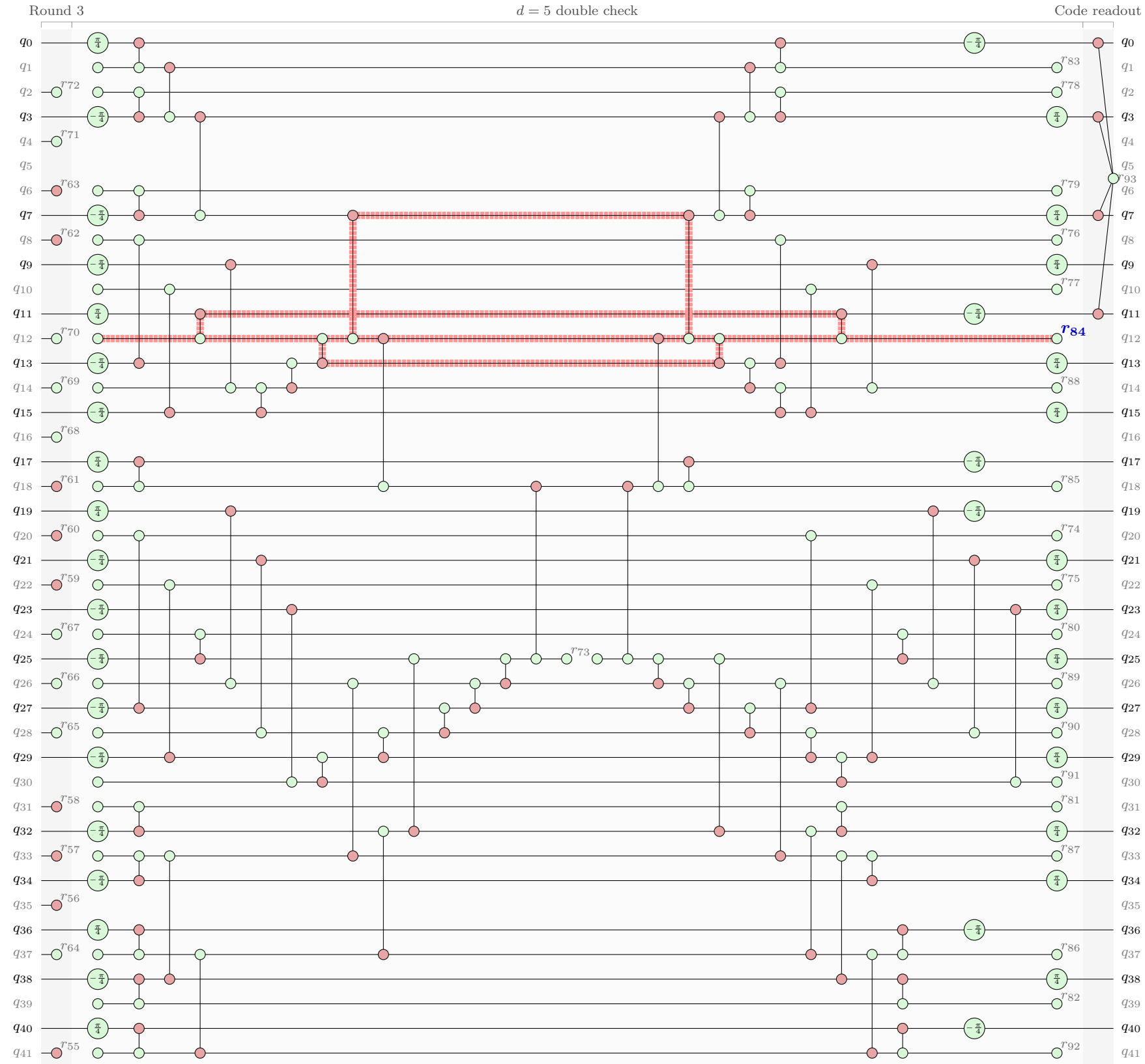


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{83}}$. So $D_{\mathbf{r}} = 0$ unless $r_{83} = 0$.

Records: r_{83} : MX q_1 , line 444.

$d = 5$: detector 80 $r_{84} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

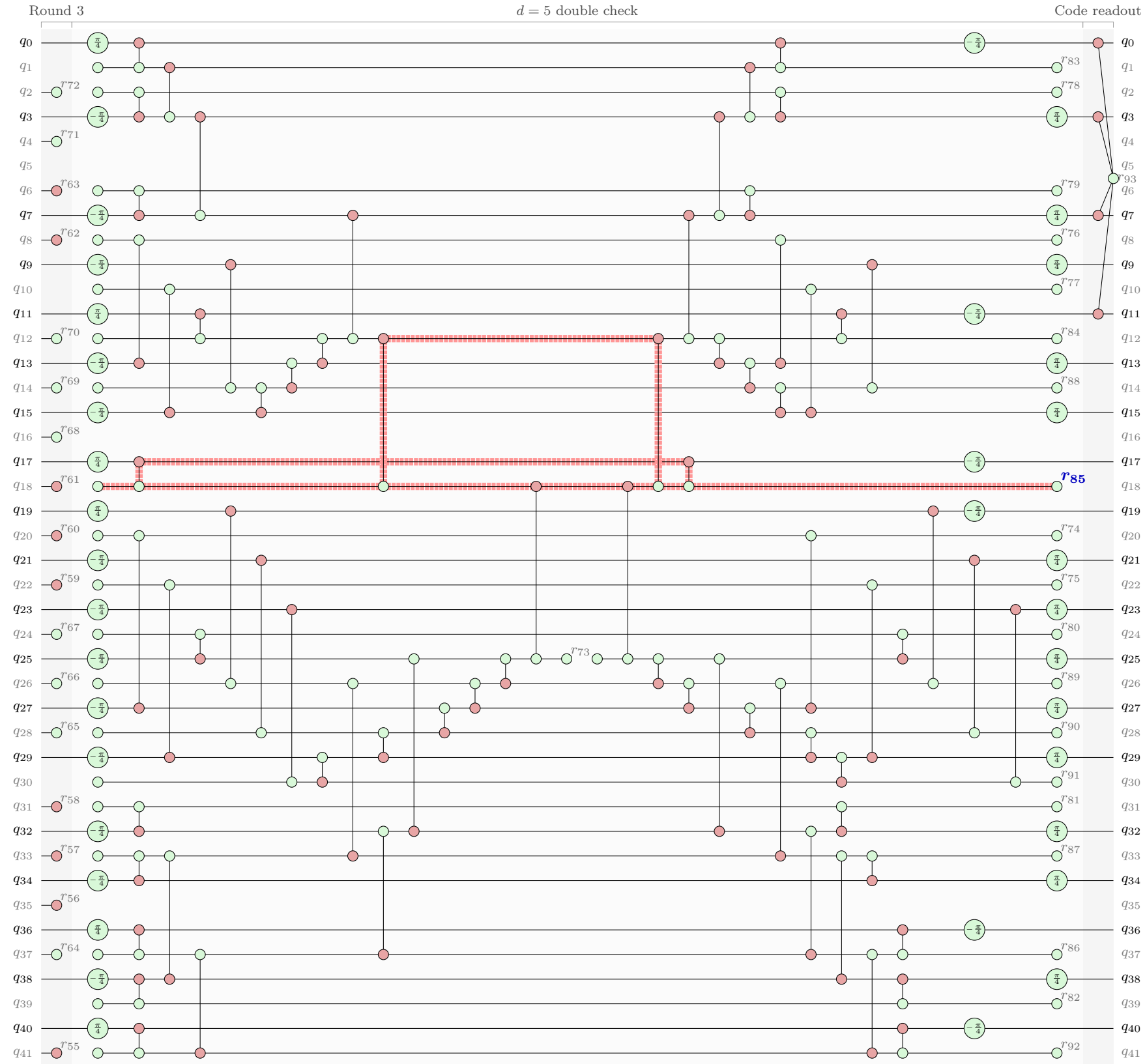


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{84}}$. So $D_{\mathbf{r}} = 0$ unless $r_{84} = 0$.

Records: r_{84} : MX q_{12} , line 444.

$d = 5$: detector 81 $r_{85} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

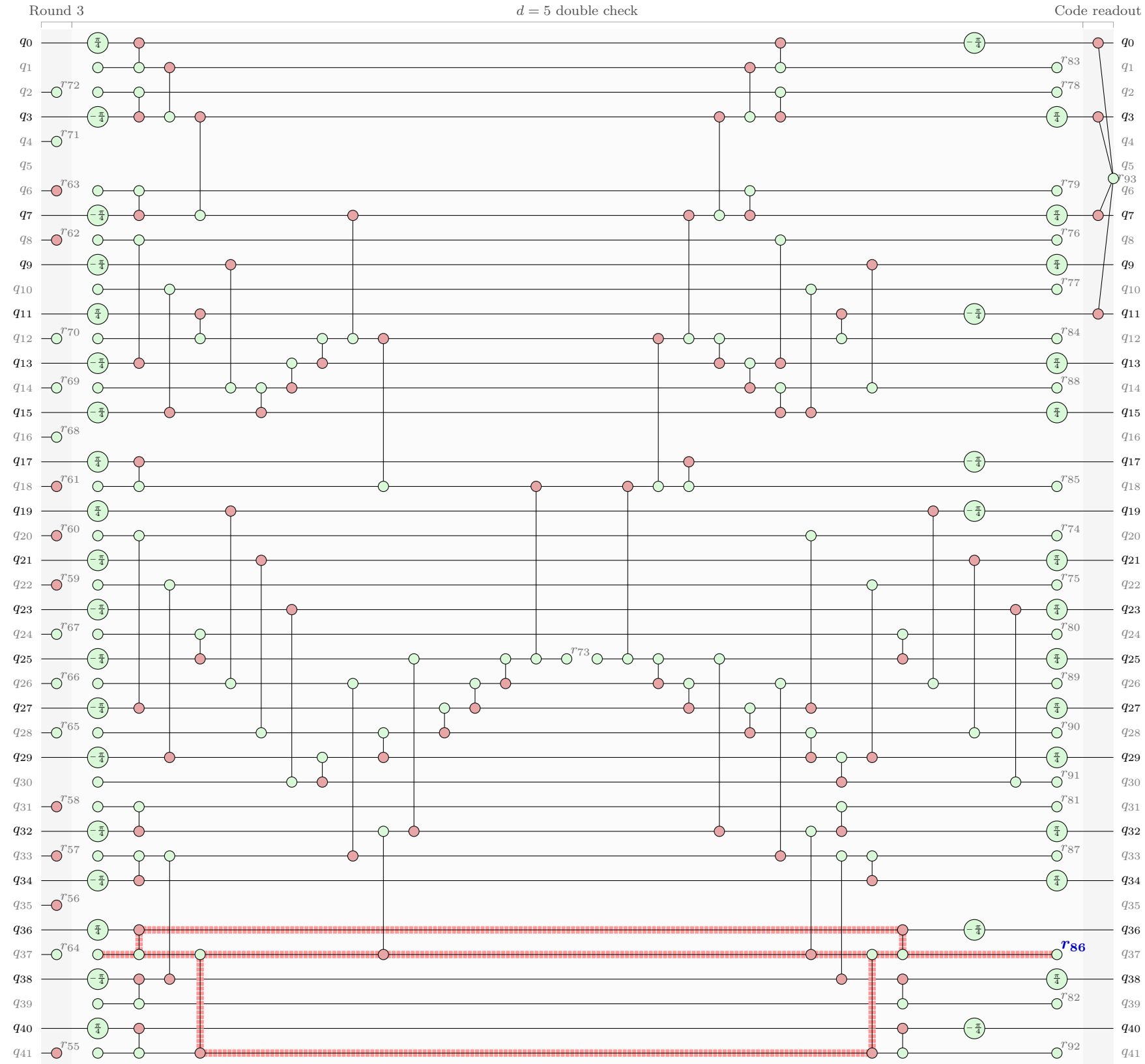


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{85}}$. So $D_{\mathbf{r}} = 0$ unless $r_{85} = 0$.

Records: r_{85} : MX q_{18} , line 444.

$d = 5$: detector 82 $r_{86} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

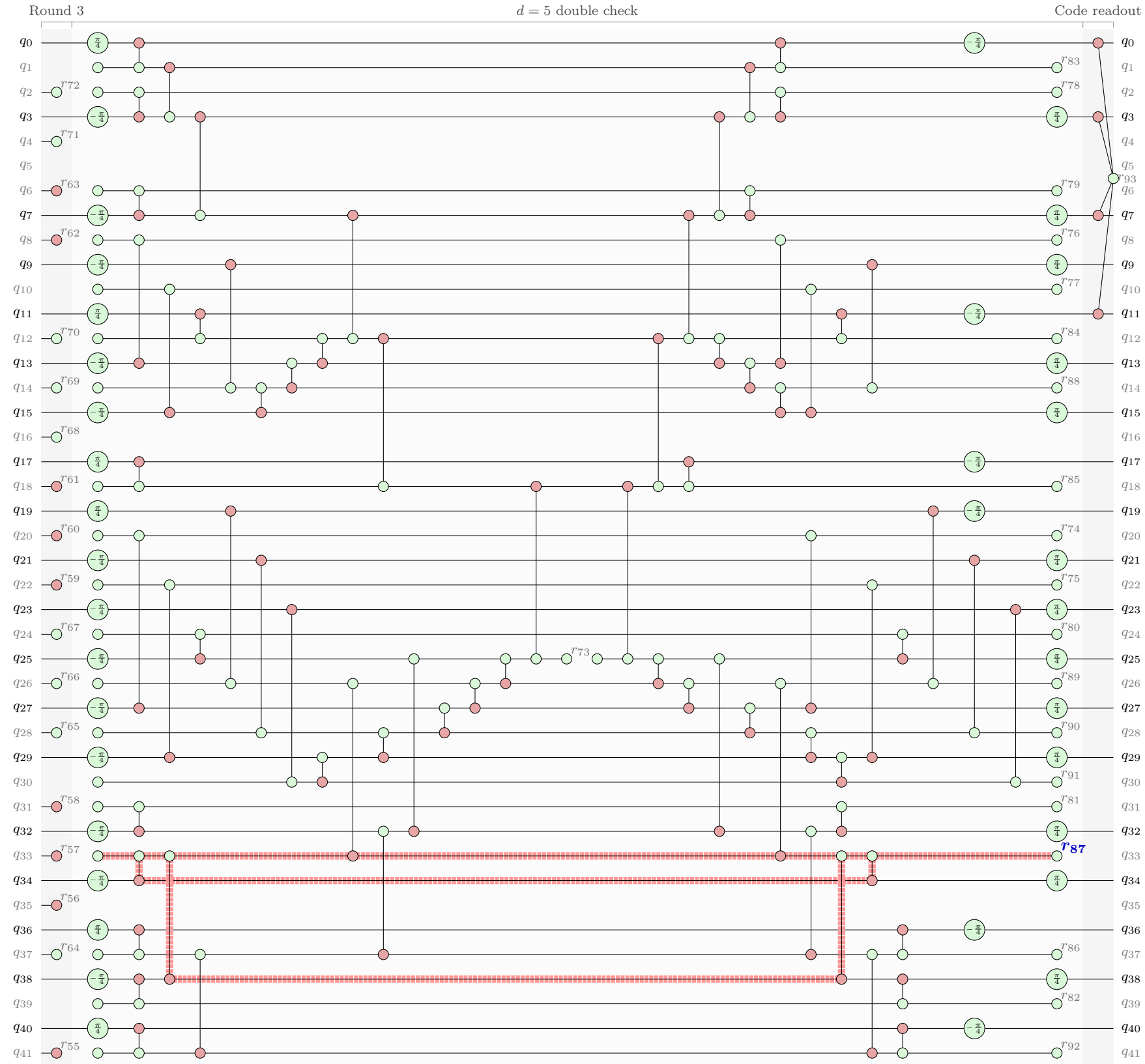


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{86}}$. So $D_{\mathbf{r}} = 0$ unless $r_{86} = 0$.

Records: r_{86} : MX q_{37} , line 444.

$d = 5$: detector 83 $r_{87} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

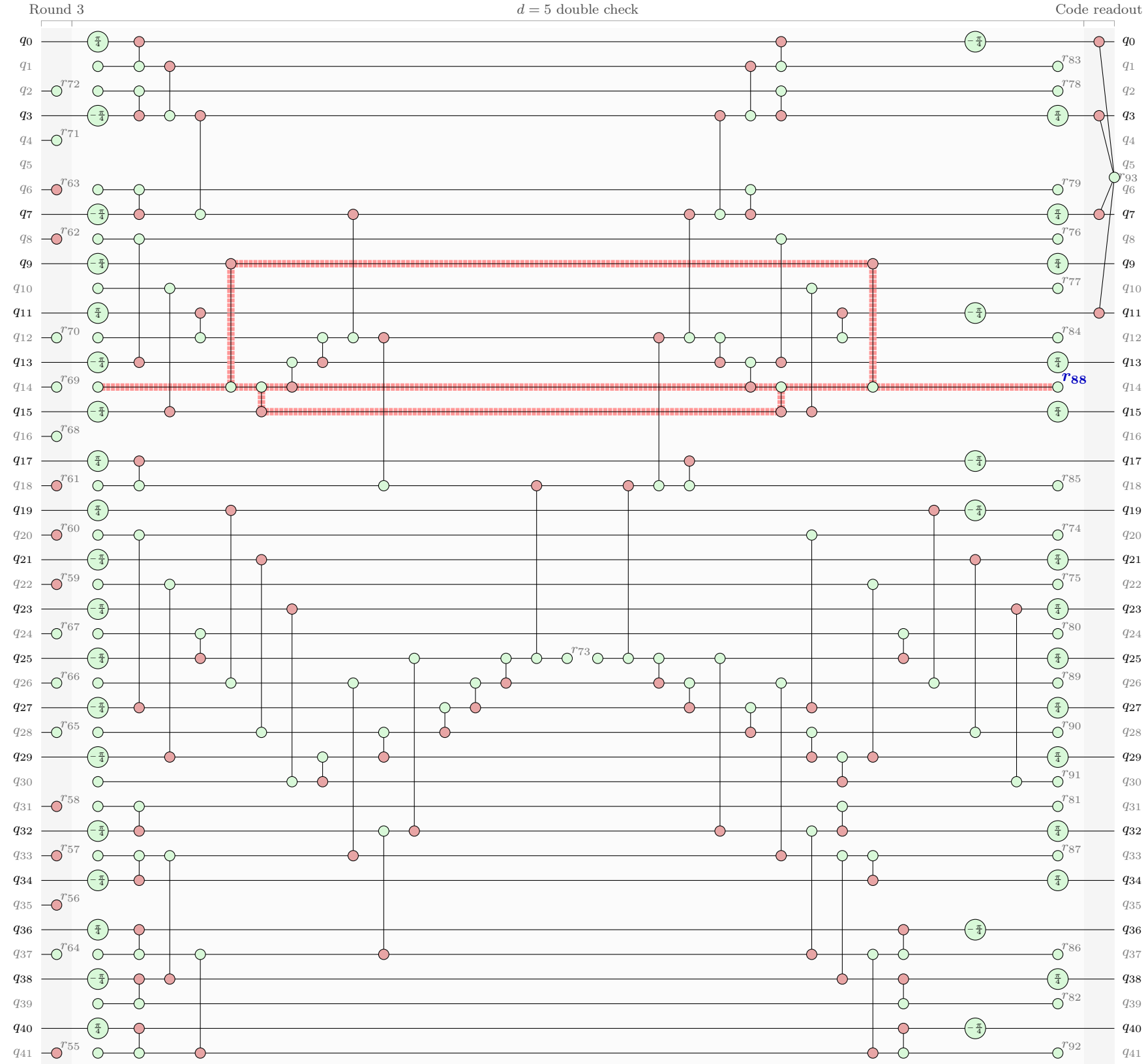


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{87}}$. So $D_{\mathbf{r}} = 0$ unless $r_{87} = 0$.

Records: r_{87} : MX q_{33} , line 444.

$d = 5$: detector 84 $r_{88} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

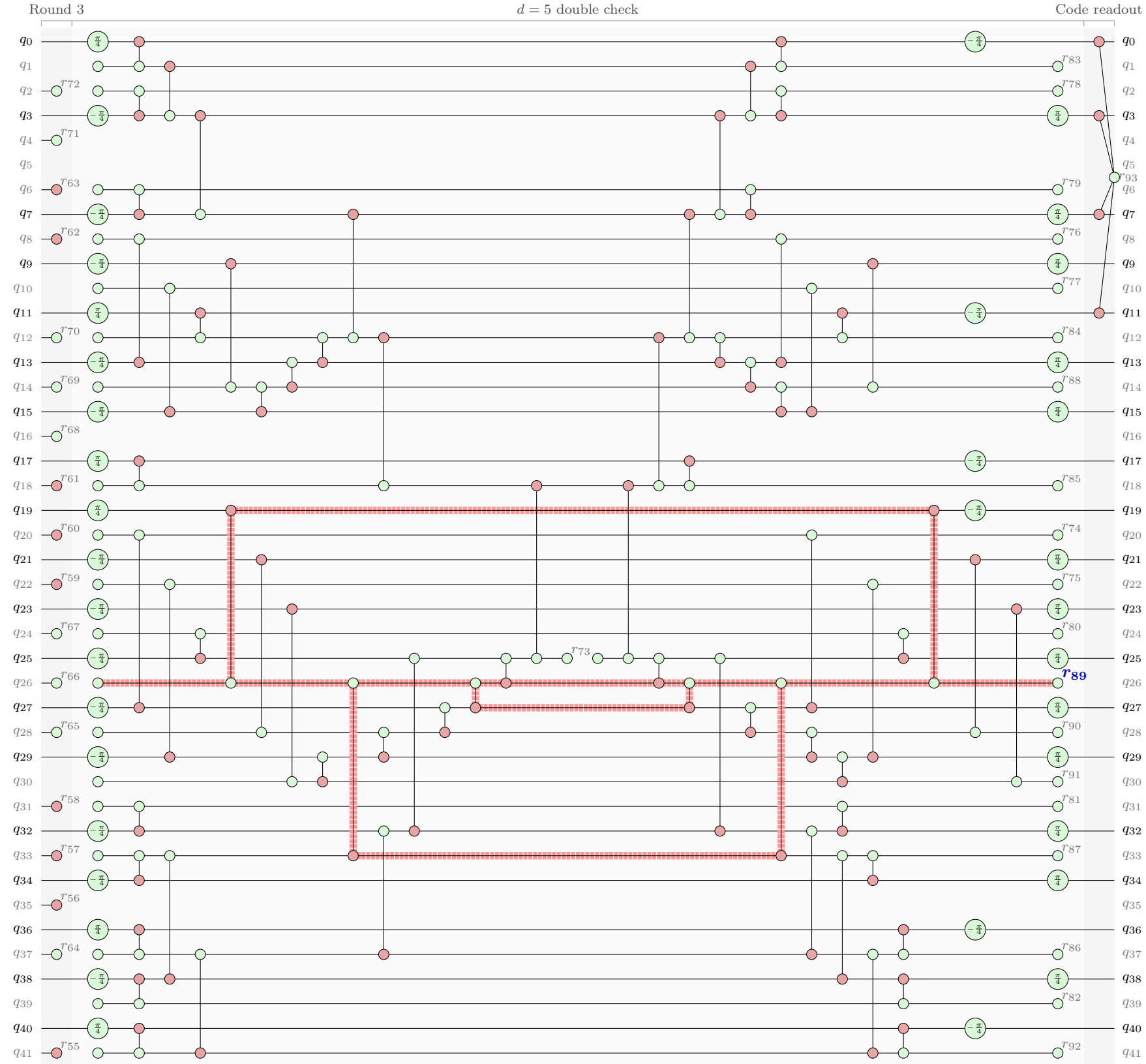


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{88}}$. So $D_{\mathbf{r}} = 0$ unless $r_{88} = 0$.

Records: r_{88} : MX q_{14} , line 444.

$d = 5$: detector 85 $r_{89} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

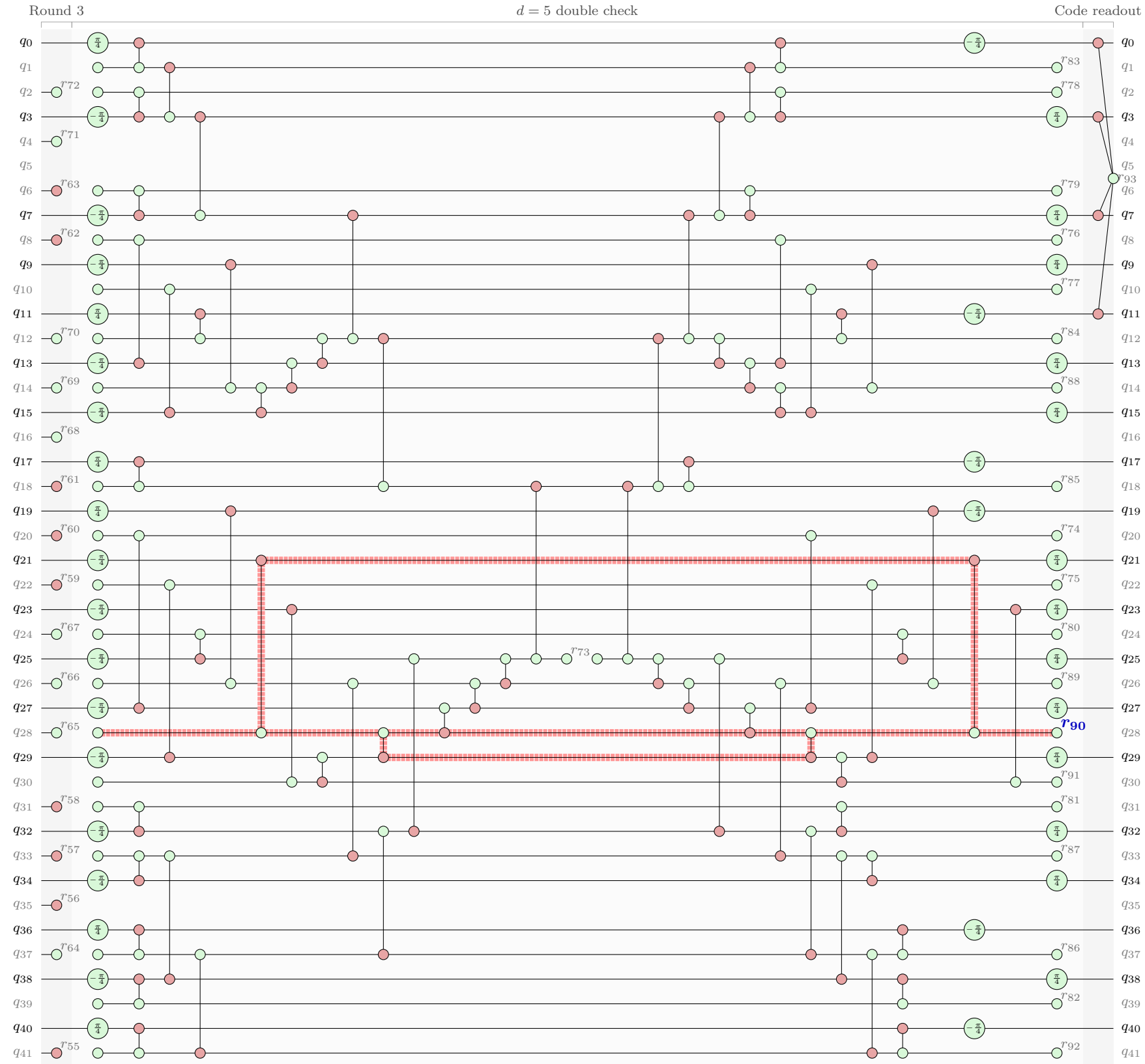


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{89}}$. So $D_{\mathbf{r}} = 0$ unless $r_{89} = 0$.

Records: r_{89} : MX q_{26} , line 444.

$d = 5$: detector 86 $r_{90} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

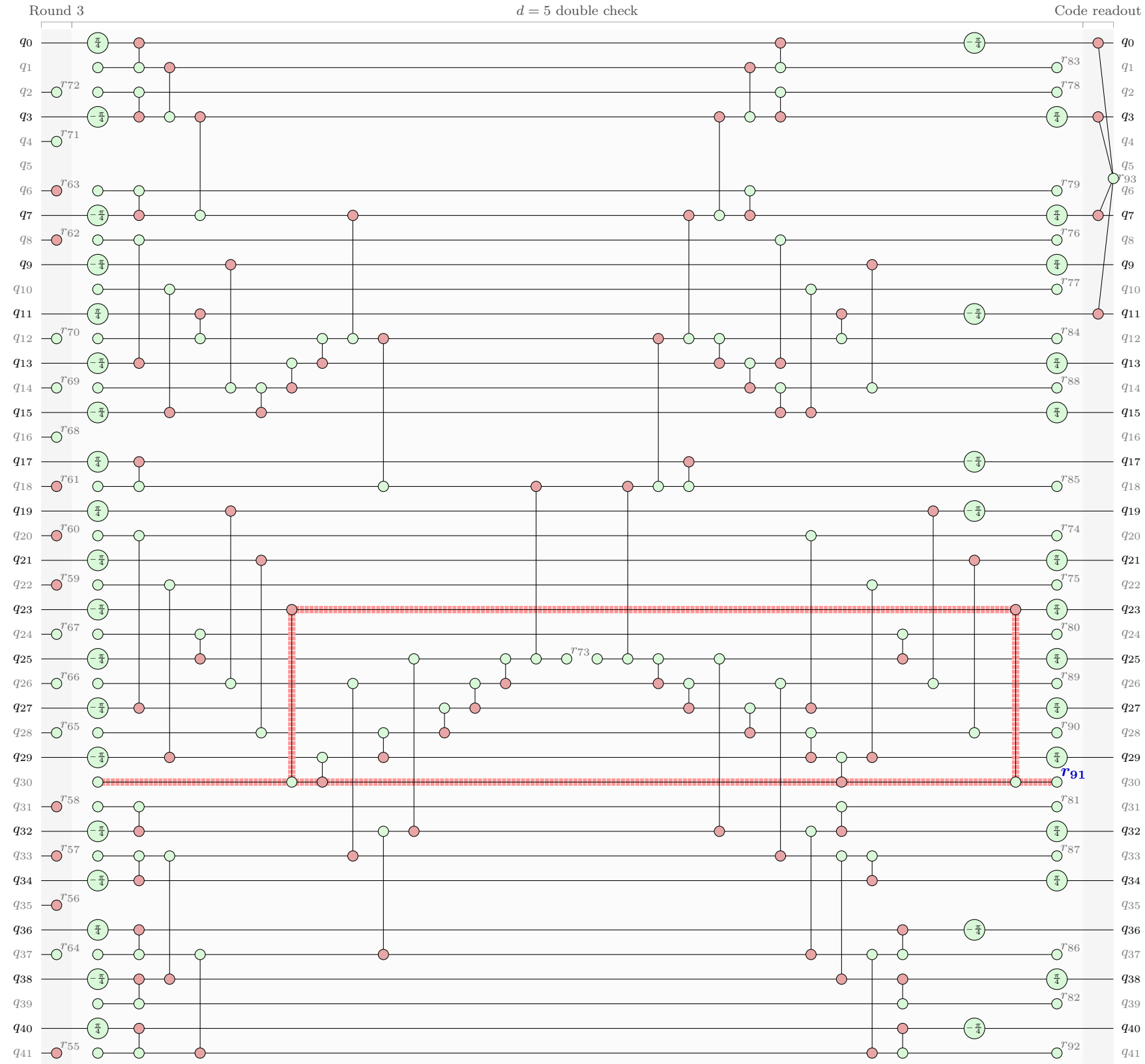


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{90}}$. So $D_{\mathbf{r}} = 0$ unless $r_{90} = 0$.

Records: r_{90} : MX q_{28} , line 444.

$d = 5$: detector 87 $r_{91} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

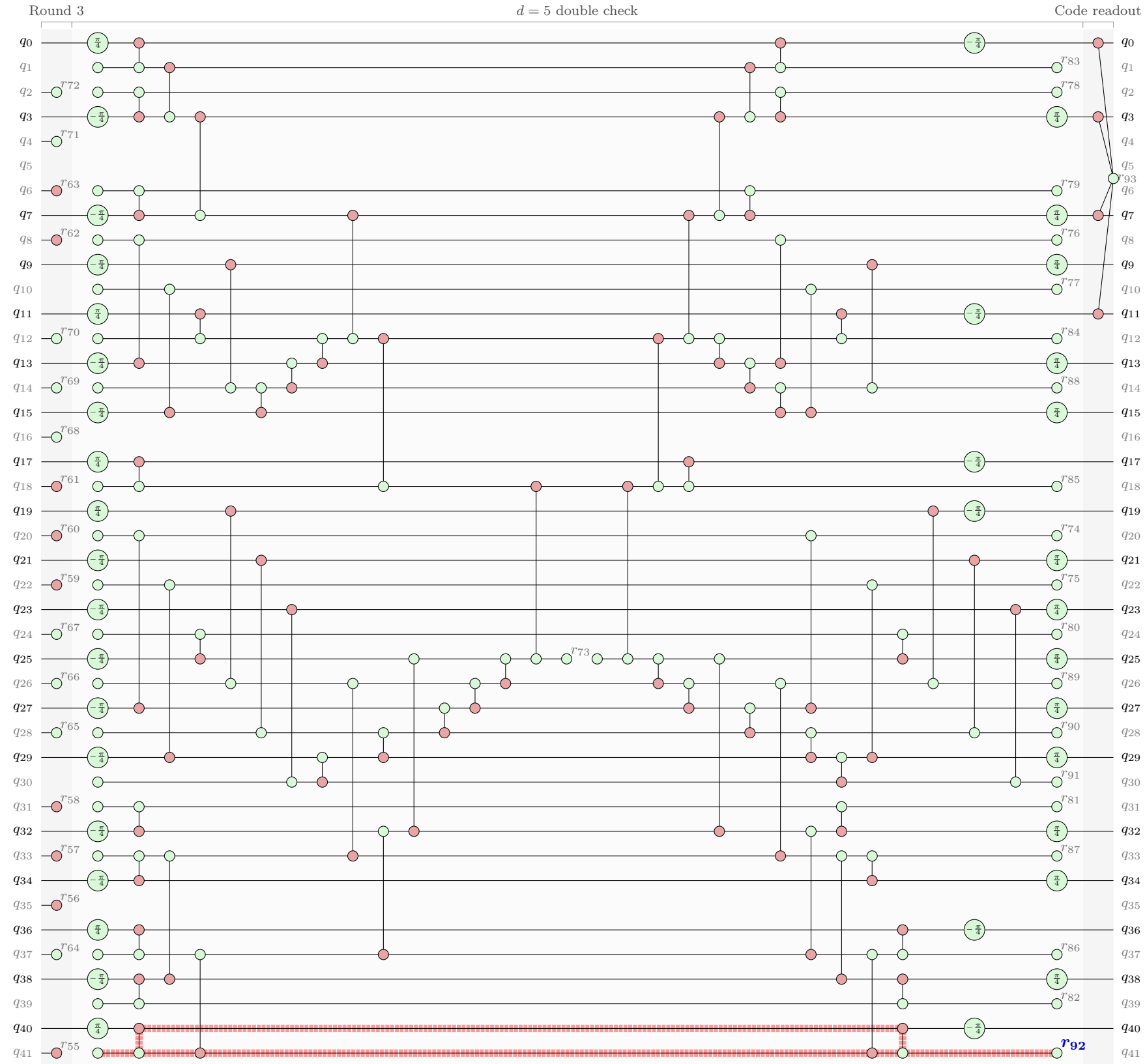


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{91}}$. So $D_{\mathbf{r}} = 0$ unless $r_{91} = 0$.

Records: r_{91} : MX q_{30} , line 444.

$d = 5$: detector 88 $r_{92} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

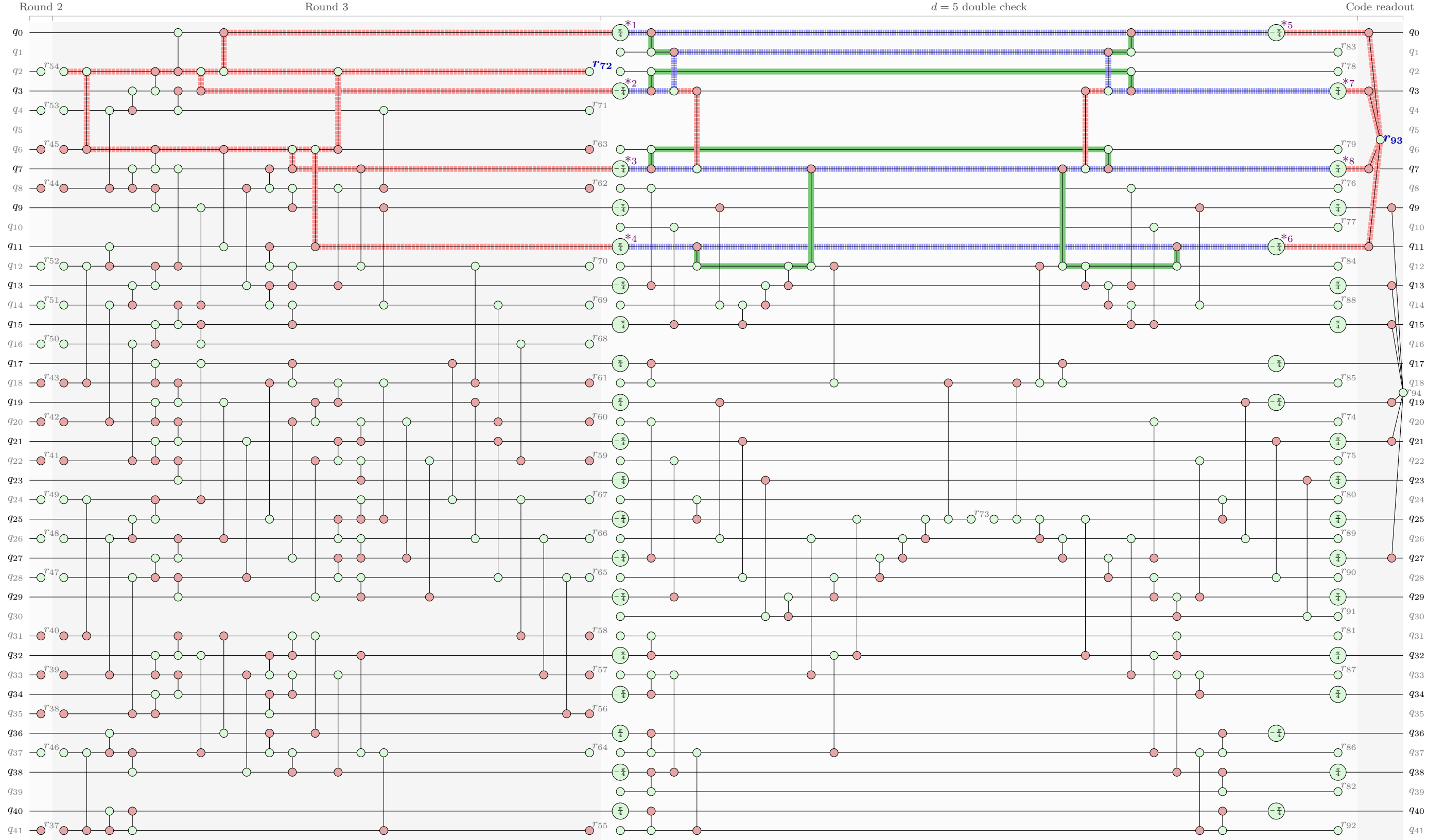


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{92}}$. So $D_{\mathbf{r}} = 0$ unless $r_{92} = 0$.

Records: r_{92} : MX q_{41} , line 444.

$d = 5$: detector 89 $r_{72} \oplus r_{93} = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



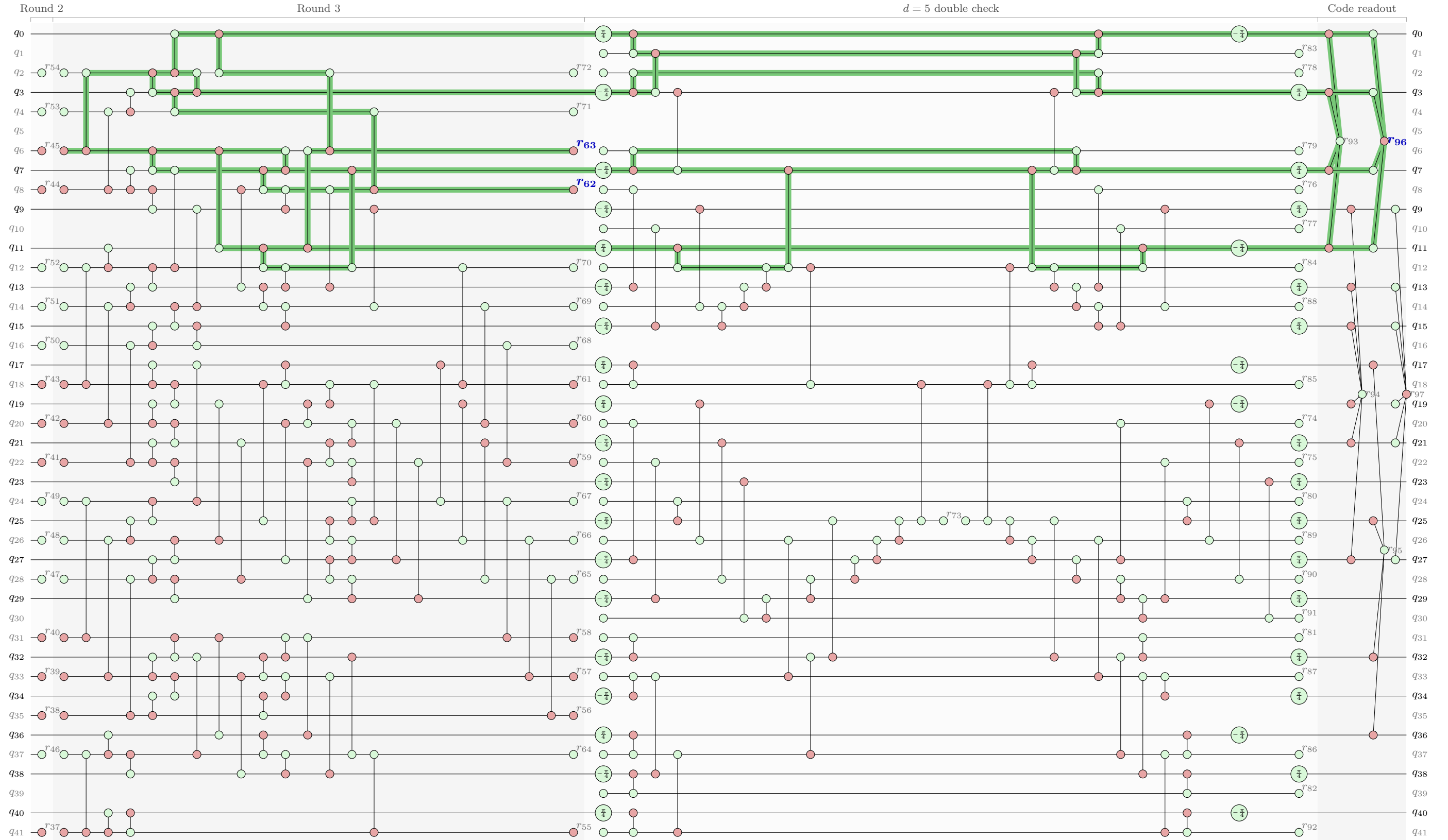
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}$, $\lambda(\mathbf{r}) = (-1)^{r_{72}+r_{93}}$. $D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_{72} : MX q_2 , line 362; r_{93} : MPP $X_0X_3X_7X_{11}$, line 470.

Phase changes: $*_1$: $\delta = \frac{\pi}{2}$ at T on q_0 , line 391; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 392; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 392; $*_4$: $\delta = \frac{\pi}{2}$ at T on q_{11} , line 391; $*_5$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_0 , line 441; $*_6$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{11} , line 441; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_3 , line 442; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 442.

$d = 5$: detector 90 $r_{62} \oplus r_{63} \oplus r_{96} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

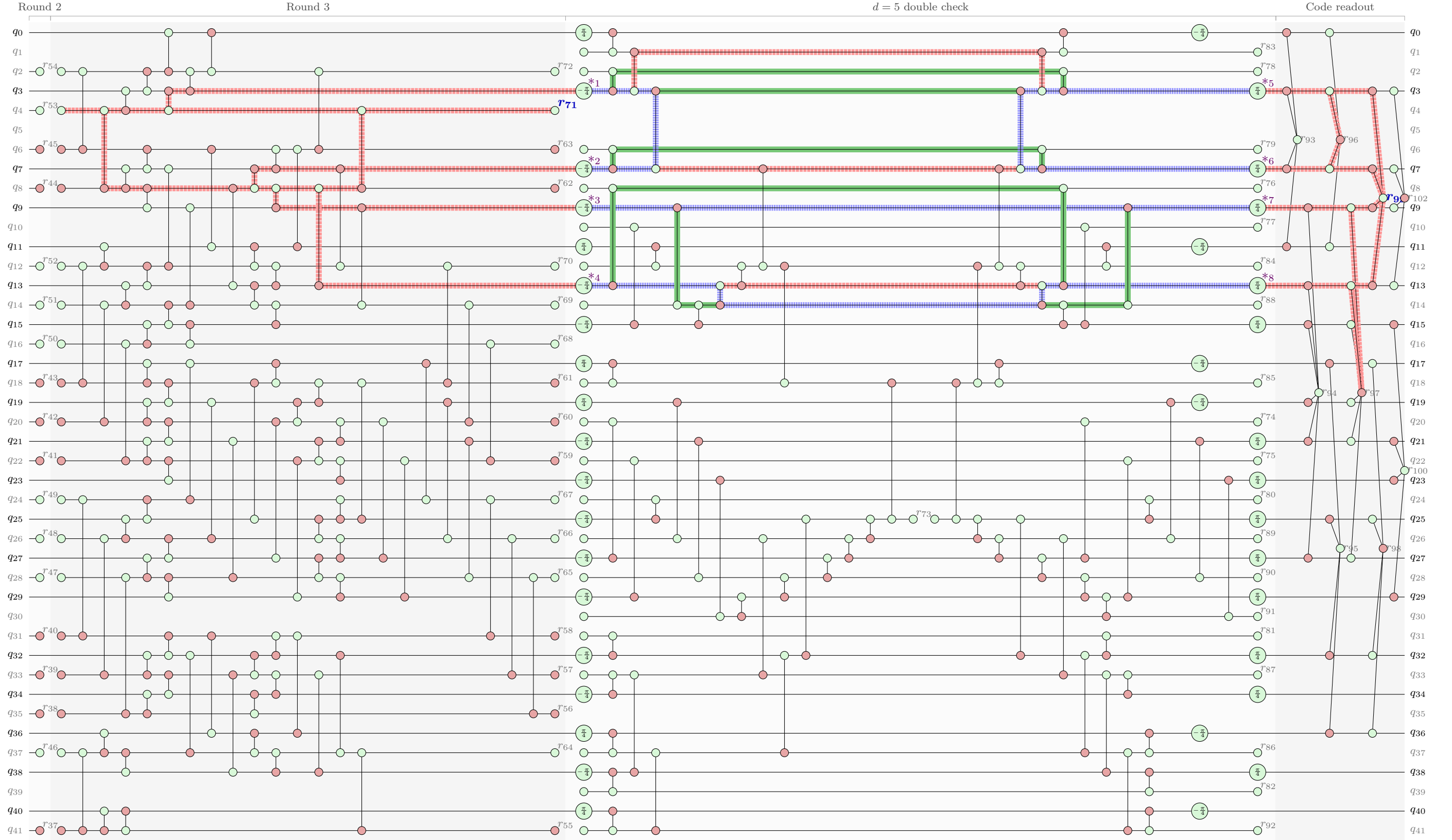


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{62} + r_{63} + r_{96}}$. So $D_{\mathbf{r}} = 0$ unless $r_{62} \oplus r_{63} \oplus r_{96} = 0$.

Records: r_{62} : M q_8 , line 361; r_{63} : M q_6 , line 361; r_{96} : MPP $Z_0 Z_3 Z_7 Z_{11}$, line 470.

$d = 5$: detector 91 $r_{71} \oplus r_{99} = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}$, $\lambda(\mathbf{r}) = (-1)^{r_{71}+r_{99}}$. $D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

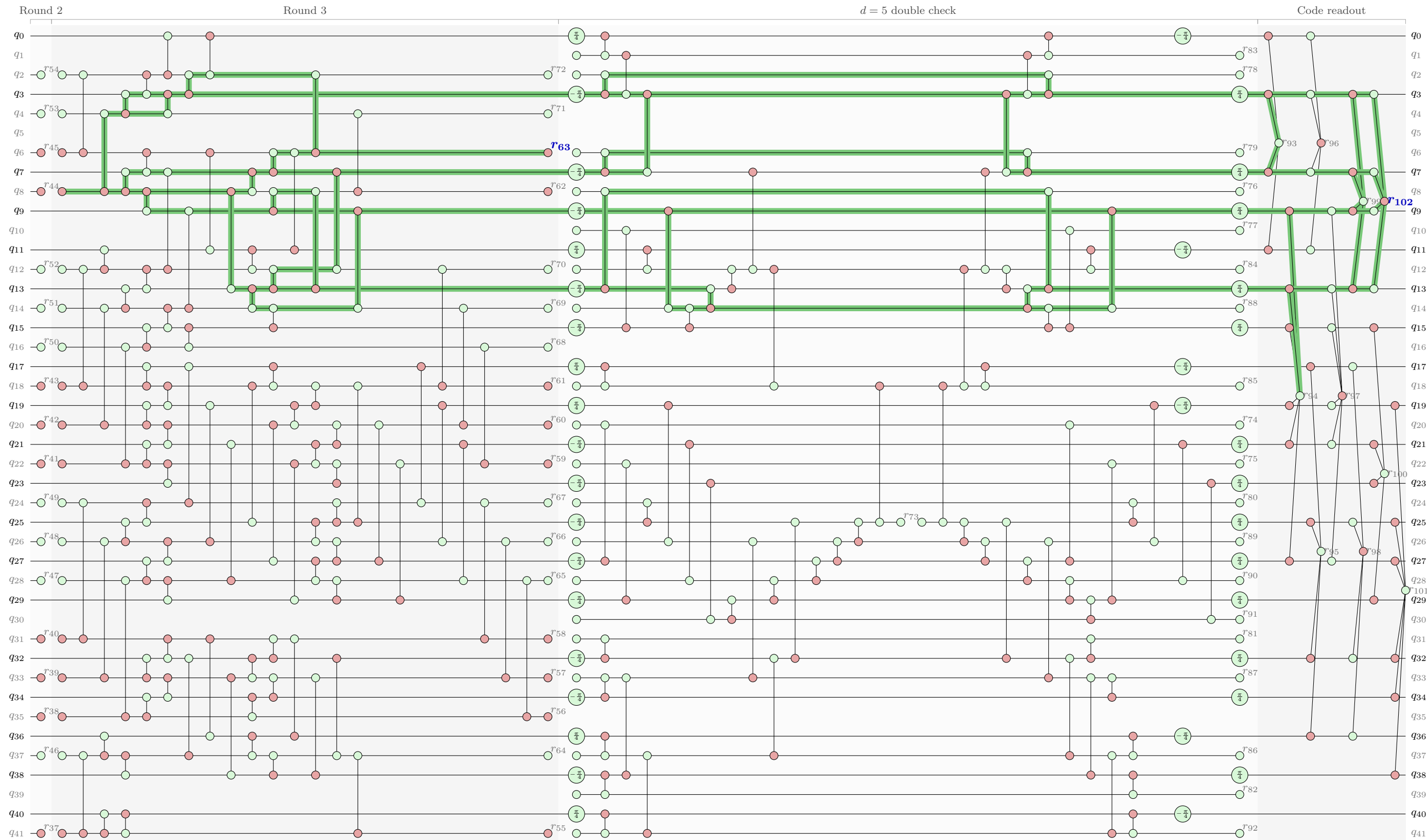
Records: r_{71} : MX q_4 , line 362; r_{99} : MPP $X_3X_7X_9X_{13}$, line 470.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_3 , line 392; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 392; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_9 , line 392; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 392; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_3 , line 442; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 442; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_9 , line 442; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{13} , line 442.

d5_det091.tikz, declared at line 474 of data/inputs/soft_cultivation_d5_p0005.stim

$d = 5$: detector 92 $r_{63} \oplus r_{102} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

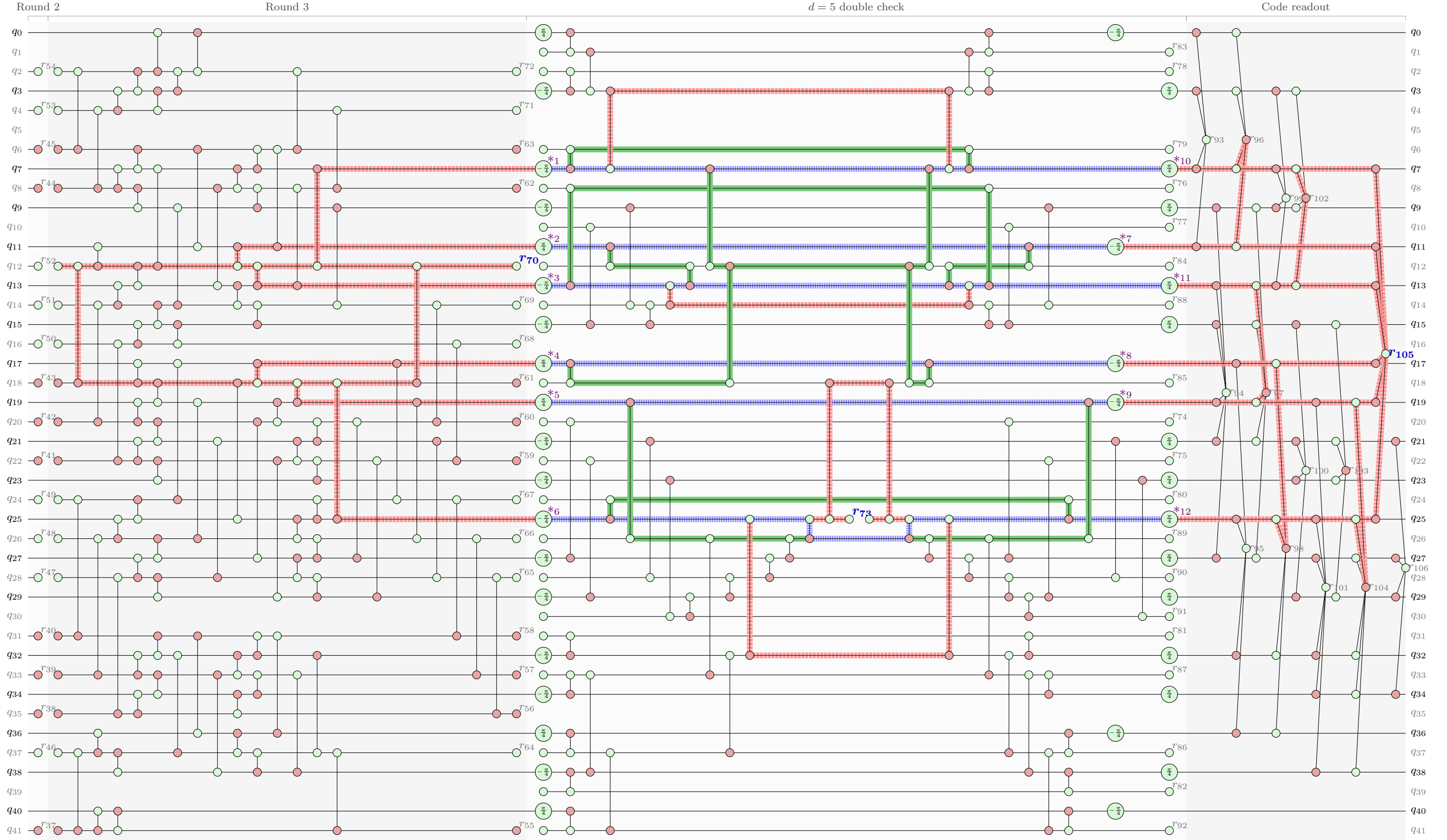


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{63}+r_{102}}$. So $D_{\mathbf{r}} = 0$ unless $r_{63} \oplus r_{102} = 0$.

Records: r_{63} : M q_6 , line 361; r_{102} : MPP $Z_3 Z_7 Z_9 Z_{13}$, line 470.

$d = 5$: detector 93 $r_{70} \oplus r_{73} \oplus r_{105} = 0$

Phase-dependent. Closed semiweb with 12 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}$, $\lambda(\mathbf{r}) = (-1)^{r_{70}+r_{73}+r_{105}}$. $D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

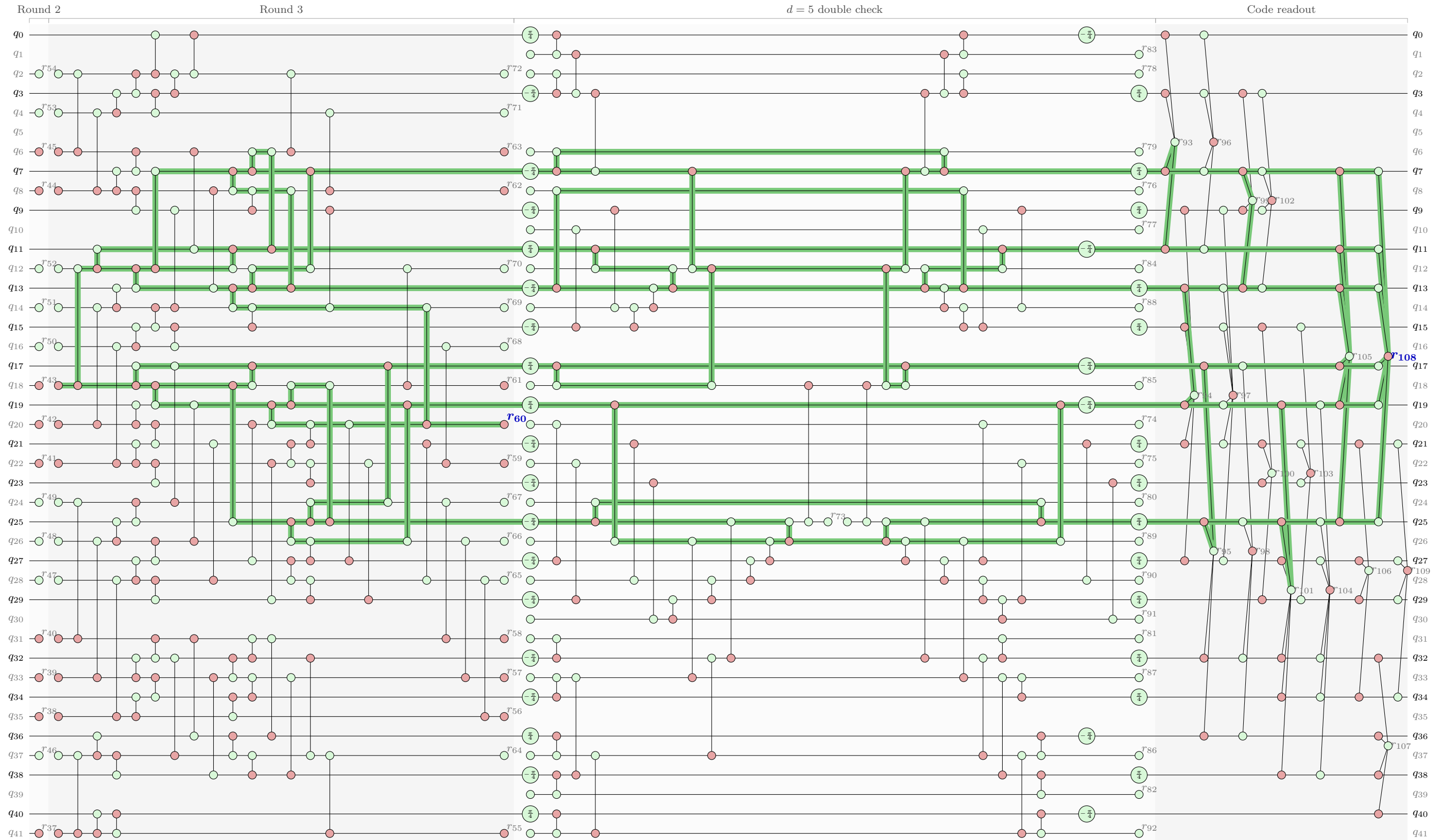
Records: r_{70} : MX q_{12} , line 362; r_{73} : MX q_{25} , line 415; r_{105} : MPP $X_7X_{11}X_{13}X_{17}X_{19}X_{25}$, line 470.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_7 , line 392; $*_2$: $\delta = \frac{\pi}{2}$ at T on q_{11} , line 391; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 392; $*_4$: $\delta = \frac{\pi}{2}$ at T on q_{17} , line 391; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_{19} , line 391; $*_6$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{25} , line 392; $*_7$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{11} , line 441; $*_8$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{17} , line 441; $*_9$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{19} , line 441; $*_{10}$: $\delta = \frac{\pi}{2}$ at T on q_7 , line 442; $*_{11}$: $\delta = \frac{\pi}{2}$ at T on q_{13} , line 442; $*_{12}$: $\delta = \frac{\pi}{2}$ at T on q_{25} , line 442.

d5_det093.tikz, declared at line 476 of data/inputs/soft_cultivation_d5_p0005.stim

$d = 5$: detector 94 $r_{60} \oplus r_{108} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

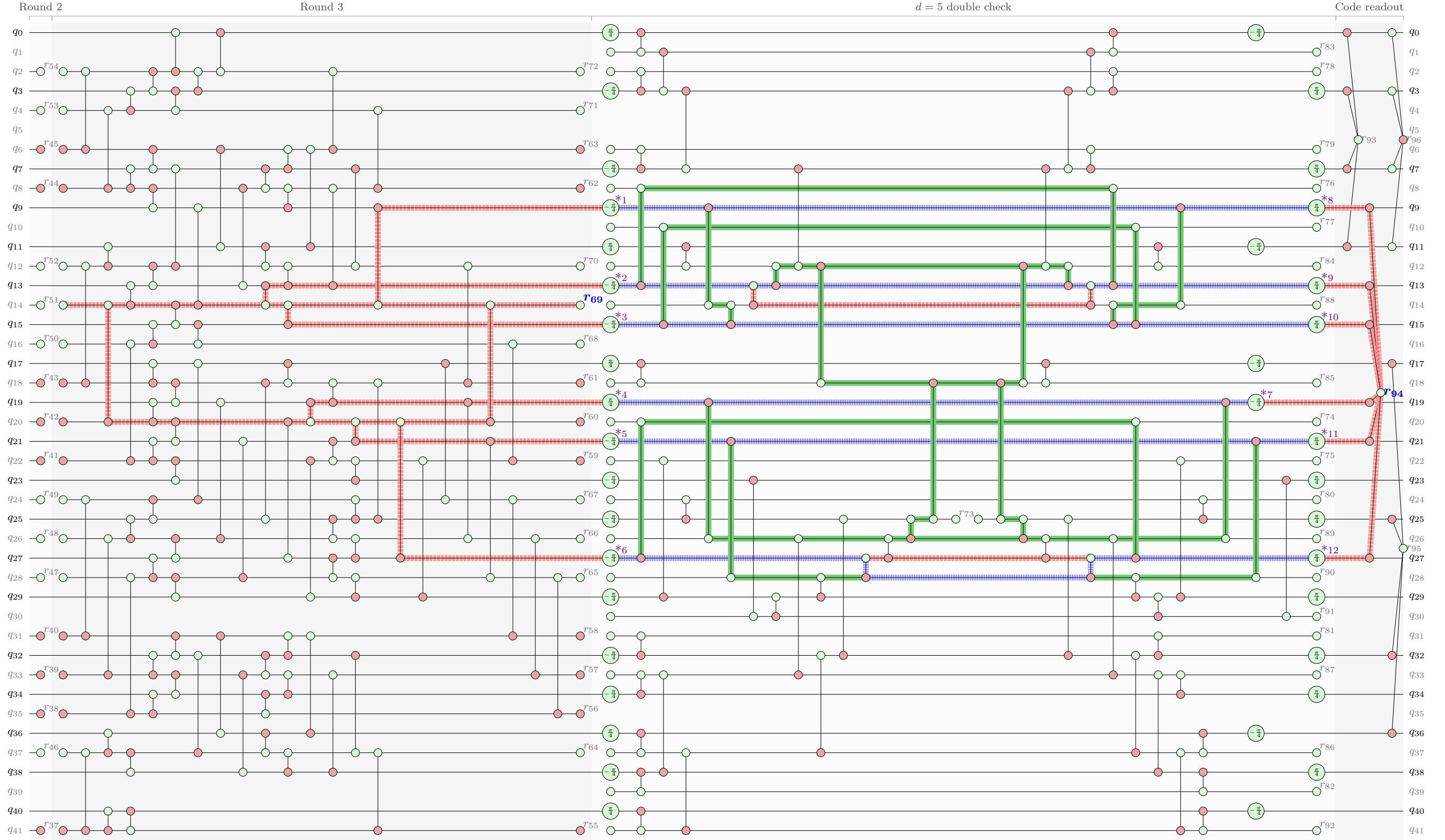


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{60} + r_{108}}$. So $D_{\mathbf{r}} = 0$ unless $r_{60} \oplus r_{108} = 0$.

Records: r_{60} : M q_{20} , line 361; r_{108} : MPP $Z_7 Z_{11} Z_{13} Z_{17} Z_{19} Z_{25}$, line 470.

$d = 5$: detector 95 $r_{69} \oplus r_{94} = 0$

Phase-dependent. Closed semiweb with 12 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



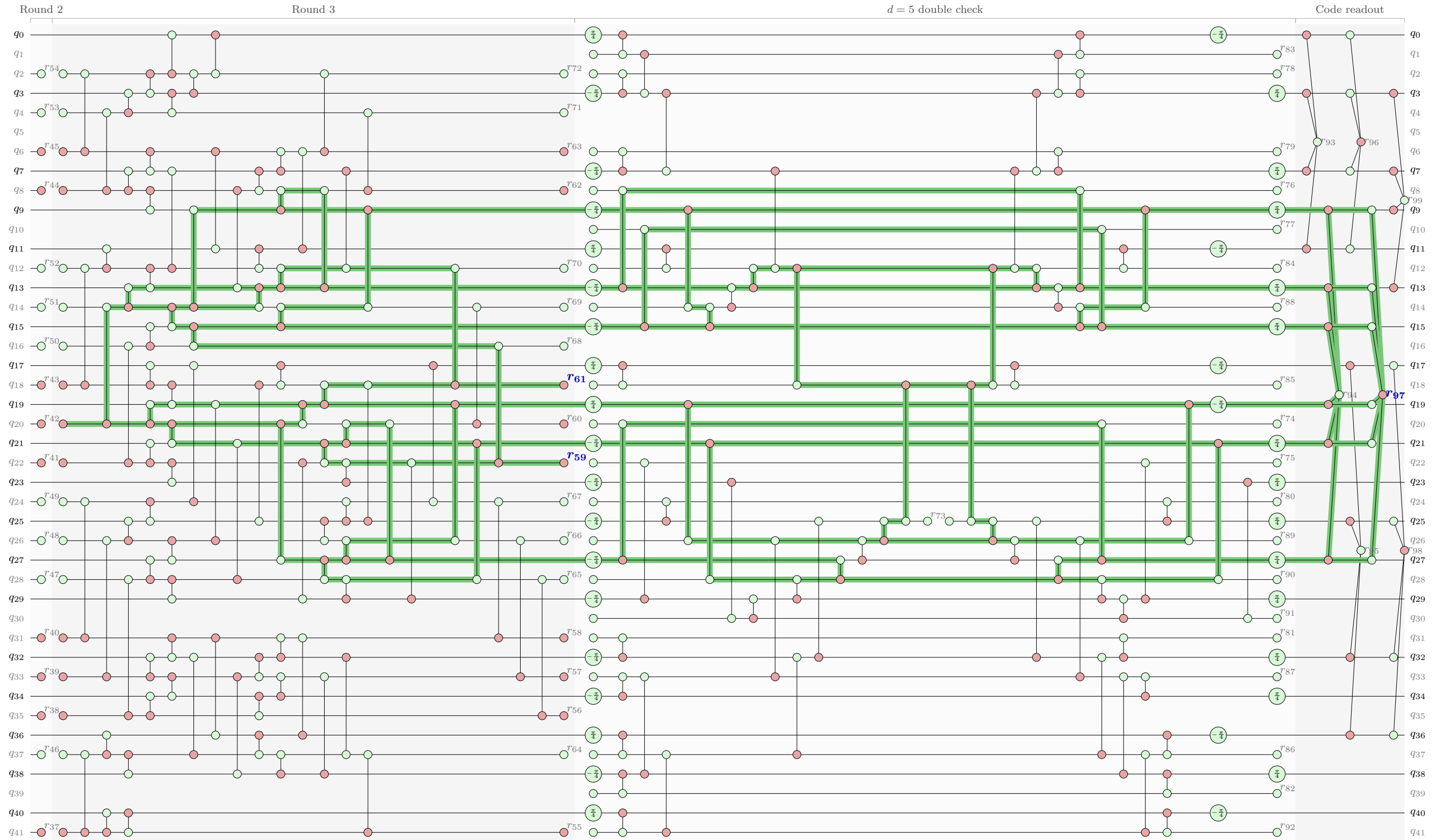
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}$, $\lambda(\mathbf{r}) = (-1)^{r_{69}+r_{94}}$. $D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_{69} : MX q_{14} , line 362; r_{94} : MPP $X_9X_{13}X_{15}X_{19}X_{21}X_{27}$, line 470.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_9 , line 392; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{13} , line 392; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{15} , line 392; $*_4$: $\delta = \frac{\pi}{2}$ at T on q_{19} , line 391; $*_5$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{21} , line 392; $*_6$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{27} , line 392; $*_7$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{19} , line 441; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_9 , line 442; $*_9$: $\delta = \frac{\pi}{2}$ at T on q_{13} , line 442; $*_{10}$: $\delta = \frac{\pi}{2}$ at T on q_{15} , line 442; $*_{11}$: $\delta = \frac{\pi}{2}$ at T on q_{21} , line 442; $*_{12}$: $\delta = \frac{\pi}{2}$ at T on q_{27} , line 442.

$d = 5$: detector 96 $r_{59} \oplus r_{61} \oplus r_{97} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

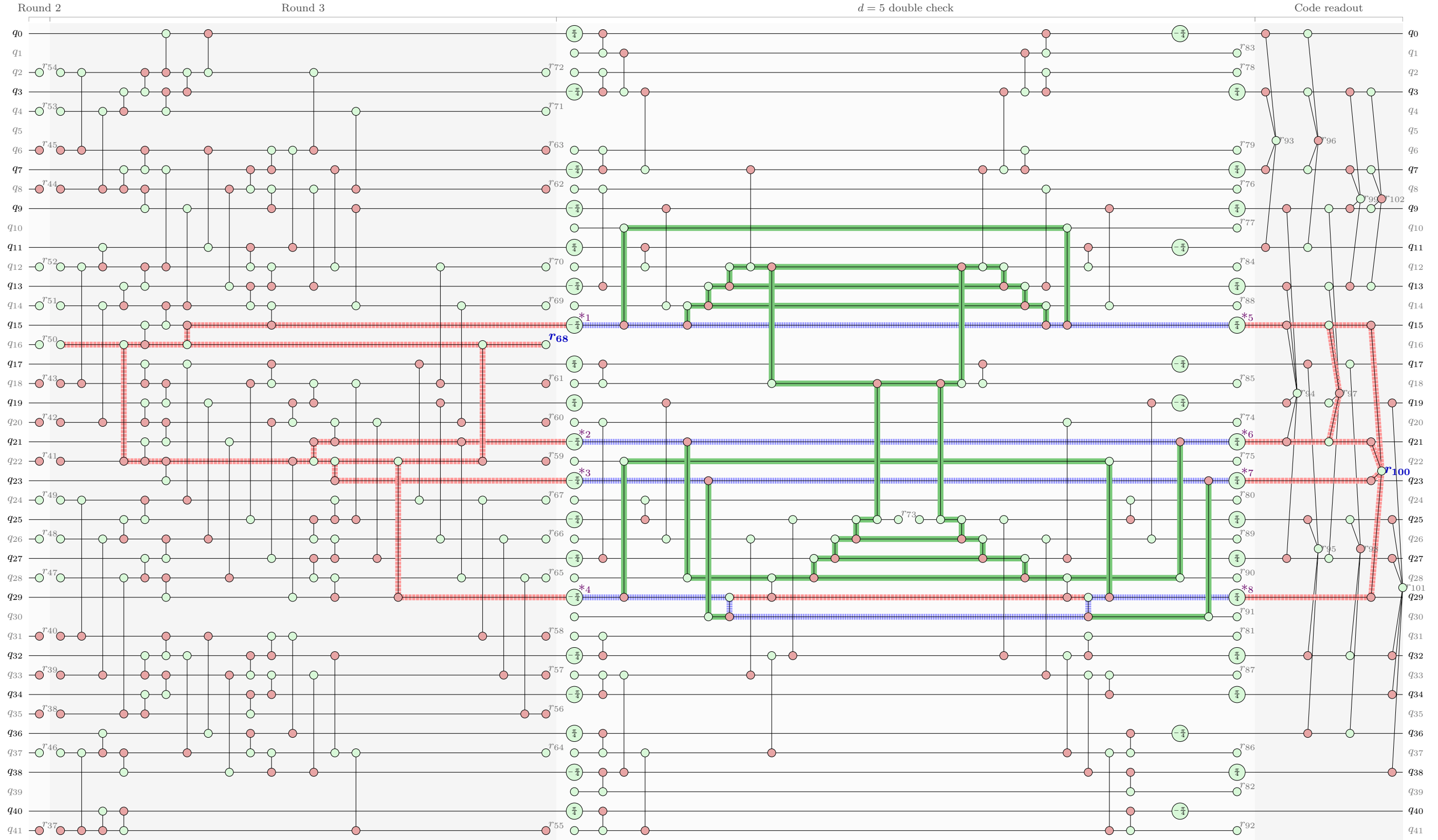


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{59} + r_{61} + r_{97}}$. So $D_{\mathbf{r}} = 0$ unless $r_{59} \oplus r_{61} \oplus r_{97} = 0$.

Records: r_{59} : M q_{22} , line 361; r_{61} : M q_{18} , line 361; r_{97} : MPP $Z_9 Z_{13} Z_{15} Z_{19} Z_{21} Z_{27}$, line 470.

$d = 5$: detector 97 $r_{68} \oplus r_{100} = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}$, $\lambda(\mathbf{r}) = (-1)^{r_{68}+r_{100}}$. $D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

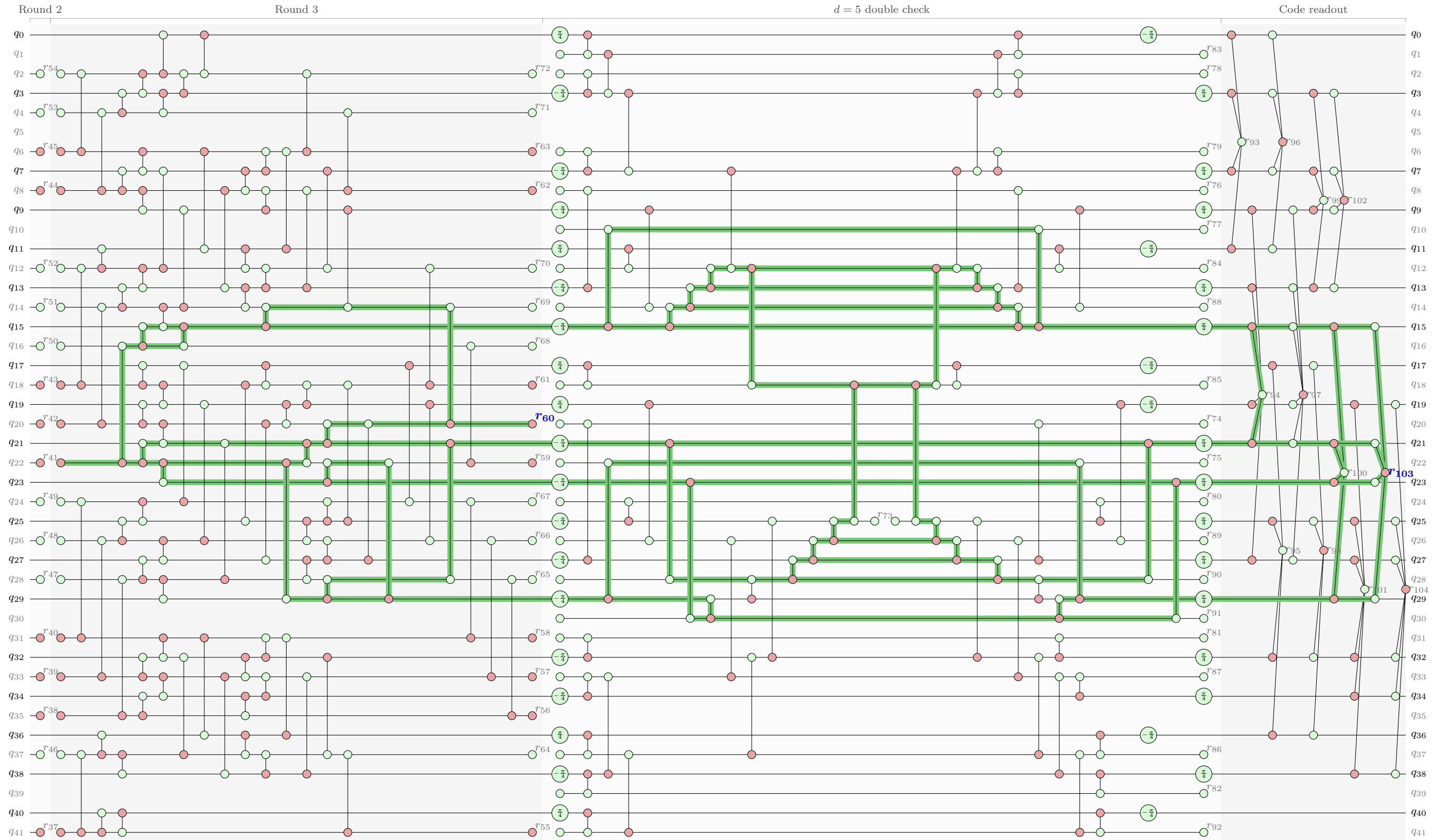
Records: r_{68} : MX q_{16} , line 362; r_{100} : MPP $X_{15}X_{21}X_{23}X_{29}$, line 470.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{15} , line 392; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{21} , line 392; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{23} , line 392; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{29} , line 392; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_{15} , line 442; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_{21} , line 442; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_{23} , line 442; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{29} , line 442.

d5_det097.tikz, declared at line 480 of data/inputs/soft_cultivation_d5_p0005.stim

$d = 5$: detector 98 $r_{60} \oplus r_{103} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

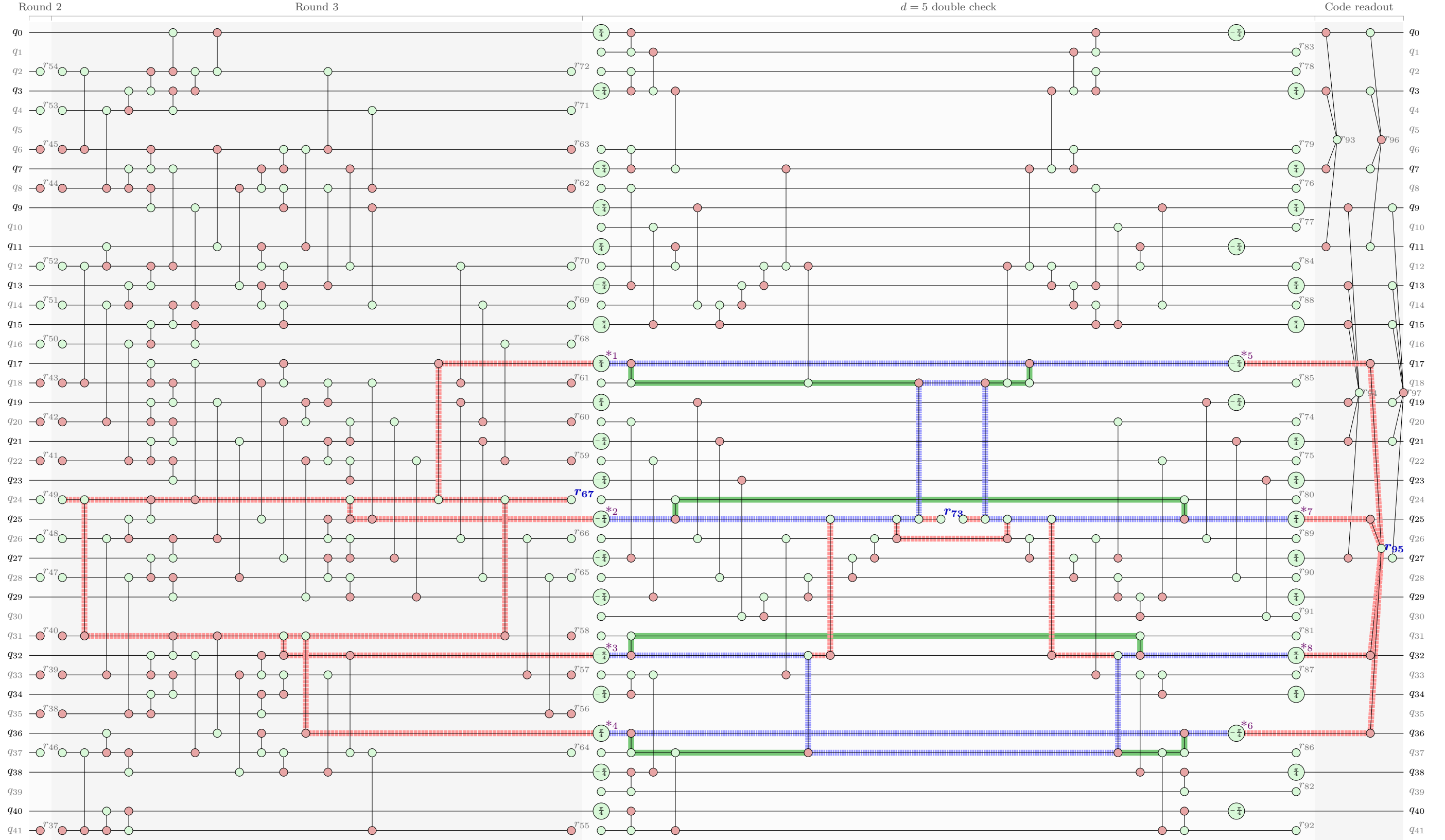


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{60} + r_{103}}$. So $D_{\mathbf{r}} = 0$ unless $r_{60} \oplus r_{103} = 0$.

Records: r_{60} : M q_{20} , line 361; r_{103} : MPP $Z_{15}Z_{21}Z_{23}Z_{29}$, line 470.

$d = 5$: detector 99 $r_{67} \oplus r_{73} \oplus r_{95} = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



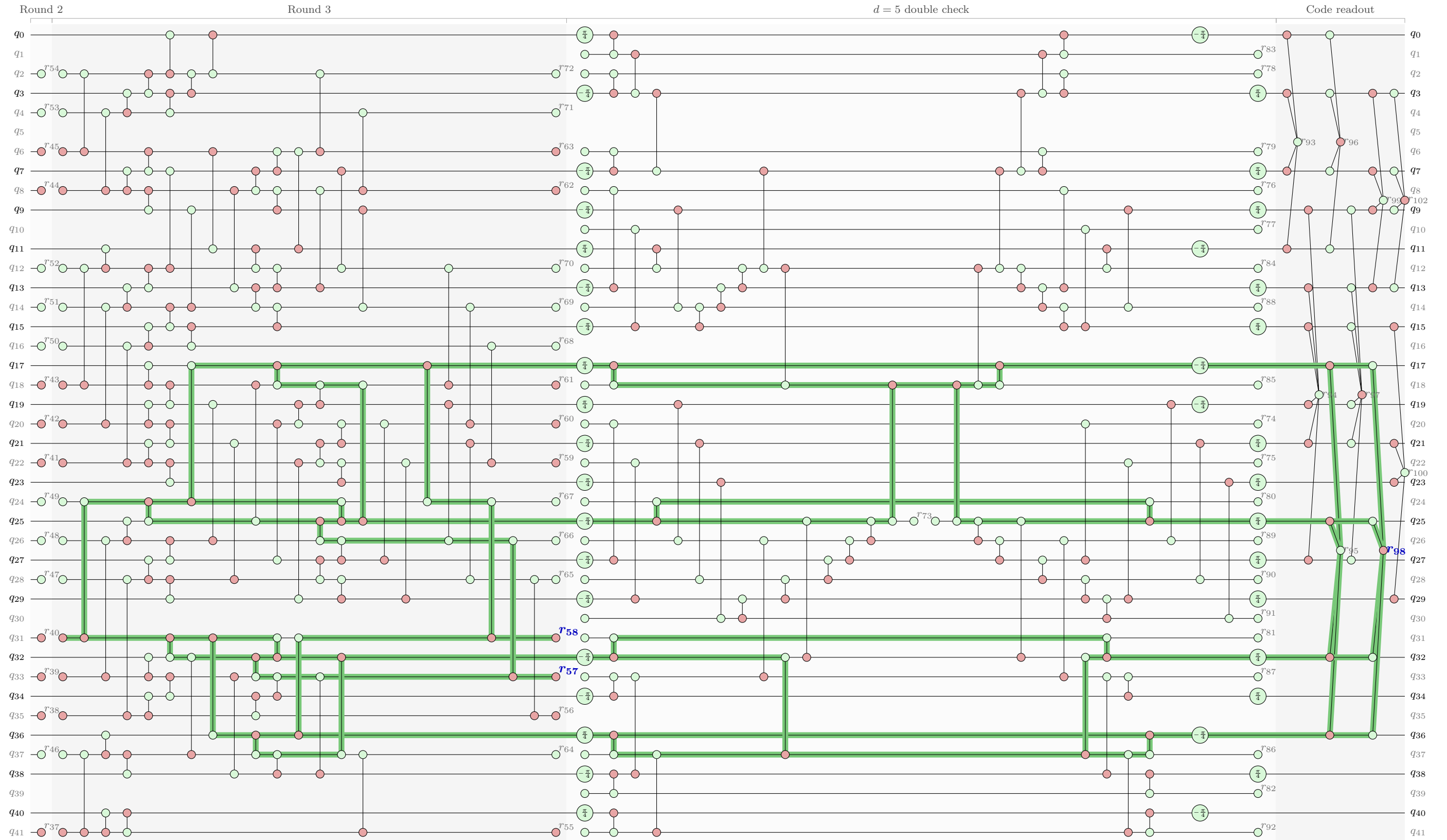
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}$, $\lambda(\mathbf{r}) = (-1)^{r_{67}+r_{73}+r_{95}}$. $D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_{67} : MX q_{24} , line 362; r_{73} : MX q_{25} , line 415; r_{95} : MPP $X_{17}X_{25}X_{32}X_{36}$, line 470.

Phase changes: $*_1$: $\delta = \frac{\pi}{2}$ at T on q_{17} , line 391; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{25} , line 392; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{32} , line 392; $*_4$: $\delta = \frac{\pi}{2}$ at T on q_{36} , line 391; $*_5$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{17} , line 441; $*_6$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{36} , line 441; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_{25} , line 442; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{32} , line 442.

$d = 5$: detector 100 $r_{57} \oplus r_{58} \oplus r_{98} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

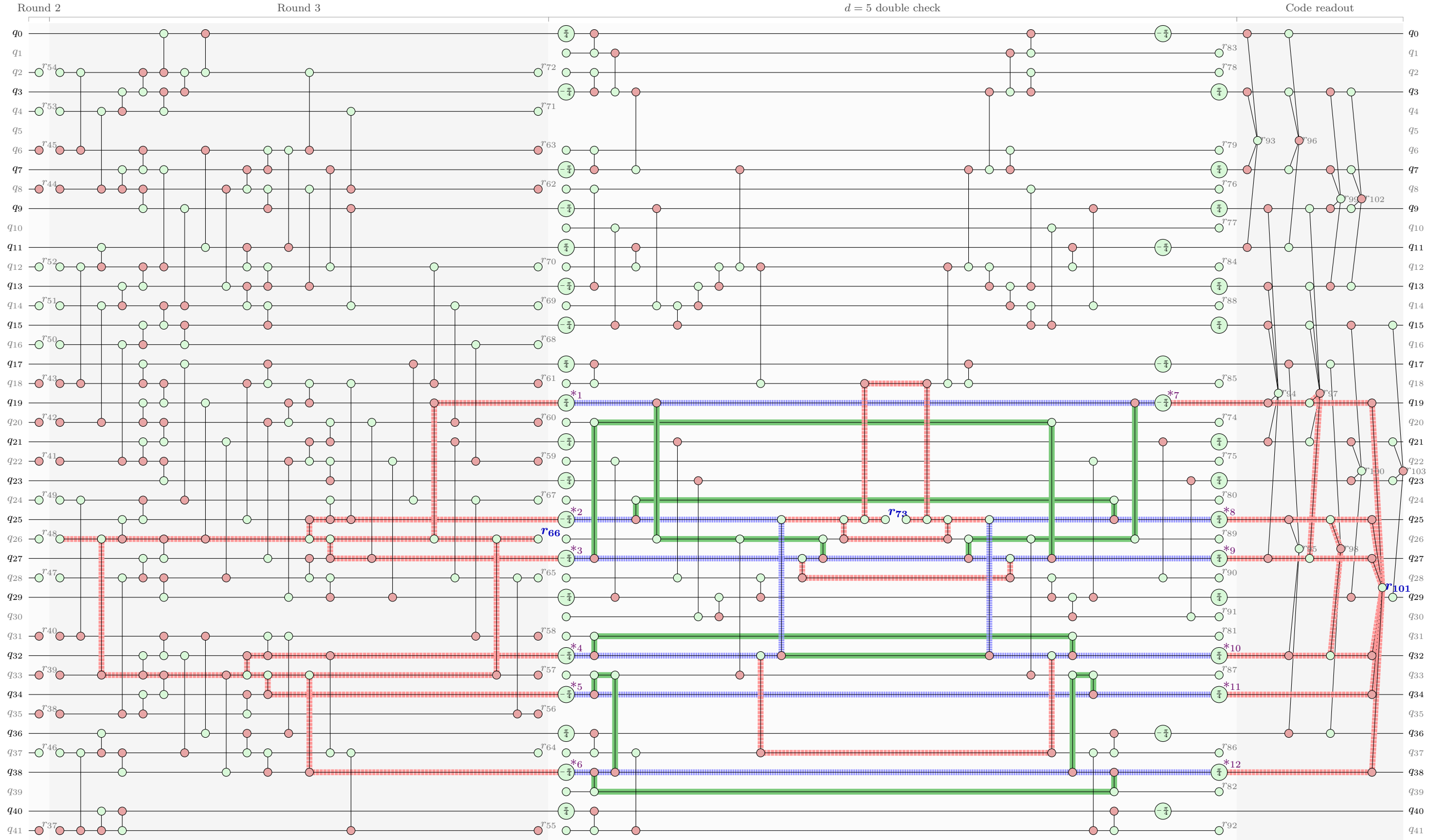


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{57}+r_{58}+r_{98}}$. So $D_{\mathbf{r}} = 0$ unless $r_{57} \oplus r_{58} \oplus r_{98} = 0$.

Records: r_{57} : M q_{33} , line 361; r_{58} : M q_{31} , line 361; r_{98} : MPP $Z_{17}Z_{25}Z_{32}Z_{36}$, line 470.

$d = 5$: detector 101 $r_{66} \oplus r_{73} \oplus r_{101} = 0$

Phase-dependent. Closed semiweb with 12 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



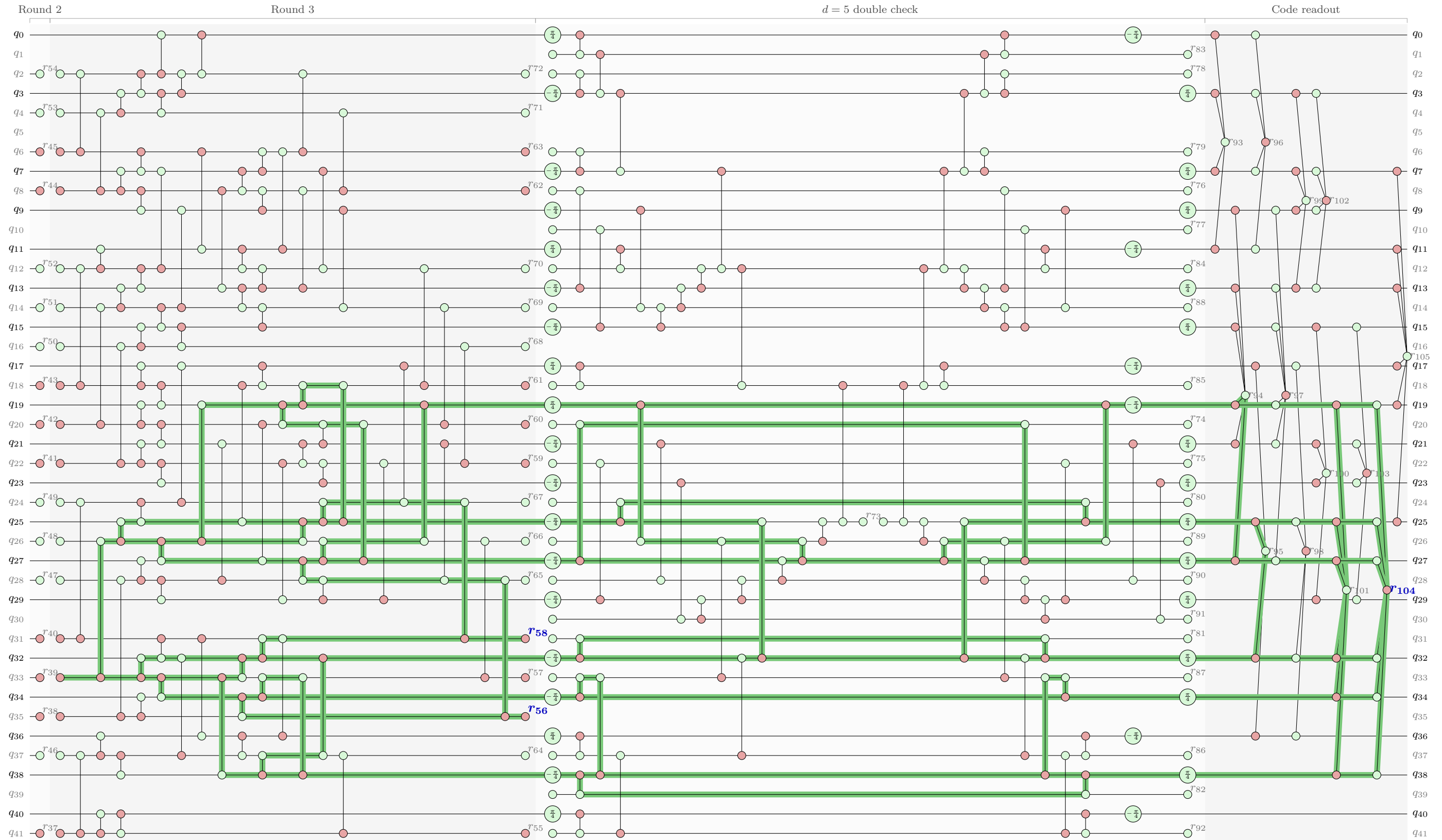
$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}$, $\lambda(\mathbf{r}) = (-1)^{r_{66}+r_{73}+r_{101}}$. $D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

Records: r_{66} : MX q_{26} , line 362; r_{73} : MX q_{25} , line 415; r_{101} : MPP $X_{19}X_{25}X_{27}X_{32}X_{34}X_{38}$, line 470.

Phase changes: $*_1$: $\delta = \frac{\pi}{2}$ at T on q_{19} , line 391; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{25} , line 392; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{27} , line 392; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{32} , line 392; $*_5$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{34} , line 392; $*_6$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{38} , line 392; $*_7$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{19} , line 441; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{25} , line 442; $*_9$: $\delta = \frac{\pi}{2}$ at T on q_{27} , line 442; $*_{10}$: $\delta = \frac{\pi}{2}$ at T on q_{32} , line 442; $*_{11}$: $\delta = \frac{\pi}{2}$ at T on q_{34} , line 442; $*_{12}$: $\delta = \frac{\pi}{2}$ at T on q_{38} , line 442.

$d = 5$: detector 102 $r_{56} \oplus r_{58} \oplus r_{104} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

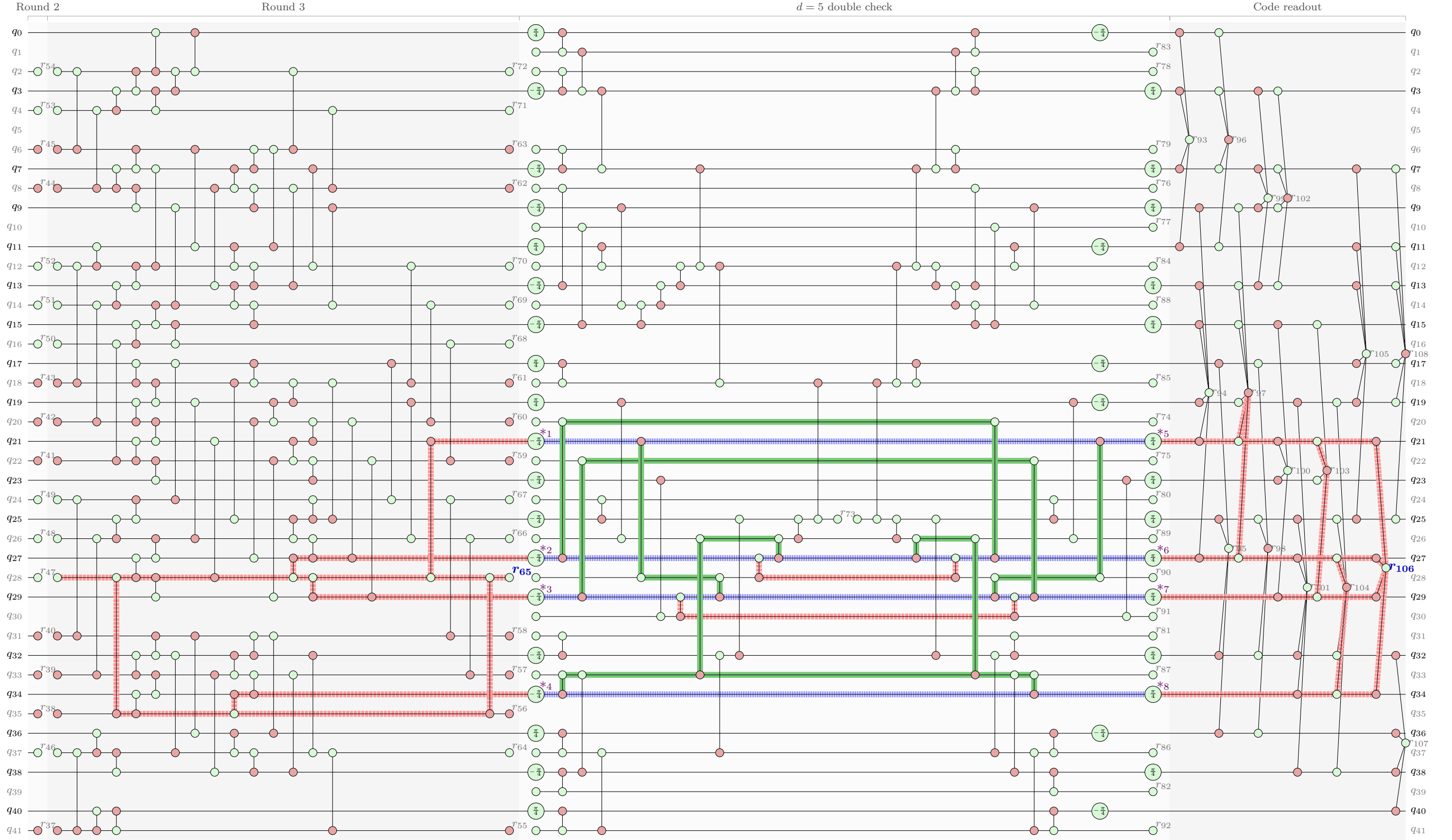


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{56} + r_{58} + r_{104}}$. So $D_{\mathbf{r}} = 0$ unless $r_{56} \oplus r_{58} \oplus r_{104} = 0$.

Records: r_{56} : M q_{35} , line 361; r_{58} : M q_{31} , line 361; r_{104} : MPP $Z_{19}Z_{25}Z_{27}Z_{32}Z_{34}Z_{38}$, line 470.

$d = 5$: detector 103 $r_{65} \oplus r_{106} = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}$, $\lambda(\mathbf{r}) = (-1)^{r_{65}+r_{106}}$. $D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

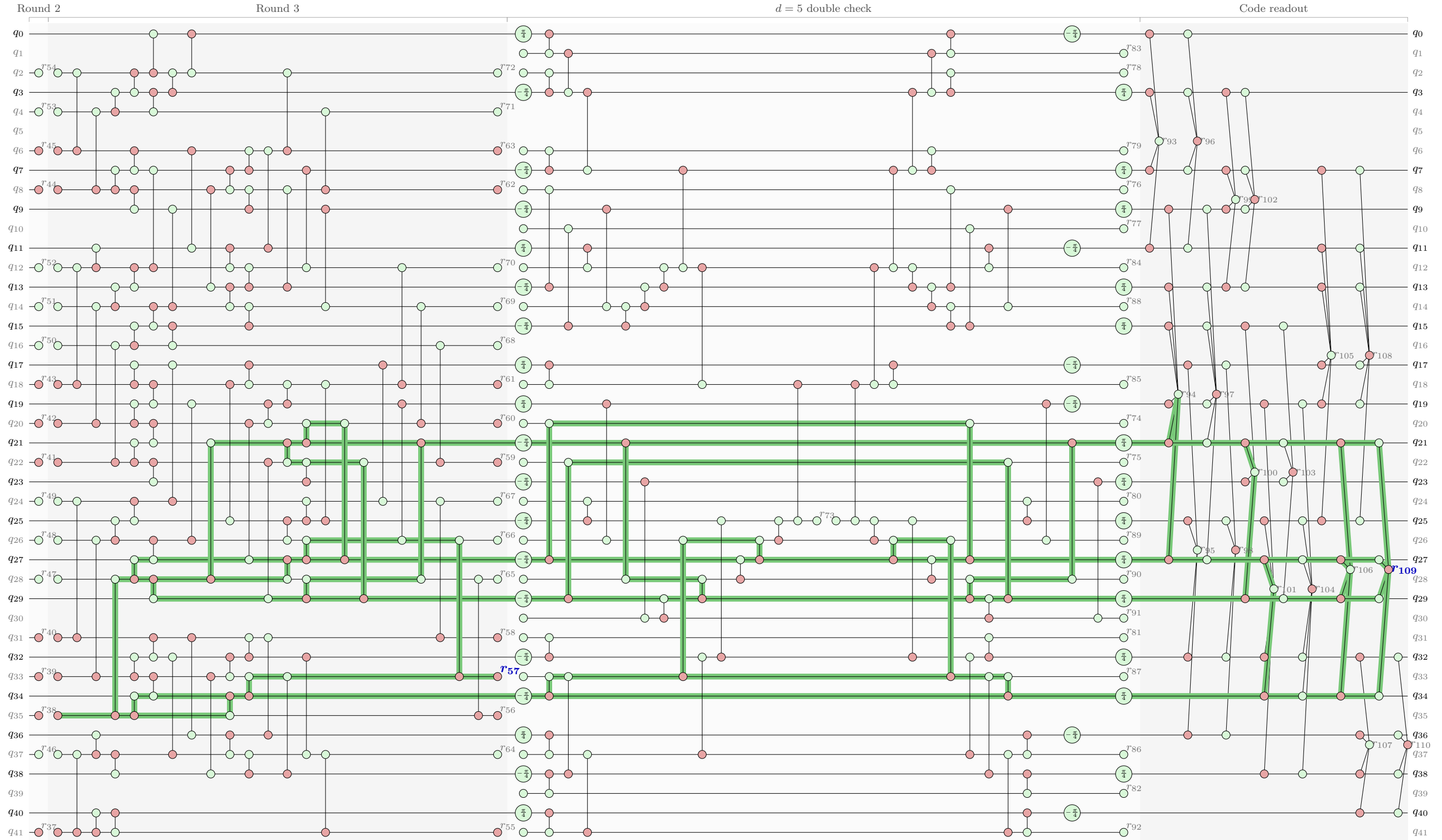
Records: r_{65} : MX q_{28} , line 362; r_{106} : MPP $X_{21}X_{27}X_{29}X_{34}$, line 470.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{21} , line 392; $*_2$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{27} , line 392; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{29} , line 392; $*_4$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{34} , line 392; $*_5$: $\delta = \frac{\pi}{2}$ at T on q_{21} , line 442; $*_6$: $\delta = \frac{\pi}{2}$ at T on q_{27} , line 442; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_{29} , line 442; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{34} , line 442.

d5_det103.tikz, declared at line 486 of data/inputs/soft_cultivation_d5_p0005.stim

$d = 5$: detector 104 $r_{57} \oplus r_{109} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.

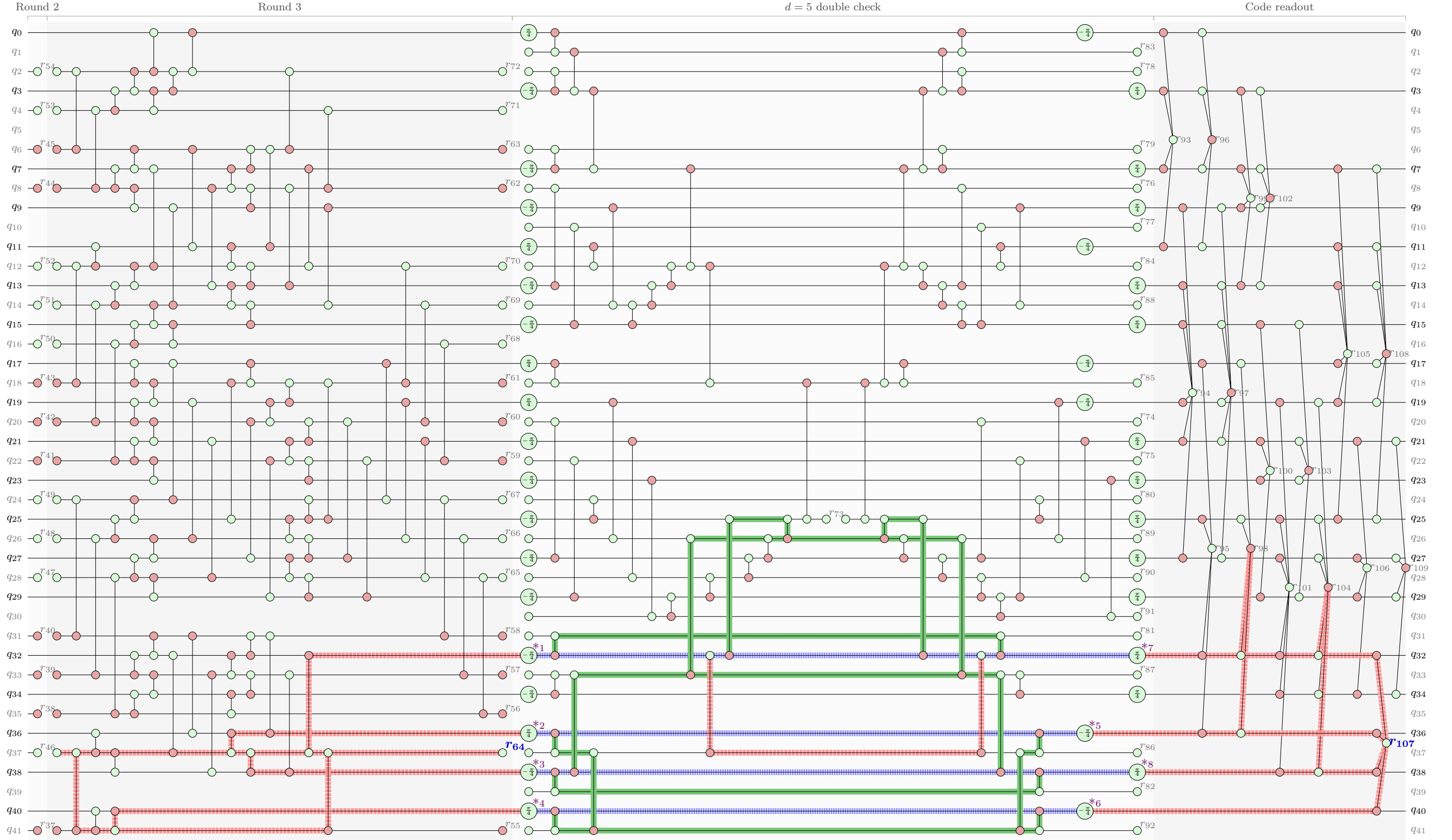


$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{57} + r_{109}}$. So $D_{\mathbf{r}} = 0$ unless $r_{57} \oplus r_{109} = 0$.

Records: r_{57} : M q_{33} , line 361; r_{109} : MPP $Z_{21}Z_{27}Z_{29}Z_{34}$, line 470.

$d = 5$: detector 105 $r_{64} \oplus r_{107} = 0$

Phase-dependent. Closed semiweb with 8 defects, every one at a T or $T^\dagger$ spider. No closed Pauli web without defects has these records. With every T replaced by S (the S proxy), the same labels are a Pauli web.



$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r},\delta}$, $\lambda(\mathbf{r}) = (-1)^{r_{64}+r_{107}}$. $D_{\mathbf{r},\delta}$ has the marked phases changed, so this identity alone does not fix the parity.

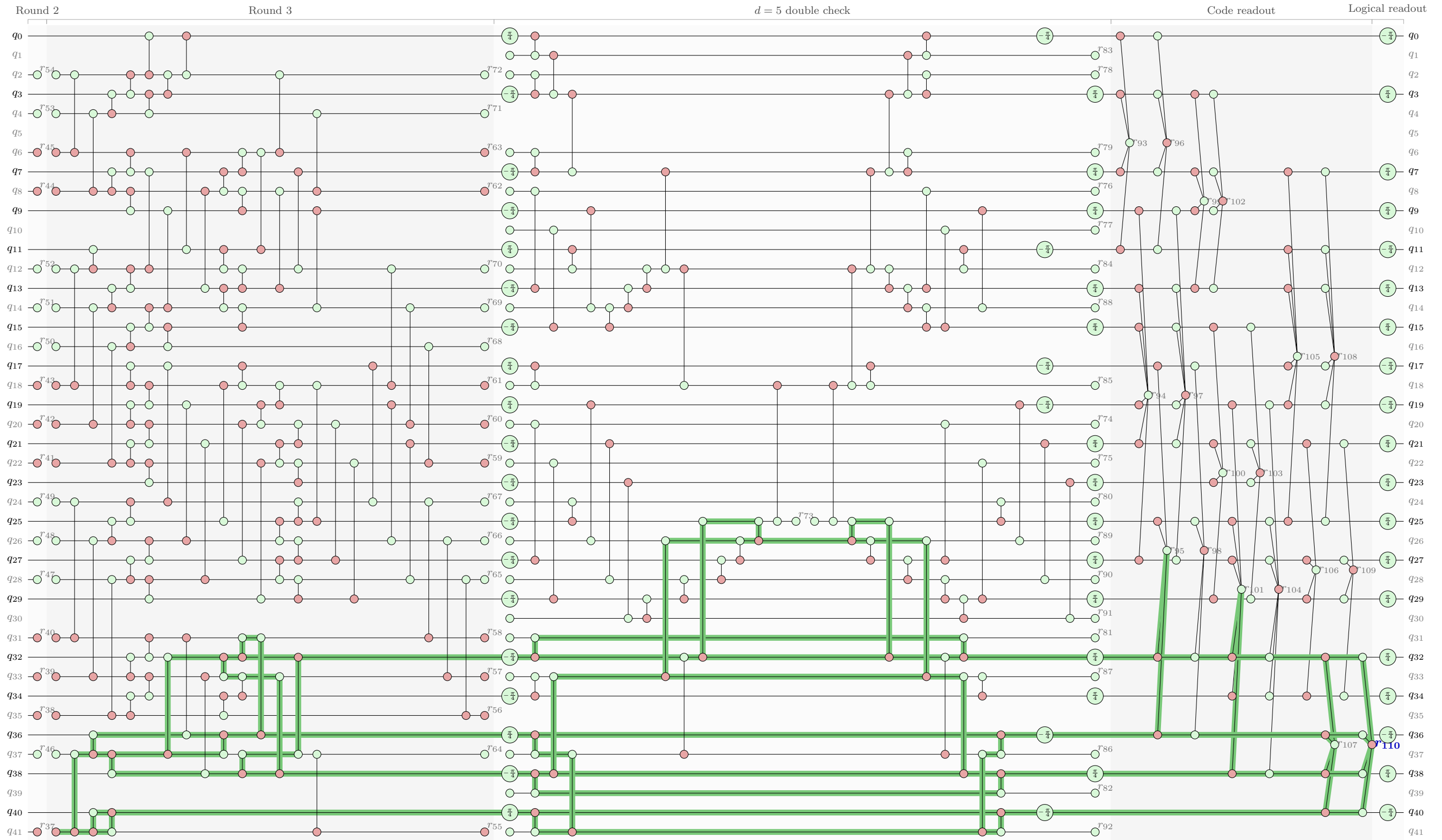
Records: r_{64} : MX q_{37} , line 362; r_{107} : MPP $X_{32}X_{36}X_{38}X_{40}$, line 470.

Phase changes: $*_1$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{32} , line 392; $*_2$: $\delta = \frac{\pi}{2}$ at T on q_{36} , line 391; $*_3$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{38} , line 392; $*_4$: $\delta = \frac{\pi}{2}$ at T on q_{40} , line 391; $*_5$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{36} , line 441; $*_6$: $\delta = -\frac{\pi}{2}$ at $T^\dagger$ on q_{40} , line 441; $*_7$: $\delta = \frac{\pi}{2}$ at T on q_{32} , line 442; $*_8$: $\delta = \frac{\pi}{2}$ at T on q_{38} , line 442.

d5_det105.tikz, declared at line 488 of data/inputs/soft_cultivation_d5_p0005.stim

$d = 5$: detector 106 $r_{110} = 0$

Phase-independent. Closed Pauli web with no defects, so the check holds for any angle at every T spider.



$D_{\mathbf{r}} = \lambda(\mathbf{r}) D_{\mathbf{r}}$, $\lambda(\mathbf{r}) = (-1)^{r_{110}}$. So $D_{\mathbf{r}} = 0$ unless $r_{110} = 0$.

Records: r_{110} : MPP $Z_{32}Z_{36}Z_{38}Z_{40}$, line 470.